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(54) **WHEAT COMPRISING MALE FERTILITY RESTORER ALLELES**

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(58) **Field of Classification Search**
None
See application file for complete search history.

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(57) **ABSTRACT**

A wheat transgenic plant carrying restorer of fertility genes specific to *T. timopheevii* CMS cytoplasm.

14 Claims, 28 Drawing Sheets

Specification includes a Sequence Listing.

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No.	Tribe	Species	Cultivar	Website	References	Number of Identified RFL proteins
1	Triticeae	*Aegilops tauschii (1)*	AL8/78	http://plants.ensembl.org/Aegilops_tauschii/Info/Index	Jia et al., 2013	46
2		*Aegilops tauschii (2)*		http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Jia et al., 2013	31
3		*Aegilops speltoides*		http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Mayer et al., 2014	20
4		*Aegilops sharonensis*		http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Mayer et al., 2014	32
5		*Triticum aestivum*		http://plants.ensembl.org/Triticum_aestivum/Info/Index	Mayer et al., 2014	98
6		*Triticum durum*	cv. Cappeli	http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Mayer et al., 2014	30
7		*Triticum durum*	cv. Strongfiled	http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Mayer et al., 2014	30
8		*Triticum monococcum*		http://wheat-urgi.versailles.inra.fr/Seq-Repository/Assemblies	Brenchley et al., 2012	12
9		*Triticum turgidum*		http://www.ncbi.nlm.nih.gov/bioproject/PRJNA191054	Krasileva et al., 2013	25
10		*Triticum urartu*		http://archive.gramene.org/Triticum_urartu/Info/Annotation/	Ling et al., 2013	24
11		*Hordeum vulgare*	Barke	http://pgsb.helmholtz-muenchen.de/plant/barley/index.jsp	Mayer et al., 2013	13
12		*Hordeum vulgare*	Morex	http://pgsb.helmholtz-muenchen.de/plant/barley/index.jsp	Mayer et al., 2013	12
13		*Hordeum vulgare*	Bowman	http://pgsb.helmholtz-muenchen.de/plant/barley/index.jsp	Mayer et al., 2013	13
14		*Hordeum vulgare*	var. distichum	http://plants.ensembl.org/Hordeum_vulgare/Info/Index	Mayer et al., 2012	0
15		*Lolium perenne*	P226/135/1	http://www.ncbi.nlm.nih.gov/nuccore/GAYX00000000	Farrel et al., 2014	2
16		*Secale cereale*		http://pgsb.helmholtz-muenchen.de/plant/rye/index.jsp	Martis et al., 2013	0
17	Oryzeae	*Oryza sativa*		http://phytozome.jgi.doe.gov/pz/portal.html#!bulk?org=Org_Osativa	Ouyang et al., 2007	14
18		*Oryza sativa japonica*		http://plants.ensembl.org/Oryza_sativa/Info/Index	Kawahara et al., 2013	14
19		*Oryza indica*	Nipponbare	http://plants.ensembl.org/Oryza_indica/Info/Index	Sakai et al., 2013	18
20		*Oryza sativa Nipponbare*		http://rapdb.dna.affrc.go.jp/index.html	Kawahara et al., 2013	14
21		rice cultivar 'Kasalath'	Kasalath	http://rapdb.dna.affrc.go.jp/index.html	Sakai et al., 2013	11
22		*Oryza barthii*		http://plants.ensembl.org/Oryza_barthii/Info/Index	Jacquemin et al., 2012	13

Figure 1A.

23		*Oryza brachyantha*		http://plants.ensembl.org/Oryza_brachyantha/Info/Index	Chen et al., 2013	4
24		*Oryza glaberrima*		http://plants.ensembl.org/Oryza_glaberrima/Info/Index	Wang et al., 2014	2
25		*Oryza glumaepatula*		http://plants.ensembl.org/Oryza_glumaepatula/Info/Index	Jacquemin et al., 2012	12
26		*Oryza meridionalis*		http://plants.ensembl.org/Oryza_meridionalis/Info/Index	Jacquemin et al., 2012	15
27		*Oryza nivara*		http://plants.ensembl.org/Oryza_nivara/Info/Index	Jacquemin et al., 2012	13
28		*Oryza punctata*		http://plants.ensembl.org/Oryza_punctata/Info/Index	Jacquemin et al., 2012	7
29		*Oryza rufipogon*		http://plants.ensembl.org/Oryza_rufipogon/Info/Index	Jacquemin et al., 2012	12
30	*Brachypodieae*	*Brachypodium distachyon*		http://phytozome.jgi.doe.gov/pz/portal.html#!bulk?org=Org_Bdistachyon	Brachypodium Initiative	11
31	*Eragrostideae*	*Eragrostis tef*		http://www.tef-research.org/index.html	Cannarozzi et al., 2014	33
32	*Poniceae*	*Setaria italica*		http://phytozome.jgi.doe.gov/pz/portal.html#!bulk?org=Org_Sitalica	Bennetzen et al., 2012	15
33	*Andropogoneae*	*Sorghum bicolor*	BTx623	http://plants.ensembl.org/Sorghum_bicolor/Info/Index	Paterson et al., 2006	20
34	*Andropogoneae*	*Zea mays*	B73	http://plants.ensembl.org/Zea_mays/Info/Index	Schnable et al., 2010	6
					Total	**633**
				Reference RFLs	Fujii et al., 2011	49
				Sorghum WGS data sets	Mace et al., 2013	517
					Total	**1188**

Figure 1B

Figure 2

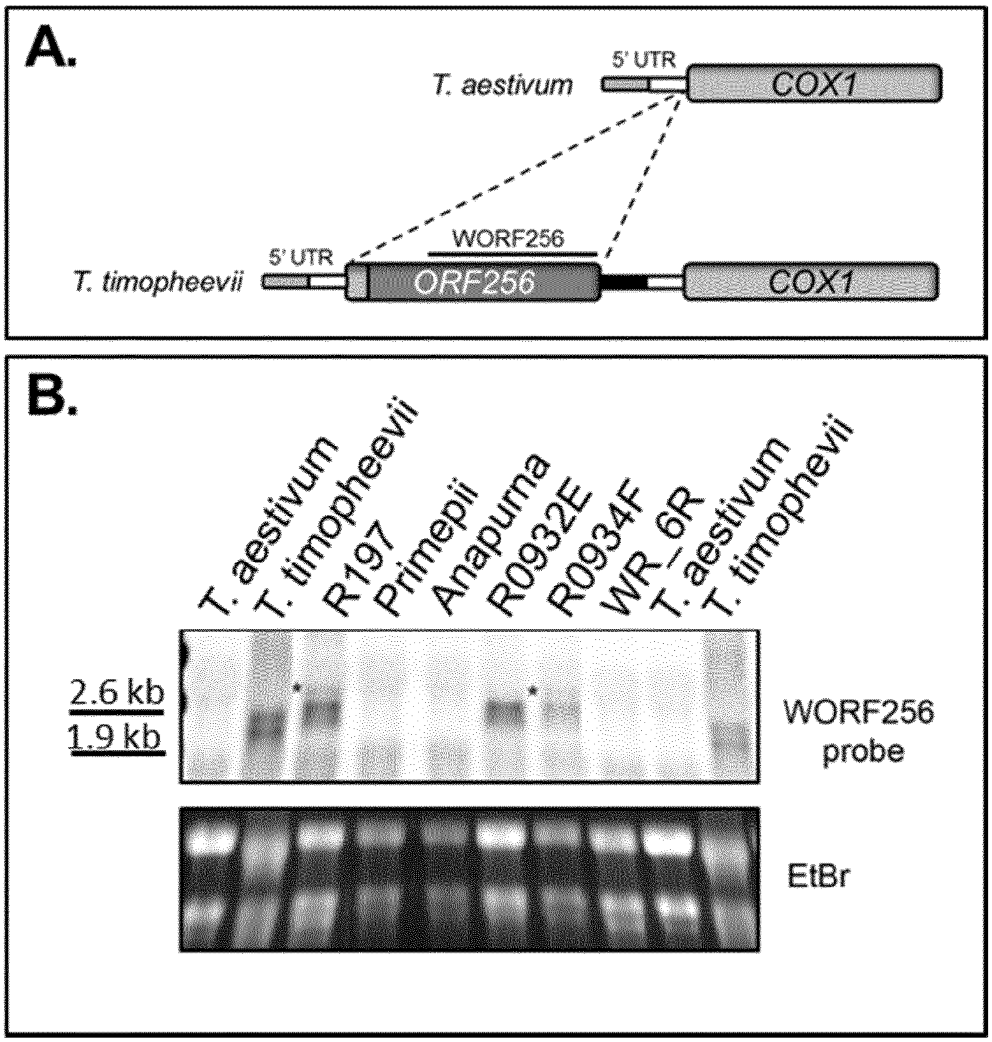
A.
T. aestivum
5' UTR
COX1
WORF256
5' UTR
T. timopheevii
ORF256
COX1
B.
T. aestivum
T. timopheevii
R197
Primepii
Anapurna
R0932E
R0934F
WR_6R
T. aestivum
T. timophevii
2.6 kb
1.9 kb
WORF256
probe
EtBr

Figure 3

	origin	T. timophevi i	A. speltoides	T. timophevi i	T. timophevi i					
	Maintainer	restorer Rf4+	restorer Rf4+	restorer Rf4+	restorer Rf4+	maintaine r	maintaine r	restorer Rf4-	restorer Rf4-	T. timophevii
RFL group	representative sequence	17F3R-0377	GSTR435	L13	R113	ANAPURN A	FIELDER	R0934F	R197	TRI13159
RFL120	Triticum-timopheevii.300k_Assembly_Contig_67_1	103	42	101	93	0	0	0	0	157
RFL211	Triticum-timopheevii.300k_Assembly_Contig_99_1	110	0	126	94	0	0	1	0	182
RFL123	Triticum-timopheevii.300k_Assembly_Contig_65_1	102	0	96	89	0	0	0	0	160
RFL204	Triticum-timopheevii.300k_Assembly_Contig_100_1	89	0	100	91	0	0	0	0	144
RFL130	Triticum-timopheevii.300k_Assembly_Contig_53_2	84	0	91	101	0	0	0	0	132
RFL207	Triticum-timopheevii.300k_Assembly_Contig_62_1	92	0	91	85	1	0	1	0	155
RFL135	Triticum-timopheevii.300k_Assembly_Contig_93_1	78	0	78	77	0	0	0	0	135
RFL162	Triticum-timopheevii.300k_Assembly_Contig_94_1	79	0	66	77	0	0	0	0	131
RFL340	Triticum-timopheevii.300k_Assembly_Contig_91_1	65	6	74	71	1	2	1	1	128
RFL160	Triticum-timopheevii.300k_Assembly_Contig_111_1	52	0	54	52	0	0	0	0	100

FIGURE 4a

```
Query  1    AAAAAAGAGAAACGTTTTTATTCAGCAAAGCCACTTCTTGAGAGAGCCGAGGTCTCCCAG  60
            ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  391  AAAAAAGAGAAACGTTTTTATTCAGCAAAGCCACTTCTTGAGAGAGCCGAGGTCTCCCAG  450
Query  61   GCCAGTGGCGGCGGCGGAAATACAGAGTGGCGGCG-------------------GTCGGG  101
            |||||||||||||||||||| ||||||||||||||                   ||||||
Sbjct  451  GCCAGTGGCGGCGGCGGAAAAACAGAGTGGCGGCGGTCGTCCTCACCGGCGGGCGTCGGG  510
Query  102  GACGCAGCGCGCGGCGACGGAGGGTCGCATTCTCTTCATCGGCGTCAGGGCCGCCGCAGC  161
            |||||||||||||||||||||||||||||||||||||||| |||||||||||||||||||
Sbjct  511  GACGCAGCGCGCGGCGACGGAGGGTCGCATTCTCTTCATCAGCGTCAGGGCCGCCGCAGC  570
Query  162  CAGCGCCGAGCACGGAGATCTCGTCCTCGCCCCCCTCAACGCTGCGGCCGTGTCGACCAG  221
            ||||||||||| ||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  571  CAGCGCCGAGCGCGGAGATCTCGTCCTCGCCCCCCTCAACGCTGCGGCCGTGTCGACCAG  630
Query  222  CTCGCCGGCGGTCAGATCTCCTGTACATCGCATTTACCTACTGCAGCAGCTGCAGCTCGA  281
            ||||||||||| ||||||||||||||| |||||||||||||||||||||||||| |||||
Sbjct  631  CTCGCCGGCGGCCAGATCTCCTGTACAGCGCATTTACCTACTGCAGCAGCTGCAACTCGA  690
Query  282  CGGTGAGCATCCTGGCATCCTGACATCTCTTCCATCTTGCTCTGCTTTCCTTCCTTCTAA  341
             ||||||||||||||        |||||| ||||||||||||||||||||||||||||||
Sbjct  691  AGGTGAGCATCCTGG--------CATCTCGTCCATCTTGCTCTGCTTTCCTTCCTTCTAA  742
Query  342  CCCCAGGTTCATGCCCAGGCTGCGCCGGTGGTGAGCGGCGGCGGCGCCATCCTATCCCCT  401
            ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  743  CCCCAGGTTCATGCCCAGGCTGCGCCGGTGGTGAGCGGCGGCGGCGCCATCCTATCCCCT  802
Query  402  CGAGGGATTCTTCCGGCCAGGTGGAGCTATGAGCCGCCGCTTTGTCCCCGTCGGCAGACG  461
            ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  803  CGAGGGATTCTTCCGGCCAGGTGGAGCTATGAGCCGCCGCTTTGTCCCCGTCGGCAGACG  862
Query  462  CATCTTAGAGCAGAACATCAAGGCTCGGTACCACGCCGGAGACATTGGCACCGAGGACGC  521
            ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  863  CATCTTAGAGCAGAACATCAAGGCTCGGTACCACGCCGGAGACATTGGCACCGAGGACGC  922
Query  522  ACTCCACCTGTTTGACGAATTGCTCCAGGTTGCTGGGCGCTCCTCGATCCATGCCATCAA  581
            ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
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FIGURE 4a
CONT-1

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Sbjct  923   ACTCCACCTGTTTGACGAATTGCTCCAGGTTGCTGGGCGCTCCTCGATCCATGCCATCAA  982

Query  582   CTGCCTCCTCACCGTTGTCGGCCGTGATTGCCCTGCGCTCGGCGTCTCCCTCTTCAACCG  641
             ||||||||||||||||||||||||||||||||||||||| ||||||||||||||||| ||
Sbjct  983   CTGCCTCCTCACCGTTGTCGGCCGTGATTGCCCTGCGCTTGGCGTCTCCCTCTTCAATCG  1042

Query  642   CGTCGCAAGGGCCAAGGTGGCACCCTGCAGTATCACCTATGCCATTCTGGTCGACTGCTG  701
             |||||||||| |||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1043  CGTCGCAAGGTCCAAGGTGGCACCCTGCAGTATCACCTATGCCATTCTGGTCGACTGCTG  1102

Query  702   CTGCCGCACTGGCCGCCAGGACCTTGGTTTCGCTGCCATGGGACACGTCATCAAGATGGG  761
             ||||||| ||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1103  CTGCCGCGCTGGCCGCCAGGACCTTGGTTTCGCTGCCATGGGACACGTCATCAAGATGGG  1162

Query  762   ATTTACTGCAGATGCTATGATCACTTTCAGCCACCTACTAAAGGCCATCTGTGCGGAGAA  821
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1163  ATTTACTGCAGATGCTATGATCACTTTCAGCCACCTACTAAAGGCCATCTGTGCGGAGAA  1222

Query  822   CAAGACCAGCTATGCAATGGACATCGTACTCCGAATAATGCCCATGTTTAACTGCATACC  881
             ||||||||||||||||||||||||||||||||||||||||||| ||||||||||||||||
Sbjct  1223  CAAGACCAGCTATGCAATGGACATCGTACTCCGAATAATGCCCGTGTTTAACTGCATACC  1282

Query  882   GGATGTCTTCTCTTACAACATTCTTTTCAAGGGTCTCTGCAACGAGAAGAGAAGCCAAGA  941
             |||||||||||| |||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1283  GGATGTCTTCTCCTACAACATTCTTTTCAAGGGTCTCTGCAACGAGAAGAGAAGCCAAGA  1342

Query  942   GGCTCTCGAGCTGATTCAGGTGATGGTTGAGCATGGAGGTCGCTGCCAACCTGATGTGGT  1001
             |||||||||||||||||||||||||||||||||||||||| |||||||||||||||||||
Sbjct  1343  GGCTCTCGAGCTGATTCAGGTGATGGTTGAGCATGGAGGTGGCTGCCAACCTGATGTGGT  1402

Query  1002  GTCCTATAGCACCGTAATTGATGGCTTGTTGAAAGAGGGTTTGGTGGGCAAGGTTTACAT  1061
             ||| |||||||| |||||||||||||||||||||||||||  ||||  ||||| ||||| 
Sbjct  1403  GTCATATAGCACTGTAATTGATGGCTTGTTGAAAGAGGGTGAGGTGAACAAGGCTTACAG  1462

Query  1062  CCTATTTTGTGAAATGATACAGAGAGGAATT-TCGCCGAATGCTGTGACCTATAACTCAA  1120
              ||||||||||||||| | | |  |||   | |||||||||| |||||||| ||  ||||
Sbjct  1463  TCTATTTTGTGAAATGCTGC-GTCAGGGGGTATCGCCGAATGTTGTGACCTGTAGTTCAA  1521

Query  1121  TCATCTCTGGCATGTGCAAGGTTCATGCGATGGACAAGGCTGAGCAGGTTCTTCAACAGA  1180
             |||||||||||||||||||| |||||||||||||||| |||||| |||||||||||||||
Sbjct  1522  TCATCTCTGGCATGTGCAAGCTTCATGCGATGGACAAAGCTGAGGAGGTTCTTCAACAGA  1581

Query  1181  TGCCTGATAGAGGAATTCTGCCAAATGTTGCCACGTATACTAGTCTAATACATGGATATT  1240
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FIGURE 4a
CONT-2

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                 ||| ||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1582  TGCTTGATAGAGGAATTCTGCCAAATGTTGCCACGTATACTAGTCTAATACATGGATATT  1641

Query  1241  TTTCATTAGGACAGTGCAAAGAGGTGGATCGGATTTTCAAAGAAATGTCTAGAAATGGTG  1300
             ||||||||||||||||||||||||| ||||||||||||||||||||||||||||||||||
Sbjct  1642  TTTCATTAGGACAGTGCAAAGAGGTTGATCGGATTTTCAAAGAAATGTCTAGAAATGGTG  1701

Query  1301  TTCAACCAAATGTTGTAACTTATAATATACAGATGGATTATCTTTGCAAGAATGGAAGAT  1360
             ||||||||||| || ||||||| |||||||||||||||||||||||||||||||||||||
Sbjct  1702  TTCAACCAAATATTTTAACTTACAATATACAGATGGATTATCTTTGCAAGAATGGAAGAT  1761

Query  1361  GCGCAGAAGCTAGGAAGATTTTTGATTCCATGGTCAGTTTGGGCCAAAAACCGACTGTTA  1420
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1762  GCGCAGAAGCTAGGAAGATTTTTGATTCCATGGTCAGTTTGGGCCAAAAACCGACTGTTA  1821

Query  1421  CTACCTACAGCATTTTGCTTCATGGGTATGCTCTGGAACGATCTTTTCATGAGATACATT  1480
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||| ||||
Sbjct  1822  CTACCTACAGCATTTTGCTTCATGGGTATGCTCTGGAACGATCTTTTCATGAGATGCATT  1881

Query  1481  GTCTCATTGATTTGATGGTGGGAAATGGTATTGCGCCAAATCATTTTCTCTACAACATAC  1540
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1882  GTCTCATTGATTTGATGGTGGGAAATGGTATTGCGCCAAATCATTTTCTCTACAACATAC  1941

Query  1541  TCATATCTGCATATGCTAAAGAAGAAATGATTGGTGAAGTAATGCATATATTTaaaaaaa  1600
             |||||||||||| |||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  1942  TCATATCTGCATGTGCTAAAGAAGAAATGATTGGTGAAGTAATGCATATATTTAAAAAAA  2001

Query  1601  TGCGGCAGCATGGATTGAACCCTAATGTAGCGACCTATGGAGCGGTAGTAAACTTACTTT  1660
             |||||||||||||||||||||||||||||| |||||||||||||||||||||||||||||
Sbjct  2002  TGCGGCAGCATGGATTGAACCCTAATGTAGTGACCTATGGAGCGGTAGTAAACTTACTTT  2061

Query  1661  CCAAGATTGGCCGAATGGATGATGCTATGTCCCAATTCAATCAAATGATAACTGAAGGGT  1720
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2062  CCAAGATTGGCCGAATGGATGATGCTATGTCCCAATTCAATCAAATGATAACTGAAGGGT  2121

Query  1721  TAGCTCCTGATATCATAGTTTTCACCCTCCTTATTAGTGGTTTCTGTTCTTGTGGCAAAT  1780
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2122  TAGCTCCTGATATCATAGTTTTCACCCTCCTTATTAGTGGTTTCTGTTCTTGTGGCAAAT  2181

Query  1781  GGGAGAAGGTTGATGAACTATTTTCTGAGATGTTGGATCGCGGCATCTGTCCCAACACAG  1840
             |||||||||||||||||||||||||||||||||||||||||||||||| |||||||||||
Sbjct  2182  GGGAGAAGGTTGATGAACTATTTTCTGAGATGTTGGATCGCGGCATCTATCCCAACACAG  2241
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FIGURE 4a
CONT-3

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Query  1841  TGTTCTTCAACATAATTATGGACTGCCTCTGCAAAAATGAAAGGGTTATGGAAGCCCAAG  1900
             |||||||||||| ||||||||||||||| |||||||||||||||||||||||||||||||
Sbjct  2242  TGTTCTTCAACACAATTATGGACTGCCTTTGCAAAAATGAAAGGGTTATGGAAGCCCAAG  2301

Query  1901  ATCTCTTCGACCTGATGGTACACATGGGTGTGAAGCCTGATGTGTGTACTTATAACACAC  1960
             |||||||||| |||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2302  ATCTCTTCGATCTGATGGTACACATGGGTGTGAAGCCTGATGTGTGTACTTATAACACAC  2361

Query  1961  TGATAGGTGGATACTTGTTCATTGGTAAGATGGATGAAGTGAGGAAGTTACTTGACAATA  2020
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2362  TGATAGGTGGATACTTGTTCATTGGTAAGATGGATGAAGTGAGGAAGTTACTTGACAATA  2421

Query  2021  TGGTCTCAATTGGCTTGAAACCAAATGTTATCACATATAGCATACTGATTGATGGTTACT  2080
             ||||||||||||||||||||||||||||||| ||||||||||||||||||||||||||||
Sbjct  2422  TGGTCTCAATTGGCTTGAAACCAAATGTTATTACATATAGCATACTGATTGATGGTTACT  2481

Query  2081  CTAAGAATGGAAGCATAGATGATGCATTGGTTGTTTCCAGGGAAATGTTGGCCGGGAAGG  2140
             ||||||||||||| ||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2482  CTAAGAATGGAAGGATAGATGATGCATTGGTTGTTTCCAGGGAAATGTTGGCCGGGAAGG  2541

Query  2141  TTAAGCCTTGTGTTATCACTTTTAATATTATGATTGGTGCATTGCTTAAATGTGGCAGGA  2200
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2542  TTAAGCCTTGTGTTATCACTTTTAATATTATGATTGGTGCATTGCTTAAATGTGGCAGGA  2601

Query  2201  AGGCAGAAGCCAAAGATTTGTTTGACGGTATCTGGGCCAACGGATTAGTGCCCCGATGTTA  2260
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||| ||||
Sbjct  2602  AGGCAGAAGCCAAAGATTTGTTTGACGGTATCTGGGCCAACGGATTAGTGCCCGACGTTA  2661

Query  2261  TTACATATAGCTTAATGATACAAAAACTTATTGAAGAAGGTTCTCTACAAGAGTCTGATG  2320
             ||||||||||||||||||||||||||||| ||||||||||||||||||||||||||||||
Sbjct  2662  TTACATATAGCTTAATGATACAAAAACTTGTTGAAGAAGGTTCTCTACAAGAGTCTGATG  2721

Query  2321  ATCTATTCCTTTCTATGGAGAAGAATGGATGTGTTGCCGACTCCCATATGCTAAATGCTA  2380
             ||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||
Sbjct  2722  ATCTATTCCTTTCTATGGAGAAGAATGGATGTGTTGCCGACTCCCATATGCTAAATGCTA  2781

Query  2381  TAGTTAGAAGCTTACTTCAGAAAGGGGAGGTGCCAAGGGCTGGGACTTATCTGTCTAAAA  2440
             |||||||||||||||||| ||||||||||||||| |||||||||||||||||||||||||
Sbjct  2782  TAGTTAGAAGCTTACTTCGGAAAGGGGAGGTGCCCAGGGCTGGGACTTATCTGTCTAAAA  2841

Query  2441  TTGATGAGAGGAGGTTCACCTTTGAAGCTTCCACAGCTAGCTTGTTGGTTGCACTTGCGT  2500
             ||||||||||||| |||||| |||||||||||||||||||||||||   ||||||||| |
```

FIGURE 4a
CONT-4

```
Sbjct  2842  TTGATGAGAGGAGCTTCACCCTTGAAGCTTCCACAGCTAGCTTGTTGACTGCACTTGCTT  2901

Query  2501  CAGGGCGGAAAAGCCAGGAATATAAGGAGTTACTCCCCGAAAAATACCACTATTTTCTGG  2560
             ||||| ||||| |||| |||||||||| ||||||||| ||||| ||||||| ||||||||
Sbjct  2902  CAGGGGGGAAAGGCCAAGAATATAAGGGGTTACTCCCTGAAAACTACCACTCTTTTCTGG  2961

Query  2561  AAGAGGGCACTGATTGAAACCTTTTCTAGAGATTTTGCCTAGATGCTCTGGAGTGCTAAT  2620
             || ||| ||||| ||||||||| ||||| |||||||||||||||||||||||||||||||
Sbjct  2962  AATAGGACACTGGTTGAAACCTGTTCTACAGATTTTGCCTAGATGCTCTGGAGTGCTAAT  3021

Query  2621  CGAGGGGTGAACTTCTCTTTGTAGGTTACTTCTTCCAACCTTTTCCTTTTTTAGTACGTT  2680
             |||||| ||||||||||||||||||||| ||||||||||||||||| |||||||||||||
Sbjct  3022  CGAGGGATGAACTTCTCTTTGTAGGTTATTTCTTCCAACCTTTTCC-TTTTTAGTACGTT  3080

Query  2681  TGAAATGTAATAATCGTTGGAGAAACAGTAGTAAACT  2717
             ||||||||||||||||||||||||| |||||||||||
Sbjct  3081  TGAAATGTAATAATCGTTGGAGAAACTGTAGTAAACT  3117
```

FIGURE 4b

```
RFL120-17F3R-0377_1     ---------------NIKARYHAGDIGTEDALHLFDELLQVAGRSSIHAINCLLTVVGRD
RFL120-L13_1            MSRRFVPVGRRILEQNIKARYHAGDIGTEDALHLFDELLQVAGRSSIHAINCLLTVVGRD
RFL120-R113_1           MSRRFVPVGRRILEQNIKARYHAGDIGTEDALHLFDELLQVAGRSSIHAINCLLTVVGRD
RFL120-timo_1           MSRRFVPVGRRILEQNIKARYHAGDIGTEDALHLFDELLQVAGRSSIHAINCLLTVVGRD
RFL120-GSTR435_1        MSRRFVPVGRRILEQNIKARYHAGDIGTEDALHLFDELLQVAGRSSIHAINCLLTVVGRD
                                       *********************************************

RFL120-17F3R-0377_1     CPALGVSLFNRVARAKVAPCSITYAILVDCCCRTGRQDLGFAAMGHVIKMGFTADAMITF
RFL120-L13_1            CPALGVSLFNRVARAKVAPCSITYAILVDCCCRTGRQDLGFAAMGHVIKMGFTADAMITF
RFL120-R113_1           CPALGVSLFNRVARAKVAPCSITYAILVDCCCRTGRQDLGFAAMGHVIKMGFTADAMITF
RFL120-timo_1           CPALGVSLFNRVARAKVAPCSITYAILVDCCCRTGRQDLGFAAMGHVIKMGFTADAMITF
RFL120-GSTR435_1        CPALGVSLFNRVARSKVAPCSITYAILVDCCCRAGRQDLGFAAMGHVIKMGFTADAMITF
                        **************:******************:**************************

RFL120-17F3R-0377_1     SHLLKAICAENKTSYAMDIVLRIMPMFNCIPDVFSYNILFKGLCNEKRSQEALELIQVMV
RFL120-L13_1            SHLLKAICAENKTSYAMDIVLRIMPMFNCIPDVFSYNILFKGLCNEKRSQEALELIQVMV
RFL120-R113_1           SHLLKAICAENKTSYAMDIVLRIMPMFNCIPDVFSYNILFKGLCNEKRSQEALELIQVMV
RFL120-timo_1           SHLLKAICAENKTSYAMDIVLRIMPMFNCIPDVFSYNILFKGLCNEKRSQEALELIQVMV
RFL120-GSTR435_1        SHLLKAICAENKTSYAMDIVLRIMPVFNCIPDVFSYNILFKGLCNEKRSQEALELIQVMV
                        *************************:**********************************

RFL120-17F3R-0377_1     EHGGRCQPDVVSYSTVIDGLLKEGLVGKVYILFCEMIQRGISPNAVTYNSIISGMCKVHA
RFL120-L13_1            EHGGRCQPDVVSYSTVIDGLLKEGLVGKVYILFCEMIQRGISPNAVTYNSIISGMCKVHA
RFL120-R113_1           EHGGRCQPDVVSYSTVIDGLLKEGLVGKVYILFCEMIQRGISPNAVTYNSIISGMCKVHA
RFL120-timo_1           EHGGRCQPDVVSYSTVIDGLLKEGLVGKVYILFCEMIQRGISPNAVTYNSIISGMCKVHA
RFL120-GSTR435_1        EHGGGCQPDVVSYSTVIDGLLKEGEVNKAYSLFCEMLRQGVSPNVVTCSSIISGMCKLHA
                        **** ******************* *.*.* *****:::*:***.** .********:**

RFL120-17F3R-0377_1     MDKAEQVLQQMPDRGILPNVATYTSLIHGYFSLGQCKEVDRIFKEMSRNGVQPNVVTYNI
RFL120-L13_1            MDKAEQVLQQMPDRGILPNVATYTSLIHGYFSLGQCKEVDRIFKEMSRNGVQPNVVTYNI
RFL120-R113_1           MDKAEQVLQQMPDRGILPNVATYTSLIHGYFSLGQCKEVDRIFKEMSRNGVQPNVVTYNI
RFL120-timo_1           MDKAEQVLQQMPDRGILPNVATYTSLIHGYFSLGQCKEVDRIFKEMSRNGVQPNVVTYNI
RFL120-GSTR435_1        MDKAEEVLQQMLDRGILPNVATYTSLIHGYFSLGQCKEVDRIFKEMSRNGVQPNILTYNI
                        *****:***** ******************************************::****

RFL120-17F3R-0377_1     QMDYLCKNGRCAEARKIFDSMVSLGQKPTVTTYSILLHGYALERSFHEIHCLIDLMVGNG
RFL120-L13_1            QMDYLCKNGRCAEARKIFDSMVSLGQKPTVTTYSILLHGYALERSFHEIHCLIDLMVGNG
RFL120-R113_1           QMDYLCKNGRCAEARKIFDSMVSLGQKPTVTTYSILLHGYALERSFHEIHCLIDLMVGNG
RFL120-timo_1           QMDYLCKNGRCAEARKIFDSMVSLGQKPTVTTYSILLHGYALERSFHEIHCLIDLMVGNG
RFL120-GSTR435_1        QMDYLCKNGRCAEARKIFDSMVSLGQKPTVTTYSILLHGYALERSFHEMHCLIDLMVGNG
                        ************************************************:***********

RFL120-17F3R-0377_1     IAPNHFLYNILISAYAKEEMIGEVMHIFKKMRQHGLNPNVATYGAVVNLLSKIGRMDDAM
```

FIGURE 4b
CONT-1

```
RFL120-L13_1           IAPNHFLYNILISAYAKEEMIGEVMHIFKKMRQHGLNPNVATYGAVVNLLSKIGRMDDAM
RFL120-R113_1          IAPNHFLYNILISAYAKEEMIGEVMHIFKKMRQHGLNPNVATYGAVVNLLSKIGRMDDAM
RFL120-timo_1          IAPNHFLYNILISAYAKEEMIGEVMHIFKKMRQHGLNPNVATYGAVVNLLSKIGRMDDAM
RFL120-GSTR435_1       IAPNHFLYNILISACAKEEMIGEVMHIFKKMRQHGLNPNVVTYGAVVNLLSKIGRMDDAM
                       ************** *************************.*******************

RFL120-17F3R-0377_1    SQFNQMITEGLAPDIIVFTLLISGFCSCGKWEKVDELFSEMLDRGICPNTVFFNIIMDCL
RFL120-L13_1           SQFNQMITEGLAPDIIVFTLLISGFCSCGKWEKVDELFSEMLDRGICPNTVFFNIIMDCL
RFL120-R113_1          SQFNQMITEGLAPDIIVFTLLISGFCSCGKWEKVDELFSEMLDRGICPNTVFFNIIMDCL
RFL120-timo_1          SQFNQMITEGLAPDIIVFTLLISGFCSCGKWEKVDELFSEMLDRGICPNTVFFNIIMDCL
RFL120-GSTR435_1       SQFNQMITEGLAPDIIVFTLLISGFCSCGKWEKVDELFSEMLDRGIYPNTVFFNTIMDCL
                       ********************************************** ******* *****

RFL120-17F3R-0377_1    CKNERVMEAQDLFDLMVHMGVKPDVCTYNTLIGGYLFIGKMDEVRKLLDNMVSIGLKPNV
RFL120-L13_1           CKNERVMEAQDLFDLMVHMGVKPDVCTYNTLIGGYLFIGKMDEVRKLLDNMVSIGLKPNV
RFL120-R113_1          CKNERVMEAQDLFDLMVHMGVKPDVCTYNTLIGGYLFIGKMDEVRKLLDNMVSIGLKPNV
RFL120-timo_1          CKNERVMEAQDLFDLMVHMGVKPDVCTYNTLIGGYLFIGKMDEVRKLLDNMVSIGLKPNV
RFL120-GSTR435_1       CKNERVMEAQDLFDLMVHMGVKPDVCTYNTLIGGYLFIGKMDEVRKLLDNMVSIGLKPNV
                       ************************************************************

RFL120-17F3R-0377_1    ITYSILIDGYSKNGSIDDALVVSREMLAGKVKPCVITFNIMIGALLKCGRKAEAKDLFDG
RFL120-L13_1           ITYSILIDGYSKNGSIDDALVVSREMLAGKVKPCVITFNIMIGALLKCGRKAEAKDLFDG
RFL120-R113_1          ITYSILIDGYSKNGSIDDALVVSREMLAGKVKPCVITFNIMIGALLKCGRKAEAKDLFDG
RFL120-timo_1          ITYSILIDGYSKNGSIDDALVVSREMLAGKVKPCVITFNIMIGALLKCGRKAEAKDLFDG
RFL120-GSTR435_1       ITYSILIDGYSKNGRIDDALVVSREMLAGKVKPCVITFNIMIGALLKCGRKAEAKDLFDG
                       ************** *********************************************

RFL120-17F3R-0377_1    IWANGLVPDVITYSLMIQKLIEEGSLQESDDLFLSMEKNGCVADSHMLNAIVRSLLQKGE
RFL120-L13_1           IWANGLVPDVITYSLMIQKLIEEGSLQESDDLFLSMEKNGCVADSHMLNAIVRSLLQKGE
RFL120-R113_1          IWANGLVPDVITYSLMIQKLIEEGSLQESDDLFLSMEKNGCVADSHMLNAIVRSLLQKGE
RFL120-timo_1          IWANGLVPDVITYSLMIQKLIEEGSLQESDDLFLSMEKNGCVADSHMLNAIVRSLLQKGE
RFL120-GSTR435_1       IWANGLVPDVITYSLMIQKLVEEGSLQESDDLFLSMEKNGCVADSHMLNAIVRSLLRKGE
                       ********************:***********************************:***

RFL120-17F3R-0377_1    VPRAGTYLSKIDERRFTFEASTASLLVALASGRKSQEYKELLPEKYHYFLEEGTD
RFL120-L13_1           VPRAGTYLSKIDERRFTFEASTASLLVALASGRKSQEYKELLPEKYHYFLEEGTD
RFL120-R113_1          VPRAGTYLSKIDERRFTFEASTASLLVALASGRKSQEYKELLPEKYHYFLEEGTD
RFL120-timo_1          VPRAGTYLSKIDERRFTFEASTASLLVALASGRKSQEYKELLPEKYHYFLEEGTD
RFL120-GSTR435_1       VPRAGTYLSKIDERSFTLEASTASLLTALASGGKGQEYKGLLPENYHSFLE----
                       ************** **:********.***** *.**** ****:** ***
```

FIGURE 5A

```
RFL29a     MPRFSSTTPMSPPRLR--LRLCARHSSSTSHPSRIWDPHAAFAAAAQRASSGTLTTEDAH
RFL29b     MPRFSSTTPMSPPRLRLRLRLCARHSSSTSHPSRIWDPHAAFAAAAQRASSGTLTTEDAH
RFL29c_1   MPRFSSTTPMSPPRLR--LRLCARHSSCTSHPSRIWDPHAAFAAAAQRASSGTLTTEDAH
RFL29c_2   ------------------------------------------------------------

RFL29a     HLFDELLRRGNPVQERPLNKFLAALARAPASASCCDGPALAVALFGRLSRDVGRRVAQPN
RFL29b     HLFDELLRRGNPVQERPLNKFLAALARAPASASCCDGPALAVTLFGRLSRDVGRRVAQPN
RFL29c_1   HLFDELLRRGNPVQERPLNKFLAALARAPASASCCDGPALAVALFGRLSQDVRRRVAQPN
RFL29c_2   ------------------------------------------------------------

RFL29a     VFTYGVLMDCCCRACRTDLVLAFFGRLLKTGLEANQVVFNTLLKGLCHTKRADEALDVLL
RFL29b     VFTYGVLMDCCCRACRTDLVLAFFGRLLKTGLEANQVVFNTLLKGLCHTKRADEALDVLL
RFL29c_1   VFTYGVLMDCCCRACRTDLALAFFGRLLKTGLEANQVVFNTLLKGLCHTKRADEALDVLL
RFL29c_2   ------------------------------------------------------------

RFL29a     HRMPELGCTPNVVAYNTVIHGFFKEGHVSKACNLFHEMAQQGVKPNVVTYNSVIDALCKA
RFL29b     HRMPELGCTPNVVAYNTVIHGFFKEGHVSKACNLFHEMAQQGVKPNVVTYNSVIDALCKA
RFL29c_1   HRMPELGCTPDVVAYNTVIHGFFKEGHVSKACNLFHEMAQQGVKPNVVTYNSVIDALCKA
RFL29c_2   ------------------------------------------------------------

RFL29a     RAMDKAEVVLRQMIDDGVGPDNVTYSSLIHGYSSSGHWKEAVRVFKEMTSRRVTADVHTY
RFL29b     RAMDKAEVVLRQMIDDGVGPDNVTYSSLIHGYSSSGHWKEAVRVFKEMTSRRVTADVHTY
RFL29c_1   RAMDKAEVVLRQMIDDGVGP----------------------------------------
RFL29c_2   ---------------MMVLDLNVTYSSLIHGYSSSGHWKEAVRVFKEMTSRRVTADVHTY

RFL29a     NMFMTFLCKHGRSKEAAGIFDTMAIKGLKPDNVSYAILLHGYAAEGCLVDMINLFNSMER
RFL29b     NMFMTFLCKHGRSKEAAGIFDTMAIKGLKPDNVSYAIRLHGYATEGCLVDMINLFNSMAT
RFL29c_1   ------------------------------------------------------------
RFL29c_2   NMFMTFLCKHGRSKEAAGIFDTMAIKGLKPDNVSYAILLHGYAAEGCLVDMINLFNSMER

RFL29a     DCILPDCRIFNILINAYAKSGKLDKAMLIFNEMQKQGVSPNAVTYSTVIHAFCKKGRLDD
RFL29b     HCILPNCHIFNILINAYAKSGKLDKAMLIFNEMQKQGVSPNAVTYSTVIHAFCKKGRLDD
RFL29c_1   ------------------------------------------------------------
RFL29c_2   DCILPDCRIFNILINAYAKSGKLDKAMLIFNEMQKQGVSPNAVTYSTVIHAFCKKGRLDD

RFL29a     AVIKFNQMIDTGVRPDASVYRPLIQGFCTHGDLVKAKEYVTEMMKKGMPPPDIMFFSSIM
RFL29b     AVIKFNQMIDTGVRPDASVYRPLIQGFCTHGDLVKAKEYVTEMMKKGMPPPDIMFFSSIM
RFL29c_1   ------------------------------------------------------------
RFL29c_2   AVIKFNQMIDTGVRPDASVYRPLIQGFCTHGDLVKAKEYVTEMMKKGMPPPDIMFFSSIM
```

FIGURE 5A
CONT-1

```
RFL29a     QNLCTEGRVTEARDILDLIVHIGMRPNVIIFNLLIGGYCLVRKMADALKVFDDMVSYGLE
RFL29b     QNLCTEGRVTEARDILDLIVHIGMRPNVIIFNLLIGGYCLVRKMADALKVFDDMVSYGLE
RFL29c_1   ------------------------------------------------------------
RFL29c_2   QNLCTEGRVTEARDILDLIVHIGMRPNVIIFNLLIGGYCLVRKMADALKVFDDMVSYGLE

RFL29a     PCNFTYGILINGYCKNRRIDDGLILFKEMLHKGLKPTTFNYNVILDGLFLAGQTVAAKEK
RFL29b     PCNFTYGILINGYCKNRRIDDGLILFKEMLHKGLKPTTFNYNVILDGLFLAGQTVAAKEK
RFL29c_1   ------------------------------------------------------------
RFL29c_2   PCNFTYGILINGYCKNRRIDDGLILFKEMLHKGLKPTTFNYNVILDGLFLAGQTVAAKEK

RFL29a     FDEMVESGVSVCIDTYSIILGGLCRNSCSSEAITLFRKLSAMNVKFDITIVNIIIGALYR
RFL29b     FDEMVESGVSVCIDTYSIILGGLCRNSCSSEAITLFRKLSAMNVKFDITIVNIIIGALYR
RFL29c_1   ------------------------------------------------------------
RFL29c_2   FDEMVESGVSVCIDTYSIILGGLCRNSCSSEAITLFRKLSAMNVKFDITIVNIIIGALYR

RFL29a     VERNQEAKDLFAAMPANGLVPNAVTYTVMMTNLIKEGSVEEADNLFLSMEKSGCTANSCL
RFL29b     VERNQEAKDLFAAMPANGLVPNAVTYTVMMTNLIKEGSVEEADNLFLSMEKSGCTANSCL
RFL29c_1   ------------------------------------------------------------
RFL29c_2   VERNQEAKDLFAAMPANGLVPNAVTYTVMMTNLIKEGSVEEADNLFLSMEKSGCTANSCL

RFL29a     LNHIIRRLLEKGEIVKAGNYMSKVDAKSYSLEAKTVSLLISLFSRKGKYREHIKLLPTKY
RFL29b     LNHIIRRLLEKGEIVKAGNYMSKVDAKSYSLEAKTVSLLISLFSRKGKYREHIKLLPTKY
RFL29c_1   ------------------------------------------------------------
RFL29c_2   LNHIIRRLLEKGEIVKAGNYMSKVDAKSYSLEAKTVSLLISLFSRKGKYREHIKLLPTKY

RFL29a     QFLEEAATVE
RFL29b     QFLEEAATVE
RFL29c_1   ----------
RFL29c_2   QFLEEAATVE
```

FIGURE 5B

```
RFL164a   MPGFSSAASMSPLRLRLRLHARHSS-ASQPSRRQGWDPHAAFAAATECARSGNLTPEDAH
RFL164b   MPGFSSAASMSPLRLRLRLHARHSSSASQPSRRQGWDPHAAFAAATECARSGNLTPEDAH

RFL164a   NLFDELLRQGNPVLGRPLNNLLAALARAPASSACRDGPALVVALFSRISQGARLRVLHPT
RFL164b   HLFDELLRQGNPVLGRPLNNLLAALARAPASSACRDGPALAVALFSRISQGARLRVLHPT

RFL164a   ACTYGILMDCSCRAHRLDLAFAFFGRLLRTGLKAGVIEVNSLLKGLCHAKRADEAMEVLL
RFL164b   ACTYGILMDCSCRAHRLNLAFAFFGRLLRTGLKAGVIEVNSLLKGLCHAKRADEAMEVLL

RFL164a   HRMPELFIGVQGTAVYRSLIQGFCTHGDLVKAKEYVTEMMKKGMPPPDIMFFSSIMQNLC
RFL164b   HRMPELFIGVQGTAVYRSLIQGFCTHGDLVKAKEYVTEMMKKGMPPPDIMFFSSIMQNLC

RFL164a   TEGRVIEARDILDLIVRIGMRPDVFIFNILIGGYCLVGKMEDASKIFDDMVSYGLEPCNF
RFL164b   TEGRVIEARDILDLIVRIGMRPDVFIFNILIGGYCLVGKMEDASKIFDDMVSYGLEPCNF

RFL164a   TYGILINGYCKNKRIDDGLILFKEMLRKGLKPTTFNYNVILDGLFLAGQTVAAKEKFDEM
RFL164b   TYGILINGYCKNKRIDDGLILFKEMLRKGLKPTTFNYNVILDGLFLAGQTVAAKEKFDEM

RFL164a   VESGVSVCIDTYSIVLGGLCRNSCSSEAITLF----------------------------
RFL164b   VESGVSVCIDTYSIVLGGLCRNSCSSEAITLFRKLSAMNVKFNITIVNTIIGAFYRVERN

RFL164a   ------------------------------------------------------------
RFL164b   QEAKDLFAAIPASGLVPNVVTYTIMIKNLIKEGSVEEADNLFLSMEKSGCSANSYLLNHI

RFL164a   ------------------------------------------------------------
RFL164b   IRRLLEKGEIVKAGNYMSKVDAKSYSLEAKTVSLLISLFSRKGKYREHIKLLPTKYQFLE

RFL164a   ------
RFL164b   EAATVE
```

FIGURE 5C

```
RFL166a  MPRLSSTTPMSPPRLRLRLRGRHSSSTSHPSRIWDPHAAFAGATQRAHSGNLTPEDAHHL
RFL166b  ------------------------------------------------------------

RFL166a  FDELLRQGNPVQERPLTNFLAALARAPASASCSDGPALAVALFGRLSRGAGRRVAQPNVF
RFL166b  ------------------------------------------------------------

RFL166a  TYGVLMDCCCRACRPDLALAFFGRLFRKGLEANRVIFCTLLKGLCHAKRTDEALDVLLHR
RFL166b  ------------------------------------------------------------

RFL166a  MPELGCTPNVVAYTTVIHGFFKEGQVGKACNLFHGMAQQGVAPNLVTYNSVIDALCKAKA
RFL166b  MPELGCTPNVVAYTTVIHGFFKEGQVGKACNLFHGMAQQGVAPNLVTYNSVIDALCKAKA

RFL166a  MDKAEYFLGQMVDDGVVPDNVTYNSLIHGYSSSGHWKEAVRVFKEMTSRRVTADVHTYNM
RFL166b  MDKAEYFLGQMVDDGVVPDNVTYNSLIHGYSSSGHWKEAVRVFKEMTSRRVTADVHTYNM

RFL166a  FMTFLCKHGRSKEAAGIFDTMAIKGLKPDNVSYAILLHGYATEGCLVDMINLFNSMERDC
RFL166b  FMTFLCKHGRSKEAAGIFDTMAMKGLKPDNVSYAILLHGYATEGCLVDMINLFNSMERDC

RFL166a  ILPDCRIFNILINAYAKSGKLDKAMLIFNEMQKQGVSPNAVTYSTVIHTFCKKGRLDDAV
RFL166b  ILPDCRIFNILINAYAKSGKLDKAMLIFNEMQKQGVSPNAVTYSTVIDAFCKKGQLDDAM

RFL166a  IKFNQMIDTGVRQGTAVYGSLIQGFCTHGDLVKAKELLTEMMNKGMLPPDIKFFHSIMQN
RFL166b  IKFNQMIDTGVRQGTAVYGSLIQGFCTHGDLVKAKELLTEMMNKGMLPPDIKFFHSIMQN

RFL166a  LCTEGRVIEARDVLGLIAHIGMRPDVCTFNILIGGYCLVGKMEDASKIFDDMMSYGLEPS
RFL166b  LCTEGRVIEARDVLGLIAHIGMRPDVCTFNILIGGYCLVGKMEDASKIFDDMMSYGLEPS

RFL166a  NC----------------------------------------------------------
RFL166b  NITYGILINGYCKNRRIDDGLILFKEMLHKGLKPTTFNYNIILDGLLLAGRTVAAKEKFN
```

FIGURE 6

```
RFL29a_5UTR   CTCCGTCCGAAAATACTTGTCGAAGAATTTGATGAAAATGGATGCATCTAGAACAAGAAT
RFL29b_5UTR   CTCCGTCCGGAAATACTTGTCGAAGAATTTGATGAAAATGGATGCATCTAGAACAAGAAT

RFL29a_5UTR   ACATCTAGATACATCAATCTCCCCGACAAGTATTTCCGAACGGAGGGAGTACTAGATAAT
RFL29b_5UTR   ACATCTAGATACATCAATCTCCCTGACAAGTATTTCCGAGCGGAGGGAGTACTAGATAAT

RFL29a_5UTR   A-----------------------------------------------------------
RFL29b_5UTR   ACTCCCTCCGTTCCTAAATAATTGTCTTTCTAGCTATCTCAAATAAACTACAACATACGG

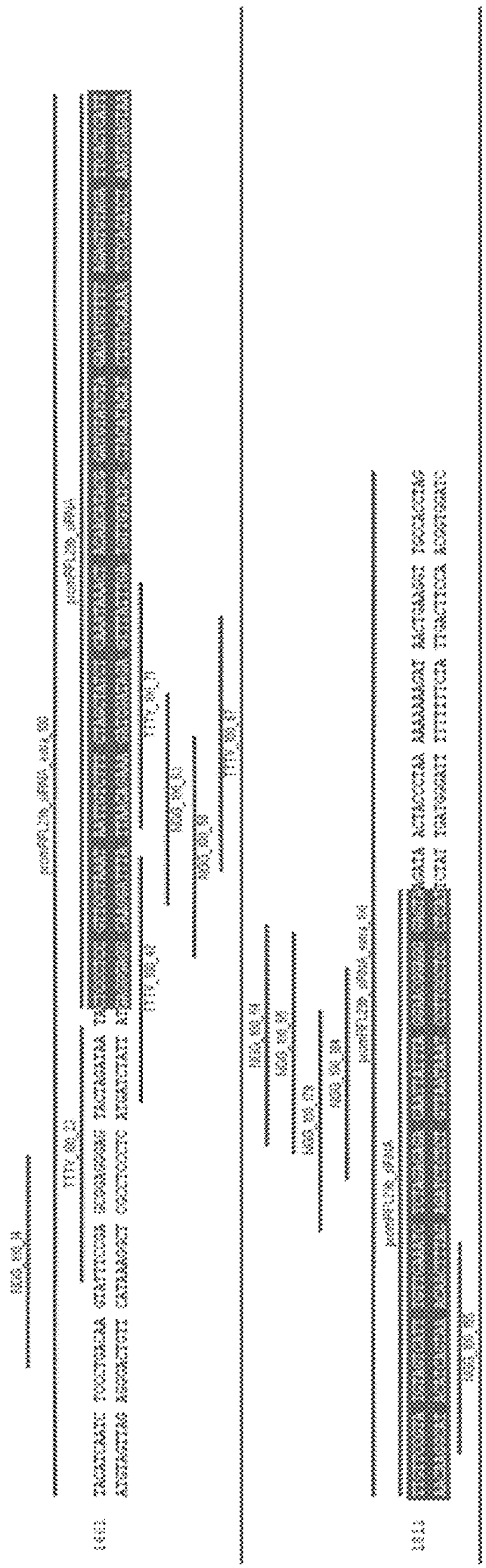
RFL29a_5UTR   ------------------------------------------------------------
RFL29b_5UTR   ATGTATGTAGACATGTTTTAGAGTGTAGATTCACTCATTTTGTTCCGTATGTAGTCATTT

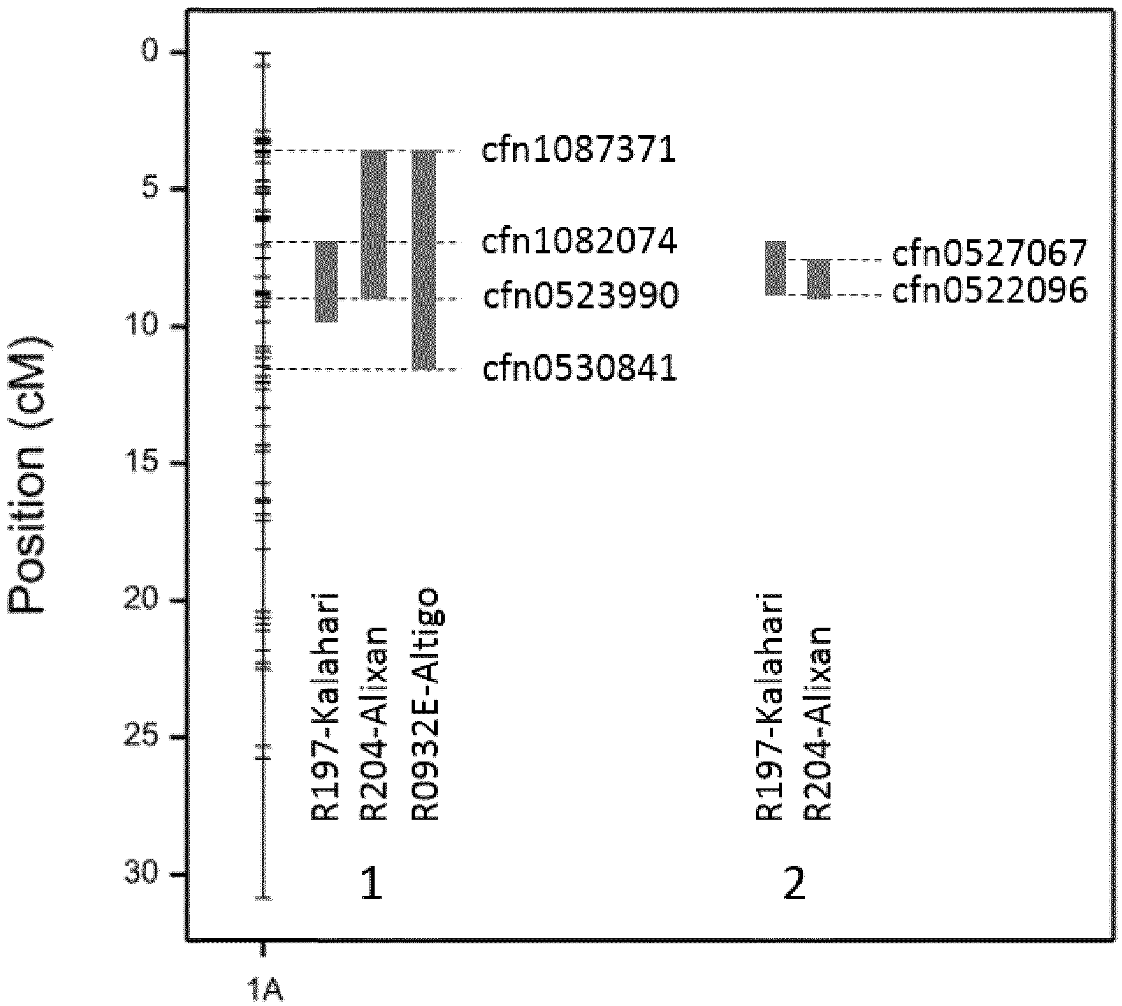
RFL29a_5UTR   --------------------------------------------AGATAACTATCCAAAA
RFL29b_5UTR   GTTGAAATCTCTAGAGAGACAATTATTTAGGAACGGAGGGAGTAAGATAACTACCC-TAA

RFL29a_5UTR   AAAAAAAGATAACTGAAGGTTGCCACCTAGCACATTCACATTGGTACAACTTGGAAAAGC
RFL29b_5UTR   AAAAAAAGATAACTGAAGGTTGCCACCTAGCACATTCACATTGGTACAACTTGGAAAAGC

RFL29a_5UTR   ACAGCCCCGTCGTCCTGCTCCCAGTTGAGTTCGCGACCTACACACCGGCC
RFL29b_5UTR   ACAGCCCCGTCGTCCTGCTCCCAGTTGAGTTCGCGACCTACACACCGGCC
```

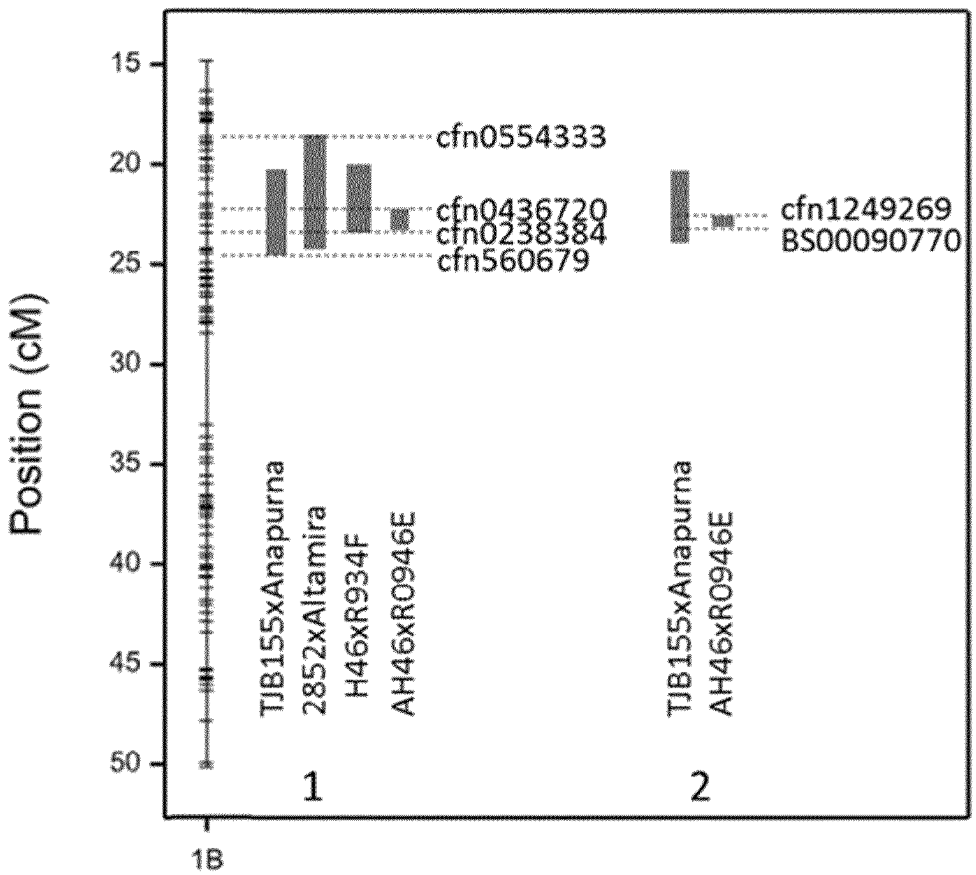
FIGURE 7





Position (cM)
0
5
10
15
20
25
30
1A
cfn1087371
cfn1082074
cfn0523990
cfn0530841
cfn0527067
cfn0522096
R197-Kalahari
R204-Alixan
R0932E-Altigo
1
R197-Kalahari
R204-Alixan
2

FIGURE 8



Position (cM)
15
20
25
30
35
40
45
50
1B
cfn0554333
cfn0436720
cfn0238384
cfn560679
cfn1249269
BS00090770
TJB155xAnapurna
2852xAltamira
H46xR934F
AH46xR0946E
TJB155xAnapurna
AH46xR0946E
1
2

FIGURE 9

Physical position	position relative to fine-mapping interval	Marker id.
13857528	Left	cfn0523072
14161545	Left	cfn0523109
14184332	Left	276I13_96B22_9 7797
14505168	Interval	Cfn0522096
14550727	Interval	cfn0527763
14603881	Interval	104A4_105172
14604601	Interval	104A4_105588
14620761	Interval	cfn0373248
15052112	Interval	cfn1097828
15346090	Interval	Cfn0527067
16373906	right	cfn0528390
16389969	right	BWS0267
16409826	right	cfn0527718
17257737	right	cfn0524469
18291418	right	cfn0524921
20205981	right	cfn1122326

FIGURE 10A

CODE	R: restorer/ M: maintenor	Rf alleles	cfn0523072	cfn0523109	_97797	cfn0527763	104A4_105172	104A4_105588	cfn0373248	cfn1097828	cfn0528390	BWS0267	cfn0527718	cfn0524469	cfn0524921	cfn1122326
R197	R	homozygous Rf1, rf3, Rf7	T	A	C	C	TG	A	T	C	G	A	T	G	G	C
R204	R	homozygous Rf1, rf3, Rf7	T	A	C	C	TG	A	T	C	G	A	T	G	G	C
R0932E	R	homozygous Rf1, rf3, ff7	T	A	C	C	TG	A	T	C	G	A	T	G	G	C
LGWR16-0016	R	homozygous Rf1, Rf3, Rf7	C	A	C	C	TG	A	T	C	G	A	C	G	G	C
LGWR16-0026	R	homozygous Rf1, Rf3, Rf7	C	A	C	C	TG	A	T	C	G	A	C	G	G	C
AIGLE	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	G	T	-	A	T
AIRBUS	M	homozygous rf1, rf3, rf7	T	A	T	C	CA	C	T	C	G	G	T	-	A	T
ALHAMBRA	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	G	T	T	A	T

FIGURE 10B

CODE	R: restorer/ M: maintenor	Rf alleles	cfn0523072	cfn0523109	_97797	cfn0527763	104A4_105172	104A4_105588	cfn0373248	cfn1097828	cfn0528390	BWS0267	cfn0527718	cfn0524469	cfn0524921	cfn1122326
ALIXAN	M	homozygous rf1, rf3, rf7	T	-	T	T	CA	C	T	T	A	G	T	-	A	T
AMADEUS	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	G	A	T	-	A	T
ANAPURNA	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	H	T	-	A	T
APACHE	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	G	G	T	-	A	T
ARKEOS	M	homozygous rf1, rf3, rf7	T	C	T	T	CA	C	T	T	A	G	T	-	A	T
ARLEQUIN	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	G	G	T	-	A	T
ARTDECO	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	G	T	T	H	T
ARTURNICK	M	homozygous rf1, rf3, rf7	T	C	T	T	CA	C	T	T	H	G	T	G	A	T

FIGURE 10C

CODE	R: restorer/ M: maintenor	Rf alleles	cfn0523072	cfn0523109	_97797	cfn0527763	104A4_105172	104A4_105588	cfn0373248	cfn1097828	cfn0528390	BWS0267	cfn0527718	cfn0524469	cfn0524921	cfn1122326
ATOMO	M	homozygous rf1, rf3, rf7	T	C	T	T	CA	C	T	C	G	A	T	G	A	T
AVENUE	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	G	T	-	A	T
CEZANNE	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	G	A	T	-	A	T
CROISADE	M	homozygous rf1, rf3, rf7	C	A	T	-	-	C	A	-	G	A	C	G	G	C
FRUCTIDOR	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	G	G	T	-	A	T
GAZUL	M	homozygous rf1, rf3, rf7	T	A	T	T	CA	C	T	T	A	G	T	T	A	T
HERMANN	M	homozygous rf1, rf3, rf7	T	C	T	C	CA	C	T	C	G	H	C	-	H	T
HORATIO	M	homozygous rf1, rf3, rf7	C	C	T	C	-	C	T	C	A	G	T	T	A	C
KALAHARI	M	homozygous rf1, rf3, rf7	T	C	T	C	-	C	T	C	A	G	T	T	A	T

FIGURE 10D

Physical position	position relative to fine-mapping interval	Marker names
1670498	Left	cfn1252000
1762323	Left	IWB14060*
1778039	Interval	Cfn1249269
1789599	Interval	219K1_166464
1789721	Interval	219K1_158251
1796283	Interval	219K1_111446
1796422	Interval	219K1_110042
1796428	Interval	219K1_110005
1796680	Interval	219K1_107461
1797457	Interval	219K1_99688
1807903	Interval	219K1_37
1816050	Interval	cfn1270524
1823908	Interval	136H5_3M5_7601
1825046	Interval	cfn1288811
1825372	Interval	136H5_3M5_89176
1825378	Interval	136H5_3M5_89263
1826466	Interval	136H5_3M5_138211
1842122	Interval	cfn0556874
1903965	Interval	136H5_3M5_64154
1904430	Interval	136H5_3M5_68807
1905289	Interval	136H5_3M5_77916
1984106	Interval	cfn1246088
1991502	Interval	cfn1287194
2012753	Interval	cfn1258380
2058798	Interval	IWB72107*
2058919	Interval	_BS00090770
3087865	Right	cfn1239345

FIGURE 11A

CODE		R: restorer/M: maintainer	Rf alleles	cfn1252000	IWB14060*	cfn1249269	219K1_166464	219K1_158251	219K1_111446	219K1_110042	219K1_110005	219K1_107461	219K1_99688	219K1_37	cfn1270524	136H5_3M5_7601	cfn1288811	136H5_3M5_89176	136H5_3M5_89263	136H5_3M5_138211	cfn0556874	136H5_3M5_64154	136H5_3M5_68807	136H5_3M5_77916	cfn1246088	cfn1287194	cfn1258380	IWB72107*	BS00090770	cfn1239345
TJB155	1	R	homozygous rf1, Rf3, rf7	A	G	G	T	G	A	T	C	A	T	C	T	T	G	A	T	T	C	C	G	A	A	G	A	A	T	A
LGWR16-0016	2	R	homozygous Rf1, Rf3, Rf7	A	G	G	T	G	A	T	C	A	T	C	T	T	G	A	T	T	C	C	G	A	A	G	A	A	T	A
LGWR16-0026	3	R	homozygous Rf1, Rf3, Rf7	A	G	G	T	G	A	T	C	A	T	C	T	T	G	A	T	T	C	C	G	A	A	G	A	A	T	A
AIGLE	4	M	homozygous rf1, rf3, rf7	G	A	A	C	G	-	T	C	A	-	-	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	A
AIRBUS	5	M	homozygous rf1, rf3, rf7	G	G	G	C	G	-	T	C	A	-	-	-	C	-	G	C	A	C	T	A	G	C	A	C	G	C	A
ALHAMBRA	6	M	homozygous rf1, rf3, rf7	A	A	A	C	G	-	T	-	A	-	C	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	A
ALIXAN	7	M	homozygous rf1, rf3, rf7	G	G	G	C	G	-	T	-	A	-	T	-	C	-	G	C	A	C	T	A	G	C	A	C	G	C	-
AMADEUS	8	M	homozygous rf1, rf3, rf7	G	G	G	C	G	-	T	-	A	-	C	-	C	-	G	C	A	C	T	A	G	C	A	C	G	C	G

FIGURE 11B

CODE		R: restorer/M: maintainer	Rf alleles	cfn1252000	IWB14060*	cfn1249269	219K1_166464	219K1_158251	219K1_111446	219K1_110042	219K1_110005	219K1_107461	219K1_99688	219K1_37	cfn1270524	136H5_3M5_7601	cfn1288811	136H5_3M5_89176	136H5_3M5_89263	136H5_3M5_138211	cfn0556874	136H5_3M5_64154	136H5_3M5_68807	136H5_3M5_77916	cfn1246088	cfn1287194	cfn1258380	IWB72107*	BS00090770	cfn1239345
ANAPURNA	9	M	homozygous rf1, rf3, rf7	H	G	G	C	G	A	T	H	A	T	C	-	C	G	G	C	A	C	T	A	H	C	H	C	G	T	G
APACHE	10	M	homozygous rf1, rf3, rf7	A	G	A	C	G	-	T	T	A	-	-	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	G
ARKEOS	11	M	homozygous rf1, rf3, rf7	G	A	-	T	G	H	H	T	A	-	-	-	C	-	G	C	A	T	C	-	-	C	G	-	G	C	G
ARLEQUIN	12	M	homozygous rf1, rf3, rf7	A	G	A	C	G	-	T	-	A	-	-	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	G
ARTDECO	13	M	homozygous rf1, rf3, rf7	H	A	A	C	-	-	T	-	-	-	-	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	A
ARTURNICK	14	M	homozygous rf1, rf3, rf7	H	G	G	T	G	A	C	T	A	-	-	A	C	G	G	C	A	C	T	A	G	C	A	C	G	H	H
ATOMO	15	M	homozygous rf1, rf3, rf7	A	G	-	T	G	A	C	T	A	-	C	A	C	G	G	C	A	C	T	A	G	C	A	C	G	C	A
AVENUE	16	M	homozygous rf1, rf3, rf7	G	G	G	C	-	-	T	-	A	-	-	-	C	-	G	C	A	C	T	A	G	C	A	C	G	C	A

FIGURE 11C

CODE		R: restorer/M: maintainer	Rf alleles	cfn1252000	IWB14060*	cfn1249269	219K1_166464	219K1_158251	219K1_111446	219K1_110042	219K1_110005	219K1_107461	219K1_99688	219K1_37	cfn1270524	136H5_3M5_7601	cfn1288811	136H5_3M5_89176	136H5_3M5_89263	136H5_3M5_138211	cfn0556874	136H5_3M5_64154	136H5_3M5_68807	136H5_3M5_77916	cfn1246088	cfn1287194	cfn1258380	IWB72107*	BS00090770	cfn1239345
CEZANNE	17	M	homozygous rf1, rf3, rf7	G	G	G	C	G	-	T	-	A	-	-	-	C	-	G	T	T	C	T	A	H	C	A	C	G	C	A
CROISADE	18	M	homozygous rf1, rf3, rf7	G	G	G	T	G	A	C	T	A	-	-	A	C	G	G	C	A	C	T	A	G	C	A	C	G	C	A
FRUCTIDOR	19	M	homozygous rf1, rf3, rf7	-	G	G	T	G	A	C	T	A	-	-	A	C	G	G	C	A	-	T	A	G	C	A	C	G	C	A
GAZUL	20	M	homozygous rf1, rf3, rf7	G	A	A	T	H	-	T	-	A	-	-	T	C	-	G	C	A	-	C	-	-	C	H	-	G	C	A
HERMANN	21	M	homozygous rf1, rf3, rf7	G	A	G	T	H	-	H	T	C	-	C	-	C	-	G	T	A	C	T	A	G	C	A	C	G	H	G
HORATIO	22	M	homozygous rf1, rf3, rf7	G	A	A	T	G	-	T	T	H	-	-	-	C	-	G	C	A	T	C	-	-	C	H	-	G	C	H
KALAHARI	23	M	homozygous rf1, rf3, rf7	G	A	G	T	G	-	T	-	H	-	-	-	C	-	G	C	A	C	T	A	G	C	A	C	G	C	A

FIGURE 11D

FIGURE 12

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RFL29a AGGAAGGCCATGTAAGCAAGGCCTGCAATCTGTTCCATGAAATGGCGCAGCAGGGCGTTA
RFL29c AGGAAGGCCATGTAAGCAAGGCCTGCAATCTGTTCCATGAAATGGCGCAGCAGGGCGTTA

RFL29a AGCCTAATGTGGTGACATATAACTCAGTTATCGATGCGCTGTGCAAGGCCAGAGCCATGG
RFL29c AGCCTAATGTGGTGACATATAACTCAGTTATCGATGCGCTGTGCAAGGCCAGAGCCATGG

RFL29a ACAAGGCAGAGGTGGTCCTTCGTCAGATGATTGATGATGGTGTTGGACCTGATAATGTGA
RFL29c ACAAGGCAGAGGTGGTCCTTCGTCAGATGATTGATGATGGTGTTGGACCT--TAATGTGA

RFL29a CGTATAGTAGCCTCATCCATGGATATTCCTCTTCAGGCCACTGGAAGGAGGCAGTTAGGG
RFL29c CGTATAGTAGCCTCATCCATGGATATTCCTCTTCAGGCCACTGGAAGGAGGCAGTTAGGG

RFL29a TATTCAAAGAGATGACAAGTCGGAGGGTTACAGCAGATGTGCATACTTACAACATGTTTA
RFL29c TATTCAAAGAGATGACAAGTCGGAGGGTTACAGCAGATGTGCATACTTACAACATGTTTA

RFL29a TGACCTTTCTTTGCAAACATGGAAGAAGCAAAGAAGCTGCAGGAATTTTTGATACCATGG
RFL29c TGACCTTTCTTTGCAAACATGGAAGAAGCAAAGAAGCTGCAGGAATTTTTGATACCATGG

RFL29a CTATCAAGGGCCTGAAACCTGACAACGTTTCATATGCTATTCTCCTTCATGGGTATGCCG
RFL29c CTATCAAGGGCCTGAAACCTGACAACGTTTCATATGCTATTCTCCTTCATGGGTATGCCG
```

WHEAT COMPRISING MALE FERTILITY RESTORER ALLELES

The present application contains a Sequence Listing that has been submitted electronically in ASCII format and is hereby incorporated by reference in its entirety. The ASCII copy, created on Dec. 8, 2020, is named Substitute Sequence Listing_ST25.txt and is 9,515,008 bytes in size.

The invention is in the field of plant genetics and plant breeding. The invention more specifically relates to wheat plants carrying restorer of fertility genes specific to *T. timopheevii* CMS cytoplasm.

BACKGROUND

Hybrid production is based on crossing two parental lines to increase heterosis and de facto, increase genetic variability to create new varieties or genotypes with higher yield and better adapted to environmental stresses. Even in a predominantly autogamous species like wheat, research studies have shown that hybrid lines exhibit improved quality and greater tolerance to environmental and biotic stresses.

In order to promote commercially viable rates of hybrid production, self-fertilization must be avoided, i.e. fertilization of the female organ by the pollen of the same plant. It is desired that the female organ of the female parent is exclusively fertilized with the pollen of the male parent. In order to obtain a reliable and efficient system for producing seeds needed for hybrid production, one generally needs three essential elements: a means to induce male sterility, a means to propagate the sterility, and a means to restore fertility. For example a fully genetically based system is composed of a male-sterile line (female parent), a fertile maintainer line (male parent allowing propagation of the male-sterile line), and a fertility restorer line (male parent for hybrid production).

Male sterility can be achieved by three different ways. Manual emasculation is the simplest one and is still used in some species where male and female flowers are separated, e.g. corn. However, it is impractical in species like wheat where flowers contain both female and male organs. Male sterility can also be induced by chemical hybridization agents (CHAs) with gametocidic effects. Currently, only a few commercial hybrid wheat cultivars are based on this technology as it can bear substantial financial risks.

Finally, male sterility can also be induced by genetic means. There are many examples of hybrid systems in corn or sorghum based on male sterility induced by genetic means showing the preponderance of this technology compared to the two mentioned previously. However, in other species which are predominantly self-pollinated like wheat, hybrid production is still a challenge (Longin et al, 2012).

The first case of male sterility in wheat was observed in 1951 (Kihara, 1951), where it was observed that sterility was caused by incompatibility between the cytoplasm of *Aegilops caudata* L. and the nucleus of *T. aestivum* var. *erythrospermum*. Subsequently research on *T. timopheevii* cytoplasm showed that this cytoplasm is able to induce sterility in bread wheat (*T. aestivum*) (Wilson and Ross, 1961, Crop Sci, 1:191-193). Orf256 was previously identified as a gene specific to the *T. timopheevii* mitochondrial genome (Rathburn and Hedgcoth, 1991; Song and Hedgcoth, 1994), however, it remains to be shown that orf256 is the genetic determinant of *T. timopheevii* CMS. It was expected that such a cytoplasm could be used in a hybrid production system. However, major limitations arose from the difficulty in finding a completely dominant and stable fertility restorer gene with no negative side effects (notably on yield).

Fertility restoration of male sterile plants harboring *T. timopheevii* CMS cytoplasm (T-CMS cytoplasm) has been reported and eight major restorer loci (designated as Rf1 to Rf8) have been identified and located approximate within the wheat genome. One of the most effective restorer loci is Rf3 (Ma and Sorrells, 1995; Kojima et al, 1997; Ahmed et al 2001; Geyer et al 2016). Two SNP markers allowed the location of the Rf3 locus within a 2 cM fragment on chromosome 1B (Geyer et al, 2016). The author notes that these markers are not diagnostic markers.

While it is understood that restoration to normal pollen fertility could require two or more Rf loci, it is also well known that modifier loci exist that have either minor effects with low penetrance (Zhou et al 2005, Stojalowski et al 2013) or inhibitory effects on fertility, depending on environmental conditions (Wilson, 1984). It is not yet understood which combination of genes or loci is needed to complete a full restoration of T-CMS in different genetic backgrounds and environmental conditions.

In this context, the development of technologies that enable a full restoration of pollen fertility is of major importance in wheat. It is therefore the object of this invention to propose suitable fertility restorer genes in wheat for the development of a hybrid production system useful for the seed industry.

SUMMARY

A first object of the present disclosure relates to an isolated Rf1 nucleic acid encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361. An example of Rf1 nucleic acid comprises SEQ ID NO:3119.

The disclosure also relates to a transgenic wheat plant comprising such Rf1 nucleic acid, and, optionally, one or more nucleic acids comprising Rf3, Rf4 and/or Rf7 restorer allele(s), as transgenic element(s).

Another aspect relates to a genetically engineered wheat plant comprising such Rf1 nucleic acid, and, optionally, one or more nucleic acids comprising Rf3, Rf4 and/or Rf7 restorer allele(s), as genetically engineered element(s).

in specific embodiments, said transgenic element(s) or genetically engineered element(s) express polypeptides which restore or improve male fertility to the plant as compared to the parent plant without such transgenic element(s) or genetically engineered element(s).

Yet another aspect relates to a wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprising such Rf1 restorer allele, and at least two fertility restorer alleles within the restorer loci chosen amongst Rf3, Rf4 and Rf7, wherein,

- the Rf3 locus is located at most 10 cM from marker cfn1249269 of SEQ ID NO: 3205 or marker BS00090770 of SEQ ID NO:3228,
- the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO: 3231, and,
- the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO: 3233.

The disclosure also provides methods for producing a transgenic wheat plant as described above, wherein the method comprises the steps of transforming a parent wheat plant with one or more Rf1 nucleic acids encoding protein restorer of *T. timopheevii* CMS cytoplasm, selecting a plant comprising said one or more nucleic acid(s) as transgene(s), regenerating and growing said wheat transgenic plant.

Also part of the present disclosure is a method for producing a genetically modified wheat plant as described above, wherein the method comprises the steps of genetically modifying a parent wheat plant to obtain in their genome one or more nucleotide sequence encoding Rf1 protein restorer of *T. timopheevii* CMS cytoplasm, preferably by genome-editing, selecting a plant comprising said one or more nucleotide sequences as genetically engineered elements, regenerating and growing said wheat genetically engineered plant.

The disclosure further relates to a method for producing a wheat plant by crossing, said method includes the following:

- providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles,
- crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first plant and the second plant,
- collecting the F1 hybrid seed,
- obtaining homozygous plants from the F1 plants,
- optionally detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or at each generation.

Preferably in such methods, the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant.

The disclosure also relates to a method for producing a transgenic or genetically engineered wheat plant, wherein the fertility level of said plant is modified, comprising the step of knocking-down Rf1 restorer allele expression, wherein said Rf1 restorer allele comprises a Rf1 nucleic acid.

The disclosure also relates to the method for producing a wheat hybrid plant comprising the steps of:

- crossing a sterile female comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant as described above;
- collecting the hybrid seed;
- optionally detecting hybridity level of the hybrid seeds.

The wheat hybrid plant as obtained by the above methods are also part of the present disclosure.

The present disclosure also relates to a method of identifying a wheat plant as described above, wherein said wheat plant is identified by detecting the presence of at least one restorer allele Rf1 and, optionally, one or more further restorer alleles selected from the group consisting of Rf3, Rf4 and Rf7.

Accordingly, nucleic acid probes or primers for the specific detection of the restorer allele Rf1 in a wheat plant, and, optionally, one or more of the Rf3, Rf4, and Rf7 restorer alleles, are also disclosed herein.

Another aspect of the disclosure relates to a recombinant nucleic acid comprising a Rf1 nucleic acid encoding a Rf1 protein restorer of *T. timopheevii* CMS cytoplasm, operably linked to regulatory elements and the vectors for use in transformation of a wheat plant, comprising such recombinant nucleic acids.

DETAILED DESCRIPTION

Nucleic Acids of the Present Disclosure

An aspect of the present disclosure relates to the cloning and characterization of genes encoding restorer of fertility proteins that act on *T. timopheevii* CMS cytoplasm (hereafter referred as Rf genes or nucleic acids) in wheat plants and the use of the corresponding Rf nucleic acids for producing transgenic wheat plants, for modifying wheat plants by genome editing, and/or for detecting such Rf genes in wheat plants.

Whenever reference to a "plant" or "plants" is made, it is understood that also plant parts (cells, tissues or organs, seed pods, seeds, severed parts such as roots, leaves, flowers, pollen, etc.), progeny of the plants which retain the distinguishing characteristics of the parents (especially, male fertility associated with the claimed Rf nucleic acids), such as seed obtained by selfing or crossing, e.g. hybrid seeds (obtained by crossing two inbred parent plants), hybrid plants and plant parts derived therefrom are encompassed herein, unless otherwise indicated.

As used herein, the term "wheat plant" refers to species of the genus *Triticum* as for example, *T. aestivum, T. aethiopicum, T. araraticum, T. boeoticum, T. carthlicum, T. compactum, T. dicoccoides, T. dicoccon, T. durum, T. ispahanicum, T. karamyschevii, T. macha, T. militinae, T. monococcum, T. polonicum, T. spelta, T. sphaerococcum, T. timopheevii, T. turanicum, T. turgidum, T. urartu, T. vavilovii, T. zhukovskyi* Faegi. Wheat plant also refers to species of the genera *Aegilops* and Triticale.

As used herein, the term "restorer of fertility of *T. timopheevii* CMS cytoplasm" refers to a protein whose expression in a wheat plant comprising *T. timopheevii* CMS cytoplasm contributes to the restoration of the production of pollen in the *Triticum timopheevii* CMS system.

As used herein, the term "allele(s)" means any of one or more alternative forms of a gene at a particular locus. In a diploid, alleles of a given gene are located at a specific location or locus on a chromosome. One allele is present on each chromosome of the pair of homologous chromosomes. The same definition is used for plants bearing a higher level of ploidy like in *Triticum* gender wherein, for example, *T. aestivum* is an hexaploid plant.

As used herein, the term "restorer allele of *T. timopheevii* CMS cytoplasm" refers to an allele which contributes to the restoration of the production of pollen in the CMS *Triticum timopheevii* system.

The restoration of pollen fertility may be partial or complete. The pollen fertility can be evaluated by the pollen fertility tests as described in the Examples below. In particular, the fertility score of F1 wheat plants having CMS-*T timopheevii* cytoplasm (from test restorer line with CMS hybrids) may be calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets and may be compared with the fertility scores of a panel of control fertile plants, for example elite inbred lines bearing a normal wheat cytoplasm, grown in the same area and under the same agro-environmental conditions. It is preferred that such panels of lines comprise a set of at least 5 elite inbred lines wherein these lines are representative of the area where the fertility test is achieved. Besides, it is preferred that at least 10 spikes from different individual F1 plants be assessed for a given experiment.

If the fertility score is not null, then the plant has acquired partial or full restoration of fertility. For each fertility score, a statistical test is calculated to obtain a p-value. Examples of statistical tests are the Anova or mean comparison tests. A p-value below a 5% threshold will indicate that the two distributions are statistically different. Therefore, a significant decrease of the fertility score of the tested wheat plant as compared to the fertility score of the fully fertile control plant is indicative that the F1 plant has not acquired full restoration of fertility (i.e. partial restoration). A similar or higher fertility score is indicative that the F1 plant has acquired full restoration of fertility. In a preferred embodiment, the wheat plant, such as transgenic or genetically engineered wheat plant, according to the present disclosure, has acquired full restoration of fertility.

The loci of the restorer alleles of *T. timopheevii* CMS cytoplasm within Rf1, Rf3, Rf4 and Rf7 have been mapped in the present disclosure. The corresponding restorer alleles are designated Rf1, Rf3, Rf4 and Rf7 restorer alleles and have been described in the art. In particular, a wheat plant source of the Rf3 restorer allele includes the commercial following lines: Allezy, Altigo, Altamira, see table 15. A wheat plant source of the Rf4 restorer allele includes the following lines: R113 or L13.

In specific embodiments, representative alleles of Rf1, Rf3, Rf4 and Rf7 restorer alleles are provided by the seed sample chosen amongst: NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817.

As used herein, the term "centimorgan" ("cM") is a unit of measure of recombination frequency. One cM is equal to a 1% chance that a marker at one genetic locus will be separated from a marker at a second locus due to crossing over in a single generation.

As used herein, the term "chromosomal interval" designates a contiguous linear span of genomic DNA that resides in planta on a single chromosome. The genetic elements or genes located on a single chromosomal interval are physically linked. The size of a chromosomal interval is not particularly limited. In some aspects, the genetic elements located within a single chromosomal interval are genetically linked, typically with a genetic recombination distance of, for example, less than or equal to 20 cM, or alternatively, less than or equal to 10 cM. That is, two genetic elements within a single chromosomal interval undergo recombination at a frequency of less than or equal to 20% or 10%.

The present disclosure provides nucleic acids and their recombinant forms comprising the coding sequence of either Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer of fertility proteins active in *T. timopheevii* CMS cytoplasm.

As used herein, a "recombinant nucleic acid" is a nucleic acid molecule, preferably a DNA molecule, comprising a combination of nucleic acid molecules that would not naturally occur together and is the result of human intervention, e.g., a DNA molecule that is comprised of a combination of at least two DNA molecules heterologous to each other, and/or a DNA molecule that is artificially synthesized and comprises a polynucleotide sequence that deviates from the polynucleotide sequence that would normally exist in nature.

Such nucleic acids encoding candidate restorer of fertility proteins of *T. timopheevii* CMS cytoplasm have been isolated as described in the Examples below. Accordingly, a first aspect of the disclosure are nucleic acids encoding a protein restorer of fertility of *T. timopheevii* having an amino acid sequence at least 95% identical, typically at least 96% identical, to an amino acid sequence chosen amongst any one of SEQ ID NO:1 to SEQ ID NO: 1554.

Percentage of sequence identity as used herein is determined by calculating the number of matched positions in aligned amino acid sequences, dividing the number of matched positions by the total number of aligned amino acids, and multiplying by 100. A matched position refers to a position in which identical amino acids occur at the same position in aligned amino acid sequences. For example, amino acid sequences may be aligned using the CD-hit (settings-c 0.96-n 5-G 0-d 0-AS 60-A 105-g 1, see http://weizhongli-lab.org/cd-hit/).

The above candidate nucleic acids encoding any one of polypeptides SEQ ID NO: 1 to 1554, can further be assessed for their capacity to restore fertility of sterile wheat plant as described below.

It is therefore disclosed herein a method for assessing the capacity of a nucleic acid to restore fertility, wherein the method comprises the steps of:

a. introducing one or more candidate Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid encoding a putative amino acid sequence of at least 95% identical to any one of SEQ ID NO 1 to SEQ ID NO 1554 into a parent wheat sterile plant and *T. timopheevii* CMS cytoplasm,
b. selecting the transgenic plant bearing one or more candidate nucleic acid as transgene(s), and
c. evaluating the fertility of the transgenic plants as compared to the parent wheat sterile plant based on a fertility restoration assay,

wherein an improvement in the fertility restoration is indicative that said nucleic acid has the capacity to restore fertility.

In a specific embodiment, the parent wheat sterile plant is the Fielder line bearing the *T. timopheevii* CMS cytoplasm.

In another specific embodiment of the above method, the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye candidate nucleic acid sequence is selected from those encoding an amino acid sequence having at least 95% identity, or at least 96% identity, for example 100% identity, to any one of SEQ ID NO 1 to SEQ ID NO 1554.

Typically, the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye candidate nucleic acids are selected among the following nucleic acids of SEQ ID NO: 1555 to SEQ ID NO:3107 and 3133.

In a further specific embodiment, where appropriate, the nucleic acid sequence may be optimized for increased expression in the transformed plant. There are a number of optimizations that can be performed at the DNA level, without changing the protein sequence, by conservative codon exchanges which replace one codon by another codon encoding the same amino acid. Besides, the nucleic acid sequence can be modified for cloning purpose. Like for optimization, such modification is achieved without changing the protein sequence.

Rf1 Nucleic Acids

In specific embodiment, the nucleic acid of the present disclosure is a Rf1 nucleic acid.

As used herein, the term "Rf1 nucleic acid" refers to a nucleic acid comprising a gene encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs 1-2, SEQ ID NOs 288-290, SEQ ID NOs 293-296, SEQ ID NOs 343-346, SEQ ID NOs 349-354, SEQ ID NOs 359, 361 and 362, SEQ ID NOs 396 and 397, SEQ ID NOs 428-430, SEQ ID NO 517 and 519, SEQ ID NOs 752-754, SEQ ID NOs 1092, 1093 and 1095, typically, SEQ ID NOs 359, 361 and 362 and SEQ ID NO 428-430. In a particularly preferred embodiment, the Rf1 nucleic acid encodes an amino acid sequence having at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361. Examples of corresponding specific Rf1 nucleic acids are referred to in Table 7.

In particular, and as shown in Example 6, the inventors have identified that RFL79 sequence of SEQ ID NO:361 (as depicted in Table 7) can restore male fertility of CMS-Fielder plants. Accordingly, in a preferred embodiment, examples of Rf1 nucleic acids comprises the disclosed Rf1 nucleic acid sequences of SEQ ID NO:1913, SEQ ID NO: 1914, SEQ ID NO: 1915, SEQ ID NO: 1916 or SEQ ID NO:3119, preferably a Rf1 nucleic acid comprises SEQ ID NO:3119.

Rf3 Nucleic Acids

In specific embodiment, the nucleic acid of the present disclosure is a Rf3 nucleic acid.

As used herein, the term "Rf3 nucleic acid" refers to a nucleic acid comprising a gene encoding a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 124 and 125, SEQ ID NO:147, SEQ ID NO:150, SEQ ID NO: 156, SEQ ID NO: 158, SEQ ID NO:297, SEQ ID NO:299, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 553 and 554, SEQ ID NOs: 557 and 558, SEQ ID NOs: 676 and 677, SEQ ID NOs: 684 and 685, SEQ ID NOs: 696 and 697, SEQ ID NOs: 938 and 939 and SEQ ID NOs: 1038 and 1039, typically, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299. Preferred Rf3 nucleic acids encode a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, with the corresponding amino acid sequence having at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO: 158, SEQ ID NO:676 and SEQ ID NO:684. Examples of corresponding specific Rf3 nucleic acids are referred to in Table 7 or further described in Example 12. Typically, examples of specific Rf3 nucleic acids comprises SEQ ID NO:1712, SEQ ID NO:2230, SEQ ID NO: 2238, SEQ ID NO:3146, SEQ ID NO:3147 or SEQ ID NO:3148.

Rf4 Nucleic Acids

In specific embodiment, the nucleic acid of the present disclosure is a Rf4 nucleic acid.

As used herein, the term "Rf4 nucleic acid" refers to a nucleic acid comprising a gene encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:477, and SEQ ID NOs 3135-3138, typically SEQ ID NO:477 and SEQ ID NOs 3136-3138. Examples of corresponding specific Rf4 nucleic acids are listed in Table 7 and further include any of SEQ ID NO: 2031, and SEQ ID NO:3140-3142.

Rf7 Nucleic Acids

In specific embodiment, the nucleic acid of the present disclosure is a Rf7 nucleic acid.

As used herein, the term "Rf7 nucleic acid" refers to a nucleic acid comprising a gene encoding a Rf7 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 240-243, SEQ ID NOs 303-305, SEQ ID NO:363, SEQ ID NOs 375-377, SEQ ID NOs 497-499, SEQ ID NO:516, SEQ ID NOs 709-711, SEQ ID NO: 768, typically SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768. Examples of corresponding specific Rf7 nucleic acids are referred to in Table 7.

Rf-Rye Nucleic Acids

In specific embodiment, the nucleic acid of the present disclosure is a Rf-rye nucleic acid.

As used herein, the term "Rf-rye nucleic acid" refers to a nucleic acid comprising a gene encoding a Rf-rye protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably, 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859. Examples of corresponding specific Rf-rye nucleic acids are referred to in Table 7.

Rf Nucleic Acids as Transgene

The present disclosure more specifically relates to DNA molecules including one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids. In particular, the disclosure relates to any DNA molecule resulting from the insertion of a transgene in the wheat plant, said transgene including one or more of the above described Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, and which insertion results in the expression of corresponding RNA and/or protein in the wheat plant.

Also part of the present disclosure is a nucleic acid that has been extracted from cells, or tissues, or homogenate from a plant or seed or plant tissue; or can be produced as an amplicon from extracted DNA or RNA from cells, or tissues, or homogenate from a plant or seed or plant tissue, any of which is derived from such materials derived from a plant comprising such nucleic acid as disclosed above.

As used herein, the term "transgene" or "transgenic element" refers to the nucleic acid (e.g. DNA molecule) incorporated into a host cell's genome. The term "transgene" or 'transgenic element" refers in particular to a sequence that is not normally present in a given host genome in the genetic context in which the sequence is currently found. In this respect, the sequence may be native to the host genome, but be rearranged with respect to other genetic sequences within the host genomic sequence. For example, the transgene is rearranged at a different locus as compared to the native gene.

Said one or more transgenic element(s) enables the expression of polypeptides which restore or improve male fertility to the plant having *T. timopheevii* CMS cytoplasm, as compared to the parent plant which do not comprise the transgenic element.

A particular transgenic element is the recombinant nucleic acid as defined above, for example Rf1 nucleic acids as defined above. In specific embodiments, a transgenic element includes an Rf nucleic acid under the control of a constitutive promoter, such as the ZmUbi promoter.

Recombinant Nucleic Acids for Use in Transforming Wheat Plants

Such Rf nucleic acids as defined above are also useful to transform or genetically modify wheat plant, in particular wheat plant which does not have one or more of the fertility restoration alleles Rf1, Rf3, Rf4, Rf7 and Rf-rye.

Another aspect of the present disclosure relates to a vector for use in transforming wheat plant, comprising one or more of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic acids as described above.

Vectors for use in transforming wheat plant includes at least the coding sequence of the corresponding protein restorer of fertility (either naturally occurring coding sequence, or improved sequence, such as codon optimized sequence), such coding sequence being operably linked to a regulatory element such as a promoter.

The term "promoter" as used herein refers to a region of DNA upstream of the coding sequence (upstream of start codon) and including DNA regions for recognition and binding of RNA polymerase and other proteins to initiate transcription at the start codon. Examples of constitutive promoters useful for expression include the 35S promoter or the 19S promoter (Kay et al, 1987), the rice actin promoter (McElroy et al, 1990), the pCRV promoter (Depigny-This et al, 1992), the CsVMV promoter (Verdaguer et al. 1996), the ubiquitin 1 promoter of maize (Christensen and Quail, 1996), the regulatory sequences of the T-DNA of *Agrobacterium tumefaciens*, including mannopine synthase, nopaline synthase, octopine synthase.

Promoters may be «tissue-preferred», i.e. initiating transcription in certain tissues or "tissue-specific", i.e. initiating transcription only in certain tissues. Examples of such promoters are DHN12, LTR1, LTP1 specific of the embryo, SS1 specific of the phloem, OSG6B specific of the tapetum (Gotz et al 2011 and Jones 2015).

Other suitable promoters could be used. It could be an inducible promoter, a developmentally regulated promoter. An "inducible" promoter initiates transcription under some environmental control or any stress-induced like for example the abiotic stress-induced RD29, COR14b (Gotz et al, 2011).

Constitutive promoters may be used, such as the ZmUbi promoter, typically the ZmUbi promoter of SEQ ID NO:3134. Finally, promoter of SEQ ID NO:3114, SEQ ID NO:3123 and SEQ ID NO:3113 corresponding to pTaRFL46, 79 and 104 can also be used.

In specific embodiments, the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids of the present disclosure are operably linked to heterologous promoters, i.e. a promoter which is not the natural promoter of the corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as found in wheat. Typical recombinant constructs of Rf3 nucleic acids with heterologous promoters include any one of SEQ ID NO:3150-3153, and any one of SEQ ID NO: 3156-SEQ ID NO:3159. Typical recombinant constructs of Rf1 nucleic acids with heterologous promoters includes SEQ ID NO:3122, or a nucleic acid of SEQ ID NO: 3119 under the regulation of the promoter of SEQ ID NO:2123.

The vector may further comprise additional elements including selection gene marker, operably linked to regulatory element, that allows to select the transformed plant cells containing the vector, comprising the nucleic acids of the present disclosure as transgene.

The vector may further comprise additional elements including counter-selection gene marker, operably linked to regulatory element that allows to counter-select the transformed plant cells which do not have maintained the counter-selection gene marker in its genome.

In a specific embodiment, the vector according to the present disclosure may be vector suitable for *Agrobacterium*-mediated transformation, in particular *Agrobacterium tumefaciens* or *Agrobacterium rhizogenes* mediated transformation, as described in the next section.

Methods for Producing a Wheat Transgenic Plant

Another aspect of the present disclosure relates to the use of the above-described nucleic acids for producing wheat transgenic plant expressing protein restorer of fertility.

The term "transgenic plant" refers to a plant comprising such a transgene. A "transgenic plant" includes a plant, plant part, a plant cell or seed whose genome has been altered by the stable integration of recombinant DNA. A transgenic plant includes a plant regenerated from an originally-transformed plant cell and progeny transgenic plants from later generations or crosses of a transformed plant. As a result of such genomic alteration, the transgenic plant is distinctly different from the related wild type plant. An example of a transgenic plant is a plant described herein as comprising one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, typically as transgenic elements. For example, the transgenic plant includes one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as transgene, inserted at loci different from the native locus of the corresponding Rf gene(s). Accordingly, it is herein disclosed a method for producing a wheat transgenic plant, wherein the method comprises the steps of

(i) transforming a parent wheat plant with Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids,
(ii) selecting a plant comprising said one or more nucleic acid(s) as transgene(s),
(iii) regenerating and
(iv) growing said wheat transgenic plant.

For transformation methods within a plant cell, one can cite methods of direct transfer of genes such as direct micro-injection into plant embryos, vacuum infiltration or electroporation, direct precipitation by means of PEG or the bombardment by gun of particles covered with the plasmidic DNA of interest.

It is preferred to transform the plant cell with a bacterial strain, in particular *Agrobacterium*, in particular *Agrobacterium tumefaciens*. In particular, it is possible to use the method described by Ishida et al. (Nature Biotechnology, 14, 745-750, 1996) for the transformation of monocotyledons.

Descriptions of *Agrobacterium* vector systems and methods for *Agrobacterium*-mediated gene transfer are provided by Moloney et al., Plant Cell Reports 8:238 (1989). See also, U.S. Pat. No. 5,591,616 issued Jan. 7, 1997.

Alternatively, direct gene transfer may be used. A generally applicable method of plant transformation is microprojectile-mediated transformation wherein DNA is carried on the surface of microprojectiles measuring 1 to 4 micron. The expression vector is introduced into plant tissues with a biolistic device that accelerates the microprojectiles to speeds of 300 to 600 m/s which is sufficient to penetrate plant cell walls and membranes. Sanford et al., Part. Sci. Technol. 5:27 (1987), Sanford, J. C., Trends Biotech. 6:299 (1988), Klein et al., BioTechnology 6:559-563 (1988), Sanford, J. C., Physiol Plant 7:206 (1990), Klein et al., BioTechnology 10:268 (1992). Several target tissues can be bombarded with DNA-coated microprojectiles in order to produce transgenic plants, including, for example, callus (Type I or Type II), immature embryos, and meristematic tissue.

Following transformation of wheat target tissues, expression of the selectable marker genes allows for preferential selection of transformed cells, tissues and/or plants, using regeneration and selection methods now well known in the art.

The foregoing methods for transformation would typically be used for producing a transgenic plant including one or more of Rf1, Rf3, Rf4, Rf7 or Rf Rye nucleic acids as transgenic element(s).

The transgenic plant could then be crossed, with another (non-transformed or transformed) inbred line, in order to produce a new transgenic line. Alternatively, a genetic trait which has been engineered into a particular line using the foregoing transformation techniques could be moved into another line using traditional backcrossing techniques that are well known in the plant breeding arts. For example, a backcrossing approach could be used to move an engineered trait from a public, non-elite inbred line into an elite inbred line, or from an inbred line containing a foreign gene in its genome into an inbred line or lines which do not contain that gene. As used herein, "crossing" can refer to a simple X by Y cross, or the process of backcrossing, depending on the context.

When the term transgenic wheat plant is used in the context of the present disclosure, this also includes any wheat plant including, as a transgenic element one or more of Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids and wherein one or more desired traits have further been introduced through backcrossing methods, whether such trait is a naturally occurring one or a transgenic one. Backcrossing methods can be used with the present invention to improve or introduce one or more characteristic into the inbred. The term backcrossing as used herein refers to the repeated crossing of a hybrid progeny back to one of the parental wheat plants. The parental wheat plant which contributes the gene or the genes for the desired characteristic is termed the nonrecurrent or donor parent. This terminology refers to the fact that the nonrecurrent parent is used one time in the backcross protocol and therefore does not recur. The parental wheat plant to which the gene or genes from the nonrecurrent parent are transferred is known as the recurrent parent as it is used for several rounds in the backcrossing protocol (Fehr et al, 1987).

In a typical backcross protocol, the recurrent parent is crossed to a second nonrecurrent parent that carries the gene or genes of interest to be transferred. The resulting progeny from this cross are then crossed again to the recurrent parent and the process is repeated until a wheat plant is obtained wherein all the desired morphological and physiological characteristics of the recurrent parent are recovered in the converted plant in addition to the gene or genes transferred from the nonrecurrent parent. It should be noted that some, one, two, three or more, self-pollination and growing of a population might be included between two successive backcrosses.

Methods for Producing a Wheat Genetically Engineered Plant

An aspect of the present disclosure relates to a DNA fragment of the corresponding protein restorer of fertility (either naturally occurring coding sequence, or improved sequence, such as codon optimized sequence) combined with genome editing tools (such TALENs, CRISPR-Cas, Cpf1 or zing finger nuclease tools) to target the corresponding Rf restorer alleles within the wheat plant genome by insertion at any locus in the genome or by partial or total allele replacement at the corresponding locus. In particular, the disclosure relates to a genetically modified (or engineered) wheat plant, wherein the method comprises the steps of genetically modifying a parent wheat plant to obtain in their genome one or more nucleotide sequence encoding protein restorer of *T. timopheevii* CMS cytoplasm Rf1, Rf3, Rf4 or Rf7 as disclosed herein, preferably by genome-editing, selecting a plant comprising said one or more nucleotide sequences as genetically engineered elements, regenerating and growing said wheat genetically engineered plant.

As used herein, the term "genetically engineered element" refers to a nucleic acid sequence present in the genome of a plant and that has been modified by mutagenesis or by genome-editing tools, preferentially by genome-editing tools. In specific embodiments, a genetically engineered element refers to a nucleic acid sequence that is not normally present in a given host genome in the genetic context in which the sequence is currently found but is incorporated in the genome of plant by use of genome-editing tools. In this respect, the sequence may be native to the host genome, but be rearranged with respect to other genetic sequences within the host genomic sequence. For example, the genetically engineered element is a gene that is rearranged at a different locus as compared to a native gene. Alternatively, the sequence is a native coding sequence that has been placed under the control of heterologous regulatory sequences. Other specific examples are described hereafter. The term "genetically engineered plant" or "genetically modified plant" refers to a plant comprising such genetically engineered element. A "genetically engineered plant" includes a plant, plant part, a plant cell or seed whose genome has been altered by the stable integration of recombinant DNA. As used herein, the term "genetically engineered plant" further includes a plant, plant part, a plant cell or seed whose genome has been altered by genome editing techniques. A genetically engineered plant includes a plant regenerated from an originally-engineered plant cell and progeny of genetically engineered plants from later generations or crosses of a genetically engineered plant. As a result of such genomic alteration, the genetically engineered plant is distinctly different from the related wild type plant. An example of a genetically engineered plant is a plant described herein as comprising one or more of the Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids. For example, the genetically engineered plant includes one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids as genetically engineered elements, inserted at loci different from the native locus of the corresponding Rf gene(s).

In specific embodiments, said genetically engineered plants do not include plants which could be obtained exclusively by means of an essentially biological process.

Said one or more genetically engineered element(s) enables the expression of polypeptides which restore or improve male fertility to the plant having *T. timopheevii* CMS cytoplasm, as compared to the parent plant which do not comprise the genetically engineered element(s).

A particular genetically engineered element is the Rf nucleic acid as defined above, for example Rf1 nucleic acids as defined above. In specific embodiments, a genetically engineered element includes an Rf nucleic acid under the control of expression elements as promoter and/or terminator. Suitable promoter can be a constitutive promoter, such as the ZmUbi promoter or an endogenous Rf promoter native or modified.

Another aspect of the disclosure relates to a genetically engineered wheat plant, which comprises the modification by point mutation insertion or deletion of one or few nucleotides of an allele sequence rf or Rf, as genetically engineered element, into the respectively Rf or rf allele, by any of the genome editing tools including base-editing tool as described in WO2015089406 or by mutagenesis.

The present disclosure further includes methods for modifying fertility level in a plant by genome editing, comprising providing a genome editing tool capable of replacing partially or totally a rf1, rf3, rf4, rf7 or rf-rye non-restorer allele sequence or form in a wheat plant by its corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer allele sequence as disclosed herein.

The term "rf non-restorer allele sequence or form" can be related to the presence in the genome of a rf non-restorer allele or to the absence in the genome of any rf or Rf allele. For example, in the rf3/Rf3 system, one wheat non-restorer line can be characterized by the presence of a rf3 allele sequence while another non-restorer line is characterized by the absence of any Rf3 or rf3 allelic sequence. The Rf3 restorer plant will be characterized by the presence of the Rf3 allele sequence in the genome. In the case of the rf4/Rf4 system, the non-restorer plant is characterized by the absence of any rf4 or Rf4 allelic forms while the Rf4 restorer plant will be characterized by the presence in the genome of the Rf4 gene sequence.

In specific embodiments, methods for modifying fertility level in a plant by genome editing comprises providing a genome editing tool capable of replacing or modifying a rf3 non-restorer allele to obtain a Rf3 restorer allele comprising SEQ ID NO:3146 (RFL29a). In other specific embodiments, rf3 non-restorer allele may comprise RFL29c characterized by a frameshift compared to RFL29a nucleotide sequence as shown in Example 22 and FIG. **12** and SEQ ID NO:3457.

The disclosure further includes methods for modifying fertility level in a plant introducing the endogenous promoter of a restorer Rf gene in order to increase the expression of the corresponding endogenous Rf gene either by genome-editing or by mutagenesis.

In a specific embodiment, the disclosure includes methods for modifying fertility level in a plant by genome editing of a weak fertile plant by modifying the 5'UTR sequence of the Rf3 "weak" RFL29b allele which 5'UTR region includes a 163 bp insertion to be deleted as shown in example 15.

In a further specific embodiment, the invention includes a method for modifying fertility level in a plant by mutagenesis or by genome editing the endogenous promoter of a restorer Rf gene in order to increase the expression of the endogenous Rf gene. As an example, the sequence of proTaRFL79 depicted in SEQ ID N°3123 could be mutated or edited to increase RFL79 protein level.

In another specific embodiment, whenever a stronger promoter is located upstream to the promoter of the Rf restorer gene, a deletion of that promoter and the region upstream can be achieved in order to juxtapose the stronger promoter to the Rf gene.

In a specific embodiment, one rf non-restorer allele is replaced partially or totally by anyone of the Rf1, Rf3, Rf4, Rf7 or Rf-rye restorer allele. In such disclosure, a non-restorer rf1 allele could be replaced, for example, by a Rf3, or a Rf4, or a Rf7, or Rf-rye allele.

In another aspect of the disclosure, at least one restorer allele from among Rf1, Rf3, Rf4, Rf7 and Rf-Rye can be integrated at one or more target sites in the wheat plant genome, typically in order to get an expression of said restorer alleles. Said expression can be achieved either by taking advantage of the presence at the targeting locus of a promoter, more specifically a strong promoter, and/or a terminator and by targeting with the Rf allele said expression elements. In a specific embodiment, the endogenous Rf allele is deleted from one first locus and further integrated downstream a suitable promoter at a second locus in the same plant genome.

In a specific embodiment of the invention, the target sites can be located in the Rf1, Rf3, Rf4 and/or Rf7 locus as defined in example 17. In a more specific embodiment, the target sites can either be the Rf1, Rf3, Rf4, Rf7 or Rf-Rye endogenous gene sequence or any other target sites different from the Rf1, Rf3, Rf4, Rf7 or Rf-Rye endogenous gene sequences.

In a preferred aspect of the disclosure, a wheat plant can comprise in its genome, at only one locus, the Rf1, Rf3 and Rf7 restorer alleles.

Such genome editing tool includes without limitation targeted sequence modification provided by double-strand break technologies such as, but not limited to, meganucleases, ZFNs, TALENs (WO2011072246) or CRISPR CAS system (including CRISPR Cas9, WO2013181440), Cpf1 or their next generations based on double-strand break technologies using engineered nucleases.

Method for Decreasing the Fertility Level in Wheat Plant:

Alternatively, the present disclosure further includes methods for modifying fertility level in a plant by reverting a restorer line comprising a Rf allele to a maintainer line comprising a rf allele. Such method could be used for any plant comprising a Rf allele, including Rf1. It is of particular interest in the case of a hybrid production system based on Rf3 restorer allele, wherein for example, the Rf3 sequence is RFL29a and the rf3 sequence is RFL29c.

The decrease in fertility could be obtained by knocking-down Rf gene or allele as described below.

More specifically, the method can correspond first to an inhibition of the expression by RNAi directed against the Rf allele of interest or by impairing the promoter function of the Rf allele either by mutagenesis or by genome editing.

In another aspect, the fertility level decrease is obtained by mutagenesis, classically induced with mutagenic agents, or by genome editing technologies to impair the protein function, by deleting totally or partially the gene, or by modifying the gene sequence reading frame to impair protein translation.

In a further aspect, the disclosure relates to the transgenic or genetically engineered wheat plant with decreased fertility level obtained by the methods described above.

The Transgenic or Genetically Engineered Wheat Plant of the Present Disclosure

Another aspect of the present disclosure relates to a transgenic or genetically engineered wheat plant comprising one or more Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acid(s) as described in the previous sections, as transgenic or genetically engineered elements respectively.

Said transgenic or genetically engineered plant may be obtained by the methods described in the previous section.

The transgenic or genetically engineered plants of the present disclosure may advantageously be used as parent plant in order to produce fertile wheat transgenic or genetically engineered plant restorer of *T. timopheevii* CMS cytoplasm. In particular, in specific embodiment, the wheat transgenic or genetically engineered plant is a fertile wheat transgenic or genetically engineered plant restorer of *T. timopheevii* CMS cytoplasm. Typically, a transgenic or genetically engineered wheat plant according to the present disclosure comprises a combination of at least two different transgenic or genetically engineered elements selected from the group consisting of Rf1, Rf3, Rf4, Rf7 and Rf-rye encoding nucleic acids.

The transgenic or genetically engineered wheat plant as disclosed herein may express such protein restorer of fertility Rf1, Rf3, Rf4, Rf7 and/or Rf-Rye together as the result of the transgene's expression or expression of the genetically engineered element and may additionally express other protein restorer of fertility, as the result of naturally occurring alleles.

In one embodiment, said combination of at least two, three or four of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic is found in the same locus in the genome of the transgenic or genetically engineered plant. In other embodiments, the corresponding nucleic acids of the combination are located in distinct loci. In one specific embodiment, said combination may be obtained by crossing transgenic plants of the present disclosure each bearing one nucleic acid of the combination as a transgene at a distinct locus.

Typically, said transgenic or genetically engineered plant includes the following combination of nucleic acids as transgenic or genetically engineered elements:

a. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96%

identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684,

b. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768,

c. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity, with any one of SEQ ID NO: 227, SEQ ID NO:378 and SEQ ID NO:859,

d. a Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768,

e. a Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859,

f. a Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859,

g. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO: 684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768;

h. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO:361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO: 684, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity, or at least 96% identity, with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859;

i. a Rf1 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity, for example at least 96% identity with any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, preferably SEQ ID NO: 361, typically a Rf1 nucleic acid comprises SEQ ID NO:3119, and Rf7 nucleic acid of claim **4**, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity, with any one of SEQ ID NO: 227, SEQ ID NO:378 and SEQ ID NO:859; or,

j. a Rf3 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid, preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid, preferably preferably encoding an amino acid sequence having at least 95% identity or at least 96% identity with any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859.

In other specific embodiments, said transgenic or genetically engineered plant further includes a Rf4 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 477, and SEQ ID NOs 3135-3138 in addition to any one of the above defined combination of Rf1, Rf3, Rf7 and/or Rf-Rye nucleic acids as defined in the previous paragraph, as transgenic elements.

The disclosure also relates to hybrid wheat plants which can be produced by crossing a transgenic or genetically engineered wheat plant restorer of fertility according to the present disclosure as described above with a second plant.

In certain embodiments, the wheat plant according to the disclosure is alloplasmic and comprises the *T. timopheevii* cytoplasm.

For example, a hybrid wheat plant may be obtained by crossing a wheat plant restorer of fertility according to the present disclosure as described above, preferably comprising Rf1, Rf3, Rf4, Rf7 or Rf-rye nucleic acids, and a wheat plant which does not express corresponding Rf1, Rf3, Rf4, Rf7 or Rf-rye protein restorer of fertility.

It is also disclosed herein a method for producing a wheat hybrid transgenic or genetically engineered plant comprising the steps of:

a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male transgenic or genetically engineered wheat plant restorer of fertility of the present disclosure as described above;
b. collecting the hybrid seed;
c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least one or more of the Rf nucleic acids chosen amongst Rf1, Rf3, Rf4, Rf7 and Rf-rye in the hybrid seed, as transgenic elements or genetically engineered elements; and,
d. optionally detecting hybridity level of the hybrid seed.

Therefore, it is also disclosed herein the wheat transgenic or genetically engineered plants or lines according to the present disclosure developed to obtain such hybrid plants. Such transgenic or genetically engineered plants or lines typically comprise the cytoplasmic elements necessary for the implementation of the corresponding hybrid system. Preferably, the transgenic or genetically engineered plants or lines comprise a combination of at least two, three or four of Rf1, Rf3, Rf4, Rf7 and Rf-rye nucleic acids, and *T. timopheevii* cytoplasm.

Alternatively, the detection of the presence of *T. timopheevii* cytoplasm and of at least one or more of the Rf nucleic acids chosen amongst Rf1, Rf3, Rf4, Rf7 and Rf-rye (step "c" of the method described above) can be performed on the parent lines in order to check their genotype before to start the cross (step "a").

In a certain embodiment of the disclosure, the male wheat plant is taller than the female wheat plant. This can be achieved by using male plant bearing Rht alleles that allows to obtain the size differences. Optionally, the disclosure further comprises the step of to applying an herbicide to the fertile plants standing above the height of the shorter female plants, and further optionally, comprises the step of harvesting the seeds and selecting the seeds to remove undesirable self-fertilized male seeds, using a morphological character and/or a phenotypic character like size, shape, color etc. . . . An example of such method for hybrid production is described in WO2015135940.

The T-CMS cytoplasm can be detected either phenotypically wherein a plant bearing rf genes and a T-CMS cytoplasm will be sterile or by molecular means able to detect the orf256 gene as described in Rathburn and Hedgcoth, 1991 and Song and Hedgcoth, 1994.

The disclosure also relates to a method for improving the level of fertility restoration of a parent wheat plant bearing a fertility level lower than a full restoration level, comprising the steps of transforming said parent wheat plant with a vector comprising a Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid as described above, or genetically engineering said Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids in said wheat plant. The method further comprises the step of selecting a transgenic or genetically engineered wheat plant comprising said Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid(s) as a transgene or genetically engineered element, preferably Rf1, Rf3 and Rf4, regenerating and growing said transgenic or genetically engineered wheat plant, wherein said transgenic or genetically engineered wheat plant has an improved fertility restoration level as compared to the parent plant.

The disclosure further provides a method for restoring fertility of a sterile wheat plant bearing the *T. timopheevii* CMS cytoplasm, comprising the step of transforming a parent sterile wheat plant bearing the *T. timopheevii* CMS cytoplasm, with a Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acid as described above or genetically engineering said sterile plant to express a Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

Use of the Nucleic Acids of the Present Disclosure for Identifying Rf1, Rf3, Rf4, Rf7 and Rf-Rye Restorer Alleles or Transgenic Elements

The present disclosure further provides methods of identifying the respective Rf1, Rf3, Rf4, Rf7 and/or Rf-rye restorer alleles and/or Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids as disclosed in the previous sections, and more generally methods of selecting or breeding wheat plants for the presence or absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye fertility restorer alleles and/or corresponding Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

Such methods of identifying, selecting or breeding wheat plants comprise obtaining one or more wheat plants and assessing their DNA to determine the presence or absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye fertility restorer alleles and/or corresponding Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

Such methods may be used, for example, to determine which progeny resulting from a cross have the required fertility restorer allele or Rf nucleic acids (or combination thereof) and accordingly to guide the preparation of plants having the required fertility restorer allele or Rf nucleic acids in combination with the presence or absence of other desirable traits.

The method will consist of identifying the presence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye allele or nucleic acids in the fertile plant, said plant being either a fertile transgenic plant or a non-transgenic plant. Optionally, the method further consists of identifying the absence of the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye allele or nucleic acids in non-restorer plants and/or sterile plants.

Accordingly, it is disclosed herein the means for specifically detecting the Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids in a wheat plant.

Such means include for example a pair of primers for the specific amplification of a fragment nucleotide sequence of Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids from plant wheat genomic DNA.

As used herein, a primer encompasses any nucleic acid that is capable of priming the synthesis of a nascent nucleic acid in a template-dependent process, such as PCR. Typically, primers are oligonucleotides from 10 to 30 nucleotides, but longer sequences can be employed. Primers may be provided in double-stranded form though single-stranded form is preferred.

Alternatively, nucleic acid probe can be used for the specific detection of any one of Rf1, Rf3, Rf4, Rf7 and/or Rf-rye nucleic acids.

As used herein, a nucleic acid probe encompass any nucleic acid of at least 30 nucleotides and which can specifically hybridizes under standard stringent conditions with a defined nucleic acid. Standard stringent conditions as used herein refers to conditions for hybridization described for example in Sambrook et al 1989 which can comprise 1) immobilizing plant genomic DNA fragments or library DNA on a filter 2) prehybridizing the filter for 1 to 2 hours at 65° C. in 6×SSC 5×Denhardt's reagent, 0.5% SDS and 20 mg/ml denatured carrier DNA 3) adding the probe (labeled) 4) incubating for 16 to 24 hours 5) washing the filter once for 30 min at 68° C. in 6×SSC, 0.1% SDS 6) washing the filter three times (two times for 30 min in 30 ml and once for 10 min in 500 ml) at 68° C. in 2×SSC 0.1% SDS. The nucleic acid probe may further comprise labeling agent, such as fluorescent agents covalently attached to the nucleic acid part of the probe.

Methods of Producing a Wheat Plant Carrying a Modified Rf3 Restorer of Fertility

The inventors have also identified two types of Rf3 restorer of fertility, one with a strong fertility restoration, for example, capable of providing plants with a fertility score above 1.0, for example comprised between 1.0 and 2.0 and another with a weak fertility restoration, for example, capable of providing plants having a fertility score below 0.1, for example comprised between between 0.5 and 1.0.

Surprisingly, the strong fertility restoration correlates with the absence in the genome of the wheat plant carrying Rf3 of a 163 bp fragment of SEQ ID NO:3174, located in the 5'UTR of Rf3 coding sequence.

Accordingly, the disclosure relates to a method for producing a wheat plant carrying a Rf3 restorer of fertility, said method comprising (i) providing a parent wheat plant comprising in its genome at least a 163 bp fragment of SEQ ID NO:3174, and (ii) deleting a fragment of at least 10 bp of said fragment of SEQ ID NO:3174, for example at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO:3174 in the genome of said wheat plant, thereby obtaining said wheat plant carrying a Rf3 restorer of fertility.

Advantageously, the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant, due to the deletion of said fragment depicted in SEQ ID NO:3174. The skilled person may select the deletion such as to obtain an increase in fertility restoration as compared to the parent wheat plant with the full fragment of SEQ ID NO:3174 in its genome.

In specific embodiments, the parent wheat plant has a fertility score below 1, for example comprised between 0.5 and 1.0 and the obtained wheat plant has a fertility score above 1.0, for example comprised between 1.0 and 2.0.

It is also possible to restore fertility from plants having rf3 non-restorer allele presenting a frameshift due to nucleotides deletion or insertion as compared to the Rf3 restorer allele RLF29a sequence of SEQ ID NO:3146 (see also Example 22).

Such deletion of genomic fragment or correction of frameshift may be obtained by any suitable methods known by the skilled person in the art, including genome editing tools such as, but not limited to, meganucleases, ZFNs, TALENs (WO2011072246) or CRISPR CAS system (including CRISPR Cas9, WO2013181440) or their next generations based on double-strand break technologies using engineered nucleases. Examples of such methods are also described in Example 15 and Example 22.

The wheat plants as obtained by the method described above are also part of the disclosure. Typically, such wheat plant as obtained by the above method, or obtainable by such method, carries a Rf3 restorer of fertility, wherein only a part but not the whole genomic fragment of SEQ ID NO:3174 is deleted in the genome of said wheat plant. Typically, a fragment between 10 bp and 162 bp of SEQ ID NO:3174 is deleted in the genome of said wheat plant. It is expected that the obtained wheat plant with the genome deletion has a fertility score higher than the fertility score as measured in a parent wheat plant with identical genome except for the full sequence of SEQ ID NO: 3174 in its genome. Alternatively, such wheat plant as obtained by the above method, or obtainable by such method, carries a Rf3 restorer of fertility, wherein nucleotides of RFL29c has been deleted or added to restore inframe translation.

Methods for Assessing Fertility Restoration in a Wheat Plant

The disclosure also includes a method for assessing fertility restoration in a wheat plant, said method comprising determining the presence or absence of a fragment of SEQ ID NO:3174 in the genome of said plant, wherein the presence of the whole fragment is indicative of a weak restoration of fertility and a deletion of at least a part of such fragment is indicative of a strong restoration of fertility. Typically said method is performed in a wheat plant which is susceptible to carry a Rf3 restorer of fertility.

It is also disclosed herein nucleic acid probes for use in the above methods for assessing fertility restoration in wheat plant, wherein said nucleic acid probe consists of a nucleic acid of at least 10 nucleotides within SEQ ID NO:3174.

Typically, said nucleic acid probe is a fragment of at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO: 3174.

The Wheat Plant Restorer of Fertility of *T. timopheevii* CMS Cytoplasm with at Least Three Specific Fertility Restorer Alleles

The inventors have shown that a combination of at least 3 specific fertility restorer alleles within the restorer loci Rf1, Rf3, Rf4 and Rf7 enable the obtention of plants with full restoration of fertility of *T. timopheevii* CMS cytoplasm.

Therefore, a first aspect of the present disclosure relates to a wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the plant comprises at least three fertility restorer alleles within the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7.

In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf3, Rf4.

In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf4, Rf7.

In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf1, Rf3, Rf7.

In specific embodiments, the plant comprises at least the three fertility restorer alleles Rf3, Rf4, Rf7.

As used herein, the Rf1 locus refers to the locus of the Rf1 restorer allele, which locus is located at most 10 cM, preferably at most 7 cM, more preferably at most 2 cM, from marker cfn0522096 of SEQ ID NO:4 and/or from marker cfn05277067 of SEQ ID NO: 10. In a specific embodiment, the wheat plant restorer of fertility according to the present disclosure includes at least one Rf1 restorer allele, said Rf1 restorer allele being located within the chromosomal interval between SNP markers cfn0522096 of SEQ ID NO:3190 and cfn05277067 of SEQ ID NO:3196. In specific embodiments, the wheat plant restorer of fertility includes one Rf1 restorer allele at the Rf1 locus characterized by the presence of one or more of the SNP allele(s) as identified by Table 1.

TABLE 1

SNP markers for mapping of Rf1 locus

SNP#	Marker Name	Marker SEQ ID NO:	Restorer Allele
SNP1	cfn523072	3187	T
SNP2	cfn0523109	3188	A
SNP3	276I13_96B22_97797	3189	C
SNP4	cfn0522096	3190	C
SNP5	cfn0527763	3191	C
SNP6	104A4_105172	3192	TG

TABLE 1-continued

SNP markers for mapping of Rf1 locus			
SNP#	Marker Name	Marker SEQ ID NO:	Restorer Allele
SNP7	104A4_105588	3193	A
SNP8	cfn0373248	3194	T
SNP9	cfn1097828	3195	C
SNP10	cfn0527067	3196	A
SNP11	cfn0528390	3197	G
SNP12	BWS0267	3198	A
SNP13	cfn0527718	3199	T
SNP14	cfn0524469	3200	G
SNP15	cfn0524921	3201	G
SNP16	cfn1122326	3202	C

Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf1 restorer allele at the Rf1 locus characterized by the presence of the SNP3 and/or SNP7 restorer alleles(s) as described in Table 1. More preferably, the wheat plant restorer of fertility is characterized by the haplotypes of the SNP3 and SNP7 restorer alleles "C" and "A". In specific embodiments, the wheat plant restorer of fertility with Rf1 restorer allele comprises a Rf1 nucleic acid of the present disclosure as described above. Examples of Rf1 nucleic acids comprises the disclosed Rf1 nucleic acid sequences of SEQ ID NO: 1913, SEQ ID NO: 1914, SEQ ID NO: 1915, SEQ ID NO: 1916 or SEQ ID NO:3119, preferably a Rf1 nucleic acid comprises SEQ ID NO: 3119.

As used herein, the Rf3 locus refers to the locus of the Rf3 restorer allele, which locus is at most 10 cM, preferably at most 7 cM, more preferably at most 2 cM, from marker cfn1249269 of SEQ ID NO:3205 and/or from marker BS00090770 of SEQ ID NO:3228. In a specific embodiment, the wheat plant restorer of fertility includes at least one Rf3 restorer allele within the Rf3 locus, said Rf3 restorer allele being located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770. In specific embodiment, the wheat plant restorer of fertility includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 2.

TABLE 2

SNP Markers for mapping of Rf3 locus			
SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP17	cfn1252000	3203	A
SNP18	IWB14060*	3204	G
SNP19	cfn1249269	3205	G
SNP20	219K1_166464	3206	T
SNP21	219K1_158251	3207	G
SNP22	219K1_111446	3208	A
SNP23	219K1_110042	3209	T
SNP24	219K1_110005	3210	C
SNP25	219K1_107461	3211	A
SNP26	219K1_99688	3212	T
SNP27	219K1_37	3213	C
SNP28	cfn1270524	3214	T
SNP29	136H5_3M5_7601	3215	T
SNP30	cfn1288811	3216	G
SNP31	136H5_3M5_89176	3217	A
SNP32	136H5_3M5_89263	3218	T
SNP33	136H5_3M5_138211	3219	T
SNP34	cfn0556874	3220	C
SNP35	136H5_3M5_64154	3221	C
SNP36	136H5_3M5_68807	3222	G
SNP37	136H5_3M5_77916	3223	A
SNP38	cfn1246088	3224	A

TABLE 2-continued

SNP Markers for mapping of Rf3 locus			
SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP39	cfn1287194	3225	G
SNP40	cfn1258380	3226	A
SNP41	IWB72107*	3227	A
SNP42	BS00090770	3228	T
SNP43	cfn1239345	3229	A

Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of the SNP29 and/or SNP31 restorer alleles(s) as described in Table 2. More preferably, the wheat plant restorer of fertility is characterized by the haplotype of the SNP29 and SNP31 restorer alleles "T" and "A" respectively.

In another particular embodiment, that may be combined with the previous embodiments, the wheat plant restorer of fertility according to the present disclosure includes one Rf3 restorer allele at the Rf3 locus characterized by the presence of the SNP38 and SNP41 restorer alleles "A" and "A" respectively.

Preferably, the wheat plant restorer of fertility according to the present disclosure includes a Rf3 nucleic acid comprising SEQ ID NO:1712, SEQ ID NO:2230, SEQ ID NO: 2238, SEQ ID NO:3146, SEQ ID NO:3147 or SEQ ID NO:3148, preferably SEQ ID NO: 3146.

As used herein, the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO:3231. In specific embodiment, the wheat plant restorer of fertility includes one Rf7 restorer allele at the Rf7 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 3:

TABLE 3

SNP markers of Rf7 locus			
SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP44	cfn0917304	3230	T
SNP45	cfn0919993	3231	G
SNP46	cfn0920459	3232	C
SNP49	cfn0915987	3445	G
SNP50	cfn0920253	3446	A
SNP51	cfn0448874	3447	T
SNP52	cfn0923814	3448	C
SNP53	cfn0924180	3449	G
SNP54	cfn0919484	3450	G

Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf7 restorer allele at the Rf7 locus characterized by the presence of the nine SNP restorer alleles SNP44-SNP46 and SNP49-54 of "restorer allele" haplotype, as described in Table 3.

As used herein, the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO:3233. In specific embodiment, the wheat plant restorer of fertility includes one Rf4 restorer allele at the Rf4 locus characterized by the presence of one or more of the SNP alleles(s) as identified by Table 4.

TABLE 4

SNP markers of Rf4 locus			
SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP47	cfn0393953	3233	C
SNP48	cfn0856945	3234	G

Preferably, the wheat plant restorer of fertility according to the present disclosure includes one Rf4 restorer allele at the Rf4 locus characterized by the presence of the two SNP restorer alleles, SNP47, and SNP48, of the haplotype "C" and "G" respectively, as described in Table 4.

In specific embodiments, the wheat plant restorer of fertility with Rf4 restorer allele comprises a Rf4 nucleic acid of the present disclosure as described above. Examples of Rf4 nucleic acids comprises the disclosed Rf4 nucleic acid sequences of SEQ ID NO: 2031, SEQ ID NO:3140 to 3142.

In a particular embodiment, the wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprises one Rf3 restorer allele and two other fertility restorer alleles selected amongst Rf1, Rf4 and Rf7 restorer alleles. Preferably, the wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm according to the present disclosure comprises the Rf1, Rf3, and Rf7 restorer alleles.

In particular, it is hereby included a wheat plant comprising Rf1, Rf3 and Rf7 restorer alleles as provided by the seed samples as deposited on 25 Sep. 2017 under deposit number NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817 at the NCIMB collection. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

The disclosure also relates to hybrid wheat plants which can be produced by crossing a wheat plant restorer of fertility according to the present disclosure as described above with a second plant.

In certain embodiments, the wheat plant according to the disclosure is alloplasmic and comprises the *T. timopheevii* cytoplasm.

For example, a hybrid wheat plant may be obtained by crossing a wheat plant restorer of fertility according to the present disclosure as described above, preferably comprising Rf1, Rf3 and Rf7 restorer alleles, and a wheat plant which does not have said fertility restorer alleles.

It is also disclosed herein a method for producing a wheat hybrid plant comprising the steps of:

a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant of the present disclosure as described above;
b. collecting the hybrid seed;
c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 in the hybrid seed;
d. optionally detecting hybridity level of the hybrid seed.

Therefore, it is also disclosed herein the wheat plants or lines according to the present disclosure developed to obtain such hybrid plants. Such plants or lines typically comprise the cytoplasmic elements necessary for the implementation of the corresponding hybrid system. Preferably, the plants or lines comprise the fertility restorer alleles Rf1, Rf3 and Rf7 and *T. timopheevii* cytoplasm. In specific embodiments, such plants or lines comprise a fertility restorer allele Rf1 comprising a Rf1 nucleic acid of the present disclosure as disclosed above.

Alternatively, the detection of the presence of *T. timopheevii* cytoplasm and of at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 (step "c" of the method described above) can be performed on the parent lines in order to check their genotype before to start the cross (step "a").

The T-CMS cytoplasm can be detected either phenotypically wherein a plant bearing rf genes and a T-CMS cytoplasm will be sterile or by molecular means able to detect the orf256 gene as described in Rathburn and Hedgcoth, 1991 and Song and Hedgcoth, 1994.

Method of Producing and Selecting a Wheat Plant of the Disclosure

The present disclosure also relates to the methods to produce the wheat plant with the fertility restorer alleles as described in the previous section.

In one embodiment, said method for producing the wheat plant includes the following step:

a. providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles,
b. crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first plant and the second plant,
c. collecting the F1 hybrid seed,
d. obtaining homozygous plants from the F1 plants,
e. optionally detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or at each generation.

Preferentially, the female plant in step b) is bearing the T-CMS cytoplasm. In this case, the presence of the restorer alleles is assessed at every generation from step b) to step d) by using the markers and optionally by further by assessing the fertility level.

Method to generate homozygous plants are generally well known from skilled person of the art. This could be either by repetitive backcross or by double haploid development or by Single Seeds Descent (SSD) methods.

The applicant has deposited a sample of seeds of the disclosed wheat plant with said Rf1, Rf3 and Rf7 restorer alleles, on 25 Sep. 2017 under the Budapest treaty, at NCIMB collection under the number NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

The present disclosure further includes and provides methods of identifying the respective Rf1, Rf3, Rf4 and/or Rf7 restorer alleles as disclosed in the previous sections, and more generally methods of selecting or breeding wheat plants for the presence or absence of the Rf1, Rf3, Rf4 and/or Rf7 fertility restorer alleles. Such methods of identifying, selecting or breeding wheat plants comprise obtaining one or more wheat plants and assessing their DNA to determine the presence or absence of the Rf1, Rf3, Rf4 and/or Rf7 fertility restorer alleles contained in the respective locus.

Such methods may be used, for example, to determine which progeny resulting from a cross have the required combination of fertility restorer alleles and accordingly to guide the preparation of plants having the required combination in combination with the presence or absence of other desirable traits.

Accordingly, plants can be identified or selected by assessing them for the presence of one or more individual SNPs appearing in the above Tables 1, 2, 3 and 4, as well as the SNPs in Table 19, for assessing the presence of restorer alleles Rf1, Rf3, Rf7 or Rf4 respectively.

More generally, it is disclosed herein the specific means for detecting the restorer alleles in a wheat plant, more specifically Rf1, Rf3, Rf4 and Rf7 restorer alleles and their combinations.

Said means thus include any means suitable for detecting the following SNP markers within one or more of the following markers: SEQ ID NOs 3187-3235.

Any method known in the art may be used in the art to assess the presence or absence of a SNP. Some suitable methods include, but are not limited to, sequencing, hybridization assays, polymerase chain reaction (PCR), ligase chain reaction (LCR), and genotyping-by-sequence (GBS), or combinations thereof.

Different PCR based methods are available to the person skilled of the art. One can use the RT-PCR method or the Kaspar method from KBioscience (LGC Group, Teddington, Middlesex, UK).

The KASP™ genotyping system uses three target specific primers: two primers, each of them being specific of each allelic form of the SNP (Single Nucleotide Polymorphism) and one other primer to achieve reverse amplification, which is shared by both allelic form. Each target specific primer also presents a tail sequence that corresponds with one of two FRET probes: one label with FAM® dye and the other with HEX® dye.

Successive PCR reactions are performed. The nature of the emitted fluorescence is used to identify the allelic form or forms present in the mix from the studied DNA.

The primers identified in Table 5 are particularly suitable for use with the KASP™ genotyping system. Of course, the skilled person may use variant primers or nucleic acid probes of the primers as identified in Table 5, said variant primers or nucleic acid probes having at least 90%, and preferably 95% sequence identity with any one of the primers as identified in Table 5, or with the DNA genomic fragment amplified by the corresponding set of primers as identified in Table 5.

Percentage of sequence identity as used herein is determined by calculating the number of matched positions in aligned nucleic acid sequences, dividing the number of matched positions by the total number of aligned nucleotides, and multiplying by 100. A matched position refers to a position in which identical nucleotides occur at the same position in aligned nucleic acid sequences. For example, nucleic acid sequences may be aligned using the BLAST 2 sequences (Bl2seq) using BLASTN algorithms (www.ncbi.nlm.nih.gov).

As used herein, a primer encompasses any nucleic acid that is capable of priming the synthesis of a nascent nucleic acid in a template-dependent process, such as PCR. Typically, primers are oligonucleotides from 10 to 30 nucleotides, but longer sequences can be employed. Primers may be provided in double-stranded form though single-stranded form is preferred. Alternatively, nucleic acid probe can be used. Nucleic acid probe encompass any nucleic acid of at least 30 nucleotides and which can specifically hybridizes under standard stringent conditions with a defined nucleic acid. Standard stringent conditions as used herein refers to conditions for hybridization described for example in Sambrook et al 1989 which can comprise 1) immobilizing plant genomic DNA fragments or library DNA on a filter 2) prehybridizing the filter for 1 to 2 hours at 65° C. in 6×SSC 5×Denhardt's reagent, 0.5% SDS and 20 mg/ml denatured carrier DNA 3) adding the probe (labeled) 4) incubating for 16 to 24 hours 5) washing the filter once for 30 min at 68° C. in 6×SSC, 0.1% SDS 6) washing the filter three times (two times for 30 min in 30 ml and once for 10 min in 500 ml) at 68° C. in 2×SSC 0.1% SDS.

In specific embodiments, said primers for detecting the SNP markers of the present disclosure (specific for each allele "X" or "Y" or common) are as listed in the following table 5:

TABLE 5

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3253	cfn0238384 AlleleX	GAAGGTGACCAAGTTCATGCTGTAAAAAGATGTCTGTGTGTCTAGC
3254	cfn0523109 AlleleX	GAAGGTGACCAAGTTCATGCTGGTGAACAAAACAGGCCTACAATCA
3255	cfn0560679 AlleleX	GAAGGTGACCAAGTTCATGCTAATGATGTTTAACATTGGAACGGTCC
3256	cfn0917304 AlleleX	GAAGGTGACCAAGTTCATGCTGTGGTGGCGCTCTACCCG
3257	cfn0919993 AlleleX	GAAGGTGACCAAGTTCATGCTAAGTCATCGACTTACATGCTTCTTTG
3258	cfn0920459 AlleleX	GAAGGTGACCAAGTTCATGCTAGCCAAGGAAGCCCAGATTTTC
3259	cfn1087371 AlleleX	GAAGGTGACCAAGTTCATGCTAGGGGAACTTTGGGTATACACCA
3260	cfn1252000 AlleleX	GAAGGTGACCAAGTTCATGCTGTTAATGCTGTAGCCATTCTTGCAA

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3261	BWS0267 AlleleX	GAAGGTGACCAAGTTCATGCTTCAGCTGCATAAAAAMCAGAATACCA
3262	cfn0524469 AlleleX	GAAGGTGACCAAGTTCATGCTGCACGTAGTAAGTATTGATTTTTCTGTG
3263	cfn0527067 AlleleX	GAAGGTGACCAAGTTCATGCTCAAATTACTTTTGTTCTTTTATTTTTTTCGAAT
3264	cfn0527718 AlleleX	GAAGGTGACCAAGTTCATGCTAATTGTTCACAACATGGACATGAGAAC
3265	cfn1082074 AlleleX	GAAGGTGACCAAGTTCATGCTTACTGATAAAATCCGGTTCAAATATATAAC
3266	cfn1239345 AlleleX	GAAGGTGACCAAGTTCATGCTGGCTTCTTTTTTCTCCCTATAATATGGA
3267	cfn0554333 AlleleX	GAAGGTGACCAAGTTCATGCTGAGAGGCATCACATAGGCATAG
3268	cfn0436720 AlleleX	GAAGGTGACCAAGTTCATGCTATTCTTCATTCCTTACAACAAATATACCAAATT
3269	cfn0522096 AlleleX	GAAGGTGACCAAGTTCATGCTAGTAGAATACCACCCAATAAATCACTG
3270	cfn0523072 AlleleX	GAAGGTGACCAAGTTCATGCTCTAGCGCATGAGGTCTATCG
3271	cfn0523990 AlleleX	GAAGGTGACCAAGTTCATGCTACATGAAGAGTGCAGGCACACG
3272	cfn0524921 AlleleX	GAAGGTGACCAAGTTCATGCTATTGTTTCCATGTTAAGCTTATATTGTGCA
3273	cfn0528390 AlleleX	GAAGGTGACCAAGTTCATGCTAAAAACATCTATTCCAAGCAAGTATTAGTAAT
3274	cfn0530841 AlleleX	GAAGGTGACCAAGTTCATGCTTCTTGTTTATATATTCTCTTATCAGAAGTC
3275	cfn1122326 AlleleX	GAAGGTGACCAAGTTCATGCTGAATCTGATTAAGACGCTGGAGAAC
3276	cfn1249269 AlleleX	GAAGGTGACCAAGTTCATGCTGATTCAAAGAGGTGACAAATATGTGTACT
3277	contig46312_253_BS00090770 AlleleX	GAAGGTGACCAAGTTCATGCTGGTCGTAGCACATAGCCGTTTAC
3278	219K1_110042 AlleleX	GAAGGTGACCAAGTTCATGCTACGGAATCGAGTCAACCAATTCCT
3279	cfn0373248 AlleleX	GAAGGTGACCAAGTTCATGCTAACAACAATTAYGAGGATCAAATGGTCA
3280	cfn0527763 AlleleX	GAAGGTGACCAAGTTCATGCTATCTAGCCACGCAAATGCCCGT
3281	cfn0556874 AlleleX	GAAGGTGACCAAGTTCATGCTAAAGAGCATGTCAGACACAATGCAG
3282	cfn1097828 AlleleX	GAAGGTGACCAAGTTCATGCTGGTTCCTGAGAGAGCAACCA
3283	cfn1246088 AlleleX	GAAGGTGACCAAGTTCATGCTGACATCTGATGAGCCAGCATACA
3284	cfn1258380 AlleleX	GAAGGTGACCAAGTTCATGCTATCTACTCATCTATTGCAGATGCTCTT

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3285	cfn1270524 AlleleX	GAAGGTGACCAAGTTCATGCTAAATGCCTAGTCTATACCTGATAAACTAAA
3286	cfn1287194 AlleleX	GAAGGTGACCAAGTTCATGCTACCTCCTCCGTATCTGATGGC
3287	cfn1288811 AlleleX	GAAGGTGACCAAGTTCATGCTTAATTTGGTTAACCAAATCCTTTTTGATTTTT
3288	cfn1291249 AlleleX	GAAGGTGACCAAGTTCATGCTTCCCAGATTTAGCATGTGCATT
3289	cfn0231871 AlleleX	GAAGGTGACCAAGTTCATGCTACTGTATTAAATTAGCTAGTGTGGCG
3290	cfn0393953 AlleleX	GAAGGTGACCAAGTTCATGCTAAAAACAAGTTGTCACCCAGATGAATC
3291	cfn0867742 AlleleX	GAAGGTGACCAAGTTCATGCTGCATCCTCGACAATGATTTCATCG
3292	cfn3126082 AlleleX	GAAGGTGACCAAGTTCATGCTAGATTTTAGCACCTAACGCCGCAAA
3293	104A4_105172 AlleleX	GAAGGTGACCAAGTTCATGCTGTCGMACCCAATGAATAATGTTT
3294	104A4_105588 AlleleX	GAAGGTGACCAAGTTCATGCTGTTCCTTGTGACATGTACTCATAA
3295	136H5_3M5_138211 AlleleX	GAAGGTGACCAAGTTCATGCTACTGGGTGCAAAGCCAAGATGATT
3296	136H5_3M5_64154 AlleleX	GAAGGTGACCAAGTTCATGCTGGCGAAACTTCGCCGCGATAAAT
3297	136H5_3M5_68807 AlleleX	GAAGGTGACCAAGTTCATGCTCAAGTTGCTCTTAATTATCTGTGCGTA
3298	136H5_3M5_7601 AlleleX	GAAGGTGACCAAGTTCATGCTCGTCCCCCATGGCACCTGT
3299	136H5_3M5_77916 AlleleX	GAAGGTGACCAAGTTCATGCTATAGCAAGTAGAGTTAACTTATCAAGTTATTA
3300	136H5_3M5_89176 AlleleX	GAAGGTGACCAAGTTCATGCTGGATTTTCTCACCGGCATCTCCA
3301	136H5_3M5_89263 AlleleX	GAAGGTGACCAAGTTCATGCTTCCCATGTTCTTTTTTTGCTCAAAAC
3302	219K1_107461 AlleleX	GAAGGTGACCAAGTTCATGCTATATTGTTTGTATTAAAAAGTTGTGTGTTTTGA
3303	219K1_110005 AlleleX	GAAGGTGACCAAGTTCATGCTGCCTTTTCTTCTTCCAGCATCTAC
3304	219K1_111446 AlleleX	GAAGGTGACCAAGTTCATGCTAGAATCGTTCTTCGAGAAGCACTCA
3305	219K1_158251 AlleleX	GAAGGTGACCAAGTTCATGCTCCTGGAGATGGATCCGGTCAG
3306	219K1_166464 AlleleX	GAAGGTGACCAAGTTCATGCTCCTGAGCTGGGCTGCACC
3307	219K1_37	GAAGGTGACCAAGTTCATGCTAAAGGGCTATCCTGGTGAACAAC
3308	219K1_99688 AlleleX	GAAGGTGACCAAGTTCATGCTGTTGCCCTGCGCAAAATCAAACTT

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3309	276113_96B22_97797 AlleleX	GAAGGTGACCAAGTTCATGCTGTACTATGGCTAT GTCTCTGAATGC
3310	CAP7_c3847_204 AlleleX	GAAGGTGACCAAGTTCATGCTCATTCGACGCGTC TTCCGCAATA
3311	Tdurum_contig50667_ 306 AlleleX	GAAGGTGACCAAGTTCATGCTGATGACATGGAGG ATTATATCGACGA
3312	cfn0856945 AlleleX	GAAGGTGACCAAGTTCATGCTGCACATGCTTTATT ACTGATCTGATTTG
3313	S100067637 AlleleX	GAAGGTGACCAAGTTCATGCTCCAAATGTCCGAA TTCAGAGCAG
3404	S100069923 AlleleX	GAAGGTGACCAAGTTCATGCTACATATACGCGAG CGCTCCTG
3315	S3045171 AlleleX	GAAGGTGACCAAGTTCATGCTGGTTCTTGGCACA CTCCCCAG
3316	S3045222 AlleleX	GAAGGTGACCAAGTTCATGCTAACCTAAGTAGTAA GCTTGCTGGGT
3317	cfn0238384 Allele Y	GAAGGTCGGAGTCAACGGATTGTAAAAAGATGTC TGTGTGTCTAGG
3318	cfn0523109 Allele Y	GAAGGTCGGAGTCAACGGATTGTGAACAAAACAG GCCTACAATCC
3319	cfn0560679 Allele Y	GAAGGTCGGAGTCAACGGATTCAATGATGTTTAA CATTGGAACGGTCT
3320	cfn0917304 Allele Y	GAAGGTCGGAGTCAACGGATTGGTGGTGGCGCT CTACCCT
3321	cfn0919993 Allele Y	GAAGGTCGGAGTCAACGGATTCAAGTCATCGACT TACATGCTTCTTTT
3322	cfn0920459 Allele Y	GAAGGTCGGAGTCAACGGATTAGCCAAGGAAGC CCAGATTTTG
3323	cfn1087371 Allele Y	GAAGGTCGGAGTCAACGGATTGGGGAACTTTGG GTATACACCG
3324	cfn1252000 Allele Y	GAAGGTCGGAGTCAACGGATTGTTAATGCTGTAG CCATTCTTGCAG
3325	BWS0267 Allele Y	GAAGGTCGGAGTCAACGGATTCAGCTGCATAAAA AMCAGAATACCG
3326	cfn0524469 Allele Y	GAAGGTCGGAGTCAACGGATTGCACGTAGTAAGT ATTGATTTTTCTGTT
3327	cfn0527067 Allele Y	GAAGGTCGGAGTCAACGGATTCAAATTACTTTTGT TCTTTTATTTTTTTCGAAC
3328	cfn0527718 Allele Y	GAAGGTCGGAGTCAACGGATTATAAATTGTTCACA ACATGGACATGAGAAT
3329	cfn1082074 Allele Y	GAAGGTCGGAGTCAACGGATTCTTACTGATAAAAT CCGGTTCAAATATATAAT
3330	cfn1239345 Allele Y	GAAGGTCGGAGTCAACGGATTGCTTCTTTTTTCTC CCTATAATATGGG
3331	cfn0554333 Allele Y	GAAGGTCGGAGTCAACGGATTGAGAGGCATCACA TAGGCATAC
3332	cfn0436720 Allele Y	GAAGGTCGGAGTCAACGGATTCTTCATTCCTTACA ACAAATATACCAAATC

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3333	cfn0522096 Allele Y	GAAGGTCGGAGTCAACGGATTAGTAGAATACCACCCAATAAATCACTC
3334	cfn0523072 Allele Y	GAAGGTCGGAGTCAACGGATTAACTCTAGCGCATGAGGTCTATCA
3335	cfn0523990 Allele Y	GAAGGTCGGAGTCAACGGATTATACATGAAGAGTGCAGGCACACT
3336	cfn0524921 Allele Y	GAAGGTCGGAGTCAACGGATTGTTTCCATGTTAAGCTTATATTGTGCG
3337	cfn0528390 Allele Y	GAAGGTCGGAGTCAACGGATTAAACATCTATTCCAAGCAAGTATTAGTAAC
3338	cfn0530841 Allele Y	GAAGGTCGGAGTCAACGGATTCTTCTTGTTTATATATTCTCTTATCAGAAGTT
3339	cfn1122326 Allele Y	GAAGGTCGGAGTCAACGGATTGGAATCTGATTAAGACGCTGGAGAAT
3340	cfn1249269 Allele Y	GAAGGTCGGAGTCAACGGATTCAAAGAGGTGACAAATATGTGTACC
3341	contig46312 Allele Y_253_BS00090770	GAAGGTCGGAGTCAACGGATTAGGTCGTAGCACATAGCCGTTTAT
3342	219K1_110042 Allele Y	GAAGGTCGGAGTCAACGGATTCGGAATCGAGTCAACCAATTCCC
3343	cfn0373248 Allele Y	GAAGGTCGGAGTCAACGGATTAACAACAATTAYGAGGATCAAATGGTCT
3344	cfn0527763 Allele Y	GAAGGTCGGAGTCAACGGATTCTAGCCACGCAAATGCCCGC
3345	cfn0556874 Allele Y	GAAGGTCGGAGTCAACGGATTGAAAGAGCATGTCAGACACAATGCAA
3346	cfn1097828 Allele Y	GAAGGTCGGAGTCAACGGATTGGTTCCTGAGAGAGCAACCG
3347	cfn1246088 Allele Y	GAAGGTCGGAGTCAACGGATTGACATCTGATGAGCCAGCATACC
3348	cfn1258380 Allele Y	GAAGGTCGGAGTCAACGGATTCTACTCATCTATTGCAGATGCTCTG
3349	cfn1270524 Allele Y	GAAGGTCGGAGTCAACGGATTAAATGCCTAGTCTATACCTGATAAACTAAT
3350	cfn1287194 Allele Y	GAAGGTCGGAGTCAACGGATTCACCTCCTCCGTATCTGATGGT
3351	cfn1288811 Allele Y	GAAGGTCGGAGTCAACGGATTAATTTGGTTAACCAAATCCTTTTTGATTTTG
3352	cfn1291249 Allele Y	GAAGGTCGGAGTCAACGGATTCTTCCCAGATTTAGCATGTGCATG
3353	cfn0231871 Allele Y	GAAGGTCGGAGTCAACGGATTCTACTGTATTAAATTAGCTAGTGTGGCT
3354	cfn0393953 Allele Y	GAAGGTCGGAGTCAACGGATTAAAAAAACAAGTTGTCACCCAGATGAATT
3355	cfn0867742 Allele Y	GAAGGTCGGAGTCAACGGATTGGCATCCTCGACAATGATTTCATCT
3356	cfn3126082 Allele Y	GAAGGTCGGAGTCAACGGATTTAGCACCTAACGCCGCAAC

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3357	104A4_105172 Allele Y	GAAGGTCGGAGTCAACGGATTCTGTCGMACCCAATGAATAATGTTC
3358	104A4_105588 Allele Y	GAAGGTCGGAGTCAACGGATTGTTCCTTGTGACATGTACTCATAC
3359	136H5_3M5_138211 Allele Y	GAAGGTCGGAGTCAACGGATTACTGGGTGCAAAGCCAAGATGATA
3360	136H5_3M5_64154 Allele Y	GAAGGTCGGAGTCAACGGATTGCGAAACTTCGCCGCGATAAAC
3361	136H5_3M5_68807 Allele Y	GAAGGTCGGAGTCAACGGATTAAGTTGCTCTTAATTATCTGTGCGTG
3362	136H5_3M5_7601 Allele Y	GAAGGTCGGAGTCAACGGATTGTCCCCCATGGCACCTGC
3363	136H5_3M5_77916 Allele Y	GAAGGTCGGAGTCAACGGATTAGCAAGTAGAGTTAACTTATCAAGTTATTG
3364	136H5_3M5_89176 Allele Y	GAAGGTCGGAGTCAACGGATTTTCTCACCGGCATCTCCG
3365	136H5_3M5_89263 Allele Y	GAAGGTCGGAGTCAACGGATTCTTCCCATGTTCTTTTTTTGCTCAAAAT
3366	219K1_107461 Allele Y	GAAGGTCGGAGTCAACGGATTATATTGTTTGTATTAAAAAGTTGTGTGTTTTGC
3367	219K1_110005 Allele Y	GAAGGTCGGAGTCAACGGATTCGCCTTTTCTTCTTCCAGCATCTAT
3368	219K1_111446 Allele Y	GAAGGTCGGAGTCAACGGATTAATCGTTCTTCGAGAAGCACTCC
3369	219K1_158251 Allele Y	GAAGGTCGGAGTCAACGGATTCCTGGAGATGGATCCGGTCAA
3370	219K1_166464 Allele Y	GAAGGTCGGAGTCAACGGATTGCCTGAGCTGGGCTGCACT
3371	219K1_37 Allele Y	GAAGGTCGGAGTCAACGGATTACAAAGGGCTATCCTGGTGAACAAT
3372	219K1_99688 Allele Y	GAAGGTCGGAGTCAACGGATTGCCCTGCGCAAAATCAAACTC
3373	276I13_96B22_97797 Allele Y	GAAGGTCGGAGTCAACGGATTAAGTACTATGGCTATGTCTCTGAATGT
3374	CAP7_c3847_204 Allele Y	GAAGGTCGGAGTCAACGGATTCGACGCGTCTTCCGCAATG
3375	Tdurum_contig50667_306 Allele Y	GAAGGTCGGAGTCAACGGATTATGACATGGAGGATTATATCGACGG
3376	cfn0856945 Allele Y	GAAGGTCGGAGTCAACGGATTGGCACATGCTTTATTACTGATCTGATTTT
3377	S100067637 Allele Y	GAAGGTCGGAGTCAACGGATTCCAAATGTCCGAATTCAGAGCAC
3378	S100069923 Allele Y	GAAGGTCGGAGTCAACGGATTGTACATATACGCGAGCGCTCCTA
3379	S3045171 Allele Y	GAAGGTCGGAGTCAACGGATTGGTTCTTGGCACACTCCCCAA
3380	S3045222 Allele Y	GAAGGTCGGAGTCAACGGATTCCTAAGTAGTAAGCTTGCTGGGC

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3381	cfn0238384 Common	AGGGGGGCGTACGGGGTGA
3382	cfn0523109 Common	GTGTGTGCTAATGTGGATATACGTAAGTT
3383	cfn0560679 Common	GACGTTGAAGGGGGCATAGATCAAA
3384	cfn0917304 Common	CAACTGCTTGGAGAAAGGCAACACAA
3385	cfn0919993 Common	CCATTAACAAGTACTGCATAGGTGCATAT
3386	cfn0920459 Common	CCTCCTCCTAATTAAGCTCCTATAGATA
3387	cfn1087371 Common	CCCCCTTCTTCTTTCACTAGGGTAA
3388	cfn1252000 Common	GTGCCCATAAGACGACTGGGACAA
3389	BWS0267 Common	CTGCGTTAAGGTTCAGGCAACTGAT
3390	cfn0524469 Common	GCCAATTTTCAAATCTAAGTCCACAGAGA
3391	cfn0527067 Common	ATATGATTCACCCTAGATCCTTCACCTTA
3392	cfn0527718 Common	GTTTCCTCCAATGTTCTTCCC
3393	cfn1082074 Common	TGTCTCGCCTCGCTCTGGTTAATTT
3394	cfn1239345 Common	ACCCTCGCTGCAGTTCCTTCTTAAA
3395	cfn0554333 Common	AAATTCACACCATCATTGATCTGGGGTAT
3396	cfn0436720 Common	GTCCACTGAGAATTAAGGATGCATTCTTT
3397	cfn0522096 Common	AAGTAGTACTCGTAGAGAGTTAACACAGA
3398	cfn0523072 Common	GCTTGACAATGATAATGCCCCCGAA
3399	cfn0523990 Common	AATAACTCTTGTACTTCAGGATGAACGTTT
3400	cfn0524921 Common	GCCCTTTGGTAATTCCATTTCAATCTTTT
3401	cfn0528390 Common	GATGAGGAAGGTCTTCATGTTGGGTT
3402	cfn0530841 Common	GAGCAGCACATCGTTAGCTGTTCTA
3403	cfn1122326 Common	CAGATGGCCTAGTCGTGACATATCTT
3404	cfn1249269 Common	TAAAAGAACACAAATGTGGCCCTAGTGAT

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)

SEQ ID NO	ID	Sequence
3405	contig46312_253_BS00090770 Common	GAAACATTCCTTCGGACAACTATGCATTA
3406	219K1_110042 Common	GCATCTTCAAGGGAGCCACTCAAAA
3407	cfn0373248 Common	ATCATTGCCACGRAAAAAATCTCACAAGAT
3408	cfn0527763 Common	CCTTGTCCACCGAGACATGTACAAA
3409	cfn0556874 Common	CCTGCTGGAAATGGGATTTCTTGTTTATT
3410	cfn1097828 Common	GCTTCCTCTCGGTAGCGATGGAT
3411	cfn1246088 Common	GGGACGTGGAATTTGGAAAGACACAT
3412	cfn1258380 Common	TATAGGAGTGATAGCACCACACAATTCAT
3413	cfn1270524 Common	TGTACCGAAACTCAACCAAATGACCATTT
3414	cfn1287194 Common	CAGAAGGCACTGGGAGGGGATT
3415	cfn1288811 Common	GCACAATGTTTGACATTCGGTTTTCTAGTT
3416	cfn1291249 Common	CTGACTGTCGTATCTTCAACATACTGATT
3417	cfn0231871 Common	CCAAGGTATATGTGCCATTATCCTCAAA
3418	cfn0393953 Common	CACTCACCGTCGACATTGACATAGTT
3419	cfn0867742 Common	AGCCTCCGCGTCGTGATGGAAT
3420	cfn3126082 Common	AAAGGGACAGCGATTTGATCTGG
3421	104A4_105172 Common	GCCATCCTCTCGGAGCCAGAA
3422	104A4_105588 Common	CAAGGATGGGGAGTATATGGCTCTT
3423	136H5_3M5_138211 Common	CCTCCCAACGGCCATCAATCAATTT
3424	136H5_3M5_64154 Common Common	GATCATCGGGGAACCTGATGATAGTT
3425	136H5_3M5_68807 Common	TTGGTTGGTTACGTCAGGTTAAGACTTA
3426	136H5_3M5_7601 Common	CTTCTCTGTGGCCGAAAACCTCTT
3427	136H5_3M5_77916 Common	GCTKTAGACTCTAAGTACCACAGAAGAA
3428	136H5_3M5_89176 Common	CCTACCATCCTTAAATACTCTTGCTCAAA

TABLE 5-continued

Primers for use in detecting fertility restorer SNP markers of the invention (as indicated in the primer name)		
SEQ ID NO	ID	Sequence
3429	136H5_3M5_89263 Common	AAGCAACTAGAAAAATATTTGGACTAGCAT
3430	219K1_107461 Common	GTTGATGCGAATTTGAAAATGACATAATAA
3431	219K1_110005 Common	TTGACTCGATTCCGTGTGAGGCTAA
3432	219K1_111446 Common	AATATGATACAGACCCAAGACAAACCATTT
3433	219K1_158251 Common	TCCTCACAAATCACGGGCCCCT
3434	219K1_166464 Common	GACCGTGGTATATGCCACCACGTT
3435	219K1_37 Common	GGCTTCATTATCAAATTCTGACCCATCTT
3436	219K1_99688 Common	GGGCGGGACCTGACTTGATGAT
3437	276I13_96B22_97797 Common	ACGACAATATAGACAAATAAAACCAAACAA
3438	CAP7_c3847_204 Common	CCGCGGCCGAAGCAGGCAA
3439	Tdurum_contig50667_306 Common	ATACATGTCGGCGTCCCAGTCC
3440	cfn0856945 Common	GGTGTAGGCAAACCTAAAATAAACAGTCAA
3441	S100067637 Common	CAACGCCAAACGCCAACGCCAT
3442	S100069923 Common	GCCTTGTACTGCAGTGAAGTGTGAT
3443	S3045171 Common	TGACGGCTGCGAGGACGAGAAT
3444	S3045222 Common	AGTCCAGAGTTACAGGACATGGCTA

Use of the Wheat Plants of the Disclosure

The plant according to the disclosure can be crossed, with any another inbred line, in order to produce a new line comprising either an increase or a decrease in the fertility level. Alternatively, a genetic trait which has been engineered into a particular line using the foregoing techniques could be moved into another line using traditional backcrossing techniques that are well known in the plant breeding arts. For example, a backcrossing approach could be used to move an engineered trait from a public, non-elite inbred line into an elite inbred line, or from an inbred line containing a foreign gene in its genome into an inbred line or lines which do not contain that gene. As used herein, "crossing" can refer to a simple X by Y cross, or the process of backcrossing, depending on the context.

The wheat plant of the disclosure is a also a wheat plant wherein one or more desired traits have further been introduced through backcrossing methods, whether such trait is a naturally occurring one or not.

The disclosure also relates to the use of the wheat plant as described above or its seeds, for food applications, preferably for flour production and for feed applications, or for breeding applications, for example for use as a parent plant in breeding for improving agronomical value of a wheat plant, line, hybrid or variety.

As used herein, breeding applications encompass pedigree breeding to improve the agronomical value of a plant, line, hybrid, or variety.

The wheat plants disclosed herein are further useful, for example, for producing flour or for feed applications.

Seeds harvested from plants described herein can be used to make flour by any available techniques in the art. The wheat plants or their flour are also useful as food compositions, for human or animal.

The Examples below are given for illustration purposes only.

Specific Embodiments

1. An isolated nucleic acid encoding a protein restorer of fertility of *T. timopheevii*, wherein the corresponding amino acid sequence has at least 95% identity to an amino acid sequence chosen amongst any one of SEQ ID NO:1 to SEQ ID NO: 1554.
2. The nucleic acid of Embodiment 1, encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs 1-2, SEQ ID NOs 288-290, SEQ ID NOs 293-296, SEQ ID NOs 343-346, SEQ ID NOs 349-354, SEQ ID NOs 359, 361 and 362, SEQ ID NOs 396 and 397, SEQ ID NOs 428-430, SEQ ID NO 517 and 519, SEQ ID NOs 752-754, SEQ ID NOs 1092, 1093 and 1095.
3. The nucleic acid of Embodiment 1, encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to SEQ ID NO:361.
4. The nucleic acid of claim **1**, encoding a Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NOs: 124 and 125, SEQ ID NO:147, SEQ ID NO: 150, SEQ ID NO: 156, SEQ ID NO:158, SEQ ID NO:297, SEQ ID NO:299, SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 553 and 554, SEQ ID NOs: 557 and 558, SEQ ID NOs: 676 and 677, SEQ ID NOs: 684 and 685, SEQ ID NOs: 696 and 697, SEQ ID NOs: 938 and 939 and SEQ ID NOs: 1038 and 1039.
5. The nucleic acid of Embodiment 4, encoding Rf3 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO:684.
6. The nucleic acid of Embodiment 1, encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 477 and SEQ ID NOs 3135-3138.
7. The nucleic acid of Embodiment 1, encoding a Rf7 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NOs: 240-243, SEQ ID NOs 303-305, SEQ ID NO:363, SEQ ID NOs 375-377, SEQ ID NOs 497-499, SEQ ID NO:516, SEQ ID NOs 709-711, SEQ ID NO:768.
8. The nucleic acid of Embodiment 1 encoding for a Rf-rye protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid sequence selected from the group consisting of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859.
9. A recombinant nucleic acid comprising a nucleic acid encoding a protein restorer of *T. timopheevii* CMS cytoplasm of any one of Embodiments 1 to 8, operably linked to regulatory elements.
10. A vector for use in transformation of a wheat plant, comprising the recombinant nucleic acid of Embodiment 9.
11. A wheat transgenic plant comprising one or more nucleic acid(s) of any one of Embodiments 1-9, as transgenic element(s).
12. The wheat transgenic plant of Embodiment 11, which is a fertile wheat plant restorer of *T. timopheevii* CMS cytoplasm and comprising a combination of at least two different transgenic elements, selected from the group consisting of Rf1, Rf3, Rf7 and Rf-rye nucleic acids of any one of Embodiments 1-9.
13. The transgenic wheat plant of Embodiment 11 or 12, wherein said transgenic plant includes the following combination of nucleic acids as transgenic elements:
    a. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684,
    b. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO: 516 and SEQ ID NO:768,
    c. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO: 378 and SEQ ID NO: 859,
    d. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to SEQ ID NO: 363, SEQ ID NO:516 and SEQ ID NO:768,
    e. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859,
    f. a Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859,
    g. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID

NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO: 768;

h. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO: 859;

i. a Rf1 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs 359, 361, 362 or SEQ ID NOs 428-430, and Rf7 nucleic acid of Embodiment 4 encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO: 363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO:859; or,

j. a Rf3 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NOs: 315-321, SEQ ID NOs: 379-381, SEQ ID NOs: 147 and 150, SEQ ID NOs: 156 and 158, SEQ ID NOs 297 and 299, SEQ ID NO:676 and SEQ ID NO:684, and Rf7 nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:363, SEQ ID NO:516 and SEQ ID NO:768, and Rf-rye nucleic acid encoding an amino acid sequence having at least 95% identity to any one of SEQ ID NO:227, SEQ ID NO:378 and SEQ ID NO: 859.

14. The transgenic wheat plant of Embodiment 13, which further contains a Rf4 nucleic acid encoding a Rf4 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, in combination with one, two, three or four of any of the restorer nucleic acid encoding Rf1, Rf3, Rf7 or Rf-rye protein, wherein the corresponding amino acid sequence has at least 95% identity, preferably at least 96%, 97%, 98%, 99% or 100% identity to an amino acid selected from the group consisting of SEQ ID NO:477 and SEQ ID NOs 3135-3138.

15. The transgenic wheat plant of any one of Embodiments 11-14, wherein said one or more transgenic element(s) express polypeptides which restore or improve male fertility to the plant as compared to the parent plant without such transgenic element(s).

16. Method for producing a wheat transgenic plant of any one of Embodiments 11-15, wherein the method comprises the steps of transforming a parent wheat plant with one or more nucleic acids encoding protein restorer of *T. timopheevii* CMS cytoplasm according to any one of Embodiments 1-9, selecting a plant comprising said one or more nucleic acid(s) as transgene(s), regenerating and growing said wheat transgenic plant.

17. A method for producing a wheat plant carrying a Rf3 restorer of fertility, said method comprising (i) providing a parent wheat plant comprising in its genome at least a 163 bp fragment of SEQ ID NO:3174, and (ii) deleting a region of at least 10 bp in said fragment of SEQ ID NO:3174, for example at least 20 bp, 30 bp, 40 bp, 50 bp, 60 bp, 70 bp, 80 bp, 90 bp, 100 bp, 110 bp, 120 bp, 130 bp, 140 bp, 150 bp, 160 bp or the whole fragment of SEQ ID NO:3174 in the genome of said wheat plant, thereby obtaining said wheat plant carrying a Rf3 restorer of fertility.

18. The method of Embodiment 17, wherein the fertility score of the obtained wheat plant has a fertility score higher than the parent wheat plant.

19. The method of Embodiment 17 or 18, wherein the parent wheat plant has a fertility score below 1, for example comprised between 0.5 and 1.0 and the obtained wheat plant has a fertility score above 1.0, for example comprised between 1.0 and 2.0.

20. A wheat plant carrying Rf3 restorer of fertility, as obtained by the method of any one of Embodiments 17-19, wherein only a part but not the whole genomic fragment of SEQ ID NO:3174 is deleted in the genome of said wheat plant.

21. A method for assessing fertility restoration in a wheat plant, said method comprising determining the presence or absence of a fragment of SEQ ID NO: 3174 in the genome of said plant, wherein the presence of the whole fragment is indicative of a weak restoration of fertility and a deletion of at least a part of such fragment or the whole fragment of SEQ ID NO:3174 is indicative of a strong restoration of fertility.

22. A nucleic acid probe for use in a method of any one of Embodiments 17-21, characterized it consists of a nucleic acid of at least 10 nucleotides within SEQ ID NO: 3174.

23. A wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the plant comprises at least three fertility restorer alleles within the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7 wherein,

a. the Rf1 locus is located at most 10 cM from marker cfn0522096 of SEQ ID NO: 3190 or marker cfn05277067 of SEQ ID NO: 3196,

b. the Rf3 locus is located at most 10 cM from marker cfn1249269 of SEQ ID NO: 3205 or marker BS00090770 of SEQ ID NO:3228,

c. the Rf7 locus is located at most 10 cM from marker cfn0919993 of SEQ ID NO: 3231, and,

d. the Rf4 locus is located at most 10 cM from marker cfn0393953 of SEQ ID NO: 3233.

24. The wheat plant of Embodiment 23, wherein the plant comprises the Rf1, Rf3 and Rf7 restorer alleles.

25. The wheat plant of any one of Embodiments 23 to 24, characterized in that it includes at least one Rf1 restorer allele within the Rf1 locus, said Rf1 restorer allele being located within the chromosomal interval between SNP markers cfn0522096 of SEQ ID NO:3190 and cfn05277067 of SEQ ID NO:3196.

26. The wheat plant of Embodiment 25, wherein said Rf1 locus is characterized by the presence of one or more of the following SNP allele(s):

SNP#	Marker Name	Marker SEQ ID NO:	Restorer Allele
SNP1	cfn523072	3187	T
SNP2	cfn0523109	3188	A
SNP3	276I13_96B22_97797	3189	C
SNP4	cfn0522096	3190	C
SNP5	cfn0527763	3191	C
SNP6	104A4_105172	3192	TG
SNP7	104A4_105588	3193	A
SNP8	cfn0373248	3194	T
SNP9	cfn1097828	3195	C
SNP10	cfn0527067	3196	A
SNP11	cfn0528390	3197	G
SNP12	BWS0267	3198	A
SNP13	cfn0527718	3199	T
SNP14	cfn0524469	3200	G
SNP15	cfn0524921	3201	G
SNP16	cfn1122326	3202	C

27. The wheat plant of Embodiment 26, wherein the Rf1 locus is characterized by the haplotype "C" and "A" of the SNP3 and SNP7 restorer alleles as described in the table of Embodiment 5.

28. The wheat plant of any one of Embodiments 23 to 27, characterized in that it includes at least one Rf3 restorer allele within the Rf3 locus, said Rf3 restorer allele being located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770.

29. The wheat plant of Embodiment 28, wherein said Rf3 locus is characterized by the presence of one or more of the following SNP allele(s):

SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP17	cfn1252000	3203	A
SNP18	IWB14060*	3204	G
SNP19	cfn1249269	3205	G
SNP20	219K1_166464	3206	T
SNP21	219K1_158251	3207	G
SNP22	219K1_111446	3208	A
SNP23	219K1_110042	3209	T
SNP24	219K1_110005	3210	C
SNP25	219K1_107461	3211	A
SNP26	219K1_99688	3212	T
SNP27	219K1_37	3213	C
SNP28	cfn1270524	3214	T
SNP29	136H5_3M5_7601	3215	T
SNP30	cfn1288811	3216	G
SNP31	136H5_3M5_89176	3217	A
SNP32	136H5_3M5_89263	3218	T
SNP33	136H5_3M5_138211	3219	T
SNP34	cfn0556874	3220	C
SNP35	136H5_3M5_64154	3221	C
SNP36	136H5_3M5_68807	3222	G
SNP37	136H5_3M5_77916	3223	A
SNP38	cfn1246088	3224	A
SNP39	cfn1287194	3225	G
SNP40	cfn1258380	3226	A
SNP41	IWB72107*	3227	A
SNP42	BS00090770	3228	T
SNP43	cfn1239345	3229	A

30. The wheat plant of Embodiment 29, wherein the Rf3 locus is characterized by the haplotype "T" and "A" of the SNP29 and SNP31 restorer alleles as described in the table of Embodiment 7.

31. The wheat plant of any one of Embodiments 23 to 30, wherein the Rf7 locus is characterized by the presence of one or more of the following restorer SNP allele(s):

SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP44	cfn0917304	3230	T
SNP45	cfn0919993	3231	G
SNP46	cfn0920459	3232	C
SNP49	cfn0915987	3445	G
SNP50	cfn0920253	3446	A
SNP51	cfn0448874	3447	T
SNP52	cfn0923814	3448	C
SNP53	cfn0924180	3449	G
SNP54	cfn0919484	3450	G

32. The wheat plant of any one of Embodiments 23 to 31, wherein the Rf4 locus is characterized by the presence of one or more of the following SNP allele(s), preferably by the haplotype "C" and "G" of the SNP47 and SNP48 restorer alleles:

SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP47	cfn0393953	3233	C
SNP48	cfn0856945	3234	G

33. The wheat plant of any one of Embodiments 23-32, wherein representative alleles of Rf1, Rf3, Rf4 and Rf7 restorer alleles are provided by the seed sample chosen amongst: NCIMB 42811, NCIMB 42812, NCIMB 42813, NCIMB 42814, NCIMB 42815, NCIMB 42816, and NCIMB 42817.

34. The wheat plant according to any one of the Embodiments 23 to 33, wherein said wheat plant is alloplasmic and comprises the *T. timopheevii* cytoplasm.

35. A method of identifying a wheat plant according to any one of Embodiments 23 to 34, wherein said wheat plant is identified by detecting the presence of at least one restorer allele genetically associated with the restorer loci chosen amongst Rf1, Rf3, Rf4 and Rf7 loci.

36. Means for detecting one or more of SNPs of SEQ ID NOs 3187-3235.

37. The means according to Embodiment 36 consisting of one or more primers including any one of the following: SEQ ID NOs 3253-3444.

38. Method of production of a wheat hybrid plant comprising the steps of:
   a. crossing a sterile female wheat plant comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant according to any one of Embodiments 23 to 34;
   b. collecting the hybrid seed;
   c. optionally detecting the presence of *T. timopheevii* cytoplasm, and/or at least three of the Rf locus chosen amongst Rf1, Rf3, Rf4 and Rf7 in the hybrid seed; and,
   d. optionally detecting hybridity level of the hybrid seed.

LEGENDS OF THE FIGURES

FIGS. 1A and 1B is a table showing a summary of plant genomes used in the study and number of identified RFLs. In total, the analyses encompassed 16 genome data sets from Triticeae and 13 from Oryzeae, respectively, as well as single data sets from Brachypodium distachyon, tef (*Eragrostis tef*), rye (*Secale cereale*), foxtail millet (*Setaria italica*), sorghum (*Sorghum bicolor*) and maize (*Zea mays*). *Lolium perenne* and *Triticum turgidum* transcriptome data sets were used as well.

FIG. 2: Processing of orf256 in T-CMS wheat mitochondria (A) Structure of orf256 identified in the *T. timopheevii* mitochondrial genome. The binding site of the WORF256 probe (Song and Hedgoth 1994) used in the Northern blot analysis is indicated. (B) Differential processing of the orf256 in wheat lines with different restoring capabilities. No orf256 transcript was detected in *T. aestivum*, Primepii, Anapurna and Wheat-Rye-6R (WR_6R) lines. An additional, third band detected in R197 and R0934F accessions is indicated by asterisks. As a control for gel loading the picture of ethidium bromide (EtBr) stained agarose gel is shown.

FIG. 3: FIG. 3 shows the list of RFL groups potentially corresponding to the Rf4 gene.

FIGS. 4*a* and 4*b*: FIGS. 4A and 4B show respectively the alignment between nucleotide sequences of RFL120-spelt (Subject) (SEQ ID NO: 3136) with RFL120-timo (Query) (SEQ ID NO: 477) and amino acid sequences RFL120-17F3R-0377_1 (SEQ ID NO: 3135), RFL120-L13_1 (SEQ ID NO: 3137), RFL120-R113_1 (SEQ ID NO: 3138), RFL120-timo_1 (SEQ ID NO: 477), and RFL120-GSTR435_1 (SEQ ID NO: 3136).

FIG. 5A: FIG. 5A shows the protein sequence alignment of RFL29a (SEQ ID NO: 3146), RFL29b (SEQ ID NO: 3149), RFL29c_1 (SEQ ID NO: 3458) and RFL29c_2 (SEQ ID NO: 3459).

FIGS. 5B and 5C: FIGS. 5B and 5C respectively, show the protein sequence alignments of RFL164a (SEQ ID NO: 3147) and RFL164b (SEQ ID NO: 3144), and RFL166a (SEQ ID NO: 3148) and RFL166b (SEQ ID NO: 3145).

FIG. 6: FIG. 6 shows an alignment of the 5'UTR regions as identified in the RFL29a (SEQ ID NO: 3460) and RFL29b genes (SEQ ID NO: 3461).

FIG. 7: FIG. 7 shows the position of the different target sequences around and within the 163 bp region (SEQ ID NO: 3155) identified for different endonucleases.

FIG. 8: The FIG. 8 shows the relative position of the Rf1 mapping intervals identified on our internal consensus genetic map.

FIG. 9: The FIG. 9 shows the relative position of the Rf3 mapping intervals identified on our internal consensus genetic map.

FIG. 10: The FIG. 10A shows position of the markers within the chromosomal interval of Rf1 locus. Left and right refer to the marker positions relative to the interval defined by cfn0522096 and cfn0527067 SNP markers. Interval refers to markers located within the mapping interval. The physical positions correspond to LG internal ordering of the scaffolds of the IWGSC Whole genome assembly, 'IWGSC WGA'.

The FIGS. 10B, 10C, 10D show a subset of the diversity panel showing the haplotypes at the Rf1 locus for the restorer lines used in the genetic mapping (R197, R204, R0932E), derived lines LGWR16-0016 and LGWR16-0026 and a collection of maintainer lines. "-": correspond to dominant markers with no amplification in several maintainer lines. "H": means heterozygote status wherein two alleles are detected.

FIG. 11: The FIG. 11A shows the position of the markers within the chromosomal interval of Rf3 locus. Left and right refer to the marker positions relative to the interval defined by cfn1249269 and BS00090770 markers. Interval refers to markers located within the mapping interval. The physical positions correspond to LG internal ordering of the scaffolds of the IWGSC Whole genome assembly, 'IWGSC WGA'. *IWB14060 and IWB72107 are described in Geyer and al, 2016.

The FIGS. 11B, 11C, 11D show a subset of the diversity panel showing the haplotypes at the Rf3 locus for the restorer lines LGWR16-0016 and LGWR16-0026, the TJB155 line used as restorer parental line in Rf3 QTL mapping and a series of maintainer lines.

"-" corresponds to dominant markers with no amplification in several maintainer lines. "H" means heterozygote status wherein two alleles are detected.

FIG. 12: The FIG. 12 shows nucleotide sequence alignment between RFL29a (SEQ ID NO: 3146) and RFL29c (SEQ ID NO: 3157) fragment sequences. The PAM motif and target sequence for CRISPR edition are respectively is in bold and underligned.

EXAMPLES

Example 1: Identification of 1188 RFL-PPR Sequences in Cereals

32 genomic and two transcriptome data sets from 27 cereal plant species and their wild relatives were downloaded from the public sequence depositories and analysed. A complete list of files and databases from which they were downloaded is presented in FIG. 1.

The DNA sequences were screened for open reading frames (ORFs) in six-frame translations with the getorf program of the EMBOSS 6.6.0 package (Rice et al., 2000). Predicted ORFs longer than 92 codons were screened for the presence of P-and PLS-class pentatricopeptide repeat (PPR) motifs using hmmsearch from the HMMER 3.1b package (hmmer.org) and hidden Markov models defined by hmmbuild (Cheng et al., 2016). The post-processing of hmmsearch results was carried out according to rules described previously (Cheng et al., 2016). Sequences containing 10 or more P-class PPR motifs were retained for further analysis, as a previous study has shown that Restorer-to-Fertility-Like (RFL) genes are primarily comprised of tandem arrays of 15 to 20 PPR motifs (Fujii et al., 2011).

For identification of RFL sequences among the P-class PPRs, the OrthoMCL algorithm (Li et al., 2003) was used via the OrthoMCL-DB website to cluster P-class PPR proteins from each data set (http://www.orthomcl.org/orthomcl/). The resulting output files were screened for groups containing reference RFLs (Fujii et al., 2011).

In total, 633 RFLs were identified in the 34 cereal data sets by OrthoMCI analysis (see Table of FIGS. 1A and 1B). In addition, WGS data sets of 44 sorghum accessions including landraces and wild relatives (Mace et al., 2013) were analyzed and 517 additional RFL sequences were identified and included in the study (see Table of FIGS. 1A and B).

Example 2: Identification of Full Length RFL PPR Genes Potentially Involved in Fertility Restoration of *T. timopheevii* CMS in Wheat by Targeted Capture of RFL Genes

A. Selection of Germplasm Accessions:

Six wheat accessions were identified as potential restorer lines of *Timopheevii*-type CMS (T-CMS) derived from the interspecific cross between *Triticum timopheevii* and *Triticum aestivum*.

The first accession is a wheat-rye addition line, "Wheat-Rye-6R", wherein the Rf gene was mapped on the additional long arm of 6R chromosome of rye *Secale cereale* (Curtis and Lukaszewski, 1993). Four other wheat accessions are characterized by the presence of at least one of the mapped restorer genes in wheat: Rf1, Rf3 and Rf7. The commercial variety, Primepii, carries the Rf3 gene, and three Limagrain lines R197, R0934F and R0932E respectively carry both Rf1 and Rf7 genes, Rf3 or Rf1.

The sixth accession, named Anapurna, is a maintainer line not able to restore T-CMS and carries no known Rf genes. Anapurna is considered as the negative control in this experiment. In addition, a *T. timopheevii* line was included in the study as it is a fertile line expected to harbor more than one Rf gene able to restore T-type CMS (Wilson and Ross, 1962). To some extent, this line is considered as a positive control in this experiment.

All six accessions were verified in regard to their restorer status by genetic analysis.

Northern blot analysis was performed with restorer and sterile accessions using an orf256-specific probe (FIG. **2**A). Orf256 was previously identified as a gene specific to the *T. timopheevii* mitochondrial genome (Rathburn and Hedgcoth, 1991; Song and Hedgcoth, 1994). The sequence from −228 to +33 of orf256 (numbering relative to the start codon) is identical to the homologous region of the coxl gene (encoding subunit 1 of mitochondrial complex IV) in *T. aestivum*, whereas the rest of orf256, including the 3' flanking region, is unrelated to coxl (FIG. **2**A). Different patterns of orf256 transcript processing in the fertile *T. timopheevii* line and fertile restorer lines carrying the *T. timopheevii* cytoplasm were observed when compared to the sterile CMS line pattern which is coherent with the genetic analysis (FIG. **2**B). The different processing patterns are consistent with (but not conclusive proof that) orf256 is involved in causing CMS.

from DNA fragments around 600 bp with the KAPA Biosystems chemistry according to the manufacturer's recommendations.

The NGS libraries were then specifically enriched in RFL sequences using the probe pool and a capture protocol. The efficiency of the capture was confirmed by a specific qPCR assay and ultimately libraries were pooled and sequenced in paired-end mode with 300 nt read length on a MiSeq platform.

C. Assembly of Full-Length Gene Sequences Encoding RFL Proteins and Identification of Putative Orthologous Groups:

Sequence reads from the RFL capture experiment were assembled into full-length contigs spanning one or more sequences encoding RFL proteins as described below. Overlapping paired reads were merged into a single sequence using bbmerge from the bbmap package (https://sourceforge.net/projects/bbmap/) with the parameters qtrim2=t trimq=10,15,20 minq=12 mininsert=150. Read pairs that could not be merged were discarded. The merged reads were downsampled to 300,000 reads using reformat.sh in the bbmap package (samplereadstarget=300000). The merged and downsampled reads were assembled with Geneious 8 (set to Medium Sensitivity/Fast) (http://www.geneious.com/). Finally, contigs composed of more than 100 merged reads were retained for further analysis, with most of these composed of over 1000 reads. In this way, a total of 1457 contigs were generated (Table 6).

Approximately 220 contigs were obtained from each accession, except for *Triticum timopheevii* for which only 138 contigs were assembled. This is consistent with the tetraploid nature of the *Triticum timopheevii* genome. The consistency of the results indicates that the RFL-capture experiment was, a priori, comprehensive.

TABLE 6

Number of RFL contigs and ORFs identified per accession and number of orthologous groups to which the ORFs were assigned with CD-hit.

Accession name	Number of assembled contigs composed of >100 reads	Number of identified RFL ORFs >210 aa*	Number of orthologous groups with at least one RFL from the accession	Number of RFL ORFs >350 aa assigned to orthologous groups
Anapurna	211	221	202	156
Primepii	226	234	215	162
R197	219	241	219	174
R0932E	221	245	221	183
R0934F	223	237	215	174
Triticum timopheevi	138	143	129	114
Wheat-Rye-6R	219	233	212	163
TOTAL	1457	1554	397 (non-redundant)	1254

B. Bait Design and RFL-Capture from Different Wheat Genotypes:

The 1188 RFL PPR sequences identified by our bioinformatics analysis underwent a pre-treatment process that included masking of the target sequences against wheat mitochondrial and chloroplastic genome sequences (accessions NC_007579.1 and AB042240) as well as repeated elements of wheat genome. The masked target sequences were used for capture probe design. Briefly, probes were designed to cover the target sequences with a frequency masking algorithm intended to rule out probes that match with high copy number sequences in the targeted genome(s). The final probes were synthetized as a probe pool.

Seeds of each accession were sown and plantlets were grown in etiolated conditions. After DNA extraction, Illumina libraries (referred to as NGS libraries) were prepared

Sequences encoding RFL proteins were identified within these contigs as follows.

Open reading frames (ORFs) within the contigs were identified with getorf from the EMBOSS package (Rice et al. 2000) using the parameters-minsize 630-find 0-reverse true. Thus only ORFs longer than 210 amino acids were used for further analysis. In total, 1554 ORFs were identified across the seven accessions (Table 6). The number of ORFs per accession ranged from 143 ORFs in *T. timopheevii* to 245 in R0934F (Table 6). The ORFs were further analyzed using hmmsearch (with parameters -E 0.1 --domE 100) from the HMMER package (v3.1b1) to detect PPR motifs using the hidden Markov models developed by Cheng et al. (2016) and the post-processing steps described in the same paper. Finally, to identify putatively orthologous RFL sequences across all seven accessions, the 1554 RFL ORFs were clustered using CD-hit (settings -c 0.96 -n 5 -G 0 -d 0 -AS 60 -A 105 -g 1). Across all accessions, 397 non-redundant RFL clusters representing putatively orthologous groups were obtained (Table 6). We define an orthologous RFL group as a set of at least one RFL ORF from at least one accession and wherein, if at least two sequences are present, these sequences share at least 96% sequence identity over the alignment length. Some highly conserved RFL genes are present in all seven accessions, others are found in only a subset of the accessions, or in a single accession. In most cases, each orthologous RFL group contains only one RFL protein from each accession.

Table 7 below is showing the different RFL groups, the corresponding ORF names and the corresponding protein sequence number and DNA encoding protein sequence number.

However, we found that genes encoding RFL-PPR proteins are often inactivated by indels creating frameshifts that break the contiguity of the ORFs, resulting in two shorter ORFs corresponding to a single longer ORF in another accession. In these cases, both shorter ORFs could be within the same orthologous group. In our analysis, only ORF encoding more than 350 amino acids were considered as possibly functional. This threshold was used as it corresponds to 10 PPR motifs (each of 35 amino acids), and all known active Rf proteins contain at least this number of motifs (usually 15-20).

Finally, a set of 397 orthologous RFL groups were identified and numbered from 1 to 397 (see Table 7 below).

TABLE 7

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 1	R0932E.300k_Assembly_Contig_60_2	1	1555
RFL 1	R197.300k_Assembly_Contig_73_2	2	1556
RFL 2	R0934F.300k_Assembly_Contig_11_2	3	1557
RFL 2	Anapurna.300k_Assembly_Contig_47_1	4	1558
RFL 2	Primepii.300k_Assembly_Contig_16_2	5	1559
RFL 2	R0932E.300k_Assembly_Contig_39_1	6	1560
RFL 2	R197.300k_Assembly_Contig_31_1	7	1561
RFL 2	Wheat-Rye-6R.300k_Assembly_Contig_13_1	8	1562
RFL 3	R0934F.300k_Assembly_Contig_34_2	9	1563
RFL 3	R0932E.300k_Assembly_Contig_41_1	10	1564
RFL 3	Wheat-Rye-6R.300k_Assembly_Contig_44_1	11	1565
RFL 3	Primepii.300k_Assembly_Contig_27_2	12	1566
RFL 3	R197.300k_Assembly_Contig_34_1	13	1567
RFL 3	Anapurna.300k_Assembly_Contig_21_1	14	1568
RFL 4	R197.300k_Assembly_Contig_38_2	15	1569
RFL 4	Anapurna.300k_Assembly_Contig_48_2	16	1570
RFL 4	Primepii.300k_Assembly_Contig_39_2	17	1571
RFL 4	R0932E.300k_Assembly_Contig_47_1	18	1572
RFL 4	R0934F.300k_Assembly_Contig_44_1	19	1573
RFL 4	Wheat-Rye-6R.300k_Assembly_Contig_32_2	20	1574
RFL 4	Triticum-timopheevii.300k_Assembly_Contig_19_2	21	1575
RFL 5	Primepii.300k_Assembly_Contig_9_3	22	1576
RFL 5	R0934F.300k_Assembly_Contig_7_3	23	1577
RFL 5	Anapurna.300k_Assembly_Contig_15_3	24	1578
RFL 5	R197.300k_Assembly_Contig_12_3	25	1579
RFL 5	R0932E.300k_Assembly_Contig_11_2	26	1580
RFL 5	Wheat-Rye-6R.300k_Assembly_Contig_9_2	27	1581
RFL 5	Triticum-timopheevii.300k_Assembly_Contig_9_2	28	1582
RFL 6	Wheat-Rye-6R.300k_Assembly_Contig_99_1	29	1583
RFL 6	Primepii.300k_Assembly_Contig_109_1	30	1584
RFL 6	Anapurna.300k_Assembly_Contig_99_2	31	1585
RFL 6	Anapurna.300k_Assembly_Contig_99_1	32	1586
RFL 7	Anapurna.300k_Assembly_Contig_82_1	33	1587
RFL 7	Primepii.300k_Assembly_Contig_25_1	34	1588
RFL 7	R0934F.300k_Assembly_Contig_86_1	35	1589
RFL 7	R197.300k_Assembly_Contig_11_1	36	1590
RFL 7	R0932E.300k_Assembly_Contig_61_1	37	1591
RFL 7	Wheat-Rye-6R.300k_Assembly_Contig_34_1	38	1592
RFL 8	R0932E.300k_Assembly_Contig_85_1	39	1593
RFL 8	Wheat-Rye-6R.300k_Assembly_Contig_79_2	40	1594
RFL 8	R0934F.300k_Assembly_Contig_87_2	41	1595
RFL 8	Primepii.300k_Assembly_Contig_64_1	42	1596
RFL 8	R197.300k_Assembly_Contig_71_2	43	1597
RFL 8	Anapurna.300k_Assembly_Contig_54_2	44	1598
RFL 9	R0932E.300k_Assembly_Contig_66_1	45	1599
RFL 9	Wheat-Rye-6R.300k_Assembly_Contig_69_1	46	1600
RFL 9	Primepii.300k_Assembly_Contig_59_1	47	1601
RFL 9	R197.300k_Assembly_Contig_51_1	48	1602
RFL 9	R0934F.300k_Assembly_Contig_89_1	49	1603
RFL 9	Anapurna.300k_Assembly_Contig_66_1	50	1604
RFL 10	Primepii.300k_Assembly_Contig_62_2	51	1605
RFL 10	Anapurna.300k_Assembly_Contig_179_1	52	1606
RFL 10	Anapurna.300k_Assembly_Contig_70_1	53	1607
RFL 10	R0932E.300k_Assembly_Contig_34_2	54	1608
RFL 10	Primepii.300k_Assembly_Contig_191_1	55	1609
RFL 10	R197.300k_Assembly_Contig_189_1	56	1610

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 10	Triticum-timopheevii.300k_Assembly_Contig_13_1	57	1611
RFL 10	R197.300k_Assembly_Contig_64_1	58	1612
RFL 10	Wheat-Rye-6R.300k_Assembly_Contig_62_2	59	1613
RFL 10	Wheat-Rye-6R.300k_Assembly_Contig_190_1	60	1614
RFL 10	R0934F.300k_Assembly_Contig_29_1	61	1615
RFL 11	Anapurna.300k_Assembly_Contig_14_2	62	1616
RFL 11	R0934F.300k_Assembly_Contig_14_2	63	1617
RFL 11	R0934F.300k_Assembly_Contig_14_1	64	1618
RFL 11	Triticum-timopheevii.300k_Assembly_Contig_22_1	65	1619
RFL 11	Primepii.300k_Assembly_Contig_38_2	66	1620
RFL 11	Triticum-timopheevii.300k_Assembly_Contig_22_2	67	1621
RFL 11	R197.300k_Assembly_Contig_33_1	68	1622
RFL 11	R0932E.300k_Assembly_Contig_27_2	69	1623
RFL 11	R197.300k_Assembly_Contig_33_2	70	1624
RFL 11	R0932E.300k_Assembly_Contig_27_1	71	1625
RFL 11	Wheat-Rye-6R.300k_Assembly_Contig_12_2	72	1626
RFL 12	R197.300k_Assembly_Contig_119_1	73	1627
RFL 12	Wheat-Rye-6R.300k_Assembly_Contig_115_1	74	1628
RFL 12	R0932E.300k_Assembly_Contig_120_2	75	1629
RFL 13	Anapurna.300k_Assembly_Contig_8_3	76	1630
RFL 13	Primepii.300k_Assembly_Contig_8_2	77	1631
RFL 13	R0932E.300k_Assembly_Contig_8_2	78	1632
RFL 13	R197.300k_Assembly_Contig_25_2	79	1633
RFL 13	Wheat-Rye-6R.300k_Assembly_Contig_21_2	80	1634
RFL 13	R0934F.300k_Assembly_Contig_18_2	81	1635
RFL 14	R0932E.300k_Assembly_Contig_24_1	82	1636
RFL 14	R0934F.300k_Assembly_Contig_10_1	83	1637
RFL 14	Wheat-Rye-6R.300k_Assembly_Contig_26_1	84	1638
RFL 14	R197.300k_Assembly_Contig_23_1	85	1639
RFL 14	Primepii.300k_Assembly_Contig_11_1	86	1640
RFL 14	Anapurna.300k_Assembly_Contig_13_1	87	1641
RFL 15	R197.300k_Assembly_Contig_29_1	88	1642
RFL 15	Anapurna.300k_Assembly_Contig_28_1	89	1643
RFL 15	R0934F.300k_Assembly_Contig_20_1	90	1644
RFL 15	Triticum-timopheevii.300k_Assembly_Contig_11_2	91	1645
RFL 15	R0932E.300k_Assembly_Contig_38_1	92	1646
RFL 15	Wheat-Rye-6R.300k_Assembly_Contig_6_1	93	1647
RFL 15	Primepii.300k_Assembly_Contig_22_1	94	1648
RFL 16	Primepii.300k_Assembly_Contig_23_2	95	1649
RFL 16	R197.300k_Assembly_Contig_43_2	96	1650
RFL 16	Wheat-Rye-6R.300k_Assembly_Contig_11_2	97	1651
RFL 16	Anapurna.300k_Assembly_Contig_10_2	98	1652
RFL 16	R0932E.300k_Assembly_Contig_18_2	99	1653
RFL 16	R0934F.300k_Assembly_Contig_15_2	100	1654
RFL 17	Triticum-timopheevii.300k_Assembly_Contig_7_1	101	1655
RFL 17	R197.300k_Assembly_Contig_58_1	102	1656
RFL 17	Wheat-Rye-6R.300k_Assembly_Contig_59_1	103	1657
RFL 17	R0934F.300k_Assembly_Contig_51_1	104	1658
RFL 17	Triticum-timopheevii.300k_Assembly_Contig_29_2	105	1659
RFL 18	R0934F.300k_Assembly_Contig_25_2	106	1660
RFL 18	R0932E.300k_Assembly_Contig_21_2	107	1661
RFL 18	R197.300k_Assembly_Contig_14_2	108	1662
RFL 18	Anapurna.300k_Assembly_Contig_6_2	109	1663
RFL 18	Triticum-timopheevii.300k_Assembly_Contig_16_3	110	1664
RFL 18	Primepii.300k_Assembly_Contig_20_2	111	1665
RFL 18	Wheat-Rye-6R.300k_Assembly_Contig_8_2	112	1666
RFL 19	Anapurna.300k_Assembly_Contig_68_1	113	1667
RFL 20	Triticum-timopheevii.300k_Assembly_Contig_47_1	114	1668
RFL 21	R197.300k_Assembly_Contig_18_1	115	1669
RFL 21	Anapurna.300k_Assembly_Contig_16_1	116	1670
RFL 21	R0932E.300k_Assembly_Contig_58_1	117	1671
RFL 21	Wheat-Rye-6R.300k_Assembly_Contig_23_3	118	1672
RFL 21	Wheat-Rye-6R.300k_Assembly_Contig_23_2	119	1673
RFL 21	Primepii.300k_Assembly_Contig_42_2	120	1674
RFL 21	R0934F.300k_Assembly_Contig_28_3	121	1675
RFL 21	Primepii.300k_Assembly_Contig_42_3	122	1676
RFL 21	R0934F.300k_Assembly_Contig_28_2	123	1677
RFL 22	Primepii.300k_Assembly_Contig_63_1	124	1678
RFL 22	R0934F.300k_Assembly_Contig_61_1	125	1679
RFL 23	R0934F.300k_Assembly_Contig_46_2	126	1680

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 23	Wheat-Rye-6R.300k__Assembly__Contig__30__1	127	1681
RFL 23	R197.300k__Assembly__Contig__15__3	128	1682
RFL 23	R0932E.300k__Assembly__Contig__40__1	129	1683
RFL 23	R0934F.300k__Assembly__Contig__46__1	130	1684
RFL 23	R197.300k__Assembly__Contig__15__2	131	1685
RFL 23	Anapurna.300k__Assembly__Contig__42__1	132	1686
RFL 23	Primepii.300k__Assembly__Contig__14__2	133	1687
RFL 24	Anapurna.300k__Assembly__Contig__17__1	134	1688
RFL 24	R0934F.300k__Assembly__Contig__22__1	135	1689
RFL 24	R0932E.300k__Assembly__Contig__44__1	136	1690
RFL 24	Primepii.300k__Assembly__Contig__46__1	137	1691
RFL 24	R197.300k__Assembly__Contig__30__1	138	1692
RFL 24	Wheat-Rye-6R.300k__Assembly__Contig__40__1	139	1693
RFL 25	R197.300k__Assembly__Contig__92__2	140	1694
RFL 25	Wheat-Rye-6R.300k__Assembly__Contig__109__2	141	1695
RFL 26	Anapurna.300k__Assembly__Contig__38__1	142	1696
RFL 26	Wheat-Rye-6R.300k__Assembly__Contig__41__1	143	1697
RFL 27	Triticum-timopheevii.300k__Assembly__Contig__63__3	144	1698
RFL 28	R0932E.300k__Assembly__Contig__23__2	145	1699
RFL 28	R0932E.300k__Assembly__Contig__23__3	146	1700
RFL 28	Primepii.300k__Assembly__Contig__13__1	147	1701
RFL 28	R197.300k__Assembly__Contig__10__3	148	1702
RFL 28	R197.300k__Assembly__Contig__10__2	149	1703
RFL 28	R0934F.300k__Assembly__Contig__17__1	150	1704
RFL 28	Anapurna.300k__Assembly__Contig__2__3	151	1705
RFL 28	Wheat-Rye-6R.300k__Assembly__Contig__7__3	152	1706
RFL 28	Anapurna.300k__Assembly__Contig__2__2	153	1707
RFL 28	Wheat-Rye-6R.300k__Assembly__Contig__7__2	154	1708
RFL 29	Wheat-Rye-6R.300k__Assembly__Contig__77__2	155	1709
RFL 29	R0934F.300k__Assembly__Contig__78__1	156	1710
RFL 29	Wheat-Rye-6R.300k__Assembly__Contig__77__1	157	1711
RFL 29	Primepii.300k__Assembly__Contig__67__1	158	1712
RFL 30	Primepii.300k__Assembly__Contig__91__1	159	1713
RFL 30	R0934F.300k__Assembly__Contig__64__1	160	1714
RFL 30	R0932E.300k__Assembly__Contig__110__1	161	1715
RFL 31	Triticum-timopheevii.300k__Assembly__Contig__14__1	162	1716
RFL 32	Triticum-timopheevii.300k__Assembly__Contig__6__2	163	1717
RFL 32	Primepii.300k__Assembly__Contig__6__2	164	1718
RFL 32	R0934F.300k__Assembly__Contig__31__2	165	1719
RFL 32	R197.300k__Assembly__Contig__7__2	166	1720
RFL 32	Anapurna.300k__Assembly__Contig__31__2	167	1721
RFL 32	Wheat-Rye-6R.300k__Assembly__Contig__5__2	168	1722
RFL 32	R0932E.300k__Assembly__Contig__29__2	169	1723
RFL 33	Triticum-timopheevii.300k__Assembly__Contig__8__1	170	1724
RFL 33	Anapurna.300k__Assembly__Contig__5__2	171	1725
RFL 33	R197.300k__Assembly__Contig__6__3	172	1726
RFL 33	Wheat-Rye-6R.300k__Assembly__Contig__25__1	173	1727
RFL 33	R0932E.300k__Assembly__Contig__19__1	174	1728
RFL 33	Primepii.300k__Assembly__Contig__21__2	175	1729
RFL 33	R0934F.300k__Assembly__Contig__104__1	176	1730
RFL 34	R0932E.300k__Assembly__Contig__5__1	177	1731
RFL 34	R0932E.300k__Assembly__Contig__2__4	178	1732
RFL 35	Triticum-timopheevii.300k__Assembly__Contig__21__2	179	1733
RFL 36	Triticum-timopheevii.300k__Assembly__Contig__20__2	180	1734
RFL 37	Anapurna.300k__Assembly__Contig__37__2	181	1735
RFL 37	Wheat-Rye-6R.300k__Assembly__Contig__19__2	182	1736
RFL 37	R0932E.300k__Assembly__Contig__12__2	183	1737
RFL 37	R197.300k__Assembly__Contig__20__2	184	1738
RFL 37	R0934F.300k__Assembly__Contig__37__2	185	1739
RFL 37	Primepii.300k__Assembly__Contig__10__2	186	1740
RFL 38	Triticum-timopheevii.300k__Assembly__Contig__43__2	187	1741
RFL 39	R0932E.300k__Assembly__Contig__53__2	188	1742
RFL 39	Anapurna.300k__Assembly__Contig__29__2	189	1743
RFL 39	R0932E.300k__Assembly__Contig__22__1	190	1744
RFL 39	Wheat-Rye-6R.300k__Assembly__Contig__28__2	191	1745
RFL 39	Triticum-timopheevii.300k__Assembly__Contig__25__2	192	1746
RFL 39	Wheat-Rye-6R.300k__Assembly__Contig__15__1	193	1747
RFL 39	R0934F.300k__Assembly__Contig__27__1	194	1748
RFL 39	R0934F.300k__Assembly__Contig__26__2	195	1749
RFL 39	Primepii.300k__Assembly__Contig__40__2	196	1750

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 39	Primepii.300k_Assembly_Contig_30_1	197	1751
RFL 39	R197.300k_Assembly_Contig_36_2	198	1752
RFL 39	R197.300k_Assembly_Contig_27_1	199	1753
RFL 39	Anapurna.300k_Assembly_Contig_43_1	200	1754
RFL 40	R0934F.300k_Assembly_Contig_12_1	201	1755
RFL 40	Primepii.300k_Assembly_Contig_18_1	202	1756
RFL 40	Triticum-timopheevii.300k_Assembly_Contig_5_1	203	1757
RFL 40	R0932E.300k_Assembly_Contig_7_1	204	1758
RFL 40	Anapurna.300k_Assembly_Contig_20_1	205	1759
RFL 40	R197.300k_Assembly_Contig_32_1	206	1760
RFL 41	R0934F.300k_Assembly_Contig_24_2	207	1761
RFL 41	Wheat-Rye-6R.300k_Assembly_Contig_45_2	208	1762
RFL 41	Anapurna.300k_Assembly_Contig_39_2	209	1763
RFL 41	R0932E.300k_Assembly_Contig_56_2	210	1764
RFL 41	Primepii.300k_Assembly_Contig_47_2	211	1765
RFL 41	R197.300k_Assembly_Contig_40_2	212	1766
RFL 42	Triticum-timopheevii.300k_Assembly_Contig_10_3	213	1767
RFL 43	R0934F.300k_Assembly_Contig_8_1	214	1768
RFL 43	R0932E.300k_Assembly_Contig_13_1	215	1769
RFL 43	Triticum-timopheevii.300k_Assembly_Contig_12_2	216	1770
RFL 43	R197.300k_Assembly_Contig_22_1	217	1771
RFL 43	Primepii.300k_Assembly_Contig_31_1	218	1772
RFL 43	Anapurna.300k_Assembly_Contig_23_1	219	1773
RFL 43	Wheat-Rye-6R.300k_Assembly_Contig_10_1	220	1774
RFL 44	R0932E.300k_Assembly_Contig_32_1	221	1775
RFL 44	Primepii.300k_Assembly_Contig_7_1	222	1776
RFL 44	Anapurna.300k_Assembly_Contig_33_1	223	1777
RFL 45	R0932E.300k_Assembly_Contig_73_2	224	1778
RFL 45	R197.300k_Assembly_Contig_70__2	225	1779
RFL 45	Wheat-Rye-6R.300k_Assembly_Contig_72_2	226	1780
RFL 46	Wheat-Rye-6R.300k_Assembly_Contig_35_1	227	1781
RFL 47	R0934F.300k_Assembly_Contig_57_2	228	1782
RFL 47	Wheat-Rye-6R.300k_Assembly_Contig_67_2	229	1783
RFL 47	Anapurna.300k_Assembly_Contig_85_2	230	1784
RFL 47	R0932E.300k_Assembly_Contig_71_2	231	1785
RFL 47	R197.300k_Assembly_Contig_62_2	232	1786
RFL 47	Primepii.300k_Assembly_Contig_86_3	233	1787
RFL 48	Primepii.300k_Assembly_Contig_51_1	234	1788
RFL 48	R197.300k_Assembly_Contig_57_1	235	1789
RFL 48	Anapurna.300k_Assembly_Contig_74_1	236	1790
RFL 48	R0934F.300k_Assembly_Contig_71_1	237	1791
RFL 48	R0932E.300k_Assembly_Contig_72_1	238	1792
RFL 48	Wheat-Rye-6R.300k_Assembly_Contig_82_1	239	1793
RFL 49	R197.300k_Assembly_Contig_83_1	240	1794
RFL 49	R197.300k_Assembly_Contig_95_2	241	1795
RFL 49	Triticum-timopheevii.300k_Assembly_Contig_37_1	242	1796
RFL 49	Triticum-timopheevii.300k_Assembly_Contig_38_1	243	1797
RFL 50	Anapurna.300k_Assembly_Contig_24_2	244	1798
RFL 50	Primepii.300k_Assembly_Contig_24_2	245	1799
RFL 50	R0932E.300k_Assembly_Contig_25_2	246	1800
RFL 50	R197.300k_Assembly_Contig_21_2	247	1801
RFL 50	Wheat-Rye-6R.300k_Assembly_Contig_22_2	248	1802
RFL 50	R0934F.300k_Assembly_Contig_39_2	249	1803
RFL 51	Anapurna.300k_Assembly_Contig_35_2	250	1804
RFL 51	R0934F.300k_Assembly_Contig_5_4	251	1805
RFL 51	Primepii.300k_Assembly_Contig_17_3	252	1806
RFL 51	Anapurna.300k_Assembly_Contig_109_1	253	1807
RFL 51	Primepii.300k_Assembly_Contig_19_2	254	1808
RFL 51	R0932E.300k_Assembly_Contig_55_1	255	1809
RFL 51	Wheat-Rye-6R.300k_Assembly_Contig_2_1	256	1810
RFL 51	R0934F.300k_Assembly_Contig_55_2	257	1811
RFL 51	Triticum-timopheevii.300k_Assembly_Contig_4_2	258	1812
RFL 51	R197.300k_Assembly_Contig_46_2	259	1813
RFL 51	Wheat-Rye-6R.300k_Assembly_Contig_37_2	260	1814
RFL 51	Anapurna.300k_Assembly_Contig_106_1	261	1815
RFL 51	R197.300k_Assembly_Contig_5_3	262	1816
RFL 51	R0932E.300k_Assembly_Contig_6_2	263	1817
RFL 52	R197.300k_Assembly_Contig_37_1	264	1818
RFL 52	R197.300k_Assembly_Contig_37_2	265	1819
RFL 52	R0932E.300k_Assembly_Contig_45_1	266	1820
RFL 52	Wheat-Rye-6R.300k_Assembly_Contig_39_1	267	1821
RFL 52	Wheat-Rye-6R.300k_Assembly_Contig_39_2	268	1822

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 52	Primepii.300k_Assembly_Contig_49_1	269	1823
RFL 52	Anapurna.300k_Assembly_Contig_22_1	270	1824
RFL 52	R0934F.300k_Assembly_Contig_49_1	271	1825
RFL 52	R0934F.300k_Assembly_Contig_49_2	272	1826
RFL 52	Triticum-timopheevii.300k_Assembly_Contig_29_1	273	1827
RFL 53	R0934F.300k_Assembly_Contig_75_2	274	1828
RFL 53	R0932E.300k_Assembly_Contig_64_2	275	1829
RFL 53	Anapurna.300k_Assembly_Contig_72_2	276	1830
RFL 53	Primepii.300k_Assembly_Contig_57_2	277	1831
RFL 53	Triticum-timopheevii.300k_Assembly_Contig_55_3	278	1832
RFL 54	Wheat-Rye-6R.300k_Assembly_Contig_80_1	279	1833
RFL 54	R0932E.300k_Assembly_Contig_209_1	280	1834
RFL 54	Anapurna.300k_Assembly_Contig_77_1	281	1835
RFL 55	R0932E.300k_Assembly_Contig_54_1	282	1836
RFL 55	R197.300k_Assembly_Contig_39_1	283	1837
RFL 55	R0934F.300k_Assembly_Contig_54_1	284	1838
RFL 55	Anapurna.300k_Assembly_Contig_41_1	285	1839
RFL 55	Primepii.300k_Assembly_Contig_44_1	286	1840
RFL 55	Wheat-Rye-6R.300k_Assembly_Contig_31_1	287	1841
RFL 56	R197.300k_Assembly_Contig_86_1	288	1842
RFL 56	R0932E.300k_Assembly_Contig_74_1	289	1843
RFL 56	Triticum-timopheevii.300k_Assembly_Contig_39_1	290	1844
RFL 58	Wheat-Rye-6R.300k_Assembly_Contig_66_1	291	1845
RFL 58	Anapurna.300k_Assembly_Contig_12_1	292	1846
RFL 59	Triticum-timopheevii.300k_Assembly_Contig_60_1	293	1847
RFL 59	R197.300k_Assembly_Contig_115_1	294	1848
RFL 59	R0932E.300k_Assembly_Contig_161_1	295	1849
RFL 59	R0932E.300k_Assembly_Contig_48_1	296	1850
RFL 60	Primepii.300k_Assembly_Contig_94_1	297	1851
RFL 60	R197.300k_Assembly_Contig_95_1	298	1852
RFL 60	R0934F.300k_Assembly_Contig_73_2	299	1853
RFL 60	Wheat-Rye-6R.300k_Assembly_Contig_48_2	300	1854
RFL 61	Triticum-timopheevii.300k_Assembly_Contig_48_1	301	1855
RFL 62	R0934F.300k_Assembly_Contig_138_1	302	1856
RFL 63	R197.300k_Assembly_Contig_84_1	303	1857
RFL 63	R197.300k_Assembly_Contig_84_2	304	1858
RFL 63	Triticum-timopheevii.300k_Assembly_Contig_34_1	305	1859
RFL 64	Primepii.300k_Assembly_Contig_50_1	306	1860
RFL 64	Wheat-Rye-6R.300k_Assembly_Contig_54_1	307	1861
RFL 64	R0934F.300k_Assembly_Contig_45_1	308	1862
RFL 64	Anapurna.300k_Assembly_Contig_61_1	309	1863
RFL 64	R197.300k_Assembly_Contig_42_1	310	1864
RFL 64	R0932E.300k_Assembly_Contig_59_1	311	1865
RFL 65	Triticum-timopheevii.300k_Assembly_Contig_42_2	312	1866
RFL 66	Triticum-timopheevii.300k_Assembly_Contig_27_1	313	1867
RFL 66	Triticum-timopheevii.300k_Assembly_Contig_80_1	314	1868
RFL 67	Primepii.300k_Assembly_Contig_213_1	315	1869
RFL 67	Primepii.300k_Assembly_Contig_80_2	316	1870
RFL 67	Primepii.300k_Assembly_Contig_80_1	317	1871
RFL 67	R0934F.300k_Assembly_Contig_111_2	318	1872
RFL 67	Primepii.300k_Assembly_Contig_2_1	319	1873
RFL 67	R0934F.300k_Assembly_Contig_111_1	320	1874
RFL 67	R0934F.300k_Assembly_Contig_4_1	321	1875
RFL 68	Wheat-Rye-6R.300k_Assembly_Contig_53_1	322	1876
RFL 68	R0932E.300k_Assembly_Contig_68_1	323	1877
RFL 68	Wheat-Rye-6R.300k_Assembly_Contig_53_3	324	1878
RFL 68	R197.300k_Assembly_Contig_19_1	325	1879
RFL 68	R0934F.300k_Assembly_Contig_66_1	326	1880
RFL 68	Triticum-timopheevii.300k_Assembly_Contig_33_1	327	1881
RFL 68	Anapurna.300k_Assembly_Contig_71_1	328	1882
RFL 68	Anapurna.300k_Assembly_Contig_71_2	329	1883
RFL 68	Primepii.300k_Assembly_Contig_37_1	330	1884
RFL 68	R0932E.300k_Assembly_Contig_68_3	331	1885
RFL 68	R197.300k_Assembly_Contig_19_3	332	1886
RFL 69	Triticum-timopheevii.300k_Assembly_Contig_45_1	333	1887
RFL 70	R0934F.300k_Assembly_Contig_68_1	334	1888
RFL 70	R197.300k_Assembly_Contig_65_1	335	1889

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 70	R0932E.300k_Assembly_Contig_94_1	336	1890
RFL 70	Primepii.300k_Assembly_Contig_55_1	337	1891
RFL 70	Anapurna.300k_Assembly_Contig_111_1	338	1892
RFL 70	Wheat-Rye-6R.300k_Assembly_Contig_74_1	339	1893
RFL 70	Anapurna.300k_Assembly_Contig_98_1	340	1894
RFL 71	R0932E.300k_Assembly_Contig_114_1	341	1895
RFL 72	R0932E.300k_Assembly_Contig_98_1	342	1896
RFL 73	R0932E.300k_Assembly_Contig_89_2	343	1897
RFL 73	R197.300k_Assembly_Contig_78_2	344	1898
RFL 73	Triticum-timopheevii.300k_Assembly_Contig_31_2	345	1899
RFL 74	R197.300k_Assembly_Contig_194_1	346	1900
RFL 74	R0934F.300k_Assembly_Contig_36_2	347	1901
RFL 74	R0934F.300k_Assembly_Contig_150_2	348	1902
RFL 74	R0932E.300k_Assembly_Contig_185_1	349	1903
RFL 74	R0932E.300k_Assembly_Contig_201_1	350	1904
RFL 74	R0932E.300k_Assembly_Contig_92_1	351	1905
RFL 74	R197.300k_Assembly_Contig_179_1	352	1906
RFL 74	R0932E.300k_Assembly_Contig_83_1	353	1907
RFL 74	R197.300k_Assembly_Contig_4_1	354	1908
RFL 74	R0934F.300k_Assembly_Contig_126_1	355	1909
RFL 75	R0932E.300k_Assembly_Contig_88_1	356	1910
RFL 77	R0932E.300k_Assembly_Contig_214_2	357	1911
RFL 78	R0932E.300k_Assembly_Contig_112_1	358	1912
RFL 79	R0932E.300k_Assembly_Contig_103_1	359	1913
RFL 79	R0934F.300k_Assembly_Contig_80_1	360	1914
RFL 79	R197.300k_Assembly_Contig_120_1	361	1915
RFL 79	Triticum-timopheevii.300k_Assembly_Contig_57_1	362	1916
RFL 80	R197.300k_Assembly_Contig_59_1	363	1917
RFL 81	R0932E.300k_Assembly_Contig_4_1	364	1918
RFL 81	R0932E.300k_Assembly_Contig_199_1	365	1919
RFL 82	Triticum-timopheevii.300k_Assembly_Contig_64_2	366	1920
RFL 82	R0932E.300k_Assembly_Contig_123_2	367	1921
RFL 83	Primepii.300k_Assembly_Contig_52_1	368	1922
RFL 83	R0934F.300k_Assembly_Contig_43_1	369	1923
RFL 83	R0932E.300k_Assembly_Contig_46_1	370	1924
RFL 83	Anapurna.300k_Assembly_Contig_62_1	371	1925
RFL 83	R197.300k_Assembly_Contig_54_1	372	1926
RFL 83	Wheat-Rye-6R.300k_Assembly_Contig_60_1	373	1927
RFL 84	R0932E.300k_Assembly_Contig_116_1	374	1928
RFL 85	R197.300k_Assembly_Contig_80_3	375	1929
RFL 85	Triticum-timopheevii.300k_Assembly_Contig_36_3	376	1930
RFL 85	R197.300k_Assembly_Contig_80_2	377	1931
RFL 87	Wheat-Rye-6R.300k_Assembly_Contig_73_1	378	1932
RFL 89	Primepii.300k_Assembly_Contig_83_1	379	1933
RFL 89	Primepii.300k_Assembly_Contig_183_1	380	1934
RFL 89	R0934F.300k_Assembly_Contig_99_1	381	1935
RFL 90	Primepii.300k_Assembly_Contig_82_2	382	1936
RFL 90	Wheat-Rye-6R.300k_Assembly_Contig_65_2	383	1937
RFL 90	R0934F.300k_Assembly_Contig_63_1	384	1938
RFL 90	R197.300k_Assembly_Contig_66_2	385	1939
RFL 90	Anapurna.300k_Assembly_Contig_51_1	386	1940
RFL 90	R0932E.300k_Assembly_Contig_90_1	387	1941
RFL 92	Wheat-Rye-6R.300k_Assembly_Contig_55_2	388	1942
RFL 92	R197.300k_Assembly_Contig_47_2	389	1943
RFL 92	Triticum-timopheevii.300k_Assembly_Contig_30_2	390	1944
RFL 92	R0932E.300k_Assembly_Contig_43_2	391	1945
RFL 92	Triticum-timopheevii.300k_Assembly_Contig_30_3	392	1946
RFL 92	Anapurna.300k_Assembly_Contig_50_2	393	1947
RFL 92	R0934F.300k_Assembly_Contig_48_2	394	1948
RFL 92	Primepii.300k_Assembly_Contig_53_2	395	1949
RFL 93	R197.300k_Assembly_Contig_117_1	396	1950
RFL 93	R0932E.300k_Assembly_Contig_113_1	397	1951
RFL 94	Wheat-Rye-6R.300k_Assembly_Contig_41_2	398	1952
RFL 94	Anapurna.300k_Assembly_Contig_38_2	399	1953
RFL 94	R197.300k_Assembly_Contig_24_1	400	1954
RFL 94	R0934F.300k_Assembly_Contig_32_1	401	1955
RFL 94	R0932E.300k_Assembly_Contig_17_1	402	1956
RFL 94	Primepii.300k_Assembly_Contig_41_2	403	1957
RFL 95	Triticum-timopheevii.300k_Assembly_Contig_51_1	404	1958
RFL 96	Primepii.300k_Assembly_Contig_72_1	405	1959
RFL 97	Anapurna.300k_Assembly_Contig_92_2	406	1960

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 97	Anapurna.300k_Assembly_Contig_19_1	407	1961
RFL 97	Wheat-Rye-6R.300k_Assembly_Contig_103_2	408	1962
RFL 97	Wheat-Rye-6R.300k_Assembly_Contig_63_1	409	1963
RFL 98	R0932E.300k_Assembly_Contig_79_1	410	1964
RFL 98	Primepii.300k_Assembly_Contig_70_1	411	1965
RFL 98	R0934F.300k_Assembly_Contig_70_1	412	1966
RFL 98	R197.300k_Assembly_Contig_63_1	413	1967
RFL 98	Wheat-Rye-6R.300k_Assembly_Contig_71_1	414	1968
RFL 99	R0932E.300k_Assembly_Contig_15_2	415	1969
RFL 99	R0932E.300k_Assembly_Contig_153_2	416	1970
RFL 100	R197.300k_Assembly_Contig_94_1	417	1971
RFL 100	Wheat-Rye-6R.300k_Assembly_Contig_113_1	418	1972
RFL 100	R0932E.300k_Assembly_Contig_100_1	419	1973
RFL 100	Anapurna.300k_Assembly_Contig_113_1	420	1974
RFL 101	Anapurna.300k_Assembly_Contig_59_1	421	1975
RFL 101	Triticum-timopheevii.300k_Assembly_Contig_54_1	422	1976
RFL 102	Triticum-timopheevii.300k_Assembly_Contig_71_1	423	1977
RFL 103	Primepii.300k_Assembly_Contig_106_1	424	1978
RFL 103	Wheat-Rye-6R.300k_Assembly_Contig_98_1	425	1979
RFL 103	R197.300k_Assembly_Contig_79_1	426	1980
RFL 104	R0934F.300k_Assembly_Contig_69_1	427	1981
RFL 104	Triticum-timopheevii.300k_Assembly_Contig_35_1	428	1982
RFL 104	R0932E.300k_Assembly_Contig_82_1	429	1983
RFL 104	R197.300k_Assembly_Contig_72_1	430	1984
RFL 105	Primepii.300k_Assembly_Contig_85_1	431	1985
RFL 106	Wheat-Rye-6R.300k_Assembly_Contig_87_2	432	1986
RFL 106	Primepii.300k_Assembly_Contig_81_2	433	1987
RFL 106	Anapurna.300k_Assembly_Contig_80_1	434	1988
RFL 106	R0932E.300k_Assembly_Contig_76_1	435	1989
RFL 106	R197.300k_Assembly_Contig_67_1	436	1990
RFL 106	R0934F.300k_Assembly_Contig_103_2	437	1991
RFL 107	R0934F.300k_Assembly_Contig_117_1	438	1992
RFL 107	Triticum-timopheevii.300k_Assembly_Contig_69_1	439	1993
RFL 108	Wheat-Rye-6R.300k_Assembly_Contig_86_1	440	1994
RFL 108	R0932E.300k_Assembly_Contig_87_1	441	1995
RFL 108	R197.300k_Assembly_Contig_77_1	442	1996
RFL 108	Anapurna.300k_Assembly_Contig_78_1	443	1997
RFL 108	R0934F.300k_Assembly_Contig_115_1	444	1998
RFL 109	Triticum-timopheevii.300k_Assembly_Contig_78_2	445	1999
RFL 110	Triticum-timopheevii.300k_Assembly_Contig_15_2	446	2000
RFL 111	R0932E.300k_Assembly_Contig_78_1	447	2001
RFL 111	R0934F.300k_Assembly_Contig_23_1	448	2002
RFL 111	Anapurna.300k_Assembly_Contig_58_1	449	2003
RFL 111	R197.300k_Assembly_Contig_69_1	450	2004
RFL 111	Primepii.300k_Assembly_Contig_35_1	451	2005
RFL 111	Wheat-Rye-6R.300k_Assembly_Contig_83_1	452	2006
RFL 112	R0932E.300k_Assembly_Contig_20_1	453	2007
RFL 112	R197.300k_Assembly_Contig_17_1	454	2008
RFL 112	Wheat-Rye-6R.300k_Assembly_Contig_17_1	455	2009
RFL 112	Anapurna.300k_Assembly_Contig_4_1	456	2010
RFL 112	Primepii.300k_Assembly_Contig_12_1	457	2011
RFL 112	R0934F.300k_Assembly_Contig_6_1	458	2012
RFL 113	R0932E.300k_Assembly_Contig_216_1	459	2013
RFL 113	R0934F.300k_Assembly_Contig_202_1	460	2014
RFL 113	Primepii.300k_Assembly_Contig_92_1	461	2015
RFL 113	R197.300k_Assembly_Contig_98_1	462	2016
RFL 114	R0932E.300k_Assembly_Contig_86_1	463	2017
RFL 115	R0934F.300k_Assembly_Contig_128_1	464	2018
RFL 115	Triticum-timopheevii.300k_Assembly_Contig_74_1	465	2019
RFL 116	Triticum-timopheevii.300k_Assembly_Contig_44_2	466	2020
RFL 118	R197.300k_Assembly_Contig_75_1	467	2021
RFL 118	Wheat-Rye-6R.300k_Assembly_Contig_93_1	468	2022
RFL 118	Anapurna.300k_Assembly_Contig_87_1	469	2023
RFL 119	R197.300k_Assembly_Contig_1_1	470	2024
RFL 119	Wheat-Rye-6R.300k_Assembly_Contig_1_1	471	2025
RFL 119	Wheat-Rye-6R.300k_Assembly_Contig_194_1	472	2026
RFL 119	R0934F.300k_Assembly_Contig_2_1	473	2027
RFL 119	Primepii.300k_Assembly_Contig_1_1	474	2028
RFL 119	Anapurna.300k_Assembly_Contig_1_1	475	2029
RFL 119	R0932E.300k_Assembly_Contig_1_1	476	2030

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 120	Triticum-timopheevii.300k_Assembly_Contig_67_1	477	2031
RFL 121	R197.300k_Assembly_Contig_96_1	478	2032
RFL 121	R0932E.300k_Assembly_Contig_93_1	479	2033
RFL 121	Triticum-timopheevii.300k_Assembly_Contig_70_1	480	2034
RFL 121	R0934F.300k_Assembly_Contig_102_1	481	2035
RFL 121	Wheat-Rye-6R.300k_Assembly_Contig_95_1	482	2036
RFL 122	Primepii.300k_Assembly_Contig_90_1	483	2037
RFL 122	R197.300k_Assembly_Contig_68_1	484	2038
RFL 122	Anapurna.300k_Assembly_Contig_63_1	485	2039
RFL 122	R0932E.300k_Assembly_Contig_91_1	486	2040
RFL 122	R0934F.300k_Assembly_Contig_97_1	487	2041
RFL 122	Triticum-timopheevii.300k_Assembly_Contig_59_1	488	2042
RFL 122	Wheat-Rye-6R.300k_Assembly_Contig_85_1	489	2043
RFL 123	Triticum-timopheevii.300k_Assembly_Contig_65_1	490	2044
RFL 124	Primepii.300k_Assembly_Contig_96_1	491	2045
RFL 124	R197.300k_Assembly_Contig_116_1	492	2046
RFL 124	R0934F.300k_Assembly_Contig_110_1	493	2047
RFL 124	Wheat-Rye-6R.300k_Assembly_Contig_92_1	494	2048
RFL 124	Anapurna.300k_Assembly_Contig_84_1	495	2049
RFL 124	R0932E.300k_Assembly_Contig_107_1	496	2050
RFL 125	Triticum-timopheevii.300k_Assembly_Contig_40_1	497	2051
RFL 125	Triticum-timopheevii.300k_Assembly_Contig_41_3	498	2052
RFL 125	R197.300k_Assembly_Contig_3_1	499	2053
RFL 126	R0932E.300k_Assembly_Contig_35_1	500	2054
RFL 126	R197.300k_Assembly_Contig_9_3	501	2055
RFL 126	R0932E.300k_Assembly_Contig_35_2	502	2056
RFL 126	Anapurna.300k_Assembly_Contig_18_3	503	2057
RFL 126	Primepii.300k_Assembly_Contig_5_3	504	2058
RFL 126	Wheat-Rye-6R.300k_Assembly_Contig_29_1	505	2059
RFL 126	R0934F.300k_Assembly_Contig_19_3	506	2060
RFL 126	Wheat-Rye-6R.300k_Assembly_Contig_14_3	507	2061
RFL 126	Triticum-timopheevii.300k_Assembly_Contig_24_1	508	2062
RFL 126	Primepii.300k_Assembly_Contig_43_1	509	2063
RFL 126	R0934F.300k_Assembly_Contig_30_2	510	2064
RFL 126	R0934F.300k_Assembly_Contig_30_1	511	2065
RFL 126	R197.300k_Assembly_Contig_16_1	512	2066
RFL 126	R0932E.300k_Assembly_Contig_16_3	513	2067
RFL 126	Anapurna.300k_Assembly_Contig_11_1	514	2068
RFL 127	Anapurna.300k_Assembly_Contig_73_1	515	2069
RFL 128	R197.300k_Assembly_Contig_113_2	516	2070
RFL 129	R0932E.300k_Assembly_Contig_104_1	517	2071
RFL 129	R0934F.300k_Assembly_Contig_91_1	518	2072
RFL 129	R197.300k_Assembly_Contig_101_1	519	2073
RFL 130	Triticum-timopheevii.300k_Assembly_Contig_53_2	520	2074
RFL 131	R0934F.300k_Assembly_Contig_106_1	521	2075
RFL 131	Anapurna.300k_Assembly_Contig_94_1	522	2076
RFL 131	Primepii.300k_Assembly_Contig_112_1	523	2077
RFL 132	R197.300k_Assembly_Contig_82_1	524	2078
RFL 132	Wheat-Rye-6R.300k_Assembly_Contig_96_1	525	2079
RFL 132	Primepii.300k_Assembly_Contig_87_1	526	2080
RFL 133	R0932E.300k_Assembly_Contig_28_2	527	2081
RFL 133	Anapurna.300k_Assembly_Contig_7_2	528	2082
RFL 133	Primepii.300k_Assembly_Contig_26_2	529	2083
RFL 134	Wheat-Rye-6R.300k_Assembly_Contig_106_1	530	2084
RFL 134	R197.300k_Assembly_Contig_100_1	531	2085
RFL 134	Primepii.300k_Assembly_Contig_102_1	532	2086
RFL 135	Triticum-timopheevii.300k_Assembly_Contig_93_1	533	2087
RFL 136	Primepii.300k_Assembly_Contig_73_1	534	2088
RFL 136	R0934F.300k_Assembly_Contig_77_1	535	2089
RFL 136	R0932E.300k_Assembly_Contig_65_1	536	2090
RFL 136	Anapurna.300k_Assembly_Contig_52_1	537	2091
RFL 136	Wheat-Rye-6R.300k_Assembly_Contig_84_1	538	2092
RFL 136	R197.300k_Assembly_Contig_81_1	539	2093
RFL 137	R0934F.300k_Assembly_Contig_21_3	540	2094
RFL 137	R0932E.300k_Assembly_Contig_10_3	541	2095
RFL 137	Primepii.300k_Assembly_Contig_48_3	542	2096
RFL 137	Anapurna.300k_Assembly_Contig_9_3	543	2097
RFL 137	Wheat-Rye-6R.300k_Assembly_Contig_20_3	544	2098
RFL 137	R197.300k_Assembly_Contig_2_3	545	2099

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 138	R0934F.300k__Assembly__Contig__201__1	546	2100
RFL 138	Triticum-timopheevii.300k__Assembly__Contig__1__1	547	2101
RFL 138	R0934F.300k__Assembly__Contig__1__1	548	2102
RFL 139	R0934F.300k__Assembly__Contig__74__2	549	2103
RFL 139	Primepii.300k__Assembly__Contig__79__2	550	2104
RFL 139	Wheat-Rye-6R.300k__Assembly__Contig__70__2	551	2105
RFL 139	Anapurna.300k__Assembly__Contig__64__2	552	2106
RFL 140	Primepii.300k__Assembly__Contig__105__2	553	2107
RFL 140	R0934F.300k__Assembly__Contig__56__2	554	2108
RFL 141	R0932E.300k__Assembly__Contig__75__1	555	2109
RFL 141	R0934F.300k__Assembly__Contig__50__1	556	2110
RFL 142	R0934F.300k__Assembly__Contig__76__1	557	2111
RFL 142	Primepii.300k__Assembly__Contig__75__1	558	2112
RFL 143	Wheat-Rye-6R.300k__Assembly__Contig__3__1	559	2113
RFL 143	R0934F.300k__Assembly__Contig__9__1	560	2114
RFL 143	Primepii.300k__Assembly__Contig__3__1	561	2115
RFL 143	R197.300k__Assembly__Contig__8__1	562	2116
RFL 143	R0932E.300k__Assembly__Contig__3__1	563	2117
RFL 143	Anapurna.300k__Assembly__Contig__3__1	564	2118
RFL 144	Anapurna.300k__Assembly__Contig__49__1	565	2119
RFL 144	Wheat-Rye-6R.300k__Assembly__Contig__58__1	566	2120
RFL 144	R197.300k__Assembly__Contig__56__1	567	2121
RFL 144	R0934F.300k__Assembly__Contig__40__1	568	2122
RFL 144	Primepii.300k__Assembly__Contig__58__1	569	2123
RFL 144	R0932E.300k__Assembly__Contig__50__1	570	2124
RFL 145	R0932E.300k__Assembly__Contig__62__1	571	2125
RFL 145	R0934F.300k__Assembly__Contig__52__1	572	2126
RFL 145	Anapurna.300k__Assembly__Contig__53__1	573	2127
RFL 145	Primepii.300k__Assembly__Contig__34__2	574	2128
RFL 145	R197.300k__Assembly__Contig__44__2	575	2129
RFL 145	Wheat-Rye-6R.300k__Assembly__Contig__36__2	576	2130
RFL 146	Anapurna.300k__Assembly__Contig__46__3	577	2131
RFL 146	R197.300k__Assembly__Contig__147__3	578	2132
RFL 146	Wheat-Rye-6R.300k__Assembly__Contig__18__3	579	2133
RFL 146	R0932E.300k__Assembly__Contig__42__3	580	2134
RFL 147	Anapurna.300k__Assembly__Contig__36__4	581	2135
RFL 147	R0934F.300k__Assembly__Contig__35__3	582	2136
RFL 147	Anapurna.300k__Assembly__Contig__36__3	583	2137
RFL 147	Wheat-Rye-6R.300k__Assembly__Contig__43__3	584	2138
RFL 147	R0932E.300k__Assembly__Contig__49__3	585	2139
RFL 147	R197.300k__Assembly__Contig__2__6	586	2140
RFL 147	Primepii.300k__Assembly__Contig__32__3	587	2141
RFL 148	R197.300k__Assembly__Contig__28__2	588	2142
RFL 148	R0934F.300k__Assembly__Contig__13__2	589	2143
RFL 148	Primepii.300k__Assembly__Contig__4__2	590	2144
RFL 148	R0932E.300k__Assembly__Contig__218__1	591	2145
RFL 149	Triticum-timopheevii.300k__Assembly__Contig__18__2	592	2146
RFL 150	Triticum-timopheevii.300k__Assembly__Contig__66__3	593	2147
RFL 151	R0934F.300k__Assembly__Contig__215__1	594	2148
RFL 151	Primepii.300k__Assembly__Contig__119__1	595	2149
RFL 151	R197.300k__Assembly__Contig__114__1	596	2150
RFL 152	Triticum-timopheevii.300k__Assembly__Contig__114__1	597	2151
RFL 152	R0934F.300k__Assembly__Contig__177__1	598	2152
RFL 153	R0932E.300k__Assembly__Contig__121__1	599	2153
RFL 153	Anapurna.300k__Assembly__Contig__124__1	600	2154
RFL 153	Primepii.300k__Assembly__Contig__126__1	601	2155
RFL 153	R0934F.300k__Assembly__Contig__206__1	602	2156
RFL 153	R0934F.300k__Assembly__Contig__147__1	603	2157
RFL 154	R0934F.300k__Assembly__Contig__174__1	604	2158
RFL 154	Primepii.300k__Assembly__Contig__143__1	605	2159
RFL 154	Primepii.300k__Assembly__Contig__120__1	606	2160
RFL 154	R0934F.300k__Assembly__Contig__134__1	607	2161
RFL 154	Primepii.300k__Assembly__Contig__125__1	608	2162
RFL 154	Wheat-Rye-6R.300k__Assembly__Contig__42__1	609	2163
RFL 154	R0934F.300k__Assembly__Contig__139__1	610	2164
RFL 154	R197.300k__Assembly__Contig__123__1	611	2165
RFL 154	R0932E.300k__Assembly__Contig__168__1	612	2166
RFL 154	R0932E.300k__Assembly__Contig__37__1	613	2167
RFL 154	Anapurna.300k__Assembly__Contig__135__1	614	2168
RFL 154	R197.300k__Assembly__Contig__140__1	615	2169
RFL 154	R197.300k__Assembly__Contig__161__1	616	2170
RFL 154	Anapurna.300k__Assembly__Contig__112__1	617	2171
RFL 154	Wheat-Rye-6R.300k__Assembly__Contig__160__1	618	2172
RFL 154	R0934F.300k__Assembly__Contig__182__1	619	2173

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 154	R0934F.300k_Assembly_Contig_200_1	620	2174
RFL 154	Wheat-Rye-6R.300k_Assembly_Contig_105_1	621	2175
RFL 154	Primepii.300k_Assembly_Contig_176_1	622	2176
RFL 154	R0934F.300k_Assembly_Contig_165_1	623	2177
RFL 154	Wheat-Rye-6R.300k_Assembly_Contig_200_1	624	2178
RFL 154	R197.300k_Assembly_Contig_13_1	625	2179
RFL 154	R197.300k_Assembly_Contig_158_1	626	2180
RFL 154	Primepii.300k_Assembly_Contig_160_1	627	2181
RFL 154	R0932E.300k_Assembly_Contig_156_1	628	2182
RFL 154	R0932E.300k_Assembly_Contig_179_1	629	2183
RFL 154	Anapurna.300k_Assembly_Contig_128_1	630	2184
RFL 154	Anapurna.300k_Assembly_Contig_104_1	631	2185
RFL 154	R0932E.300k_Assembly_Contig_177_1	632	2186
RFL 155	R197.300k_Assembly_Contig_127_1	633	2187
RFL 155	R0934F.300k_Assembly_Contig_109_1	634	2188
RFL 155	Wheat-Rye-6R.300k_Assembly_Contig_121_1	635	2189
RFL 155	Anapurna.300k_Assembly_Contig_108_1	636	2190
RFL 155	R0932E.300k_Assembly_Contig_122_1	637	2191
RFL 155	Primepii.300k_Assembly_Contig_110_1	638	2192
RFL 156	Wheat-Rye-6R.300k_Assembly_Contig_130_1	639	2193
RFL 156	Anapurna.300k_Assembly_Contig_117_2	640	2194
RFL 156	R0934F.300k_Assembly_Contig_176_1	641	2195
RFL 156	Anapurna.300k_Assembly_Contig_171_1	642	2196
RFL 156	R0932E.300k_Assembly_Contig_154_2	643	2197
RFL 156	Primepii.300k_Assembly_Contig_134_2	644	2198
RFL 157	Primepii.300k_Assembly_Contig_108_1	645	2199
RFL 157	R197.300k_Assembly_Contig_97_1	646	2200
RFL 157	Wheat-Rye-6R.300k_Assembly_Contig_117_1	647	2201
RFL 157	Anapurna.300k_Assembly_Contig_105_1	648	2202
RFL 157	R0932E.300k_Assembly_Contig_126_1	649	2203
RFL 157	R0934F.300k_Assembly_Contig_116_1	650	2204
RFL 158	R197.300k_Assembly_Contig_105_3	651	2205
RFL 158	Anapurna.300k_Assembly_Contig_89_3	652	2206
RFL 158	Wheat-Rye-6R.300k_Assembly_Contig_88_3	653	2207
RFL 158	R0934F.300k_Assembly_Contig_83_3	654	2208
RFL 158	Primepii.300k_Assembly_Contig_77_3	655	2209
RFL 158	R0932E.300k_Assembly_Contig_80_3	656	2210
RFL 159	Wheat-Rye-6R.300k_Assembly_Contig_177_1	657	2211
RFL 159	Wheat-Rye-6R.300k_Assembly_Contig_185_1	658	2212
RFL 159	Anapurna.300k_Assembly_Contig_176_1	659	2213
RFL 159	R197.300k_Assembly_Contig_191_1	660	2214
RFL 160	Triticum-timopheevii.300k_Assembly_Contig_111_1	661	2215
RFL 161	R197.300k_Assembly_Contig_28_3	662	2216
RFL 161	R0934F.300k_Assembly_Contig_13_3	663	2217
RFL 161	Anapurna.300k_Assembly_Contig_27_3	664	2218
RFL 161	R0932E.300k_Assembly_Contig_9_3	665	2219
RFL 161	Primepii.300k_Assembly_Contig_4_3	666	2220
RFL 161	R0932E.300k_Assembly_Contig_218_2	667	2221
RFL 161	Wheat-Rye-6R.300k_Assembly_Contig_4_4	668	2222
RFL 162	Triticum-timopheevii.300k_Assembly_Contig_94_1	669	2223
RFL 163	Primepii.300k_Assembly_Contig_128_1	670	2224
RFL 163	Wheat-Rye-6R.300k_Assembly_Contig_133_1	671	2225
RFL 163	R197.300k_Assembly_Contig_133_2	672	2226
RFL 163	R0932E.300k_Assembly_Contig_133_2	673	2227
RFL 163	R0934F.300k_Assembly_Contig_125_1	674	2228
RFL 163	Anapurna.300k_Assembly_Contig_122_2	675	2229
RFL 164	Primepii.300k_Assembly_Contig_127_1	676	2230
RFL 164	R0934F.300k_Assembly_Contig_124_1	677	2231
RFL 165	Anapurna.300k_Assembly_Contig_145_1	678	2232
RFL 165	R197.300k_Assembly_Contig_149_1	679	2233
RFL 165	Primepii.300k_Assembly_Contig_139_1	680	2234
RFL 165	R0932E.300k_Assembly_Contig_140_1	681	2235
RFL 165	Wheat-Rye-6R.300k_Assembly_Contig_139_1	682	2236
RFL 165	R0934F.300k_Assembly_Contig_132_1	683	2237
RFL 166	Primepii.300k_Assembly_Contig_130_1	684	2238
RFL 166	R0934F.300k_Assembly_Contig_122_2	685	2239
RFL 167	R197.300k_Assembly_Contig_118_1	686	2240
RFL 167	Anapurna.300k_Assembly_Contig_93_1	687	2241
RFL 167	R0932E.300k_Assembly_Contig_96_1	688	2242
RFL 167	R0934F.300k_Assembly_Contig_100_1	689	2243
RFL 167	Wheat-Rye-6R.300k_Assembly_Contig_107_1	690	2244
RFL 167	Primepii.300k_Assembly_Contig_100_1	691	2245
RFL 168	Triticum-timopheevii.300k_Assembly_Contig_88_1	692	2246
RFL 168	Triticum-timopheevii.300k_Assembly_Contig_28_1	693	2247

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 169	Triticum-timopheevii.300k__Assembly__Contig__86__2	694	2248
RFL 170	R197.300k__Assembly__Contig__94__2	695	2249
RFL 170	R0934F.300k__Assembly__Contig__67__2	696	2250
RFL 170	Primepii.300k__Assembly__Contig__60__2	697	2251
RFL 170	Wheat-Rye-6R.300k__Assembly__Contig__113__2	698	2252
RFL 170	Anapurna.300k__Assembly__Contig__174__3	699	2253
RFL 171	Triticum-timopheevii.300k__Assembly__Contig__84__1	700	2254
RFL 171	Triticum-timopheevii.300k__Assembly__Contig__46__1	701	2255
RFL 172	R0932E.300k__Assembly__Contig__67__1	702	2256
RFL 172	Wheat-Rye-6R.300k__Assembly__Contig__57__1	703	2257
RFL 172	R0934F.300k__Assembly__Contig__82__1	704	2258
RFL 172	Primepii.300k__Assembly__Contig__68__1	705	2259
RFL 172	Anapurna.300k__Assembly__Contig__32__1	706	2260
RFL 172	R197.300k__Assembly__Contig__61__1	707	2261
RFL 173	Triticum-timopheevii.300k__Assembly__Contig__83__1	708	2262
RFL 174	Triticum-timopheevii.300k__Assembly__Contig__41__1	709	2263
RFL 174	R197.300k__Assembly__Contig__102__1	710	2264
RFL 174	Triticum-timopheevii.300k__Assembly__Contig__41__2	711	2265
RFL 175	Primepii.300k__Assembly__Contig__200__1	712	2266
RFL 175	Primepii.300k__Assembly__Contig__162__1	713	2267
RFL 175	R0932E.300k__Assembly__Contig__144__1	714	2268
RFL 175	R0934F.300k__Assembly__Contig__156__1	715	2269
RFL 175	Anapurna.300k__Assembly__Contig__123__1	716	2270
RFL 176	R197.300k__Assembly__Contig__93__1	717	2271
RFL 176	R0934F.300k__Assembly__Contig__90__1	718	2272
RFL 176	Wheat-Rye-6R.300k__Assembly__Contig__100__1	719	2273
RFL 176	Anapurna.300k__Assembly__Contig__86__1	720	2274
RFL 176	R0932E.300k__Assembly__Contig__108__1	721	2275
RFL 177	R197.300k__Assembly__Contig__128__1	722	2276
RFL 177	Wheat-Rye-6R.300k__Assembly__Contig__126__1	723	2277
RFL 177	Anapurna.300k__Assembly__Contig__127__1	724	2278
RFL 177	R0932E.300k__Assembly__Contig__135__1	725	2279
RFL 178	Primepii.300k__Assembly__Contig__97__1	726	2280
RFL 179	Triticum-timopheevii.300k__Assembly__Contig__89__1	727	2281
RFL 180	R0932E.300k__Assembly__Contig__160__1	728	2282
RFL 180	Triticum-timopheevii.300k__Assembly__Contig__97__1	729	2283
RFL 180	Primepii.300k__Assembly__Contig__141__1	730	2284
RFL 180	R197.300k__Assembly__Contig__141__1	731	2285
RFL 180	Wheat-Rye-6R.300k__Assembly__Contig__149__1	732	2286
RFL 180	Anapurna.300k__Assembly__Contig__142__1	733	2287
RFL 180	R0934F.300k__Assembly__Contig__146__1	734	2288
RFL 181	Wheat-Rye-6R.300k__Assembly__Contig__104__1	735	2289
RFL 181	Primepii.300k__Assembly__Contig__104__1	736	2290
RFL 182	Wheat-Rye-6R.300k__Assembly__Contig__53__2	737	2291
RFL 182	R197.300k__Assembly__Contig__19__2	738	2292
RFL 182	R0932E.300k__Assembly__Contig__68__2	739	2293
RFL 183	R0932E.300k__Assembly__Contig__31__2	740	2294
RFL 183	R197.300k__Assembly__Contig__49__2	741	2295
RFL 183	R0934F.300k__Assembly__Contig__41__2	742	2296
RFL 183	Wheat-Rye-6R.300k__Assembly__Contig__27__2	743	2297
RFL 183	Triticum-timopheevii.300k__Assembly__Contig__26__1	744	2298
RFL 183	Anapurna.300k__Assembly__Contig__40__2	745	2299
RFL 183	Primepii.300k__Assembly__Contig__65__2	746	2300
RFL 184	R0932E.300k__Assembly__Contig__63__1	747	2301
RFL 184	R197.300k__Assembly__Contig__35__1	748	2302
RFL 184	Anapurna.300k__Assembly__Contig__170__1	749	2303
RFL 184	Wheat-Rye-6R.300k__Assembly__Contig__61__1	750	2304
RFL 185	R0934F.300k__Assembly__Contig__92__3	751	2305
RFL 185	Triticum-timopheevii.300k__Assembly__Contig__52__3	752	2306
RFL 185	R0932E.300k__Assembly__Contig__109__3	753	2307
RFL 185	R197.300k__Assembly__Contig__90__3	754	2308
RFL 186	Wheat-Rye-6R.300k__Assembly__Contig__131__1	755	2309
RFL 186	R0932E.300k__Assembly__Contig__141__1	756	2310
RFL 186	R197.300k__Assembly__Contig__143__1	757	2311
RFL 186	R0934F.300k__Assembly__Contig__133__1	758	2312
RFL 186	Anapurna.300k__Assembly__Contig__139__1	759	2313
RFL 186	Primepii.300k__Assembly__Contig__144__1	760	2314
RFL 187	R0934F.300k__Assembly__Contig__53__3	761	2315

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 187	Primepii.300k_Assembly_Contig_28_3	762	2316
RFL 187	R197.300k_Assembly_Contig_53_3	763	2317
RFL 187	R0932E.300k_Assembly_Contig_14_3	764	2318
RFL 187	Wheat-Rye-6R.300k_Assembly_Contig_49_3	765	2319
RFL 187	Anapurna.300k_Assembly_Contig_55_3	766	2320
RFL 189	Triticum-timopheevii.300k_Assembly_Contig_23_1	767	2321
RFL 191	R197.300k_Assembly_Contig_136_1	768	2322
RFL 192	R0934F.300k_Assembly_Contig_119_1	769	2323
RFL 192	R197.300k_Assembly_Contig_134_1	770	2324
RFL 192	Wheat-Rye-6R.300k_Assembly_Contig_125_1	771	2325
RFL 192	Primepii.300k_Assembly_Contig_133_1	772	2326
RFL 192	Anapurna.300k_Assembly_Contig_125_1	773	2327
RFL 192	R0932E.300k_Assembly_Contig_136_1	774	2328
RFL 193	R0932E.300k_Assembly_Contig_130_1	775	2329
RFL 193	R0934F.300k_Assembly_Contig_95_1	776	2330
RFL 193	Anapurna.300k_Assembly_Contig_107_1	777	2331
RFL 193	R197.300k_Assembly_Contig_110_1	778	2332
RFL 194	Primepii.300k_Assembly_Contig_107_1	779	2333
RFL 194	R197.300k_Assembly_Contig_99_1	780	2334
RFL 194	Wheat-Rye-6R.300k_Assembly_Contig_156_1	781	2335
RFL 194	R197.300k_Assembly_Contig_132_1	782	2336
RFL 194	Primepii.300k_Assembly_Contig_121_1	783	2337
RFL 194	Wheat-Rye-6R.300k_Assembly_Contig_51_1	784	2338
RFL 195	Wheat-Rye-6R.300k_Assembly_Contig_165_1	785	2339
RFL 195	R0934F.300k_Assembly_Contig_163_1	786	2340
RFL 195	R0932E.300k_Assembly_Contig_95_1	787	2341
RFL 195	Primepii.300k_Assembly_Contig_171_1	788	2342
RFL 195	Anapurna.300k_Assembly_Contig_2_4	789	2343
RFL 195	R197.300k_Assembly_Contig_176_1	790	2344
RFL 196	Anapurna.300k_Assembly_Contig_81_2	791	2345
RFL 196	R197.300k_Assembly_Contig_88_2	792	2346
RFL 196	Wheat-Rye-6R.300k_Assembly_Contig_114_2	793	2347
RFL 196	R0934F.300k_Assembly_Contig_112_2	794	2348
RFL 196	Primepii.300k_Assembly_Contig_103_2	795	2349
RFL 196	R0932E.300k_Assembly_Contig_106_2	796	2350
RFL 197	Wheat-Rye-6R.300k_Assembly_Contig_89_1	797	2351
RFL 197	R0932E.300k_Assembly_Contig_84_1	798	2352
RFL 197	R197.300k_Assembly_Contig_89_1	799	2353
RFL 197	Primepii.300k_Assembly_Contig_78_1	800	2354
RFL 197	Anapurna.300k_Assembly_Contig_65_1	801	2355
RFL 197	R0934F.300k_Assembly_Contig_105_1	802	2356
RFL 198	Wheat-Rye-6R.300k_Assembly_Contig_155_1	803	2357
RFL 198	R0934F.300k_Assembly_Contig_151_1	804	2358
RFL 198	R197.300k_Assembly_Contig_157_1	805	2359
RFL 198	Anapurna.300k_Assembly_Contig_150_1	806	2360
RFL 198	Primepii.300k_Assembly_Contig_157_1	807	2361
RFL 198	R0932E.300k_Assembly_Contig_169_1	808	2362
RFL 199	R0934F.300k_Assembly_Contig_170_1	809	2363
RFL 199	Anapurna.300k_Assembly_Contig_166_1	810	2364
RFL 199	R0932E.300k_Assembly_Contig_181_1	811	2365
RFL 199	R197.300k_Assembly_Contig_178_1	812	2366
RFL 199	Primepii.300k_Assembly_Contig_175_2	813	2367
RFL 199	Primepii.300k_Assembly_Contig_175_1	814	2368
RFL 199	Triticum-timopheevii.300k_Assembly_Contig_110_1	815	2369
RFL 199	Wheat-Rye-6R.300k_Assembly_Contig_171_1	816	2370
RFL 200	Wheat-Rye-6R.300k_Assembly_Contig_154_1	817	2371
RFL 200	Primepii.300k_Assembly_Contig_149_1	818	2372
RFL 200	R197.300k_Assembly_Contig_148_1	819	2373
RFL 200	Anapurna.300k_Assembly_Contig_151_1	820	2374
RFL 200	R0932E.300k_Assembly_Contig_158_1	821	2375
RFL 200	R0934F.300k_Assembly_Contig_135_1	822	2376
RFL 201	Anapurna.300k_Assembly_Contig_156_1	823	2377
RFL 201	Wheat-Rye-6R.300k_Assembly_Contig_152_1	824	2378
RFL 201	R0934F.300k_Assembly_Contig_162_1	825	2379
RFL 201	Triticum-timopheevii.300k_Assembly_Contig_95_1	826	2380
RFL 201	Primepii.300k_Assembly_Contig_153_1	827	2381
RFL 201	R197.300k_Assembly_Contig_162_1	828	2382
RFL 201	R0932E.300k_Assembly_Contig_159_1	829	2383
RFL 202	Anapurna.300k_Assembly_Contig_157_1	830	2384
RFL 202	Wheat-Rye-6R.300k_Assembly_Contig_164_1	831	2385
RFL 202	R197.300k_Assembly_Contig_166_1	832	2386
RFL 202	R0932E.300k_Assembly_Contig_165_1	833	2387
RFL 202	R0934F.300k_Assembly_Contig_154_1	834	2388

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 202	Primepii.300k__Assembly__Contig__167__1	835	2389
RFL 203	Triticum-timopheevii.300k__Assembly__Contig__105__1	836	2390
RFL 203	Triticum-timopheevii.300k__Assembly__Contig__119__1	837	2391
RFL 203	Wheat-Rye-6R.300k__Assembly__Contig__147__1	838	2392
RFL 203	R0932E.300k__Assembly__Contig__173__2	839	2393
RFL 203	R0932E.300k__Assembly__Contig__173__1	840	2394
RFL 203	Triticum-timopheevii.300k__Assembly__Contig__3__1	841	2395
RFL 203	Triticum-timopheevii.300k__Assembly__Contig__120__1	842	2396
RFL 203	R197.300k__Assembly__Contig__150__1	843	2397
RFL 203	Anapurna.300k__Assembly__Contig__153__1	844	2398
RFL 203	R0934F.300k__Assembly__Contig__157__2	845	2399
RFL 203	Primepii.300k__Assembly__Contig__156__1	846	2400
RFL 203	R0934F.300k__Assembly__Contig__157__1	847	2401
RFL 204	Triticum-timopheevii.300k__Assembly__Contig__103__1	848	2402
RFL 204	Triticum-timopheevii.300k__Assembly__Contig__100__1	849	2403
RFL 205	Triticum-timopheevii.300k__Assembly__Contig__96__1	850	2404
RFL 205	R0932E.300k__Assembly__Contig__155__1	851	2405
RFL 205	Anapurna.300k__Assembly__Contig__140__1	852	2406
RFL 205	Primepii.300k__Assembly__Contig__148__1	853	2407
RFL 205	R197.300k__Assembly__Contig__151__1	854	2408
RFL 205	R0934F.300k__Assembly__Contig__159__1	855	2409
RFL 205	Wheat-Rye-6R.300k__Assembly__Contig__151__1	856	2410
RFL 206	R0932E.300k__Assembly__Contig__117__1	857	2411
RFL 207	Triticum-timopheevii.300k__Assembly__Contig__62__1	858	2412
RFL 208	Wheat-Rye-6R.300k__Assembly__Contig__136__1	859	2413
RFL 209	Wheat-Rye-6R.300k__Assembly__Contig__143__1	860	2414
RFL 209	R0934F.300k__Assembly__Contig__140__1	861	2415
RFL 209	R0932E.300k__Assembly__Contig__174__1	862	2416
RFL 209	R197.300k__Assembly__Contig__154__1	863	2417
RFL 209	Primepii.300k__Assembly__Contig__152__1	864	2418
RFL 209	Anapurna.300k__Assembly__Contig__141__1	865	2419
RFL 210	Wheat-Rye-6R.300k__Assembly__Contig__163__1	866	2420
RFL 210	Anapurna.300k__Assembly__Contig__149__1	867	2421
RFL 210	R0934F.300k__Assembly__Contig__152__1	868	2422
RFL 210	R197.300k__Assembly__Contig__168__1	869	2423
RFL 210	R0932E.300k__Assembly__Contig__172__1	870	2424
RFL 210	Primepii.300k__Assembly__Contig__166__1	871	2425
RFL 211	Triticum-timopheevii.300k__Assembly__Contig__99__1	872	2426
RFL 212	R0932E.300k__Assembly__Contig__99__1	873	2427
RFL 212	R197.300k__Assembly__Contig__85__2	874	2428
RFL 212	R0932E.300k__Assembly__Contig__97__2	875	2429
RFL 212	Triticum-timopheevii.300k__Assembly__Contig__73__1	876	2430
RFL 212	R0934F.300k__Assembly__Contig__94__2	877	2431
RFL 212	Wheat-Rye-6R.300k__Assembly__Contig__91__1	878	2432
RFL 212	Primepii.300k__Assembly__Contig__76__2	879	2433
RFL 212	R197.300k__Assembly__Contig__109__1	880	2434
RFL 212	Primepii.300k__Assembly__Contig__99__1	881	2435
RFL 212	R0934F.300k__Assembly__Contig__107__1	882	2436
RFL 212	Wheat-Rye-6R.300k__Assembly__Contig__68__2	883	2437
RFL 212	Anapurna.300k__Assembly__Contig__96__1	884	2438
RFL 212	Triticum-timopheevii.300k__Assembly__Contig__132__1	885	2439
RFL 212	Anapurna.300k__Assembly__Contig__75__2	886	2440
RFL 213	Primepii.300k__Assembly__Contig__61__2	887	2441
RFL 213	Wheat-Rye-6R.300k__Assembly__Contig__64__2	888	2442
RFL 213	Anapurna.300k__Assembly__Contig__69__2	889	2443
RFL 213	R0934F.300k__Assembly__Contig__65__2	890	2444
RFL 213	R197.300k__Assembly__Contig__48__2	891	2445
RFL 213	R0932E.300k__Assembly__Contig__70__2	892	2446
RFL 215	Triticum-timopheevii.300k__Assembly__Contig__98__1	893	2447
RFL 216	R197.300k__Assembly__Contig__96__2	894	2448
RFL 216	R0932E.300k__Assembly__Contig__93__2	895	2449
RFL 216	Anapurna.300k__Assembly__Contig__126__1	896	2450
RFL 216	R0934F.300k__Assembly__Contig__102__2	897	2451
RFL 217	Triticum-timopheevii.300k__Assembly__Contig__16__2	898	2452
RFL 218	Primepii.300k__Assembly__Contig__95__2	899	2453

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 218	Wheat-Rye-6R.300k_Assembly_Contig_166_1	900	2454
RFL 218	R197.300k_Assembly_Contig_112_2	901	2455
RFL 219	R0932E.300k_Assembly_Contig_105_1	902	2456
RFL 219	Wheat-Rye-6R.300k_Assembly_Contig_110_1	903	2457
RFL 219	R197.300k_Assembly_Contig_87_1	904	2458
RFL 219	Anapurna.300k_Assembly_Contig_91_1	905	2459
RFL 219	R0934F.300k_Assembly_Contig_101_1	906	2460
RFL 219	Primepii.300k_Assembly_Contig_98_1	907	2461
RFL 220	Primepii.300k_Assembly_Contig_115_2	908	2462
RFL 220	Wheat-Rye-6R.300k_Assembly_Contig_108_2	909	2463
RFL 221	R0934F.300k_Assembly_Contig_47_2	910	2464
RFL 221	R0932E.300k_Assembly_Contig_33_2	911	2465
RFL 221	Wheat-Rye-6R.300k_Assembly_Contig_16_2	912	2466
RFL 221	Anapurna.300k_Assembly_Contig_30_2	913	2467
RFL 221	R197.300k_Assembly_Contig_52_2	914	2468
RFL 221	Primepii.300k_Assembly_Contig_15_2	915	2469
RFL 222	Wheat-Rye-6R.300k_Assembly_Contig_97_1	916	2470
RFL 222	R197.300k_Assembly_Contig_108_1	917	2471
RFL 222	Anapurna.300k_Assembly_Contig_101_1	918	2472
RFL 223	R0934F.300k_Assembly_Contig_58_2	919	2473
RFL 223	Anapurna.300k_Assembly_Contig_25_3	920	2474
RFL 223	Wheat-Rye-6R.300k_Assembly_Contig_46_3	921	2475
RFL 223	R197.300k_Assembly_Contig_45_3	922	2476
RFL 223	R0932E.300k_Assembly_Contig_219_3	923	2477
RFL 223	Primepii.300k_Assembly_Contig_36_2	924	2478
RFL 223	R0932E.300k_Assembly_Contig_57_3	925	2479
RFL 224	R0932E.300k_Assembly_Contig_212_1	926	2480
RFL 225	R0932E.300k_Assembly_Contig_117_2	927	2481
RFL 226	R0932E.300k_Assembly_Contig_67_2	928	2482
RFL 226	Primepii.300k_Assembly_Contig_118_1	929	2483
RFL 226	Wheat-Rye-6R.300k_Assembly_Contig_101_1	930	2484
RFL 226	Wheat-Rye-6R.300k_Assembly_Contig_57_2	931	2485
RFL 226	R0934F.300k_Assembly_Contig_82_2	932	2486
RFL 226	Anapurna.300k_Assembly_Contig_95_1	933	2487
RFL 226	R0934F.300k_Assembly_Contig_98_1	934	2488
RFL 226	Primepii.300k_Assembly_Contig_68_2	935	2489
RFL 226	Anapurna.300k_Assembly_Contig_32_2	936	2490
RFL 226	R197.300k_Assembly_Contig_61_2	937	2491
RFL 227	R0934F.300k_Assembly_Contig_62_1	938	2492
RFL 227	Primepii.300k_Assembly_Contig_196_1	939	2493
RFL 228	R197.300k_Assembly_Contig_85_1	940	2494
RFL 228	Wheat-Rye-6R.300k_Assembly_Contig_68_1	941	2495
RFL 228	R0934F.300k_Assembly_Contig_94_1	942	2496
RFL 228	R0932E.300k_Assembly_Contig_97_1	943	2497
RFL 228	Anapurna.300k_Assembly_Contig_75_1	944	2498
RFL 228	Primepii.300k_Assembly_Contig_76_1	945	2499
RFL 229	R0932E.300k_Assembly_Contig_30_3	946	2500
RFL 229	R197.300k_Assembly_Contig_26_5	947	2501
RFL 229	Anapurna.300k_Assembly_Contig_26_4	948	2502
RFL 229	Wheat-Rye-6R.300k_Assembly_Contig_24_5	949	2503
RFL 229	Primepii.300k_Assembly_Contig_33_3	950	2504
RFL 229	R0934F.300k_Assembly_Contig_16_5	951	2505
RFL 230	R0932E.300k_Assembly_Contig_209_2	952	2506
RFL 231	R0932E.300k_Assembly_Contig_36_1	953	2507
RFL 231	R0934F.300k_Assembly_Contig_42_1	954	2508
RFL 231	R197.300k_Assembly_Contig_55_1	955	2509
RFL 231	Anapurna.300k_Assembly_Contig_44_1	956	2510
RFL 231	Wheat-Rye-6R.300k_Assembly_Contig_33_1	957	2511
RFL 231	Primepii.300k_Assembly_Contig_54_1	958	2512
RFL 232	R0932E.300k_Assembly_Contig_171_1	959	2513
RFL 232	Primepii.300k_Assembly_Contig_173_1	960	2514
RFL 232	Wheat-Rye-6R.300k_Assembly_Contig_168_1	961	2515
RFL 232	R197.300k_Assembly_Contig_177_1	962	2516
RFL 232	Anapurna.300k_Assembly_Contig_165_1	963	2517
RFL 232	R0934F.300k_Assembly_Contig_158_1	964	2518
RFL 234	Triticum-timopheevii.300k_Assembly_Contig_101_1	965	2519
RFL 235	Triticum-timopheevii.300k_Assembly_Contig_75_1	966	2520
RFL 236	R0934F.300k_Assembly_Contig_33_2	967	2521
RFL 236	R0932E.300k_Assembly_Contig_52_1	968	2522
RFL 236	Primepii.300k_Assembly_Contig_29_3	969	2523
RFL 236	Wheat-Rye-6R.300k_Assembly_Contig_38_1	970	2524
RFL 236	R197.300k_Assembly_Contig_50_2	971	2525
RFL 236	Anapurna.300k_Assembly_Contig_34_2	972	2526
RFL 237	R0934F.300k_Assembly_Contig_118_1	973	2527
RFL 237	Triticum-timopheevii.300k_Assembly_Contig_82_1	974	2528

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 238	Primepii.300k_Assembly_Contig_29_4	975	2529
RFL 238	R0934F.300k_Assembly_Contig_33_3	976	2530
RFL 238	Anapurna.300k_Assembly_Contig_34_3	977	2531
RFL 238	R0932E.300k_Assembly_Contig_52_2	978	2532
RFL 238	Wheat-Rye-6R.300k_Assembly_Contig_38_2	979	2533
RFL 238	R197.300k_Assembly_Contig_50_3	980	2534
RFL 239	Anapurna.300k_Assembly_Contig_92_1	981	2535
RFL 239	Wheat-Rye-6R.300k_Assembly_Contig_103_1	982	2536
RFL 240	Primepii.300k_Assembly_Contig_140_1	983	2537
RFL 240	R197.300k_Assembly_Contig_156_1	984	2538
RFL 240	R0932E.300k_Assembly_Contig_148_1	985	2539
RFL 240	R0934F.300k_Assembly_Contig_136_1	986	2540
RFL 241	Primepii.300k_Assembly_Contig_138_1	987	2541
RFL 241	Wheat-Rye-6R.300k_Assembly_Contig_134_2	988	2542
RFL 241	R197.300k_Assembly_Contig_164_2	989	2543
RFL 241	R0932E.300k_Assembly_Contig_143_1	990	2544
RFL 241	Anapurna.300k_Assembly_Contig_136_1	991	2545
RFL 241	R0934F.300k_Assembly_Contig_137_1	992	2546
RFL 242	Anapurna.300k_Assembly_Contig_72_1	993	2547
RFL 243	Anapurna.300k_Assembly_Contig_8_2	994	2548
RFL 244	Anapurna.300k_Assembly_Contig_164_1	995	2549
RFL 245	R0934F.300k_Assembly_Contig_58_1	996	2550
RFL 245	Anapurna.300k_Assembly_Contig_25_2	997	2551
RFL 245	R0932E.300k_Assembly_Contig_57_2	998	2552
RFL 245	Wheat-Rye-6R.300k_Assembly_Contig_46_2	999	2553
RFL 245	R197.300k_Assembly_Contig_45_2	1000	2554
RFL 245	Primepii.300k_Assembly_Contig_36_1	1001	2555
RFL 245	R0932E.300k_Assembly_Contig_219_2	1002	2556
RFL 246	Primepii.300k_Assembly_Contig_116_2	1003	2557
RFL 246	Anapurna.300k_Assembly_Contig_90_3	1004	2558
RFL 246	R197.300k_Assembly_Contig_125_2	1005	2559
RFL 246	Wheat-Rye-6R.300k_Assembly_Contig_102_2	1006	2560
RFL 246	R0934F.300k_Assembly_Contig_96_2	1007	2561
RFL 246	R0932E.300k_Assembly_Contig_125_2	1008	2562
RFL 247	R0932E.300k_Assembly_Contig_53_1	1009	2563
RFL 247	R0934F.300k_Assembly_Contig_26_1	1010	2564
RFL 247	Primepii.300k_Assembly_Contig_40_1	1011	2565
RFL 247	Anapurna.300k_Assembly_Contig_29_1	1012	2566
RFL 247	R197.300k_Assembly_Contig_36_1	1013	2567
RFL 247	Wheat-Rye-6R.300k_Assembly_Contig_28_1	1014	2568
RFL 247	Triticum-timopheevii.300k_Assembly_Contig_25_1	1015	2569
RFL 248	R0934F.300k_Assembly_Contig_183_1	1016	2570
RFL 248	Wheat-Rye-6R.300k_Assembly_Contig_176_1	1017	2571
RFL 248	R0932E.300k_Assembly_Contig_180_1	1018	2572
RFL 249	Anapurna.300k_Assembly_Contig_130_1	1019	2573
RFL 249	Wheat-Rye-6R.300k_Assembly_Contig_128_1	1020	2574
RFL 249	R197.300k_Assembly_Contig_144_1	1021	2575
RFL 249	R0934F.300k_Assembly_Contig_131_1	1022	2576
RFL 249	R0932E.300k_Assembly_Contig_137_1	1023	2577
RFL 249	Primepii.300k_Assembly_Contig_135_1	1024	2578
RFL 250	Wheat-Rye-6R.300k_Assembly_Contig_47_2	1025	2579
RFL 250	Primepii.300k_Assembly_Contig_69_2	1026	2580
RFL 250	R0932E.300k_Assembly_Contig_69_2	1027	2581
RFL 250	Anapurna.300k_Assembly_Contig_56_2	1028	2582
RFL 250	R0934F.300k_Assembly_Contig_59_2	1029	2583
RFL 250	R197.300k_Assembly_Contig_91_2	1030	2584
RFL 251	R197.300k_Assembly_Contig_26_6	1031	2585
RFL 251	Triticum-timopheevii.300k_Assembly_Contig_108_1	1032	2586
RFL 251	Anapurna.300k_Assembly_Contig_26_5	1033	2587
RFL 251	R0932E.300k_Assembly_Contig_30_4	1034	2588
RFL 251	Wheat-Rye-6R.300k_Assembly_Contig_24_6	1035	2589
RFL 251	Primepii.300k_Assembly_Contig_33_4	1036	2590
RFL 251	R0934F.300k_Assembly_Contig_16_6	1037	2591
RFL 252	Primepii.300k_Assembly_Contig_189_1	1038	2592
RFL 252	R0934F.300k_Assembly_Contig_169_1	1039	2593
RFL 253	Wheat-Rye-6R.300k_Assembly_Contig_112_2	1040	2594
RFL 253	R0932E.300k_Assembly_Contig_111_2	1041	2595
RFL 253	R0934F.300k_Assembly_Contig_130_2	1042	2596
RFL 253	Anapurna.300k_Assembly_Contig_88_2	1043	2597
RFL 254	R0932E.300k_Assembly_Contig_119_2	1044	2598
RFL 254	R0934F.300k_Assembly_Contig_108_2	1045	2599
RFL 254	Triticum-timopheevii.300k_Assembly_Contig_77_2	1046	2600
RFL 254	R197.300k_Assembly_Contig_122_2	1047	2601
RFL 254	Wheat-Rye-6R.300k_Assembly_Contig_116_2	1048	2602
RFL 254	Anapurna.300k_Assembly_Contig_103_2	1049	2603

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 254	Primepii.300k_Assembly_Contig_101_2	1050	2604
RFL 255	R0934F.300k_Assembly_Contig_212_1	1051	2605
RFL 256	Wheat-Rye-6R.300k_Assembly_Contig_76_1	1052	2606
RFL 256	R0934F.300k_Assembly_Contig_79_1	1053	2607
RFL 256	Triticum-timopheevii.300k_Assembly_Contig_32_1	1054	2608
RFL 256	R0932E.300k_Assembly_Contig_81_1	1055	2609
RFL 256	Anapurna.300k_Assembly_Contig_67_1	1056	2610
RFL 256	Primepii.300k_Assembly_Contig_89_1	1057	2611
RFL 256	R197.300k_Assembly_Contig_74_1	1058	2612
RFL 257	Primepii.300k_Assembly_Contig_71_1	1059	2613
RFL 257	Wheat-Rye-6R.300k_Assembly_Contig_78_1	1060	2614
RFL 257	R0932E.300k_Assembly_Contig_101_1	1061	2615
RFL 257	R0934F.300k_Assembly_Contig_85_1	1062	2616
RFL 257	R197.300k_Assembly_Contig_76_1	1063	2617
RFL 257	Anapurna.300k_Assembly_Contig_79_1	1064	2618
RFL 258	Primepii.300k_Assembly_Contig_41_1	1065	2619
RFL 259	Triticum-timopheevii.300k_Assembly_Contig_76_2	1066	2620
RFL 260	Anapurna.300k_Assembly_Contig_155_1	1067	2621
RFL 260	R0932E.300k_Assembly_Contig_163_2	1068	2622
RFL 260	R0934F.300k_Assembly_Contig_160_1	1069	2623
RFL 260	Wheat-Rye-6R.300k_Assembly_Contig_162_1	1070	2624
RFL 260	R197.300k_Assembly_Contig_171_1	1071	2625
RFL 260	Primepii.300k_Assembly_Contig_146_1	1072	2626
RFL 261	Triticum-timopheevii.300k_Assembly_Contig_90_2	1073	2627
RFL 262	Primepii.300k_Assembly_Contig_172_1	1074	2628
RFL 262	Wheat-Rye-6R.300k_Assembly_Contig_159_1	1075	2629
RFL 262	R197.300k_Assembly_Contig_167_1	1076	2630
RFL 263	R197.300k_Assembly_Contig_126_1	1077	2631
RFL 263	Wheat-Rye-6R.300k_Assembly_Contig_157_1	1078	2632
RFL 263	Wheat-Rye-6R.300k_Assembly_Contig_189_1	1079	2633
RFL 263	R0932E.300k_Assembly_Contig_132_1	1080	2634
RFL 263	Anapurna.300k_Assembly_Contig_114_1	1081	2635
RFL 264	Triticum-timopheevii.300k_Assembly_Contig_64_1	1082	2636
RFL 265	R197.300k_Assembly_Contig_129_1	1083	2637
RFL 265	R0934F.300k_Assembly_Contig_120_1	1084	2638
RFL 265	Wheat-Rye-6R.300k_Assembly_Contig_118_1	1085	2639
RFL 265	Anapurna.300k_Assembly_Contig_102_1	1086	2640
RFL 265	Triticum-timopheevii.300k_Assembly_Contig_79_1	1087	2641
RFL 265	Primepii.300k_Assembly_Contig_123_1	1088	2642
RFL 265	R0932E.300k_Assembly_Contig_127_1	1089	2643
RFL 266	Triticum-timopheevii.300k_Assembly_Contig_17_2	1090	2644
RFL 267	Triticum-timopheevii.300k_Assembly_Contig_90_3	1091	2645
RFL 268	R197.300k_Assembly_Contig_90_4	1092	2646
RFL 268	Triticum-timopheevii.300k_Assembly_Contig_52_4	1093	2647
RFL 268	R0934F.300k_Assembly_Contig_92_4	1094	2648
RFL 268	R0932E.300k_Assembly_Contig_109_4	1095	2649
RFL 269	Triticum-timopheevii.300k_Assembly_Contig_84_2	1096	2650
RFL 270	Triticum-timopheevii.300k_Assembly_Contig_56_1	1097	2651
RFL 271	R0932E.300k_Assembly_Contig_167_1	1098	2652
RFL 272	Triticum-timopheevii.300k_Assembly_Contig_17_1	1099	2653
RFL 273	Primepii.300k_Assembly_Contig_84_2	1100	2654
RFL 273	R0932E.300k_Assembly_Contig_212_2	1101	2655
RFL 273	R0934F.300k_Assembly_Contig_62_2	1102	2656
RFL 274	Triticum-timopheevii.300k_Assembly_Contig_87_1	1103	2657
RFL 275	Anapurna.300k_Assembly_Contig_208_1	1104	2658
RFL 276	R0934F.300k_Assembly_Contig_42_2	1105	2659
RFL 276	R197.300k_Assembly_Contig_55_2	1106	2660
RFL 276	R0932E.300k_Assembly_Contig_36_2	1107	2661
RFL 276	Primepii.300k_Assembly_Contig_54_2	1108	2662
RFL 276	Wheat-Rye-6R.300k_Assembly_Contig_33_2	1109	2663
RFL 276	Anapurna.300k_Assembly_Contig_44_2	1110	2664
RFL 277	Triticum-timopheevii.300k_Assembly_Contig_2_1	1111	2665
RFL 277	R0934F.300k_Assembly_Contig_3_1	1112	2666
RFL 278	Triticum-timopheevii.300k_Assembly_Contig_85_2	1113	2667

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 279	Wheat-Rye-6R.300k_Assembly_Contig_56_2	1114	2668
RFL 279	R0934F.300k_Assembly_Contig_60_2	1115	2669
RFL 279	Anapurna.300k_Assembly_Contig_60_3	1116	2670
RFL 279	Primepii.300k_Assembly_Contig_56_3	1117	2671
RFL 279	R197.300k_Assembly_Contig_60_2	1118	2672
RFL 279	R0932E.300k_Assembly_Contig_2_2	1119	2673
RFL 280	R0934F.300k_Assembly_Contig_129_1	1120	2674
RFL 280	R197.300k_Assembly_Contig_139_1	1121	2675
RFL 280	Anapurna.300k_Assembly_Contig_129_1	1122	2676
RFL 280	Wheat-Rye-6R.300k_Assembly_Contig_127_1	1123	2677
RFL 280	Primepii.300k_Assembly_Contig_136_2	1124	2678
RFL 280	R0932E.300k_Assembly_Contig_138_1	1125	2679
RFL 281	Triticum-timopheevii.300k_Assembly_Contig_23_2	1126	2680
RFL 282	R197.300k_Assembly_Contig_181_1	1127	2681
RFL 282	Wheat-Rye-6R.300k_Assembly_Contig_186_1	1128	2682
RFL 282	R0932E.300k_Assembly_Contig_95_2	1129	2683
RFL 282	Primepii.300k_Assembly_Contig_177_1	1130	2684
RFL 282	R0934F.300k_Assembly_Contig_155_1	1131	2685
RFL 282	Wheat-Rye-6R.300k_Assembly_Contig_119_1	1132	2686
RFL 282	R197.300k_Assembly_Contig_106_1	1133	2687
RFL 282	Primepii.300k_Assembly_Contig_161_1	1134	2688
RFL 282	Anapurna.300k_Assembly_Contig_160_1	1135	2689
RFL 282	R197.300k_Assembly_Contig_219_1	1136	2690
RFL 282	Anapurna.300k_Assembly_Contig_198_1	1137	2691
RFL 282	Anapurna.300k_Assembly_Contig_100_1	1138	2692
RFL 282	Wheat-Rye-6R.300k_Assembly_Contig_170_1	1139	2693
RFL 283	R0934F.300k_Assembly_Contig_79_2	1140	2694
RFL 283	Wheat-Rye-6R.300k_Assembly_Contig_76_2	1141	2695
RFL 283	Triticum-timopheevii.300k_Assembly_Contig_32_2	1142	2696
RFL 283	Primepii.300k_Assembly_Contig_89_2	1143	2697
RFL 283	R197.300k_Assembly_Contig_74_2	1144	2698
RFL 283	R0932E.300k_Assembly_Contig_81_2	1145	2699
RFL 283	Anapurna.300k_Assembly_Contig_67_2	1146	2700
RFL 284	Primepii.300k_Assembly_Contig_84_1	1147	2701
RFL 285	Primepii.300k_Assembly_Contig_111_2	1148	2702
RFL 286	Triticum-timopheevii.300k_Assembly_Contig_55_2	1149	2703
RFL 287	Anapurna.300k_Assembly_Contig_154_2	1150	2704
RFL 287	Wheat-Rye-6R.300k_Assembly_Contig_142_2	1151	2705
RFL 287	R197.300k_Assembly_Contig_153_2	1152	2706
RFL 287	Primepii.300k_Assembly_Contig_158_2	1153	2707
RFL 288	Primepii.300k_Assembly_Contig_60_1	1154	2708
RFL 288	R0934F.300k_Assembly_Contig_67_1	1155	2709
RFL 289	R0932E.300k_Assembly_Contig_206_2	1156	2710
RFL 290	R0932E.300k_Assembly_Contig_166_2	1157	2711
RFL 290	R0934F.300k_Assembly_Contig_166_2	1158	2712
RFL 290	Primepii.300k_Assembly_Contig_174_2	1159	2713
RFL 290	R197.300k_Assembly_Contig_174_2	1160	2714
RFL 290	Anapurna.300k_Assembly_Contig_161_2	1161	2715
RFL 290	Wheat-Rye-6R.300k_Assembly_Contig_161_2	1162	2716
RFL 291	Anapurna.300k_Assembly_Contig_27_2	1163	2717
RFL 291	R0932E.300k_Assembly_Contig_9_2	1164	2718
RFL 291	Wheat-Rye-6R.300k_Assembly_Contig_4_3	1165	2719
RFL 292	Triticum-timopheevii.300k_Assembly_Contig_68_2	1166	2720
RFL 293	R197.300k_Assembly_Contig_73_1	1167	2721
RFL 294	Primepii.300k_Assembly_Contig_116_1	1168	2722
RFL 294	Wheat-Rye-6R.300k_Assembly_Contig_102_1	1169	2723
RFL 294	Anapurna.300k_Assembly_Contig_90_2	1170	2724
RFL 294	R197.300k_Assembly_Contig_125_1	1171	2725
RFL 294	R0934F.300k_Assembly_Contig_96_1	1172	2726
RFL 294	R0932E.300k_Assembly_Contig_125_1	1173	2727
RFL 295	Triticum-timopheevii.300k_Assembly_Contig_112_1	1174	2728
RFL 296	R197.300k_Assembly_Contig_159_1	1175	2729
RFL 296	Wheat-Rye-6R.300k_Assembly_Contig_146_1	1176	2730
RFL 296	Primepii.300k_Assembly_Contig_154_1	1177	2731
RFL 296	R0934F.300k_Assembly_Contig_144_1	1178	2732
RFL 296	Anapurna.300k_Assembly_Contig_152_1	1179	2733
RFL 296	R0932E.300k_Assembly_Contig_149_1	1180	2734
RFL 297	R0932E.300k_Assembly_Contig_129_1	1181	2735
RFL 297	Anapurna.300k_Assembly_Contig_120_1	1182	2736
RFL 297	Primepii.300k_Assembly_Contig_129_1	1183	2737
RFL 297	R197.300k_Assembly_Contig_145_1	1184	2738
RFL 297	R0934F.300k_Assembly_Contig_121_1	1185	2739
RFL 297	Wheat-Rye-6R.300k_Assembly_Contig_129_1	1186	2740

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 298	Wheat-Rye-6R.300k__Assembly__Contig__120__1	1187	2741
RFL 298	Anapurna.300k__Assembly__Contig__110__1	1188	2742
RFL 298	R197.300k__Assembly__Contig__124__1	1189	2743
RFL 298	R0932E.300k__Assembly__Contig__124__1	1190	2744
RFL 298	Primepii.300k__Assembly__Contig__124__1	1191	2745
RFL 298	R0934F.300k__Assembly__Contig__127__1	1192	2746
RFL 299	R197.300k__Assembly__Contig__195__2	1193	2747
RFL 299	R0934F.300k__Assembly__Contig__192__1	1194	2748
RFL 299	Wheat-Rye-6R.300k__Assembly__Contig__192__1	1195	2749
RFL 300	Wheat-Rye-6R.300k__Assembly__Contig__111__2	1196	2750
RFL 300	R197.300k__Assembly__Contig__121__2	1197	2751
RFL 300	Primepii.300k__Assembly__Contig__113__2	1198	2752
RFL 301	Anapurna.300k__Assembly__Contig__69__3	1199	2753
RFL 301	Primepii.300k__Assembly__Contig__61__3	1200	2754
RFL 301	R197.300k__Assembly__Contig__48__3	1201	2755
RFL 301	R0934F.300k__Assembly__Contig__65__3	1202	2756
RFL 301	Wheat-Rye-6R.300k__Assembly__Contig__64__3	1203	2757
RFL 301	R0932E.300k__Assembly__Contig__70__3	1204	2758
RFL 302	Triticum-timopheevii.300k__Assembly__Contig__24__2	1205	2759
RFL 303	Wheat-Rye-6R.300k__Assembly__Contig__199__1	1206	2760
RFL 303	Anapurna.300k__Assembly__Contig__195__1	1207	2761
RFL 304	Anapurna.300k__Assembly__Contig__120__2	1208	2762
RFL 304	R0932E.300k__Assembly__Contig__129__2	1209	2763
RFL 304	Primepii.300k__Assembly__Contig__129__2	1210	2764
RFL 304	R197.300k__Assembly__Contig__145__2	1211	2765
RFL 304	Wheat-Rye-6R.300k__Assembly__Contig__129__2	1212	2766
RFL 304	R0934F.300k__Assembly__Contig__121__2	1213	2767
RFL 305	Wheat-Rye-6R.300k__Assembly__Contig__135__1	1214	2768
RFL 306	R197.300k__Assembly__Contig__107__2	1215	2769
RFL 306	R0934F.300k__Assembly__Contig__88__2	1216	2770
RFL 306	Anapurna.300k__Assembly__Contig__76__2	1217	2771
RFL 306	Primepii.300k__Assembly__Contig__74__2	1218	2772
RFL 306	R0932E.300k__Assembly__Contig__115__2	1219	2773
RFL 306	Wheat-Rye-6R.300k__Assembly__Contig__81__2	1220	2774
RFL 307	Triticum-timopheevii.300k__Assembly__Contig__104__1	1221	2775
RFL 308	R197.300k__Assembly__Contig__198__1	1222	2776
RFL 308	R0932E.300k__Assembly__Contig__178__2	1223	2777
RFL 308	Wheat-Rye-6R.300k__Assembly__Contig__193__1	1224	2778
RFL 308	Anapurna.300k__Assembly__Contig__183__1	1225	2779
RFL 308	R0934F.300k__Assembly__Contig__175__1	1226	2780
RFL 308	Primepii.300k__Assembly__Contig__198__1	1227	2781
RFL 309	Wheat-Rye-6R.300k__Assembly__Contig__52__2	1228	2782
RFL 309	Primepii.300k__Assembly__Contig__194__1	1229	2783
RFL 309	R0932E.300k__Assembly__Contig__51__1	1230	2784
RFL 309	R0934F.300k__Assembly__Contig__191__2	1231	2785
RFL 309	R197.300k__Assembly__Contig__190__2	1232	2786
RFL 310	Triticum-timopheevii.300k__Assembly__Contig__19__1	1233	2787
RFL 311	Wheat-Rye-6R.300k__Assembly__Contig__123__1	1234	2788
RFL 311	Anapurna.300k__Assembly__Contig__118__1	1235	2789
RFL 311	R197.300k__Assembly__Contig__131__1	1236	2790
RFL 311	R0932E.300k__Assembly__Contig__128__1	1237	2791
RFL 312	Anapurna.300k__Assembly__Contig__146__1	1238	2792
RFL 312	R0932E.300k__Assembly__Contig__175__1	1239	2793
RFL 312	Wheat-Rye-6R.300k__Assembly__Contig__158__1	1240	2794
RFL 312	R197.300k__Assembly__Contig__155__1	1241	2795
RFL 312	R0934F.300k__Assembly__Contig__153__1	1242	2796
RFL 312	Primepii.300k__Assembly__Contig__145__1	1243	2797
RFL 313	R0934F.300k__Assembly__Contig__59__1	1244	2798
RFL 313	R0932E.300k__Assembly__Contig__69__1	1245	2799
RFL 313	Wheat-Rye-6R.300k__Assembly__Contig__47__1	1246	2800
RFL 313	Primepii.300k__Assembly__Contig__69__1	1247	2801
RFL 313	Anapurna.300k__Assembly__Contig__56__1	1248	2802
RFL 313	R197.300k__Assembly__Contig__91__1	1249	2803
RFL 314	Anapurna.300k__Assembly__Contig__143__2	1250	2804
RFL 314	R197.300k__Assembly__Contig__169__2	1251	2805
RFL 314	R0932E.300k__Assembly__Contig__150__2	1252	2806
RFL 314	Wheat-Rye-6R.300k__Assembly__Contig__148__2	1253	2807
RFL 314	Primepii.300k__Assembly__Contig__147__2	1254	2808
RFL 314	R0934F.300k__Assembly__Contig__148__2	1255	2809
RFL 315	R197.300k__Assembly__Contig__129__2	1256	2810
RFL 315	R0934F.300k__Assembly__Contig__120__2	1257	2811
RFL 315	Wheat-Rye-6R.300k__Assembly__Contig__118__2	1258	2812
RFL 315	Primepii.300k__Assembly__Contig__123__2	1259	2813

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 315	Anapurna.300k_Assembly_Contig_102_2	1260	2814
RFL 315	Triticum-timopheevii.300k_Assembly_Contig_79_2	1261	2815
RFL 315	R0932E.300k_Assembly_Contig_127_2	1262	2816
RFL 316	R197.300k_Assembly_Contig_208_2	1263	2817
RFL 317	R197.300k_Assembly_Contig_9_2	1264	2818
RFL 317	Anapurna.300k_Assembly_Contig_18_2	1265	2819
RFL 317	Primepii.300k_Assembly_Contig_5_2	1266	2820
RFL 317	R0934F.300k_Assembly_Contig_19_2	1267	2821
RFL 317	Wheat-Rye-6R.300k_Assembly_Contig_14_2	1268	2822
RFL 317	R0932E.300k_Assembly_Contig_16_2	1269	2823
RFL 318	Anapurna.300k_Assembly_Contig_57_1	1270	2824
RFL 319	Primepii.300k_Assembly_Contig_117_1	1271	2825
RFL 319	R0934F.300k_Assembly_Contig_123_1	1272	2826
RFL 320	R0932E.300k_Assembly_Contig_105_2	1273	2827
RFL 320	Wheat-Rye-6R.300k_Assembly_Contig_110_2	1274	2828
RFL 320	Anapurna.300k_Assembly_Contig_91_2	1275	2829
RFL 320	R197.300k_Assembly_Contig_87_2	1276	2830
RFL 320	R0934F.300k_Assembly_Contig_101_2	1277	2831
RFL 320	Primepii.300k_Assembly_Contig_98_2	1278	2832
RFL 321	R0934F.300k_Assembly_Contig_105_2	1279	2833
RFL 321	R0932E.300k_Assembly_Contig_84_2	1280	2834
RFL 321	Wheat-Rye-6R.300k_Assembly_Contig_89_2	1281	2835
RFL 321	R197.300k_Assembly_Contig_89_2	1282	2836
RFL 321	Primepii.300k_Assembly_Contig_78_2	1283	2837
RFL 321	Anapurna.300k_Assembly_Contig_65_2	1284	2838
RFL 322	R0934F.300k_Assembly_Contig_47_1	1285	2839
RFL 322	Primepii.300k_Assembly_Contig_15_1	1286	2840
RFL 323	R0932E.300k_Assembly_Contig_21_1	1287	2841
RFL 323	R0934F.300k_Assembly_Contig_25_1	1288	2842
RFL 323	R197.300k_Assembly_Contig_14_1	1289	2843
RFL 323	Anapurna.300k_Assembly_Contig_6_1	1290	2844
RFL 323	Primepii.300k_Assembly_Contig_20_1	1291	2845
RFL 323	Wheat-Rye-6R.300k_Assembly_Contig_8_1	1292	2846
RFL 324	Anapurna.300k_Assembly_Contig_178_1	1293	2847
RFL 324	Triticum-timopheevii.300k_Assembly_Contig_116_1	1294	2848
RFL 324	R197.300k_Assembly_Contig_193_1	1295	2849
RFL 324	R0932E.300k_Assembly_Contig_191_1	1296	2850
RFL 324	Wheat-Rye-6R.300k_Assembly_Contig_184_1	1297	2851
RFL 324	Primepii.300k_Assembly_Contig_186_1	1298	2852
RFL 324	R0934F.300k_Assembly_Contig_188_1	1299	2853
RFL 325	Primepii.300k_Assembly_Contig_74_1	1300	2854
RFL 325	R197.300k_Assembly_Contig_107_1	1301	2855
RFL 325	Wheat-Rye-6R.300k_Assembly_Contig_81_1	1302	2856
RFL 325	R0934F.300k_Assembly_Contig_88_1	1303	2857
RFL 325	R0932E.300k_Assembly_Contig_115_1	1304	2858
RFL 325	Anapurna.300k_Assembly_Contig_76_1	1305	2859
RFL 326	R0932E.300k_Assembly_Contig_77_1	1306	2860
RFL 326	Wheat-Rye-6R.300k_Assembly_Contig_75_1	1307	2861
RFL 326	R197.300k_Assembly_Contig_103_1	1308	2862
RFL 326	R0934F.300k_Assembly_Contig_72_1	1309	2863
RFL 327	Primepii.300k_Assembly_Contig_192_1	1310	2864
RFL 327	R197.300k_Assembly_Contig_1921	1311	2865
RFL 327	Wheat-Rye-6R.300k_Assembly_Contig_187_1	1312	2866
RFL 327	R0932E.300k_Assembly_Contig_190_1	1313	2867
RFL 327	R0934F.300k_Assembly_Contig_185_1	1314	2868
RFL 327	Anapurna.300k_Assembly_Contig_182_1	1315	2869
RFL 328	Primepii.300k_Assembly_Contig_95_1	1316	2870
RFL 328	R197.300k_Assembly_Contig_112_1	1317	2871
RFL 328	Wheat-Rye-6R.300k_Assembly_Contig_173_1	1318	2872
RFL 329	Triticum-timopheevii.300k_Assembly_Contig_88_2	1319	2873
RFL 329	Wheat-Rye-6R.300k_Assembly_Contig_124_1	1320	2874
RFL 329	Anapurna.300k_Assembly_Contig_115_1	1321	2875
RFL 329	Primepii.300k_Assembly_Contig_122_1	1322	2876
RFL 329	R197.300k_Assembly_Contig_130_1	1323	2877
RFL 329	R0934F.300k_Assembly_Contig_114_1	1324	2878
RFL 330	Wheat-Rye-6R.300k_Assembly_Contig_141_2	1325	2879
RFL 330	R197.300k_Assembly_Contig_146_2	1326	2880
RFL 330	Primepii.300k_Assembly_Contig_155_1	1327	2881
RFL 330	Anapurna.300k_Assembly_Contig_138_2	1328	2882
RFL 330	Triticum-timopheevii.300k_Assembly_Contig_92_1	1329	2883
RFL 330	R0934F.300k_Assembly_Contig_142_2	1330	2884
RFL 330	R0932E.300k_Assembly_Contig_146_2	1331	2885
RFL 331	R0932E.300k_Assembly_Contig_26_1	1332	2886
RFL 331	R197.300k_Assembly_Contig_41_1	1333	2887

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 331	R0934F.300k_Assembly_Contig_38_1	1334	2888
RFL 331	Wheat-Rye-6R.300k_Assembly_Contig_50_1	1335	2889
RFL 331	Primepii.300k_Assembly_Contig_45_1	1336	2890
RFL 331	Anapurna.300k_Assembly_Contig_45_1	1337	2891
RFL 332	Triticum-timopheevii.300k_Assembly_Contig_72_1	1338	2892
RFL 333	R0934F.300k_Assembly_Contig_53_2	1339	2893
RFL 333	Primepii.300k_Assembly_Contig_28_2	1340	2894
RFL 333	Wheat-Rye-6R.300k_Assembly_Contig_49_2	1341	2895
RFL 333	R0932E.300k_Assembly_Contig_14_2	1342	2896
RFL 333	R197.300k_Assembly_Contig_53_2	1343	2897
RFL 333	Anapurna.300k_Assembly_Contig_55_2	1344	2898
RFL 334	R197.300k_Assembly_Contig_208_1	1345	2899
RFL 335	Wheat-Rye-6R.300k_Assembly_Contig_112_1	1346	2900
RFL 335	R0934F.300k_Assembly_Contig_130_1	1347	2901
RFL 335	R0932E.300k_Assembly_Contig_111_1	1348	2902
RFL 335	Anapurna.300k_Assembly_Contig_88_1	1349	2903
RFL 335	R0932E.300k_Assembly_Contig_206_1	1350	2904
RFL 336	Triticum-timopheevii.300k_Assembly_Contig_18_1	1351	2905
RFL 337	R197.300k_Assembly_Contig_184_1	1352	2906
RFL 337	Triticum-timopheevii.300k_Assembly_Contig_115_1	1353	2907
RFL 337	R0932E.300k_Assembly_Contig_188_1	1354	2908
RFL 337	Primepii.300k_Assembly_Contig_185_1	1355	2909
RFL 337	R0934F.300k_Assembly_Contig_186_1	1356	2910
RFL 337	Anapurna.300k_Assembly_Contig_173_1	1357	2911
RFL 337	Wheat-Rye-6R.300k_Assembly_Contig_182_1	1358	2912
RFL 338	Primepii.300k_Assembly_Contig_111_1	1359	2913
RFL 339	Anapurna.300k_Assembly_Contig_36_2	1360	2914
RFL 339	Wheat-Rye-6R.300k_Assembly_Contig_43_2	1361	2915
RFL 339	R0934F.300k_Assembly_Contig_35_2	1362	2916
RFL 339	R197.300k_Assembly_Contig_2_5	1363	2917
RFL 339	R0932E.300k_Assembly_Contig_49_2	1364	2918
RFL 339	Primepii.300k_Assembly_Contig_32_2	1365	2919
RFL 340	Triticum-timopheevii.300k_Assembly_Contig_91_1	1366	2920
RFL 341	R0932E.300k_Assembly_Contig_118_1	1367	2921
RFL 341	R197.300k_Assembly_Contig_104_1	1368	2922
RFL 341	Anapurna.300k_Assembly_Contig_97_1	1369	2923
RFL 341	Wheat-Rye-6R.300k_Assembly_Contig_94_1	1370	2924
RFL 342	Triticum-timopheevii.300k_Assembly_Contig_11_1	1371	2925
RFL 343	Primepii.300k_Assembly_Contig_179_1	1372	2926
RFL 344	Triticum-timopheevii.300k_Assembly_Contig_128_1	1373	2927
RFL 345	R0932E.300k_Assembly_Contig_120_1	1374	2928
RFL 346	Wheat-Rye-6R.300k_Assembly_Contig_142_1	1375	2929
RFL 346	R197.300k_Assembly_Contig_153_1	1376	2930
RFL 346	Primepii.300k_Assembly_Contig_158_1	1377	2931
RFL 346	Anapurna.300k_Assembly_Contig_154_1	1378	2932
RFL 347	Anapurna.300k_Assembly_Contig_9_2	1379	2933
RFL 347	R0932E.300k_Assembly_Contig_10_2	1380	2934
RFL 347	R0934F.300k_Assembly_Contig_21_2	1381	2935
RFL 347	Primepii.300k_Assembly_Contig_48_2	1382	2936
RFL 347	R197.300k_Assembly_Contig_2_2	1383	2937
RFL 347	Wheat-Rye-6R.300k_Assembly_Contig_20_2	1384	2938
RFL 348	Triticum-timopheevii.300k_Assembly_Contig_58_2	1385	2939
RFL 349	Primepii.300k_Assembly_Contig_45_2	1386	2940
RFL 349	R197.300k_Assembly_Contig_41_2	1387	2941
RFL 349	R0932E.300k_Assembly_Contig_26_2	1388	2942
RFL 349	R0934F.300k_Assembly_Contig_38_2	1389	2943
RFL 349	Anapurna.300k_Assembly_Contig_45_2	1390	2944
RFL 349	Wheat-Rye-6R.300k_Assembly_Contig_50_2	1391	2945
RFL 350	Anapurna.300k_Assembly_Contig_177_1	1392	2946
RFL 350	R0934F.300k_Assembly_Contig_184_1	1393	2947
RFL 350	R0932E.300k_Assembly_Contig_186_1	1394	2948
RFL 350	R197.300k_Assembly_Contig_187_1	1395	2949
RFL 350	Wheat-Rye-6R.300k_Assembly_Contig_181_1	1396	2950
RFL 351	R0934F.300k_Assembly_Contig_93_1	1397	2951
RFL 351	Primepii.300k_Assembly_Contig_88_1	1398	2952
RFL 352	Triticum-timopheevii.300k_Assembly_Contig_68_1	1399	2953
RFL 353	R197.300k_Assembly_Contig_138_2	1400	2954
RFL 353	Wheat-Rye-6R.300k_Assembly_Contig_122_2	1401	2955
RFL 353	Anapurna.300k_Assembly_Contig_119_2	1402	2956
RFL 353	Primepii.300k_Assembly_Contig_131_2	1403	2957

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 353	R0932E.300k_Assembly_Contig_131_2	1404	2958
RFL 353	R0934F.300k_Assembly_Contig_113_2	1405	2959
RFL 354	R0932E.300k_Assembly_Contig_106_1	1406	2960
RFL 354	Anapurna.300k_Assembly_Contig_81_1	1407	2961
RFL 354	R197.300k_Assembly_Contig_88_1	1408	2962
RFL 354	Wheat-Rye-6R.300k_Assembly_Contig_114_1	1409	2963
RFL 354	R0934F.300k_Assembly_Contig_112_1	1410	2964
RFL 354	Primepii.300k_Assembly_Contig_103_1	1411	2965
RFL 355	Anapurna.300k_Assembly_Contig_133_1	1412	2966
RFL 356	Triticum-timopheevii.300k_Assembly_Contig_91_2	1413	2967
RFL 357	Wheat-Rye-6R.300k_Assembly_Contig_56_1	1414	2968
RFL 357	Anapurna.300k_Assembly_Contig_60_2	1415	2969
RFL 357	Primepii.300k_Assembly_Contig_56_2	1416	2970
RFL 357	R0934F.300k_Assembly_Contig_60_1	1417	2971
RFL 357	R0932E.300k_Assembly_Contig_2_1	1418	2972
RFL 357	R197.300k_Assembly_Contig_60_1	1419	2973
RFL 358	R0934F.300k_Assembly_Contig_193_1	1420	2974
RFL 358	Wheat-Rye-6R.300k_Assembly_Contig_188_1	1421	2975
RFL 358	R197.300k_Assembly_Contig_197_1	1422	2976
RFL 358	Primepii.300k_Assembly_Contig_190_1	1423	2977
RFL 358	R0932E.300k_Assembly_Contig_194_1	1424	2978
RFL 358	Anapurna.300k_Assembly_Contig_184_1	1425	2979
RFL 359	Primepii.300k_Assembly_Contig_210_1	1426	2980
RFL 359	R0934F.300k_Assembly_Contig_208_1	1427	2981
RFL 360	Wheat-Rye-6R.300k_Assembly_Contig_153_1	1428	2982
RFL 360	R197.300k_Assembly_Contig_160_1	1429	2983
RFL 361	Wheat-Rye-6R.300k_Assembly_Contig_97_2	1430	2984
RFL 361	Primepii.300k_Assembly_Contig_91_2	1431	2985
RFL 361	R0932E.300k_Assembly_Contig_110_2	1432	2986
RFL 361	R0934F.300k_Assembly_Contig_86_2	1433	2987
RFL 361	Anapurna.300k_Assembly_Contig_101_2	1434	2988
RFL 361	R197.300k_Assembly_Contig_108_2	1435	2989
RFL 362	Triticum-timopheevii.300k_Assembly_Contig_61_1	1436	2990
RFL 363	R0932E.300k_Assembly_Contig_31_1	1437	2991
RFL 363	R197.300k_Assembly_Contig_49_1	1438	2992
RFL 363	R0934F.300k_Assembly_Contig_41_1	1439	2993
RFL 363	Anapurna.300k_Assembly_Contig_40_1	1440	2994
RFL 363	Wheat-Rye-6R.300k_Assembly_Contig_27_1	1441	2995
RFL 363	Primepii.300k_Assembly_Contig_65_1	1442	2996
RFL 364	Wheat-Rye-6R.300k_Assembly_Contig_141_3	1443	2997
RFL 364	R197.300k_Assembly_Contig_146_3	1444	2998
RFL 364	Triticum-timopheevii.300k_Assembly_Contig_92_2	1445	2999
RFL 364	Anapurna.300k_Assembly_Contig_138_3	1446	3000
RFL 364	R0934F.300k_Assembly_Contig_142_3	1447	3001
RFL 364	R0932E.300k_Assembly_Contig_146_3	1448	3002
RFL 364	Primepii.300k_Assembly_Contig_155_2	1449	3003
RFL 365	R0934F.300k_Assembly_Contig_5_3	1450	3004
RFL 365	Primepii.300k_Assembly_Contig_17_2	1451	3005
RFL 365	R197.300k_Assembly_Contig_5_2	1452	3006
RFL 366	Wheat-Rye-6R.300k_Assembly_Contig_144_1	1453	3007
RFL 366	Anapurna.300k_Assembly_Contig_148_1	1454	3008
RFL 366	Primepii.300k_Assembly_Contig_142_1	1455	3009
RFL 366	R0932E.300k_Assembly_Contig_147_1	1456	3010
RFL 366	R197.300k_Assembly_Contig_170_1	1457	3011
RFL 366	R0934F.300k_Assembly_Contig_168_1	1458	3012
RFL 367	R197.300k_Assembly_Contig_138_1	1459	3013
RFL 367	Wheat-Rye-6R.300k_Assembly_Contig_122_1	1460	3014
RFL 367	Anapurna.300k_Assembly_Contig_119_1	1461	3015
RFL 367	R0932E.300k_Assembly_Contig_131_1	1462	3016
RFL 367	Primepii.300k_Assembly_Contig_131_1	1463	3017
RFL 367	R0934F.300k_Assembly_Contig_113_1	1464	3018
RFL 368	R0932E.300k_Assembly_Contig_63_2	1465	3019
RFL 368	R197.300k_Assembly_Contig_35_2	1466	3020
RFL 368	Anapurna.300k_Assembly_Contig_170_2	1467	3021
RFL 368	Wheat-Rye-6R.300k_Assembly_Contig_61_2	1468	3022
RFL 369	Anapurna.300k_Assembly_Contig_159_1	1469	3023
RFL 369	R0932E.300k_Assembly_Contig_162_1	1470	3024
RFL 369	R197.300k_Assembly_Contig_165_1	1471	3025
RFL 369	Wheat-Rye-6R.300k_Assembly_Contig_138_1	1472	3026
RFL 369	R0934F.300k_Assembly_Contig_164_1	1473	3027
RFL 369	Primepii.300k_Assembly_Contig_168_1	1474	3028
RFL 370	R0932E.300k_Assembly_Contig_28_1	1475	3029
RFL 370	Primepii.300k_Assembly_Contig_26_1	1476	3030
RFL 370	Anapurna.300k_Assembly_Contig_7_1	1477	3031
RFL 371	Anapurna.300k_Assembly_Contig_117_1	1478	3032

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 371	R0932E.300k_Assembly_Contig_154_1	1479	3033
RFL 371	Primepii.300k_Assembly_Contig_134_1	1480	3034
RFL 372	R0932E.300k_Assembly_Contig_131_3	1481	3035
RFL 372	R0934F.300k_Assembly_Contig_113_3	1482	3036
RFL 373	Primepii.300k_Assembly_Contig_211_1	1483	3037
RFL 373	R197.300k_Assembly_Contig_206_1	1484	3038
RFL 373	R0934F.300k_Assembly_Contig_205_1	1485	3039
RFL 374	Triticum-timopheevii.300k_Assembly_Contig_103_2	1486	3040
RFL 375	R0932E.300k_Assembly_Contig_42_2	1487	3041
RFL 375	Anapurna.300k_Assembly_Contig_46_2	1488	3042
RFL 375	R197.300k_Assembly_Contig_147_2	1489	3043
RFL 375	Wheat-Rye-6R.300k_Assembly_Contig_18_2	1490	3044
RFL 376	R0932E.300k_Assembly_Contig_119_3	1491	3045
RFL 376	R197.300k_Assembly_Contig_122_3	1492	3046
RFL 376	R0934F.300k_Assembly_Contig_108_3	1493	3047
RFL 376	Wheat-Rye-6R.300k_Assembly_Contig_116_3	1494	3048
RFL 376	Anapurna.300k_Assembly_Contig_103_3	1495	3049
RFL 376	Primepii.300k_Assembly_Contig_101_3	1496	3050
RFL 377	R0932E.300k_Assembly_Contig_74_2	1497	3051
RFL 378	R197.300k_Assembly_Contig_202_1	1498	3052
RFL 378	R0932E.300k_Assembly_Contig_197_1	1499	3053
RFL 378	Wheat-Rye-6R.300k_Assembly_Contig_196_1	1500	3054
RFL 378	Anapurna.300k_Assembly_Contig_187_1	1501	3055
RFL 378	R0934F.300k_Assembly_Contig_199_1	1502	3056
RFL 378	Primepii.300k_Assembly_Contig_205_1	1503	3057
RFL 379	Primepii.300k_Assembly_Contig_223_1	1504	3058
RFL 379	Wheat-Rye-6R.300k_Assembly_Contig_179_1	1505	3059
RFL 379	Anapurna.300k_Assembly_Contig_193_1	1506	3060
RFL 379	R0932E.300k_Assembly_Contig_178_1	1507	3061
RFL 380	Triticum-timopheevii.300k_Assembly_Contig_85_1	1508	3062
RFL 381	R197.300k_Assembly_Contig_169_1	1509	3063
RFL 381	Wheat-Rye-6R.300k_Assembly_Contig_148_1	1510	3064
RFL 381	R0932E.300k_Assembly_Contig_150_1	1511	3065
RFL 381	Primepii.300k_Assembly_Contig_147_1	1512	3066
RFL 381	R0934F.300k_Assembly_Contig_148_1	1513	3067
RFL 381	Anapurna.300k_Assembly_Contig_143_1	1514	3068
RFL 382	Triticum-timopheevii.300k_Assembly_Contig_109_1	1515	3069
RFL 383	Wheat-Rye-6R.300k_Assembly_Contig_111_1	1516	3070
RFL 383	R197.300k_Assembly_Contig_121_1	1517	3071
RFL 383	Primepii.300k_Assembly_Contig_113_1	1518	3072
RFL 384	Triticum-timopheevii.300k_Assembly_Contig_117_1	1519	3073
RFL 385	Anapurna.300k_Assembly_Contig_167_1	1520	3074
RFL 385	R0932E.300k_Assembly_Contig_182_1	1521	3075
RFL 385	Wheat-Rye-6R.300k_Assembly_Contig_175_1	1522	3076
RFL 385	R0934F.300k_Assembly_Contig_172_1	1523	3077
RFL 385	Primepii.300k_Assembly_Contig_170_1	1524	3078
RFL 385	R197.300k_Assembly_Contig_175_1	1525	3079
RFL 386	Primepii.300k_Assembly_Contig_105_1	1526	3080
RFL 386	R0934F.300k_Assembly_Contig_56_1	1527	3081
RFL 387	Primepii.300k_Assembly_Contig_159_2	1528	3082
RFL 388	Wheat-Rye-6R.300k_Assembly_Contig_175_2	1529	3083
RFL 388	Anapurna.300k_Assembly_Contig_167_2	1530	3084
RFL 388	R0934F.300k_Assembly_Contig_172_2	1531	3085
RFL 388	Primepii.300k_Assembly_Contig_170_2	1532	3086
RFL 388	R0932E.300k_Assembly_Contig_182_2	1533	3087
RFL 388	R197.300k_Assembly_Contig_175_2	1534	3088
RFL 389	R0934F.300k_Assembly_Contig_74_1	1535	3089
RFL 389	Anapurna.300k_Assembly_Contig_64_1	1536	3090
RFL 389	Wheat-Rye-6R.300k_Assembly_Contig_70_1	1537	3091
RFL 389	Primepii.300k_Assembly_Contig_79_1	1538	3092
RFL 390	Wheat-Rye-6R.300k_Assembly_Contig_217_1	1539	3093
RFL 390	Anapurna.300k_Assembly_Contig_203_1	1540	3094
RFL 391	R197.300k_Assembly_Contig_159_2	1541	3095
RFL 391	Wheat-Rye-6R.300k_Assembly_Contig_146_2	1542	3096
RFL 391	Primepii.300k_Assembly_Contig_154_2	1543	3097
RFL 391	R0934F.300k_Assembly_Contig_144_2	1544	3098
RFL 391	Anapurna.300k_Assembly_Contig_152_2	1545	3099
RFL 391	R0932E.300k_Assembly_Contig_149_2	1546	3100
RFL 392	R0934F.300k_Assembly_Contig_118_2	1547	3101
RFL 392	Triticum-timopheevii.300k_Assembly_Contig_82_2	1548	3102
RFL 393	Primepii.300k_Assembly_Contig_115_3	1549	3103
RFL 393	Wheat-Rye-6R.300k_Assembly_Contig_108_3	1550	3104
RFL 394	R0934F.300k_Assembly_Contig_222_1	1551	3105

TABLE 7-continued

RFL-Name	Name	SEQID-PRT1	SEQID-DNA
RFL 395	Primepii.300k_Assembly_Contig_159_1	1552	3106
RFL 396	Triticum-timopheevii.300k_Assembly_Contig_58_1	1553	3107
RFL 397	Triticum-timopheevii.300k_Assembly_Contig_46_2	1554	3133

Example 3: Mapping of the Genes Encoding Candidate Rf Proteins in the Chromosomal Intervals Associated with Fertility Restoration

A. Fine-Mapping of the Genomic Region Containing Rf1 Genetic Determinants

Three F2 mapping populations segregating for Rf1 (R197xKalahari, R204xAlixan and R0932ExAltigo) encompassing 210, 218 and 212 individuals respectively were phenotyped and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform.

Fertility tests were conducted indoors under controlled growth conditions, either in growth chambers or in greenhouses, enabling normal fertility of the tested wheat plants. The fertility scores indicated have been calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets. The t-tests conducted were done by comparing the fertility scores of F1s made with a restorer and the fertility scores of a panel of elite inbred lines grown under the same conditions.

Rf1 was first mapped on the short arm of the chromosome 1A between 4 cM and 10.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1087371 and cfn0530841. These two SNP markers delimit the largest possible interval defined by the three mapping populations.

Subsequently, joint analysis of the three mapping populations and phenotyping of the individual F2 recombinant plants on derived F3 families validated the QTL position and delimited the Rf1 interval between 7 CM and 8.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1082074 and cfn0523990. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the wheat genome reference physical map. The left border (cfn1082074) was anchored on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the right border (cfn0523990) was anchored on the IWGSCWGAV02_1AS_scaffold47238 scaffold.

Next, the locus was fine-mapped by screening 2976 and 3072 F3 lines from R197xKalahari and R204xAlixan derived from F2 plants heterozygous at the locus. Phenotyping and analysis of recombinant plant progenies within the interval redefined a smaller mapping interval between 7.5 and 8.8 CM delimited by cfn0522096 and cfn0527067 SNP markers on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the IWGSCWGAV02_1AS_scaffold47238 scaffold, respectively.

B. Fine-Mapping of the Genomic Region Containing Rf3 Genetic Determinants

Three F2 mapping populations (TJB155xAnapurna, 2852xAltamira, and AH46xR0946E) encompassing 217, 135, and 246 individuals respectively and a doubled-haploid (DH) population (H46xR934F) consisting of 140 individual plants segregating for Rf3 were phenotyped as described in example 1, and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Rf3 was first mapped on the short arm of the chromosome 1B between 18.9 CM and 24.2 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn0554333 and cfn0560679. These two SNP markers delimit the largest possible interval defined by the four mapping populations.

Subsequently, joint analysis of the four mapping populations and validation of the phenotype of the individual F2/DH recombinant plants on derived F3 families validated the QTL, genetically delimited the locus between 22.2 cM and 22.7 cM on Limagrain's internal consensus map, and physically delimited the Rf3 interval between SNP markers cfn0436720 and cfn0238384. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the physical map. The left border (cfn0436720) was anchored on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the right border (cfn0238384) was anchored on the IWGSCWGAV02_1BS_scaffold5117 scaffold.

Next, the locus was fine-mapped by screening 2496 and 672 plants from TJB155xAnapurna and AH46xR0946E F2 plants heterozygous at the locus. Analysis of recombinant F3 plant progenies within the interval redefined a smaller mapping interval between 22.5 and 22.7 cM delimited by cfn1249269 and BS00090770 SNP markers on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the IWGSCWGAV02_1BS_scaffold5117 scaffold, respectively.

C. Mapping of the Genomic Region Containing Rf7 Genetic Determinants

We crossed R197 and Primepii and then derived a population of 176 plants from individuals that were rf1 and rf3, i.e. not carrying the restorer alleles at the loci Rf1 and Rf3. The plants were genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. We mapped the Rf7 locus on chromosome 7BL. Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generations. We identified a large QTL ranging from 45 CM to 88 cM on chromosome 7B on Limagrain's internal consensus map with a peak on 46.7 CM (cfn0919993 with LOD score of 3.37E-40). Initial analysis of the recombinant plants suggests the Rf7 gene could be located between cfn3407185 and W90K_RAC875_c33564_120 markers delimiting a mapping interval of 0.3 cM between. 46.7 CM and 47 cM on Limagrain's internal consensus map.

Example 4: Identification of Candidate Orthologous RFL Groups

For each of the 282 RFL groups identified in example 2, the captured RFL ORFs (in the following referred to as protein sequences) were identified and the total number of RFL protein sequences is reported for each accession (Table 7, Table 8).

Only the following RFL clusters will be taken into consideration:

1. RFL cluster contains protein representatives for all seven accessions and the sequences show polymorphism and/or length differences.
2. RFL cluster contains representatives for only these accessions for which genetic characterization indicated that they may contain the same Rf gene or genes.

Tables 8A, 9A, 10 and 11 show the lists of the orthologous RFL groups correlating respectively with Rf1, Rf3, Rf7 and Rf-Rye-6R genes after the first screen. *T. timopheevii* is known to be fertile and, as a consequence, to restore T-CMS. This line is added here as it could contain any of the target Rf1, Rf3, Rf7 or Rf-rye genes.

Finally, only the orthologous RFL groups mapping in the chromosomal interval in wheat *Triticum aestivum* Chinese Spring reference genome are considered as candidates for further analyses. These orthologous RFL groups will be selected as "candidate Rf groups".

Mapping was achieved using the tool tblastn from the BLAST+ Suite (https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastDocs&DOC_TYPE=Download). Specific parameters (-evalue 1e-25-best_hit_score_edge 0.05-best_hit_overhang 0.25) were used in order to keep all best hits.

A. Results for Rf1 Accessions:

Table 8A shows the orthologous RFL groups comprising at least one sequence captured from an accession characterized as bearing the Rf1 gene (R197, R0932E and *T. timopheevii*). The mapping described in example 3 allows us to discard orthologous RFL groups mapped outside of the chromosomal interval genetically associated with Rf1 fertility on the short arm of chromosome 1A. In this way, four RFLs clusters (79, 104, 185 and 268) were identified as potentially corresponding to the Rf protein encoded by the Rf1 gene.

All protein sequences from group RFL185 contain ~500 amino acids and only 8.5 PPR motifs. Typically, full-length functional RFL proteins are expected to contain 15-20 PPR motifs. In addition, the last PPR motif of RFL185 is composed of only 15 amino acids. This indicates that RFL185 is truncated. RFL268 is also truncated (382 amino acids). Detailed sequence analysis has shown that RFL185 and RFL268 are remnants of the same gene that was split by a frameshift. Thus, both proteins are unlikely to be functional.

Hence RFL79 and RFL104 are considered as being the best candidate Rf groups for Rf1.

It can be noted from Table 8A that the only accession lacking Rf1 containing sequences in these candidate Rf groups is the accession R0934F.

TABLE 8B

presents the proteins in the candidate Rf groups 79 and 104.

RFL 79	Cluster 79	length	ORF name
0	0	808aa	R197.300k_Assembly_Contig_120_1
1	1	808aa	R0932E.300k_Assembly_Contig_103_1
2	2	808aa	R0934F.300k_Assembly_Contig_80_1
3	3	808aa	Triticum-timopheevii.300k_Assembly_Contig_57_1
RFL 104	**Cluster 104**	**length**	**ORF name**
0	0	757aa	R197.300k_Assembly_Contig_72_1
1	1	757aa	R0932E.300k_Assembly_Contig_82_1
2	2	757aa	R0934F.300k_Assembly_Contig_69_1
3	3	757aa	Triticum timopheevii.300k_Assembly_Contig_35_1

The DNA sequences derived from the contigs identified in example 2 and encoding RFL proteins from candidate Rf groups 79 and 104 were aligned with BWA-MEM software (Li H. and Durbin R., 2010). It was observed that these sequences differ in the 5' UTR region in R0934F compared to R0932E and R197. One hypothesis is that the DNA sequences in R0934F were generated by a recombination event between the DNA sequences from candidate Rf groups 79 and 104. This recombined sequence may not be functional in R0934F as this line is only known to carry Rf3.

B. Results for Rf3 Accessions:

The same rationale as for the Rf1 accessions was applied to the accessions (Primepii and R0934F) characterized as carrying the Rf3 gene. Table 9A shows that orthologous RFL groups 67, 89, 140, 166 and 252 are candidate Rf groups for the Rf protein encoded by Rf3 as they contain proteins identified in Primepii and R0934, the two accessions characterized as carrying Rf3 restorer gene and are located in the mapped genetic interval on the short arm on chromosome 1B.

TABLE 8A

Selection of RFL based on accession CMS information for Rf1.

CMS genotype RFLGene	MAINTAINER ANAPURNA	Rf3 PRIMEPII	Rf1 + Rf7 R197	Rf1 R0932E	Rf3 R0934F	Restorer from Rye introgression Wheat-Rye-6R	Restorer *Triticum timopheevii*	Mapping Positionned in Rf1 mapping interval
RFL1	0	0	1	1	0	0	0	NO
RFL56	0	0	1	1	0	0	1	NO
RFL59	0	0	1	2	0	0	1	NO
RFL73	0	0	1	1	0	0	1	NO
RFL74	0	0	3	4	3	0	0	NO
RFL79	0	0	1	1	1	0	1	YES
RFL93	0	0	1	1	0	0	0	NO
RFL104	0	0	1	1	1	0	1	YES
RFL129	0	0	1	1	1	0	0	NO
RFL185	0	0	1	1	1	0	1	YES
RFL268	0	0	1	1	1	0	1	YES

TABLE 9A

selection of RFL clusters based on germplasm CMS information for Rf3								
CMS genotype RFLGene	MAINTAINER ANAPURNA	Rf3 PRIMEPII	Rf1 + Rf7 R197	Rf1 R0932E	Rf3 R0934F	Restorer from Rye introgression Wheat-Rye-6R	Restorer *Triticum timopheevii*	Mapping Positionned in Rf3 mapping interval
RFL22	0	1	0	0	1	0	0	NO
RFL67	0	4	0	0	3	0	0	YES*
RFL89	0	2	0	0	1	0	0	YES
RFL140	0	1	0	0	1	0	0	YES
RFL142	0	1	0	0	1	0	0	NO
RFL164	0	1	0	0	1	0	0	NO
RFL166	0	1	0	0	1	0	0	YES
RFL227	0	1	0	0	1	0	0	NO
RFL252	0	1	0	0	1	0	0	YES

In order to achieve an exhaustive analysis for selection of Rf3 candidates, all 282 RFL clusters were carefully analyzed with regard to number of RFLs, their length and their origin in relation to Rf3 genotype information. RFL clusters composed of RFL sequences from multiple accessions were screened for full-length protein sequences originating only from Primepii and R0934F genotypes and partial/shorter sequences from the non-Rf3-carrying genotypes. This analysis allowed the identification of four additional Rf3-candidate RFL clusters: RFL28, RFL29, RFL60 and RFL170.

TABLE 9B

details of the proteins present in RFL cluster 28, 29, 60 and 170 including protein length (aa = amino acids) and name.

RFL 28	Cluster 28	Rf3	
0	0	323aa	Anapurna.300k_Assembly_Contig_2_3
1	1	479aa	Anapurna.300k_Assembly_Contig_2_2
2	2	857aa	Primepii.300k_Assembly_Contig_13_1
3	3	323aa	R197.300k_Assembly_Contig_10_3
4	4	479aa	R197.300k_Assembly_Contig_10_2
5	5	479aa	R0932E.300k_Assembly_Contig_23_2
6	6	323aa	R0932E.300k_Assembly_Contig_23_3
7	7	857aa	R0934F.300k_Assembly_Contig_17_1
8	8	323aa	Wheat-Rye-6R.300k_Assembly_Contig_7_3
9	9	479aa	Wheat-Rye-6R.300k_Assembly_Contig_7_2

RFL 29	Cluster 29	Rf3	
0	0	828aa	Primepii.300k_Assembly_Contig_67_1
1	1	828aa	R0934F.300k_Assembly_Contig_78_1
3	2	536aa	Wheat-Rye-6R.300k_Assembly_Contig_77_2
4	3	295aa	Wheat-Rye-6R.300k_Assembly_Contig_77_1

RFL 60	Cluster 60	Rf3	
0	0	828aa	Primepii.300k_Assembly_Contig_94_1
1	1	287aa	R197.300k_Assembly_Contig_95_1
2	2	828aa	R0934F.300k_Assembly_Contig_73_2
4	3	809aa	Wheat-Rye-6R.300k_Assembly_Contig_48_2

RFL 170	Cluster 170	Rf3	
0	0	219aa	Anapurna.300k_Assembly_Contig_174_3
1	1	560aa	Primepii.300k_Assembly_Contig_60_2
2	2	369aa	R197.300k_Assembly_Contig_94_2

TABLE 9B-continued

details of the proteins present in RFL cluster 28, 29, 60 and 170 including protein length (aa = amino acids) and name.

3	3	560aa	R0934F.300k_Assembly_Contig_67_2
4	4	369aa	Wheat-Rye-6R.300k_Assembly_Contig_113_2

Table 9B shows that:

In candidate group RFL 28, the sequences from Rf3 genotypes (Primepii and R0934F) are 857 amino acids long whereas sequences from all other germplasms (non-Rf3) are truncated (sizes ranging from 323-479 amino acids) and thus are most probably nonfunctional.

In candidate group RFL 29, the sequences from Rf3 accessions (Primepii and R0934F) are 828 amino acids long whereas sequences from all other (non-Rf3) germplasms are truncated and thus most probably nonfunctional.

In candidate group RFL 60, the sequences in non-Rf3 genotypes (R197 and R0932E) are either absent (R0932E) or deemed nonfunctional due to their amino acid length (287 amino acids). The sequences from Rf3 genotypes Primepii and R0934F based on their sequence length appear to be full length and functional. In addition, our mapping analysis positioned RFL60 cluster within the Rf3 interval.

In cluster RFL 170, the sequences from Rf3 genotypes (Primepii and R0934F) are significantly larger (560 amino acids) than sequences from non Rf3 genotypes that appear truncated (below 370 amino acids) and are considered as being nonfunctional. Our detailed sequence analysis has shown that RFL170 is actually a second ORF, in addition to RFL 288, encoded by the same contig. Both ORFs originate from the same RFL gene in which contiguity was disrupted by a frameshift.

C. Rf7 and Rf-Rye Accessions:

The same rationale as for the analysis of Rf1 and Rf3 carrying accessions was applied to the accessions carrying the Rf7 restorer gene (R197 and *T. timopheevii*). The Rf7 gene was mapped on chromosome 7BL.

Table 10 shows that RFL 80, 128 and 191 are candidate Rf groups for the Rf protein encoded by the Rf7 gene as the proteins assigned to those clusters were found only in either R197 or *T. timopheevii*.

In regard to a restorer gene that originates from the introgression of rye chromosome 6R into wheat genome, clusters composed of single proteins originating from the Wheat-Rye-6R restorer line are considered as good candidates for a Rye-6R-specific restorer gene. Those criteria are true for the RFL 46, 87 and 208 orthologous groups listed in

Table 6. Due to their high sequence divergence compared to *Triticum* sequences these genes are great candidates for restorer genes originating from rye.

TABLE 10

selection of RFL based on germplasm CMS information for Rf7								
CMS genotype RFLGene	MAINTAINER ANAPURNA	Rf3 PRIMEPII	Rf1 + Rf7 R197	Rf1 R0932E	Rf3 R0934F	Restorer from Rye introgression Wheat-Rye-6R	Restorer *Triticum timopheevii*	Mapping Positionned in Rf7 mapping interval
RFL49	0	0	2	0	0	0	2	NO
RFL63	0	0	2	0	0	0	1	NO
RFL80	0	0	1	0	0	0	0	Not mapped
RFL85	0	0	2	0	0	0	1	NO
RFL125	0	0	1	0	0	0	2	NO
RFL128	0	0	1	0	0	0	0	Not mapped
RFL174	0	0	1	0	0	0	2	NO
RFL191	0	0	1	0	0	0	0	Not mapped

TABLE 11

selection of RFL based on germplasm CMS information for Rf-rye							
CMS genotype RFLGene	MAINTAINER ANAPURNA	Rf3 PRIMEPII	Rf1 + Rf7 R197	Rf1 R0932E	Rf3 R0934F	Restorer from Rye introgression Wheat-Rye-6R	Restorer *Triticum timopheevii*
RFL46	0	0	0	0	0	1	0
RFL87	0	0	0	0	0	1	0
RFL208	0	0	0	0	0	1	0

Example 5: Cloning of Candidate Genes for Fertility Restoration of *T. timopheevii* CMS

The nucleic acid sequence encoding any of the RFL proteins from a candidate Rf group could be used for cloning and transformation. However, in the present experiment for cloning and transformation purposes, a DNA sequence encoding the longest RFL protein which was characterized as having a start codon, mitochondrial targeting sequence and number of PPR motifs between 15 and 20 was preferentially used. If at least two longest RFL proteins happen to have the same length, the nucleic acid encoding such RFL protein, and presenting the longest 5'-UTR sequence will be preferentially chosen to perform the cloning and transformation steps.

Wheat-Rye-6R RFL46 sequence derived from Wheat-Rye-6R.300k_Assembly_Contig_35_1 was optimized to provide SEQ ID N°3115. This sequence was cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880); The termination sequence, named terSbHSP, is depicted in SEQ ID N° 3110. The sequence of the recombinant construct is depicted in SEQ ID N°3125.

Similarly as above, TaRFL104 sequence (derived from R0932E.300k_Assembly_Contig_82_1), TaRFL67 sequence (derived from Primepii.300k_Assembly_Contig_2_1), TaRFL79 sequence (derived from R197.300k_Assembly_Contig_120_1) and TaRFL89 sequence (derived from R0934F.300k_Assembly_Contig_99_1) were adapted for cloning purpose as depicted respectively in SEQ ID N°3117 to 3120. These sequences were cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746. The sequences of the recombinant constructs are respectively depicted in SEQ ID N°3131, 3128, 3129 and 3130.

TaRFL104 sequence (depicted in SEQ ID N°3117) was cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL104, SEQ ID N° 3113) and the 3' termination sequence of *Triticum aestivum* RFL104 encoding gene (terTaRFL104, depicted in SEQ ID N° 3112), into the destination binary plasmid pBIOS10747. The sequence of the recombinant construct is depicted in SEQ ID N°3126.

TaRFL79 sequence (depicted in SEQ ID N°3119) was cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL79, SEQ ID N°3123) and the 3' termination sequence of *T. aestivum* RFL79 encoding gene (terTaRFL79, depicted in SEQ ID N° 3124), into the destination binary plasmid pBIOS10747. The sequence of the recombinant construct is depicted in SEQ ID N°3122.

Wheat-Rye RFL46 sequence was also cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL46, SEQ ID N° 3114) and the 3' termination sequence of *Triticum aestivum* RFL46 encoding gene (terTaRFL46, depicted in SEQ ID N° 3111), into the destination binary plasmid pBIOS10747. For this construct, Wheat-Rye RFL46 sequence is derived from Wheat-Rye-6R.300k_Assembly_Contig_35_1 coding sequence and modified for cloning purpose without any optimization steps. The coding sequence is depicted in SEQ ID N°3116. The sequence of the recombinant construct is depicted in SEQ ID N°3127.

The binary destination vectors pBIOS10746 and pBIOS10747 are a derivative of the binary vector pMRT (WO2001018192A3).

All the binary plasmids described above were transformed into *Agrobacterium* EHA105.

Example 6: Transformation & Fertility Restoration Phenotyping Assays

In order to screen for candidate genes involved in fertility restoration, the BGA_Fielder_CMS wheat cultivar harboring both cytoplasmic male sterility and strong transformability and regeneration potential was developed. BGA_Fielder_CMS wheat cultivars were transformed with *Agrobacterium* strains obtained in example 5 essentially as described by WO 2000/063398. Wheat transgenic events were generated for each construct described above.

For construct comprising RFL46, transformation was also performed with the cultivar Fielder.

All wheat transgenic plants generated in example 6 and control fertile plants were grown in a glasshouse under standard wheat growth conditions (16 h of light period at 20° C. and 8 h of dark period at 15° C. with constant 60% humidity) until control grains of the wild type Fielder cultivar reached maturity stage.

Fertility of the transgenic plants was evaluated by counting the number of seeds and empty glumes per spikes on each plant and comparing with the wild type Fielder and BGA_Fielder_CMS control plants. Plants are also evaluated by observing anther extrusion.

16 transformed CMS-Fielder plants overexpressing the RFL79 sequence recited in SEQ ID N°361 (as listed in table 7) under the ZmUbi promoter derived from 11 independent transformation events were analyzed.

All the plants present restoration of male fertility while 100% of untransformed CMS-Fielder plants grown in parallel are fully sterile with no anther extrusion and no seed produced, and 100% of WT-Fielder plants are fertile.

These results confirm that RFL79 can restore fertility of a CMS-T plant and that genetic transformation of CMS-Fielder is an efficient system to test the function of restorer-of-fertility genes.

Example 7: Cloning of Rf Gene Promoter, Transformation and GUS Assays

The *E. coli* beta-glucuronidase (EcGUS) sequence was optimized (as depicted in SEQ ID N°3121) and cloned via restriction enzyme reaction, between the native *Triticum aestivum* promoter (proTaRFL46, SEQ ID N° 3114) and the 3' termination sequence of *T. aestivum* gene encoding RFL46 (terTaRFL46, depicted in SEQ ID N° 3111), into the destination binary plasmid pBIOS10743 forming pBIOS11468.

Fielder wheat cultivars were transformed with these *Agrobacterium* strains essentially as described by WO 2000/063398. Wheat transgenic events were generated for each construct described above.

After booting stage until anthesis, heads and floral organs were dissected and incubated in X-Gluc solution (Jefferson, 1987) at 37° C. for 16 hours, to assess GUS expression.

Example 8: Identification of Full Length RFL PPR Genes Potentially Involved in Rf4 Fertility Restoration

A second capture was achieved using a set of different accessions compared to the capture performed in example 2. The following accessions were used:

- Two Maintainer lines (Anapurna, Fielder)
- The *T. timopheevii* accession as described in example 2
- Four accessions identified as restorer lines of *T. timopheevii*-type CMS and characterized by the presence of the Rf4 restorer locus: L13, R113, 17F3R-0377 and GSTR435.
- Rf1, Rf3 and Rf7 restorer accessions which are characterized by the absence of the Rf4 restorer locus: R197, R0934F.

GSTR435 is derived from an introgression of *Aegilops* speltoïdes into *Triticum aestivum* and at is available USDA (https://npgsweb.ars-grin.gov/gringlobal/search.aspx). The three other accessions, R113 (available via Australian Grains Genebank: 90819), L13 (available via Australian Grains Genebank: 90821) and 17F3R-0377 (derived from R113) are all derived from *Triticum* timopheevi introgressions into *Triticum aestivum.*

The bait design and hybridization with DNA fragments from the accessions were performed as in example 2. Then, a subset of 100K read pairs from each accession were mapped to the RFL groups identified in table 7 using Novoalign (version 3.04.06, http://www.novocraft.com/products/novoalign/) with settings allowing multiple hits with approximately 97% of identity (options: -r all -t 240). The average coverage per RFL was calculated using Bedtools utilities (version 2.26.0, then http://github.com/arq5x/bedtools2) coverageBed (option: -d) and groupBy (options: -o mean).

The relative coverage of all RFLs with the reads from each accession was assessed. A first ranking of the RFLs according to their coverage with reads from accessions derived from *T. timopheevii* was assessed. Only RFL groups showing no coverage (value from 0 to 10) with reads from "non-Rf" (maintainer) or non-Rf4 accessions but showing significant coverage (value>30) with reads from Rf4 accessions were considered. FIG. **3** shows the list of these RFL groups potentially corresponding to the Rf4 gene.

The coverage with accession GSTR435 was also assessed. FIG. **3** shows that only RFL120 shows significant coverage in accession GSTR435, although lower than for the other accessions. This could be explained by a greater phylogenetic distance between *T. aestivum* and *Aegilops* speltoïdes than between *T. aestivum* and *T. timopheevii.*

In order to investigate further the sequences related to the RFL120 group, the reads mapping to RFL120 for each Rf4 accession were then assembled in two steps. The first step consisted of merging overlapping read pairs with the utility bbmerge.sh from the BBMAP package (version 36.59 https://sourceforge.net/projects/bbmap/) and assembling them with the utility tadwrapper.sh from the same package (options: k=150,180,210,240,270,300,330,360,390,420,450 bisect=t). The contigs from the first step were deduplicated with the tool dedupe.sh also from the same package and given to another assembler, SPADES (version 3.10.1 http://bioinf.spbau.ru/spades), as "trusted contigs" along with all read pairs from the same accession (options: --cov-cutoff 5-careful) to generate the final assembly of the accession. Then for each accession the protein sequence RFL120 (SEQ ID N° 477 as listed in table 7) best hit was searched using the tblastn utility from the BLAST+ package (version 2.2.30 https://blast.ncbi.nlm.nih.gov/Blast.cgi?PAGE_TYPE=BlastDocs&DOC_TYPE=Download) using default settings.

The following sequences were finally identified to be included in the RFL120 group:

RFL120-R113 which is depicted in SEQ ID N°3138 and is encoded by SEQ ID N°3142, RFL120-L13 which is depicted in SEQ ID N°3137 and is encoded by SEQ ID N°3141, RFL120-17F3R-0377 is depicted in SEQ ID N°3135 and is encoded by SEQ ID N°3139 and finally RFL120-GSTR 435 (called here RFL120-spelt) is depicted in SEQ ID N°3136 and is encoded by SEQ ID N°3140.

Alignment between the above amino acid or nucleotide sequences with the corresponding sequences from RFL120, called here "RFL120_timo", and recited in SEQ ID N°477 and SEQ ID N°2031 (see table 7) shows that RFL120-17F3R-0377 is truncated. Regarding RFL120-R113 and RFL120-L13, the nucleotide sequences are both identical to RFL120-timo except that they are respectively 357 and 349 nucleotides longer on the 5'UTR region.

FIGS. **4***a* and **4***b* shows respectively the alignment between nucleotide and amino acid sequences of RFL120-spelt with RFL120-timo. This shows that RFL120-timo nucleotide sequence is 95% identical to the "RFL120_spelt" one which explains the low coverage previously observed with GSTR435 in FIG. **3**.

In conclusion, the results confirm that the RFL120 group is the strongest candidate for Rf4.

Example 9: Cloning, Transformation and Fertility Restoration Assays

Following the same methods as described in Examples 5 and 6, the nucleotide sequences of RFL120-timo, RFL120-spelt, RFL120-R113 and RFLK120-L13 are, when appropriate, optimized for ensuring proper expression in wheat and adapted for cloning purposes.

These sequences are cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880); The termination sequence, named terSbHSP, is depicted in SEQ ID N° 3110.

Transformation and fertility assays are performed as in Example 6.

Example 10: Evaluation of the CMS *T. timopheevii* Rf3 Restorer Lines Fertility

Eleven wheat elite lines classified as carrying Rf3 restorer gene known to be involved in restoration of the T-type cytoplasmic male sterility in *Triticum timopheevii* (T-CMS) were assessed for their capacity to restore the T-CMS cytoplasm and characterized for their fertility genotype. Hybrids resulting from crosses between a sterile CMS wheat line, used as a female parent, and a given wheat restorer line, used as a pollen donor, were studied regarding their ability to produce grain. The elite lines included both commercially available lines such as Altigo, Aristote, Cellule, Altamira, Rubisko, Primepii or Premio as well as Limagrain proprietary lines (Table 12). In addition, Chinese Springlines were included in the study (Table 12).

Fertility tests have been conducted indoors, either in growth chamber or in greenhouse, under controlled growth conditions enabling a normal expression of the fertility phenotype of the tested wheat plants. The plants were grown under 16 hours light period and temperature between 2° and 25° C. and 8 hours dark period at a temperature between 15° C. and 20° C., with humidity between 50 and 70%. The observed restoration of pollen fertility may be partial or complete.

The fertility score of F1 wheat plants carrying T-CMS cytoplasm may be calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets and may be compared with the fertility scores of a panel of control fertile plants which in this study consists of elite inbred lines bearing a normal wheat cytoplasm, grown in the same area and under the same agro-environmental conditions. It is preferred that such panel of lines comprises a set of at least 5 elite inbred lines. Besides, it is preferred that at least 10 spikes from different F1 individual plant will be assessed for a given experiment.

Fertility score i higher than zero (>0) indicates that the plant has acquired partial or full fertility restoration. For each fertility score, a statistical test is achieved to obtain a p-value. Examples of statistical tests are the Anova or mean comparison tests. A p-value below a 5% threshold will indicate that the two distributions are statistically different. Therefore, a significantly lower fertility score of the tested wheat plant as compared to the fertility score of the fertile control plant is indicative that the F1 plant has not acquired full restoration fertility (i.e. partial restoration). A significantly similar or higher fertility score is indicative that the F1 plant has acquired full restoration of fertility. The fertility scores indicated were calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets.

The t-tests were conducted by comparing the fertility scores of F1 plants carrying Rf3 restorer gene and the fertility scores of a panel of elite inbred lines grown under the same conditions (633 spikes from 37 winter and spring elite lines; $\mu=2.36$, $\alpha=0.59$).

The results presented in Table 12 show that there are two types of partial Rf3 restorer of fertility in the CMS hybrids. One that can be referred to as "Rf3" and which fertility scores are comprised between 1 and 1.8, and a second type referred to as "Rf3 weak" which fertility scores are less than 1. For example, the CMS-hybrids made with Primepii produced an average fertility score of 1.7 grains/spikelet over 10 individual F1 spikes ("Rf3" phenotype) while hybrids made with Altigo produced an average fertility score of 0.7 grains/spikelet over 47 individual F1 spikes ("Rf3 weak" phenotype) (Table 12). Different Chinese Spring lines were shown by genetic mapping and marker assisted selection to harbor an Rf3 restorer locus (data not shown). CMS-hybrids made with Chinese Spring lines were evaluated for fertility score. They show a mean estimated fertility score of 0.6 grains/spikelet and a "Rf3 weak" phenotype.

TABLE 12

Wheat elite lines used in the study and the fertility scores of CMS-hybrids generated with their pollen.

Elite variety	Genotype	Phenotype	Number of analysed spikes	Fertility score	STD
CHINESE SPRING	RFL29b	Rf3 weak	20	0.6	0.7
ALTIGO	RFL29b	Rf3 weak	47	0.7	0.5
ARISTOTE	RFL29b	Rf3 weak	43	0.8	0.8
CELLULE	RFL29a	Rf3	10	1.2	1.0
ALTAMIRA	RFL29a	Rf3	20	1.5	1.1
PREMIO	RFL29a	Rf3	12	1.2	1.0
PRIMEPII	RFL29a	Rf3	10	1.7	0.8
R0946E	RFL29a	Rf3	35	1.8	0.6
RUBISKO	RFL29a	Rf3	7	1.2	0.6

TABLE 12-continued

Wheat elite lines used in the study and the fertility scores of CMS-hybrids generated with their pollen.					
Elite variety	Genotype	Phenotype	Number of analysed spikes	Fertility score	STD
TJB155	RFL29a	Rf3	27	1.7	0.7
ATOMO	RFL29c	Maintainer	22	0.0	0.0
CONTROL ELITES			633	2.4	0.6

*NA: Not available, STD: Standard Deviation

Example 11: Comparison of Genotypes Between the Analyzed Rf3 Restorer Lines

The RFL gene capture was achieved with accessions listed in Table 12 as described in Example 2. For each RFL identified in Example 4B, the corresponding protein sequences from each accession were aligned for comparison.

The results show that for RFL29, RFL164 and RFL166, strong association between the phenotype and the genotype exist. For RFL29, three different alleles referred to as "a", "b" and "c" were identified while two different alleles "a" and "b" are identified either for RFL164 or RFL166. All accessions with an "Rf3" phenotype carry RFL29a, RFL164a and RFL 166a alleles while all the "Rf3 weak" accessions carry RFL29b, RFL166b and RFL164b alleles in their genotype

For RFL29, the maintainer line Atomo is characterized by the presence of two truncated ORFs, probably due to a frameshift mutation, RFL29c_1 and RFL29c_2, encoding proteins consisting of 258 and 535 amino acids, respectively. This genotype form is only present in maintainer lines (data not shown).

FIG. **5**A shows the protein sequence alignment of RFL29a, RFL29b, RFL29c_1 and RFL29c_2. FIGS. **5**B and **5**C, respectively, show the protein sequence alignments of RFL164a and RFL164b (depicted in SEQ ID N°3144), and RFL166a and RFL166b (depicted in SEQ ID N°3145).

Example 12: Cloning, Transformation and Fertility Restoration Assays

Following the same methods as described in Examples 5 and 6, the nucleotide sequences of RFL29a (depicted in SEQ ID N° 3146 or SEQ ID N°1712 and encoding a sequence identical to SEQ ID N°158), RFL29b (depicted in SEQ ID N° 3149 and encoding a sequence identical to SEQ ID N°3143), RFL164a (depicted in SEQ ID N° 3147 or SEQ ID NO° 2230 and encoding a sequence identical to SEQ ID N°676), and RFL166a (depicted in SEQ ID N° 3148 or SEQ ID NO:2238 and encoding a sequence identical to SEQ ID N°684), were cloned via a Golden Gate reaction into destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and a 3' termination sequence of the gene encoding a sorghum heat shock protein (accession number: Sb03g006880 terSbHSP depicted in SEQ ID N° 3110). The sequences of each of the recombinant constructs are respectively depicted in SEQ ID N°3150, SEQ ID N°3151, 3152 and 3153.

Similarly, the RFL29a and RFL29b sequences were cloned downstream of their endogenous promoter pRFL29a (depicted in SEQ ID N°3154) and pRFL29b (depicted in SEQ ID N°3155) and the terminator sequence terSbHSP. The sequences of each of the recombinant construct are respectively depicted in SEQ ID N°3156 and SEQ ID N°3157.

Finally, two other cassettes were made identically as described previously with the only exception that the corresponding endogenous terminator sequences terRFL29a (depicted in SEQ ID N°3160) and terRFL29b (depicted in SEQ ID N°3161) are used. The corresponding expression cassettes are respectively depicted in SEQ ID N°3158 and SEQ ID N°3159.Transformation and fertility assays are performed as in Example 6.

Example 13: Comparison of the 5'UTR Sequence of the RFL29a and RFL29b Encoding Gene

In order to analyze whether the variation of the level of fertility could be explained by a variation in the gene expression level, the 5'UTR sequence of RFL29a gene was isolated from BACs generated from TJB155 line (Table 12) classified as "Rf3" and the 5'UTR sequence of RFL29b was identified from the Chinese Spring classified as "Rf3weak" line (IWGSC RefSeq v1.0 assembly).

The alignment of the 5'UTR regions identified in the RFL29a and RFL29b genes is shown in FIG. **6**.

Sequence comparison shows that the 5'UTR sequence of the RFL29a gene comprises a deletion of the 163 bp-long region identified in the 5'UTR of RFL29b corresponding sequence (SEQ ID NO: 3174). Part of this region has been identified in the patent application WO2018015403 as being putatively involved in miRNA-mediated repression of the expression of the PPR gene identified downstream.

Sequence comparison between the different accessions listed in Table 12 shows that all "Rf3weak" accessions harbor the 163 bp and all the "Rf3" accessions harbor the 163 bp deletion.

Because of the 163 bp deletion in the 5'UTR sequence of RFL29a gene, it is expected that the 163 bp region impairs the expression of RFL29b gene such that the fertility level is weak in lines harboring the RFL29b allele compared to lines harboring the RFL29a allele.

Example 14: Cloning, Transformation and Fertility Assays with a Deleted TaRFL29b Promoter

Following the same methods as described in Example 5, the nucleotide sequence of RFL29b was expressed under the modified promoter pRFL29bdel (depicted in SEQ ID N°3162) which is bearing a deletion of the 163 bp region, from the nucleotide 1876 to nucleotide 2038 of the RFL29b promoter sequence depicted in SEQ ID N°3155. The termination sequence, terSbHSP, depicted in SEQ ID N°3110 is used. The recombinant construct is depicted in SEQ ID N°3163.

Transformation of CMS*Fielder wheat line (which is either not "Rf3" or "Rf3 weak") is performed as in Example 6 except that the following controls are added: all of the "Rf3" phenotype lines as listed in Table 12, all of the "Rf3 weak" lines as listed in Table 12 and finally the transformed lines with the cassette harboring the pTaRFL29b promoter upstream to RFL29b as described in example 12.

Example 15: Modification of the Endogenous Promoter of RFL29b by CRISPR Technology to Revert "Rf3Weak" Lines to "Rf3" Lines

In order to increase the expression of RL29b, different deletions of the 163 bp are performed in the promoter of RFL29b using endonuclease for site-directed mutagenesis.

Table 13 provides the different endonucleases with the associated PAM motif and the corresponding target sequences.

FIG. **7** shows the position of the different target sequences around and within the 163 bp region identified for different endonucleases. The designed guide sequences that can be used in combination to perform a deletion in the 163 bp region are listed in Table 13.

TABLE 13

Endonucleases	Target_ID	Target name	guide sequence
LbCPF1	23	LbCpf1-100-Target-23	TAATTTCTACTAAGTGTAGATCGAGCGGAGGGAGTACTAGATAA(SEQ ID NO: 3175)
LbCPF1	42	LbCpf1-100-Target-42	TAATTTCTACTAAGTGTAGATGGAACGGAGGGAGTATTATCTAG(SEQ ID NO: 3176)
LbCPF1	67	LbCpf1-100-Target-67	TAATTTCTACTAAGTGTAGATAGATAGCTAGAAAGACAATTATT(SEQ ID NO: 3177)
LbCPF1	71	LbCpf1-100-Target-71	TAATTTCTACTAAGTGTAGATTTGAGATAGCTAGAAAGACAAT(SEQ ID NO: 3178)
SpCAS9	14	SpCas9-100-Target-14	TGACAAGTATTTCCGAGCGGGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3179)
SpCAS9	54	SpCas9-100-Target-54	GACAATTATTTAGGAACGGAGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3180)
SpCAS9	55	SpCas9-100-Target-55	AGACAATTATTTAGGAACGGGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3181)
SpCAS9	58	SpCas9-100-Target-58	GAAAGACAATTATTTAGGAAGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3182)
SpCAS9	63	SpCas9-100-Target-63	AGCTAGAAAGACAATTATTTGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3183)
SpCAS9	155	SpCas9-100-Target-155	TTTCAACAAATGACTACATAGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3184)
SpCAS9	179	SpCas9-100-Target-179	CTCTAGAGAGACAATTATTTGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3185)
SpCAS9	184	SpCas9-100-Target-184	GAGAGACAATTATTTAGGAAGTTTTAGAGCTAGAAATAGCAAGTTAAAATAAGGCTAGTCCGTTATCAACTTGAAAAAGTGGCACCGAGTCGGTGCTTTT(SEQ ID NO: 3186)

The nucleotide sequence encoding for LbCpf1 endonuclease is optimized (as depicted in SEQ ID N°3164) and cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, between the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and in the 3' region, the mouse nuclear import NLS sequence (as depicted in SEQ ID N° 3172) and the 3' termination sequence terZmHSP depicted in SEQ ID N° 3170.

The nucleotide sequence encoding for SpCas9 endonuclease is optimized (as depicted in SEQ ID N° 3165) and cloned via a Golden Gate reaction into the destination binary plasmid pBIOS10746, downstream to the constitutive *Zea mays* ubiquitin promoter (proZmUbi depicted in SEQ ID N° 3134) with the *Zea mays* ubiquitin intron (intZmUbi, depicted in SEQ ID N° 3109, Christensen et al 1992) and the SV40NLS sequence (as depicted in SEQ ID N° 3173) and upstream of the mouse nuclear import NLS sequence and the 3' termination sequence terAtNos depicted in SEQ ID N° 3171.

Each guide sequence is cloned between the pTaU6 promoter (depicted in SEQ ID N° 3168) and the termination sequence TerRNApolIII (depicted in SEQ ID N°3169).

Each cassette expressing the endonuclease is cloned consecutively to the cassette expressing the corresponding guide sequence. A recombinant cassette expressing LbCpf1 and guide sequences directed to target-23 and target-71 is depicted in SEQ ID N°3167. A recombinant cassette expressing SpCas9 and guide sequences directed to target-58 and target-54 is depicted in SEQ ID N°3166.

Transformation is performed as in Example 6 except that all “Rf3 weak” lines as listed in Table 12 are transformed. Fertility assays to select lines with an “Rf3” phenotype are performed as in example 14.

Example 16: Evaluation of the CMS *T. timopheevii* Restorer Lines Fertility

Some wheat elite lines possess an ability to partially restore the fertility of the cytoplasmic male sterility *Triticum timopheevii* (T-CMS). The hybrid formed between a sterile CMS wheat line taken as female and the wheat restorer line taken as male can produce grain. Commercial lines as Allezy, Altamira, Altigo, Aristote, Osado, Cellule or Premio and Limagrain proprietary lines have been tested for their capacity to restore T-CMS. All lines have also been characterized for their fertility genotype.

Fertility tests have been conducted indoor, either in growth chamber or in greenhouse, under controlled growth conditions enabling a normal expression of fertility of the tested wheat plants. The fertility scores indicated have been calculated by dividing the total number of seeds threshed from a spike by the number of counted spikelets. The t-tests conducted were done by comparing the fertility scores of F1s made with a restorer and the fertility scores of a panel of elite inbred lines grown under the same conditions (633 spikes from 37 winter and spring elite lines; μ=2.36, α=0.59).

The results in Table 14 show that these lines act as partial restorer of fertility in a CMS hybrid. For example, the hybrids made with Allezy produce an average fertility of 0.99 grains/spikelets over 33 individual F1 spikes. This native ability to partially restore the fertility of the CMS *T. timopheevii* was previously identified in wheat accessions as Primepii or Maris Hunstmann (Bahl and Maan, 1973), and it is confirmed by our own internal data (See Table 14). Beside this supposedly wheat intrisic source of restoration, the hybridizations between *Triticum timopheevii* and *Triticum aestivum* conducted by Wilson (Wilson and Ross 1962) contributed to the introduction into wheat of further restorer genes which chromosome arm localization was identified through monosomic analysis (Bahl and Maan, 1973; Maan and al, 1984).

Following these initial works many breeding programs implemented worldwide aimed at creating restorer lines for the T-CMS such as TJB155 (BBSRC Small Grain Cereals Collection: 2072 to 2075), which ability to restore is also confirmed by our own internal data (Table 14).

L13 (available via Australian Grains Genebank: 90821), is a wheat restorer line selected from R113 (available via Australian Grains Genebank: 90819) which carries the restorer allele Rf4. It produces a low level of fertility restoration when crossed with CMS lines (0.85 seeds/spikelet on average). When compared with the fertility scores of a panel of elite lines grown under the same conditions it appears that none of the tested putative restorer make possible a full restoration of fertility of the hybrid (p-values<0.05. table 14).

Table 14 also shows that restorer lines bearing two different Rf loci, are not able to fully restore the sterility induced by T-CMS. Therefore, all restorer lines tested, with single or two Rf loci are only partial T-CMS restorer lines.

TABLE 14

mean fertility scores and standard deviation from the indicated number of spikes. The first column indicates the name of the male line taken as pollinator for the F1 cross. All the data (except for Elite inbred lines) presented are from the created F1 crosses between the listed lines and a CMS tester. “Elite inbred lines” data refers to a panel consisting of 37 winter and spring elite lines. The t-test was implemented comparing the mean of fertility for each of the F1 crosses with the mean of the fertility scores of a panel of 633 spikes of elite inbred lines bearing fertile cytoplasm.

	Haplotypes	SPIKES	FERTILITY	STD DEV	P-VALUE
ALLEZY	Rf3	33	0.99	0.89	3.00E−33
ALTAMIRA	Rf3	20	1.48	1.1	4.00E−10
ALTIGO	Rf3	47	0.71	0.55	1.00E−63
ARISTOTE	Rf3	37	1	0.73	3.00E−37
CELLULE	Rf3	10	1.25	0.99	5.00E−09
MARISHUNSTMANN	Rf3	16	1.26	0.88	7.00E−13
OSADO	Rf3	9	1.52	0.68	2.00E−05
PREMIO	Rf3	12	1.21	1.03	8.00E−11
PRIMEPII	Rf3	51	1.55	0.75	2.00E−19

TABLE 14-continued

mean fertility scores and standard deviation from the indicated number of spikes. The first column indicates the name of the male line taken as pollinator for the F1 cross. All the data (except for Elite inbred lines) presented are from the created F1 crosses between the listed lines and a CMS tester. "Elite inbred lines" data refers to a panel consisting of 37 winter and spring elite lines. The t-test was implemented comparing the mean of fertility for each of the F1 crosses with the mean of the fertility scores of a panel of 633 spikes of elite inbred lines bearing fertile cytoplasm.

	Haplotypes	SPIKES	FERTILITY	STD DEV	P-VALUE
TJB155	Rf3	27	1.7	0.66	2.00E−08
R204	Rf1 + Rf7	59	2.09	0.51	6.00E−04
L13	Rf4	12	0.85	0.38	4.00E−18
R0929D	Rf3 + Rf7	18	1.83	0.62	1.63E−04
R0936T	Rf1 + Rf3	4	1.76	0.11	4.06E−02
ELITE inbred lines		633	2.36	0.59	

Example 17: Fine Mapping of the Genomic Region Containing Rf1, Rf3 and Genetic Mapping of Rf4 and Rf7 Genetic Determinants

A. Fine-Mapping of the Genomic Region Containing Rf1 Genetic Determinants

Three F2 mapping populations segregating for Rf1 (R197xKalahari, R204xAlixan and R0932ExAltigo) encompassing 210, 218 and 212 individuals respectively were phenotyped as described in example 1 and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Fertility of R204 and R197 lines is genetically associated to Rf1 and Rf7 locus. Fertility in R0932E line is associated to Rf1 locus.

Rf1 was first mapped on the short arm of the chromosome 1A between 4 cM and 10.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1087371 and cfn0530841. These two SNP markers delimit the largest possible interval defined by the three mapping populations (see FIG. **8**).

Following, joint analysis of the three mapping populations and phenotyping of the individual F2 recombinant plants on derived F3 families validated the QTL position and delimited Rf1 interval between 7 cM and 8.9 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn1082074 and cfn0523990. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the wheat genome reference physical map. The left border (cfn1082074) was anchored on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the right border (cfn0523990) was anchored on the IWGSCWGAV02_1AS_scaffold47238 scaffold.

Next, we decided to enlarge the population sizes to fine-map the locus and screened 2976 and 3072 F3 lines from R197xKalahari and R204xAlixan derived from F2 plants heterozygote at the locus respectively. Phenotyping and analysis of recombinant plant progenies within the interval redefined a smaller mapping interval between 7.5 and 8.8 cM delimited by cfn0522096 and cfn0527067 SNP markers on the IWGSCWGAV02_1AS_scaffold44309 scaffold and the IWGSCWGAV02_1AS_scaffold47238 scaffold respectively.

B. Fine-Mapping of the Genomic Region Containing Rf3 Genetic Determinants

Three F2 mapping populations (TJB155xAnapurna, 2852xAltamira, and AH46xR0946E) encompassing 217, 135, and 246 individuals respectively and a doubled-haploid (DH) population (H46xR934F) consisting of 140 individual plants segregating for Rf3 were phenotyped as described in example 1 and genotyped with 18100 SNP markers using Limagrain's internal genotyping platform. Sources of Rf3 locus are TJB155, Altamira and R0946E.

Rf3 was first mapped on the short arm of the chromosome 1B between 18.9 cM and 24.2 cM on Limagrain's internal consensus map and physically delimited by SNP markers cfn0554333 and cfn0560679. These two SNP markers delimit the largest possible interval defined by the four mapping populations (see FIG. **9**).

Following, joint analysis of the four mapping populations and validation of the phenotype of the individual F2/DH recombinant plants on derived F3 families validated the QTL, genetically delimited the locus between 22.2 cM and 22.7 CM on Limagrain's internal consensus map and physically delimited the Rf3 interval between SNP markers cfn0436720 and cfn0238384. We used the genomic resources of the IWGSC Whole genome assembly, 'IWGSC WGA' (available from June 2016 from the URGI IWGSC repository) to anchor the locus to the physical map. The left border (cfn0436720) was anchored on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the right border (cfn0238384) was anchored on the IWGSCWGAV02_1BS_scaffold5117 scaffold.

Next, we decided to enlarge the population sizes to fine-map the locus and screened 2496 and 672 plants from TJB155xAnapurna and AH46xR0946E derived F2 plants heterozygote at the locus. Analysis of recombinant F3 plant progenies within the interval redefined a smaller mapping interval between 22.5 and 22.7 cM delimited by cfn1249269 and BS00090770 SNP markers on the IWGSCWGAV02_1BS_scaffold35219 scaffold and the IWGSCWGAV02_1BS_scaffold5117 scaffold respectively.

C. Mapping of the Genomic Region Containing Rf7 Genetic Determinants

We crossed R197 (harboring Rf1 and Rf7 locus) and Primepii (harboring Rf3 locus) and then derived a population of 176 plants from individuals that were rf1 and rf3, which means not carrying the restorer alleles at the loci Rf1 and Rf3. The plants were genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. We mapped the Rf7 locus on chromosome 7BL. Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generation. We identified a large QTL ranging from 45 CM to 88 cM on chromosome 7B on Limagrain's internal consensus map with a peak on 46.7 CM (cfn0919993 with LOD score of 3.37E-40). First expertise of the recombinant plants suggests the Rf7 gene could be located between cfn3407185 and

W90K_RAC875_c33564_120 markers delimiting a mapping interval of 0.3 cM between 46.7 CM and 47 cM on Limagrain's internal consensus map.

D. Mapping of the Genomic Region Containing Rf4 Genetic Determinant

A mapping population from the cross between AH46 and L13 (harboring Rf4 locus) consisting of 124 individual plants segregating for Rf4 was genotyped with 18100 SNP markers using Limagrain's internal genotyping platform and phenotyped as described in example 1. A QTL that we named Rf4 was identified on chromosome 6B between 0 cM to 65 cM on Limagrain's internal consensus map with a peak on 43.3 cM (cfn0393953 with LOD score of 1.08E-13). Moreover, internal genotyping data showing a strong genetic divergence suggests the presence of an exotic chromosomal fragment which is stably transmitted through generation. It is expected that the rate of recombination be very low within this chromosomal region and, consequently, any marker in linkage with cfn0393953 is considered to be associated with Rf4 locus.

Example 18: Identification of SNP Associated with the Rf Genes and its Use in MAS

We constructed a BAC library with a DH line comprising Rf1, Rf3 and Rf7 alleles. This library was used for the identification of BAC clones within the Rf1 and Rf3 QTL intervals. More specifically, we identified and sequenced and genetically validated 3 BAC clones within the Rf1 region and 3 BAC clones within the Rf3 region. The BAC sequences were compared to Chinese Spring reference genome for SNP discovery. We then saturated Rf1 and Rf3 mapping intervals by mining available SNPs from public genomic resources and the newly discovered SNPs from sequenced BAC clones.

Screening for polymorphic and informative SNP markers on a diversity panel consisting of 83 wheat elite plants and known restorers for Rf1, Rf3, Rf4 and Rf7 identified a set of tightly linked markers localized within or in the immediate flanking regions of the Rf1, Rf3, Rf4 and Rf7 mapping intervals. These SNP markers can be used alone and/or in haplotype to select for Rf or rf plants in MAS breeding schemes (FIG. **10**A and FIG. **11**A). Allele information, marker sequences and primer information are provided in Table 15. Following, smaller sets of 2-3 markers of high quality were chosen to follow the traits in MAS breeding schemes.

As an example, the identification of the presence of the Rf1 locus in the genome of a plant is achieved by using either the marker 276113_96B22_97797 or 104A4_105588 (FIGS. **10**A and B).

Similarly, the identification of the presence of the Rf3 locus in the genome of a plant is achieved by using either the marker 136H5_3M5_7601 or 136H5_3M5_89176 (FIG. **11**A, **11**B, **11**C, **11**D and Table 15). However, any marker listed on the FIG. **11**A can be used to distinguish the maintainer lines from the restorer lines if they are polymorphic in the germplasm. As an example, in the Table 16, the presence vs absence of the Rf3 locus is achieved by using either the marker cfn1246088 or IWB72107

The identification of the presence of the Rf7 locus in the genome of a plant is achieved by using the markers cfn0917304, cfn0919993 and cfn0920459 (Table 17). In this case the 3 markers might be used in haplotype to distinguish restorer plants from maintainer plants. Thus the haplotype TGC would identify the restorer plants.

Finally, the identification of the presence of the Rf4 locus in the genome of a plant is achieved by using the markers cfn0393953 and cfn0856945 (Table 18).

TABLE 15

Haplotypes of a series of restorer lines and maintainer lines at the locus Rf3 for 2 SNP markers. Those 2 SNP markers makes possible to fully distinguish the maintainer lines from the restorer lines (including the elite lines Altamira, Cellule, Premio and the accessions TJB155 and Primepi).

CODE	R: restorer/ M: maintainer	Rf alleles	136H5_3M5_7601	136H5_3M5_89176
LGWR16-0016	R	homozygous Rf3	T	A
LGWR16-0026	R	homozygous Rf3	T	A
ALTAMIRA	R	homozygous Rf3	T	A
CELLULE	R	homozygous Rf3	T	A
TJB155	R	homozygous Rf3	T	A
ALLEZY	R	homozygous Rf3	T	A
PREMIO	R	homozygous Rf3	T	A
PRIMEPI	R	homozygous Rf3	T	A
AIGLE	M	homozygous rf3	C	G
AIRBUS	M	homozygous rf3	C	G
ALHAMBRA	M	homozygous rf3	C	G
ALIXAN	M	homozygous rf3	C	G
AMADEUS	M	homozygous rf3	C	G
ANAPURNA	M	homozygous rf3	C	G
APACHE	M	homozygous rf3	C	G
ARKEOS	M	homozygous rf3	C	G
ARLEQUIN	M	homozygous rf3	C	G
ARTDECO	M	homozygous rf3	C	G
ARTURNICK	M	homozygous rf3	C	G
ATOMO	M	homozygous rf3	C	G
AVENUE	M	homozygous rf3	C	G
CEZANNE	M	homozygous rf3	C	G
CROISADE	M	homozygous rf3	C	G
FRUCTIDOR	M	homozygous rf3	C	G
GAZUL	M	homozygous rf3	C	G
HERMANN	M	homozygous rf3	C	G

TABLE 15-continued

Haplotypes of a series of restorer lines and maintainer lines at the locus Rf3 for 2 SNP markers. Those 2 SNP markers makes possible to fully distinguish the maintainer lines from the restorer lines (including the elite lines Altamira, Cellule, Premio and the accessions TJB155 and Primepi).

CODE	R: restorer/ M: maintainer	Rf alleles	136H5_3M5_7601	136H5_3M5_89176
HORATIO	M	homozygous rf3	C	G
KALAHARI	M	homozygous rf3	C	G

TABLE 16

Haplotypes of a series of restorer lines and maintainer lines at the locus Rf3 for 2 SNP markers. Those 2 SNP markers makes possible to fully distinguish the maintainer lines from the restorer lines.

CODE	R: restorer/ M: maintainer	Rf alleles	cfn1246088	IWB72107
ALTIGO	R	homozygous Rf3	A	A
ARISTOTE	R	homozygous Rf3	A	A
OSADO	R	homozygous Rf3	A	A
AIGLE	M	homozygous rf3	C	G
AIRBUS	M	homozygous rf3	C	G
ALHAMBRA	M	homozygous rf3	C	G
ALIXAN	M	homozygous rf3	C	G
AMADEUS	M	homozygous rf3	C	G
ANAPURNA	M	homozygous rf3	C	G
APACHE	M	homozygous rf3	C	G
ARKEOS	M	homozygous rf3	C	G
ARLEQUIN	M	homozygous rf3	C	G
ARTDECO	M	homozygous rf3	C	G
ARTURNICK	M	homozygous rf3	C	G
ATOMO	M	homozygous rf3	C	G
AVENUE	M	homozygous rf3	C	G
CEZANNE	M	homozygous rf3	C	G
CROISADE	M	homozygous rf3	C	G
FRUCTIDOR	M	homozygous rf3	C	G
GAZUL	M	homozygous rf3	C	G
HERMANN	M	homozygous rf3	C	G
HORATIO	M	homozygous rf3	C	G
KALAHARI	M	homozygous rf3	C	G

TABLE 17

Haplotypes at the Rf7 locus for the restorer lines LGWR16-0016 and LGWR16-0026 and for a series of maintainer lines. "—" scores correspond to dominant markers with no amplification in several maintainer lines.

CODE	R: restorer/ M: maintainer	Rf alleles	cfn0917304	cfn0919993	cfn0920459
LGWR16-0016	R	homozygous Rf1, Rf3, Rf7	T	G	C
LGWR16-0026	R	homozygous Rf1, Rf3, Rf7	T	G	C
AIGLE	M	homozygous rf1, rf3, rf7	G	—	G
AIRBUS	M	homozygous rf1, rf3, rf7	T	G	G
ALHAMBRA	M	homozygous rf1, rf3, rf7	G	T	G
ALIXAN	M	homozygous rf1, rf3, rf7	T	T	C
AMADEUS	M	homozygous rf1, rf3, rf7	G	G	C
ANAPURNA	M	homozygous rf1, rf3, rf7	G	T	G
APACHE	M	homozygous rf1, rf3, rf7	G	T	C
ARKEOS	M	homozygous rf1, rf3, rf7	G	T	G
ARLEQUIN	M	homozygous rf1, rf3, rf7	G	G	C
ARTDECO	M	homozygous rf1, rf3, rf7	G	—	G
ARTURNICK	M	homozygous rf1, rf3, rf7	T	T	G
ATOMO	M	homozygous rf1, rf3, rf7	T	T	G
AVENUE	M	homozygous rf1, rf3, rf7	T	T	C
CEZANNE	M	homozygous rf1, rf3, rf7	G	T	G
CROISADE	M	homozygous rf1, rf3, rf7	G	T	C
FRUCTIDOR	M	homozygous rf1, rf3, rf7	G	T	G

TABLE 17-continued

Haplotypes at the Rf7 locus for the restorer lines LGWR16-0016 and LGWR16-0026 and for a series of maintainer lines. "—" scores correspond to dominant markers with no amplification in several maintainer lines.

CODE	R: restorer/ M: maintainer	Rf alleles	cfn0917304	cfn0919993	cfn0920459
GAZUL	M	homozygous rf1, rf3, rf7	—	G	C
HERMANN	M	homozygous rf1, rf3, rf7	T	G	G
HORATIO	M	homozygous rf1, rf3, rf7	G	—	G
KALAHARI	M	homozygous rf1, rf3, rf7	G	—	G

TABLE 18

Haplotypes at the Rf4 locus for the restorer lines L13 and R113 and maintainer lines.

CODE	R: restorer/ M: maintainer	Rf alleles	cfn0393953	cfn0856945
LGWR16-0016	M	homozygous rf4	T	T
LGWR16-0026	M	homozygous rf4	T	T
AIGLE	M	homozygous rf4	T	T
AIRBUS	M	homozygous rf4	T	T
ALHAMBRA	M	homozygous rf4	T	T
ALIXAN	M	homozygous rf4	T	T
AMADEUS	M	homozygous rf4	T	T
ANAPURNA	M	homozygous rf4	T	T
APACHE	M	homozygous rf4	T	T
ARKEOS	M	homozygous rf4	T	T
ARLEQUIN	M	homozygous rf4	T	T
ARTDECO	M	homozygous rf4	T	T
ARTURNICK	M	homozygous rf4	T	T
ATOMO	M	homozygous rf4	C	T
AVENUE	M	homozygous rf4	T	T
Cezanne	M	homozygous rf4	T	T
CROISADE	M	homozygous rf4	T	T
FRUCTIDOR	M	homozygous rf4	T	T
GAZUL	M	homozygous rf4	—	T
HERMANN	M	homozygous rf4	T	T
HORATIO	M	homozygous rf4	C	T
KALAHARI	M	homozygous rf4	T	T
L13	R	homozygous Rf4	C	G
R113	R	homozygous Rf4	C	G

Sequences of Snp Markers:

TABLE 19

Marker sequences and SNP position in sequence.
SEQUENCES OF SNP MARKERS

Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
cfn0523072	C	T	CAAAGGCTTGACAATGATAATGCCCCCGAATC TTG[C/T]GATAGACCTCATGCGCTAGAGTTGTT TTCCTCAAT	3187
cfn0523109	A	C	GACAAAGTTGAGGTGAACAAAACAGGCCTAC AATC[A/C]GCTAACTTACGTATATCCACATTAG CACACACCAC	3188
276I13_96B22_97797	C	T	AAATTCGACAAGTACTATGGCTATGTCTCTGA ATG[C/T]TTGTTTGGTTTTATTTGTCTATATTGT CGTTGTAT	3189
cfn0522096	C	G	ATGCAAAGTAGTACTCGTAGAGAGTTAACACA GAC[C/G]AGTGATTTATTGGGTGGTATTCTACT TGATATTTG	3190
cfn0527763	T	C	ATAAAGAAAAGTAGAGGAAGCTTATGAATAA AATGGAAAAGGAATTCAAAATTGCCGATAAA TATAAAACTCATAACAAATCTAGCCACGCAAA	3191

TABLE 19-continued

Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
			Marker sequences and SNP position in sequence. SEQUENCES OF SNP MARKERS	
			TGCCCG[T/C]GCCGCTCTGCTCGTTTGTACATG TCTCGGTGGACAAGGAAGAACCCAACAATTGC ACAGGTCAATCTTATCCAGCAAAACAAGGAA GCAAACCAAACAGG	
104A4_105172	TG	CA	ATGTTGCCTCTCGCTAGCCGCTGTCGMACCCA ATGAATAATGTT[TG/CA]TGGGTTCTGGCTCCG AGAGGATGGCCGGCTYCCC	3192
104A4_105588	A	C	GTTCCTTGTGACATGTACTCATA[A/C]ACAAGA GCCATATACTCCCCATCCTTGCA	3193
cfn0373248	T	A	GACATAATGTGTAATAACAGCCCATAATGCAA TAAATATCAATATAAAAGCATGATGCAAAATG GACGTATCATTGCCACGRAAAAAATCTCACAA GATG[T/A]GACCATTTGATCCTCRTAATTGTTG TTCTAGACCCACTCCTAAGTMTAACATTCTTT ATGTCTATYCTTCAAATCCCGAAGAGTAATGA AAACTATCGAA	3194
cfn1097828	T	C	CCATGAGTACCCGCTACTATCGATCTCCCTCCT CCCTGTAGGAGGCCTACGAACGATGCCCTCAG GTCCTGCTTCCTCTCGGTAGCGATGGATCCAC CTG[T/C]GGTTGCTCTCTCAGGAACCAGTGTTG GCGGCGGCTCATCCGGGGCGCTGGATCTTGGT GATGTGCTGGAACAACTCAACTTGGAAGACGA AGAATTTGAT	3195
cfn0527067	A	G	GACAATATGATTCACCCTAGATCCTTCACCTT ACA[A/G]TTCGAAAAAAATAAAAGAACAAAAG TAATTTGACA	3196
cfn0528390	A	G	ACGAAGATGAGGAAGGTCTTCATGTTGGGTTT ATG[A/G]TTACTAATACTTGCTTGGAATAGATG TTTTTGATC	3197
BWS0267	A	G	GTTACCCCAATATGCTCCCTCCTTGCACATTTT CTTCAGCTGCATAAAAAMCAGAATACC[A/G]C ATCAGTTGCCTGAACCTTAACGCAGGTGCAGA AATAAGGCGACATAATTTYCACTAATC	3198
cfn0527718	C	T	AGGAAAATAAATTGTTCACAACATGGACATGA GAA[C/T]GGGGCAACCAAAAAGGGAAGAACAT TGGAGGAAAC	3199
cfn0524469	G	T	TTTGTACTGCACGTAGTAAGTATTGATTTTTCT GT[G/T]TGCTCTCTGTGGACTTAGATTTGAAAA TTGGCCTT	3200
cfn0524921	A	G	ATGCACATTGTTTCCATGTTAAGCTTATATTGT GC[A/G]TAACTCAAAAGATTGAAATGGAATTA CCAAAGGGC	3201
cfn1122326	C	T	ACTGACTGTTGGAATCTGATTAAGACGCTGGA GAA[C/T]CCGAGCCAAGATATGTCACGACTAG GCCATCTGGA	3202
cfn1252000	A	G	AATCAGATCCTGTTAATGCTGTAGCCATTCTTG CA[A/G]GCGACACCTTGTCCCAGTCGTCTTATG GGCACTTA	3203
IWB14060	A	G	GGCAGAGCCGGTCGACGGAGAGGAGCGCCAT TCGACGCGTCTTCCGCAAT[A/G]TGTTTGCCTG CTTCGGCCGCGGCCATTCGGCGAGCTCCCACG CTTCGTCC	3204
cfn1249269	A	G	CGTTTAAAAGAACACAAATGTGGCCCTAGTGA TCA[A/G]GTACACATATTTGTCACCTCTTTGAA TCTTACTTA	3205
219K1_166464	C	T	CGGGCTGATGAGGCTCTCGACGTGCTGCTTCA CAGGATGCCTGAGCTGGGCTGCAC[C/T]CCCA ACGTGGTGGCATATACCACGGTCATCCACGGC TTCTTTAAGGAAGGC	3206

TABLE 19-continued

Marker sequences and SNP position in sequence. SEQUENCES OF SNP MARKERS				
Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
219K1_158251	G	A	GCGCTATCCGGCGTCGTGTTCCCTCTTGGGGG AATCGTCCTGGAGATGGATCCGGTCA[G/A]AG GGGCCCGTGATTTGTGAGGATGTGTGTTGT TTCCCGAAAGGCG	3207
219K1_111446	A	C	CTTTGACCTTAAATTCTTGTACTAATTTAGCAG AATCGTTCTTCGAGAAGCACTC[A/C]AAAAAT GGTTTGTCTTGGGTCTGTATCATATTTTCTCTG AACAAACAGGCGTGA	3208
219K1_110042	T	C	GACTTAGCCTCACACGGAATCGAGTCAACCAA TTCC[T/C]GTCGGTTTTGAGTGGCTCCCTTGAA GATGCAATCGTTTTCAGCATGGTCAGATTAAT CAGCGAGCGTGC	3209
219K1_110005	C	T	CATGTAGTGGCTGGCGTCTAAGCGCCTTTTCTT CTTCCAGCATCTA[C/T]GACTTAGCCTCACACG GAATCGAGTCAACCAATTCCTGTCGGTTTTGA GTGGCTCCCTTGAAGATG	3210
219K1_107461	A	C	GTCGTATATATTGTTTGTATTAAAAAGTTGTGT GTTTTG[A/C]GTCATAATTTTTAAAATATTATT ATGTCATTTTCAAATTCGCATCAAC	3211
219K1_99688	T	C	AATCTTCTTGACTTCATCCATCCGCCTTGTTGC CCTGCGCAAAATCAAACT[T/C]CCCCGTCCTTA TCATCAAGTCAGGTCCCGCCCTGGGCAGAGAG AG	3212
219K1_37	C	T	CGGCAGATATCACAAAGGGCTATCCTGGTGAA CAA[C/T]AAGATGGGTCAGAATTTGATAATGA AGCCTCAAGCCC	3213
cfn1270524	A	T	AATAGATGCACGCATCGGCGACCATTTTTTAG TACTTTTTGCCTTTTTTGAAAATTTTGTCATTA AAAGACAAATGCCTAGTCTATACCTGATAAAC TAA[A/T]ATCATACATAGAGAAAATGGTCATTT GGTTGAGTTTCGGTACATGCTGAGATGGTTGC ACTTCGGTGCATCTGCTTTGCTTCCATCACATC ATAATGTCT	3214
136H5_3M5_7601	T	C	GCTGCTTGTAGCGTCCCCCATGGCACCTG[T/C] GAAGAGGTTTTCGGCCACAGAGAAGGGGAAG GCTC	3215
cfn1288811	T	G	AAAATTACTTTTCACGCGCTTCGTTGGTCTGAC AGTGCGAGCATAATTTTACTTTTTCTCAGTTTT ACTTAATTTGGTTAACCAAATCCTTTTTGATTT T[T/G]AACTAGAAAACCGAATGTCAAACATTG TGCAAATTTGGAAACTGAAACTGAAACCAAA AACCTAAAAAAATGATTAGTTTGTTTTTTTGTT CTTGTTTCG	3216
136H5_3M5_89176	A	G	gtatttCTTAGGATTTTCTCACCGGCATCTCC[A/G] TTTTTTGAGCAAGAGTATTTAAGGATGGTAGG C	3217
136H5_3M5_89263	C	T	AACAAAGATGCTAGTAAGAACATGAACCTAG TTGCTCATTTTTAACAACAATTGCCCACCAACC TGACATGCTCTTCCCATGTTCTTTTTTTGCTCA AAA[C/T]AGAGATGCTAGTCCAAATATTTTTCT AGTTGCTTACATTTTAAACAACAATTGCCTAC CATCCTTAAATACTCTTGCTCAAAAAACGGAG ATGCCGGTGA	3218
136H5_3M5_138211	T	A	AATACAGACTGGGTGCAAAGCCAAGATGAT[T/A] GTAAAATTGATTGATGGCCGTTGGGAGGT	3219
cfn0556874	C	T	TGTAAAGAAGCTTAACCAGGAAAGCTATCAG GGCCATAGGGAATGGCTGGTTAGTGACAATTT TGCCTGCTGGAAATGGGATTTCTTGTTTATTTC AGTT[C/T]TGCATTGTGTCTGACATGCTCTTTCT TTTGGGCGCAGGCTGAAGTGAATTACCTTGGA CAACTATCGCACCCGAATCTTGTAAAGCTCGT TGGGTACTGT	3220

TABLE 19-continued

Marker sequences and SNP position in sequence. SEQUENCES OF SNP MARKERS				
Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
136H5_3M5_64154	T	C	TGGCGGAGCTGGGGCTGTTCCTCCTACGCAGG CGAAACTTCGCCGCGATAAA[T/C]GGAACTAT CATCAGGTTCCCCGATGATCCATACG	3221
136H5_3M5_68807	A	G	ACAAGCAACCGAGACAAGTTGCTCTTAATTAT CTGTGCGT[A/G]CACCTCTAAGTCTTAACCTGA CGTAACCAACCAACCGTGT	3222
136H5_3M5_77916	A	G	GATGGTTACAAGGCATGCATAGCAAGTAGAGT TAACTTATCAAGTTATT[A/G]GTATTTTTCTTCT GTGGTACTTAGAGTCTAMAGCTTGAGC	3223
cfn1246088	A	C	AATGGAAGCTGATGTGCGTTAGCGATAAAGCA ACAGCGATAACGACGCATGGATCACCATGCTA CTTGGGGAAGCAGGGACATCTGATGAGCCAG CATAC[A/C]CCCAGATATGTGTCTTTCCAAATT CCACGTCCCAACAGATGAGCTATAAATTAATG CCACCTTCCTCCTACAGCTAAATACTCCATCCG TTTCATAATGT	3224
cfn1287194	G	A	CAGAGGCATTCGTGAATTGGGCGAAATCAGA AGCAAGGAGCAGCGATGTTCAGCGCAGAAGG CACTGGGAGGGGATTCCAGGGAGGCTGCCCA CCAGCCC[G/A]CCATCAGATACGGAGGAGGTG GATCCATGGCCCTACCTGTGTCCTGCGCCGAA TCTGGACTGTGGTAACTACAGCGCCTGAATCT AGAGGTTCAGCCTGG	3225
cfn1258380	A	C	GATCCATCTCCCTTAATAATTTTGCTATTGGTA TTGGGTATGGACATCTGAAGTGAAGGTTACGG CCGATTTATAGGAGTGATAGCACCACACAATT CAT[A/C]AGAGCATCTGCAATAGATGAGTAGA TGTAAAACTACTTAACTTTTACATCTCCGGGCC TAAAAACGCATCTGTAATAAGATAATGTAGAT GTAAAGAAAA	3226
IWB72107	A	G	CGACGACGACGAGGATGCCGAGTTTGATGAC ATGGAGGATTATATCGACG[A/G]cgcggactgggacg ccgacatgtatgatgatgtgttcgatgtctgaagga	3227
BS00090770	C	T	TAGCCGTAGGTCGTAGCACATAGCCGTTTA[C/T] GTAATGCATAGTTGTCCGAAGGAATGTTTC	3228
cfn1239345	A	G	TAACCTGGGGCTTCTTTTTCTCCCTATAATAT GG[A/G]CTGCCCTTTTAAGAAGGAACTGCAGC GAGGGTGCA	3229
cfn0917304	G	T	GACTACGCGTTCCTCCCGGTGGTGGCGCTCTA CCC[G/T]TTGTGTTGCCTTTCTCCAAGCAGTTGT GCCCTTCG	3230
cfn0919993	G	T	ATATCTTTACAAGTCATCGACTTACATGCTTCT TT[G/T]TATTATATGCACCTATGCAGTACTTGTT AATGGGT	3231
cfn0920459	C	G	GATGATATAACCGTAGCCAAGGAAGCCCAGA TTTT[C/G]TTCTGTGTATCTATAGGAGCTTAATT AGGAGGAGG	3232
cfn0393953	C	T	AGTATATAAAAAAACAAGTTGTCACCCAGATG AAT[C/T]CGAAACTATGTCAATGTCGACGGTGA GTGTGGACC	3233
cfn0856945	G	T	GGACATCGGCACATGCTTTATTACTGATCTGA TTT[G/T]TTGACTGTTTATTTTAGGTTTGCCTAC ACCACTGA	3234
cfn1291249	T	G	ATGGTTGAATATGTGACTGCATTTGGACTCAC TCCTTGTTTCTGCATTTCATTGAAGATAAGCAT GGCCTTATCAAGCTTCCCAGATTTAGCATGTG CAT[T/G]AATCAGTATGTTGAAGATACGACAGT CAGGTAGAATGCAGTATCTTTCCATTGAATTG AAGAGATTAATCATATCAACTAAGCATCCTTC GGTGGCATAC	3462

TABLE 19-continued

Marker sequences and SNP position in sequence.
SEQUENCES OF SNP MARKERS

Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
cfn0231871	G	T	GGAGGCGCCTACTGTATTAAATTAGCTAGTGTGGC[G/T]TGTTTGAGGATAATGGCACATATACCTTGGCGGTG	3463
cfn0867742	G	T	TCCAGGCAGGGGGCATCCTCGACAATGATTTCATC[G/T]AAGCTGCATCCCCATTCCATCACGACGCGGAGGCT	3464
cfn0523990	G	T	TGTAAGCTAACTATACATGAAGAGTGCAGGCACAC[G/T]AAAACGTTCATCCTGAAGTACAAGAGTTATTTTGG	3465
cfn3126082	A	C	GGAGAAAGGCGAGATTTTAGCACCTAACGCCGCAA[A/C]CCAGATCAAATCGCTGTCCCTTT	3466
W90K_RAC875_c33564_120	A	G	AACCTTGGAAGCTATCATTGCGCACTTGAAGAGCA[A/G]TAGTGTGGACATTCCTGTTTATGCTTGGAGCTTAG	3467
cfn3407185	A	G	CGCAGGCAGCGGGCATGTATCCTCGTCTGACGGAT[A/G]CCCAGATTATTAAACTGTCACCCTGCACGCCTGCA	3468
S100069923	G	A	GAGGCTAATCCCGACGTGCCACATTGAGCACGTGTGTTCTTGCTGTGGCCTGGTCGAAAGACATGACGCATGCACGTGCCCCACGCCTCACACGGCTTGGGTTCTCGCCTGTCCGGTGCTCGACGGACCAGTACATATACGCGAGCGCTCCT[G/A]GCCACCTCAGTTCATCACACTTCACTGCAGTACAAGGCCTCGGCTCTCGGCAGACTCCTCATTGCTGCTTCTGCTAGTGAAAAGAGAGATTCTTCAGCGCTGCTCCTGAAAGAGATAAGAAATACGATGGCAACAATGGTCAGAG	3469
S3045171	G	A	ACTTACTGCGCGCAGACGTTGCAGCTCTTCCTGCAGAAGCCAGGAGCTTCCTTGGTGCCCACCATATAGTTGGGGTTCTTGGCACACTCCCCA[G/A]CGGCAGCCCACTGCGAGCAGAGGACATTCTCGTCCTCGCAGCCGTCACCGGAGCCT	3470
S3045222	T	C	AAGCAAGCTACGCGTTGCTCAAAAAAAAAAAAAGCAAGCAAGCTACGCTGATCAAAGGCTGAATAGTCCAGAGTTACAGGACATGGCTACTCTGCAGC[A/G]CCCAGCAAGCTTACTACTTAGGTTGGTGGAGAAGCAGCACCCACTCGAGACTCGACAAGCAACCTTGGACGTTCTACTCGCCAGTGCATTGCTGCTTTACC	3471
cfn1087371	A	G	ACCAGAGAAAGAGAGGGGAACTTTGGGTATACACC[A/G]CATTACCCTAGTGAAAGAAGAAGGGGGTATTATGT	3472
cfn0436720	A	G	GAAGGGTCCACTGAGAATTAAGGATGCATTCTTTC[A/G]ATTTGGTATATTTGTTGTAAGGAATGAAGAATCGG	3473
S100067637	G	C	TTTCGTGGCGGGGGATCTCGTGCCGGTCGAGGAGGTCCACCTCCAGCGAATTCTGCAGCAACCAACACAAACAGGCCCAAATGTCCGAATTCAGAGCA[C/G]AGCCCGACCGACCGACCGCGAAATCGCGCGGCATGGCGTTGGCGTTTGGCGTTGGCGAGAAGGAAAAAGGCACTCTATGCAGACCTTAGCTTGGTTATGGC	3474
cfn0554333	C	G	TTGCCAAATTCACACCATCATTGATCTGGGGTATC[C/G]TATGCCTATGTGATGCCTCTCACCTCTTTCTTCCC	3475
cfn0238384	C	G	CCGTGAAACCTGTAAAAAGATGTCTGTGTGTCTAG[C/G]AAAGCCCTAATTTTAATCACCCCGTACGCCCCCCT	3476

TABLE 19-continued

Marker sequences and SNP position in sequence. SEQUENCES OF SNP MARKERS				
Marker ID	AlleleX	AlleleY	Sequence	SEQ ID NO:
cfn0530841	C	T	GCAGCTTCTTGTTTATATATTCTCTTATCAGAAGT[C/T]GGGTAGAACAGCTAACGATGTGCTGCTCATTTCCT	3477
cfn1082074	C	T	TGATCTTACTGATAAAATCCGGTTCAAATATATAA[C/T]GGTGAGAAAAAATTAACCAGAGCGAGGCGAGACAT	3478
cfn0560679	C	T	GTTCATGTACAATGATGTTTAACATTGGAACGGTC[C/T]GGGATCTGTTTGATCTATGCCCCCTTCAACGTCTT	3479
cfn0915987	G	T	AAGTCTGCCATCCAGATCATTACCCAACGGCCAAT[G/T]GAGCCATGAGGTTTGCCTCGTTGCACGTTTTGGCT	3445
cfn0920253	A	C	GCAACAAAGCTGGTCATCCAAACATTTACATCGTT[A/C]GGCAGGCTTTCCGCCCAAACCATGCGGCCGACCTG	3446
cfn0448874	C	T	TATGTAAAACCTCTTTGTTTCTAAATAGCTGCGGC[C/T]CGCTACCTAAATTTATGTTGAACCTAGAGGCACCC	3447
cfn0923814	A	C	GTTCGGCAGAATCCAAGTCGCAAATGTAAGGTCAG[A/C]AAATGAATGATGATCATGATAATGAAAATCATAAG	3448
cfn0924180	A	G	ACGTATGGAGCTTCCTCTTTTCATCATGCACCATT[A/G]TGATCTCCCTCTTATTTTGTCTGAAGCCATTCATG	3449
cfn0919484	A	G	AGGTCATGAAAATGCAAGTGGCGAATCTTATCTCT[A/G]TTATACCATTTGGCAAAACAAAGGCGAGAGTTCTG	3450

Example 19: Test of the Cumulative Effect of Rf Genes

To test the hypothesis of the cumulative effect of the restorer alleles and to identify the combination(s) of Rf genes that can produce a full fertility in the CMS hybrid, the SNPs linked to the 4 major loci mapped are used to create restorer lines combining the different restoring alleles. The 2 Rf alleles, 3 Rf alleles and 4 Rf alleles combinations (Rf1+Rf3, Rf1+Rf4, Rf1+Rf7, Rf3+Rf4, Rf3+Rf7, Rf4+Rf7, Rf1+Rf3+Rf4, Rf1+Rf3+Rf7, Rf3+Rf4+Rf7 and Rf1+Rf3+Rf4+Rf7) can be created employing different breeding techniques such as pedigree breeding, backcross, single-seed descent or double haploid.

In the example exposed 2 double-haploid lines from the cross TJB155/R204 were created: LGWR16-0016 and LGWR16-0026. Those two lines carry the restorer alleles Rf1, Rf7 (donor R204) and Rf3 (donor TJB155) and are alloplasmic for the cytoplasm from *T. timopheevii* (donor TJB155).

LGWR16-0016 and LGWR16-0026 have winter growth habits and show a normal fertility (LGWR16-0016:2.54 seeds/spikelet, average over 29 individual spikes. LGWR16-0026:2.33 seeds/spikelet, average over 40 individual spikes). LGWR16-0016 and LGWR16-0026 were used as pollinators in a series of crosses onto a series of A-line with elite background (14 A-lines for LGWR16-0016 and 16 A-lines for LGWR16-0026). The F1 plants originating from those crosses were assessed indoor for their fertility.

Fertility Assessment with the Main Tillers

The spikes from the hybrids produced with the restorer lines LGWR16-0016 yielded on average 45.4 grains and did show an average fertility of 2.44 grains/spikelet. The spikes from the hybrids produced with the restorer lines LGWR16-0026 yielded on average 44.7 grains and did show an average fertility of 2.37 grains/spikelet. Neither of these two groups differ significantly in their fertility distribution from the group formed by the elite lines (Table 20. T-test, P-values<0.05). The distribution of the fertility scores are, as a consequence, relatively similar between groups (see Table 20)

TABLE 20

fertility (expressed as number of seeds/spikelet) of a series of spikes from hybrids produced with the restorers lines LGWR16-0016 and LGWR16-0026 and from a panel of elite lines. The average number of seeds per spike are indicated in the column "SEEDS". The standard deviation "STD. DEV." are given for each of the three groups. The T-test p-value "P-Value" is given for the two groups of hybrids produced with LGWR16-0016 and LGWR16-0026: it indicates the statistical significance of the difference for the value "Fertility" between the group of hybrids ad the group of elite lines The spikes originate from the tallest tiller(s).

	SPIKES	SEEDS	FERTILITY	STD. DEV.	P-VALUE
LGWR16-0016	164	45.4	2.44	0.43	0.090
LGWR16-0026	206	44.7	2.37	0.45	0.692
ELITES inbred	637	44.1	2.36	0.59	

Fertility Assessment with all Tillers from F1 Plants

To be considered as complete, the restoration of fertility has to be observed in all the spikes formed by the hybrid plant. For that purpose, the integrality of the spikes of 56 F1 plants produced with LRWG16-0016, of 61 F1 plants produced with LGWR16-0026 and of 52 elite lines were assessed for their fertility and represent, per group, 294, 262 and respectively 288 individual spikes (Table 21). The 56 F1 plants produced with LGWR16-0016 produced on average 41.4 grains/spikes and did show an average fertility of 2.21 grains per spikelet (standard deviation 0.32). The 61 F1 plants produced with LGWR16-0026 produced on average 42.8 grains/spikes and did show an average fertility of 2.27 grains per spikelet (Table 5. Standard deviation 0.32). The fertility distribution of these two groups differ significantly from the fertility distribution of the groups formed by the elite lines (average number of seeds per spike: 38.4. Average fertility: 2.01. Standard deviation 0.44. see Table 21).

TABLE 21

average spike fertility (expressed as number of seeds/spikelet) of individual plants from hybrids produced with the restorers lines LGWR16-0016 and LGWR16-0026 and from a panel of elite lines. The average number of seeds per spikes are indicated in the column "SEEDS". The standard deviation "STD. DEV." are given for each of the three groups. The T-test p-value "P-Value" is given for the two groups of hybrids produced with LGWR16-0016 and LGWR16-0026: it indicates the significance of the statistical difference for the value "Fertility" between the group of hybrids ad the group of elite lines. On average the F1 plants produced with the restorer lines LGWR16-0016 and LGWR16-0026 formed 5.25 and respectively 4.3 spikes and the plants from the elite lines group formed on average 4.35 spikes.

	PLANTS	SEED	FERTILITY	STD. DEV.	P-VALUE
LGWR16-0016	56	41.4	2.21	0.32	0.003
LGWR16-0026	61	42.8	2.27	0.32	0.000
ELITES inbred	52	38.4	2.01	0.44	

Example 20: Development of Restorer Lines from Different Sources

With the help of the markers developed in the previous examples, new restorer lines comprising Rf1, Rf3 and Rf7 loci were obtained using different sources of restoration locus. Table 22 shows the list of these restorer lines and their characteristics.

LGWR17-0015 is a winter wheat double haploid line produced from the F1 plants from a complex cross: R0934F was first pollinated by Altigo, the created F1 R0934F/Altigo was then crossed with the F1 R197/Apache taken as male and the resulting four ways F1 "R0934F/ALTIGO//R197/APACHE" was then pollinated with the line Altamira. LGWR17-0015 inherited the T-CMS cytoplasm from the restorer line R0934F. LGWR17-0015 has been selected under local environment and is a fully fertile wheat line.

LGWR17-0022 and LGWR17-0157 are winter wheat double haploid lines arising from a complex cross: the F1 formed from the cross between the restorer lines R204 and R213 was pollinated with Aristote, the resulting 3-ways cross "R204/R213//Aristote" was then pollinated with the line NIC07-5520. LGWR17-0022 and LGWR17-0157 inherited the T-CMS cytoplasm from the restorer line R204. They have both been selected under local environment and are fully fertile wheat lines.

LGWR17-0096 and LGWR17-0154 are winter wheat lines developed by conventional pedigree breeding technique from the 3-ways cross R204/R213//ARISTOTE. LGWR17-0096 and LGWR17-0154 inherited the T-CMS cytoplasm from the restorer line R204. They have both been selected under local environment and are fully fertile wheat lines.

All lines have been genotyped with the markers as described above to check for the presence of Rf1, Rf3 and Rf7 haplotypes.

Representative seed samples for each line have been deposited before NCIMB Collection on 25 Sep. 2017. The NCIMB is located at Ferguson Building, Craibstone Estate, Bucksburn, Aberdeen, Scotland, UK, AB21 9YA.

TABLE 22

Full restorer lines data regarding pedigree, the process of selection, Rf haplotype identified by markers and NCIMB deposit number"HD": homozygous plant obtained after a Double Haploid process, "F6": homozygous plant obtained through six generations of self-crosses.

CODE	PEDIGREE	SELECTION	HAPLOTYPE Rf	NCIMB deposit number
LGWR16-0016	TJB155/R204	HD	Rf1 + Rf3 + Rf7	NCIMB 42811
LGWR16-0026	TJB155/R204	HD	Rf1 + Rf3 + Rf7	NCIMB 42812
LGWR17-0015	R0934F/ALTIGO//R197/APACHE///ALTAMIRA	HD	Rf1 + Rf3 + Rf7	NCIMB 42813
LGWR17-0022	R204/R213//ARISTOTE///NIC07-5520	HD	Rf1 + Rf3 + Rf7	NCIMB 42814
LGWR17-0096	R204/R213//ARISTOTE	F6	Rf1 + Rf3 + Rf7	NCIMB 42815
LGWR17-0154	R204/R213//ARISTOTE	F6	Rf1 + Rf3 + Rf7	NCIMB 42816
LGWR17-0157	R204/R213//ARISTOTE///NIC07-5520	HD	Rf1 + Rf3 + Rf7	NCIMB 42817

Example 21: Test of the Cumulative Effect of Rf1, Rf3 and Rf7 Genes Outdoor

Fertility from 4 hybrids produced with the restorer lines LGWR17-0022, LGWR17-0153, LGWR17-0154 and LGWR17-0157 was assessed in field as described in example 19 for the fertility assessment of the main tillers. The restorer lines have the pedigree and haplotypes as described in table 23.

The assays are made with agronomically adapted hybrids in three different countries France, Germany and United Kingdom. In total 28 hybrids are tested and compared to 26 control Elite inbreds. The result show that, in the field, the hybrids comprising the combination of the three alleles Rf1, Rf3 and Rf7 restorers performed as well as the elite inbreds (table 24). It is also worth to be noted that, in this combination, the Rf3 weak or Rf3 do not have any impact on the fertility level.

TABLE 23

pedigree and haplotype of the restorer lines

CODE	PEDIGREE	HAPLOTYPE Rf	NCIMB deposit
LGWR17-0022	R204/R213//ARISTOTE///NIC07-5520	Rf1 + Rf3 + Rf7	yes
LGWR17-0153	R204/R213//ARISTOTE	Rf1 + Rf3weak + Rf7	no
LGWR17-0154	R204/R213//ARISTOTE	Rf1 + Rf3weak + Rf7	yes
LGWR17-0157	R204/R213//ARISTOTE///NIC07-5520	Rf1 + Rf3 + Rf7	yes

TABLE 24

fertility (expressed as number of seeds/spikelet) of a series of spikes from hybrids produced with the restorer lines LGWR17-0022, LGWR17-0153, LGWR17-0154 and LGWR17-0157 and from a panel of elite lines. Each spike originates from the tallest tiller. The standard deviation "STD. DEV." are given for each of the three groups.

			Fertility	
		SPIKES	average	STD DEV.
Hybrids	Total	169	2.71	0.47
	Rf1 + Rf3weak + Rf7	43	2.72	0.38
	Rf1 + Rf3 + Rf7	126	2.71	0.49
Elite inbreds		226	2.73	0.43

Example 22: Modification of the Endogenous RFL29c Gene Sequence by CRISPR Technologies to Revert a Rf3 Allele to a Rf3 Allele

As shown in example 11, FIG. **5**A and FIG. **12**, RFL29c nucleotide sequence is characterized by a deletion of 2 nucleotides compared to RFL29a nucleotide sequence creating a frameshift and resulting in an inactive truncated protein. The sequence alignment in FIG. **12** shows that one "T" nucleotide in the RFL29c gene sequence could be removed or 2 nucleotides added in order to reframe the RFL29c gene sequence into a complete and functional RFL29 protein.

Such modification could be achieved in Fielder, which comprises a RFL29c as depicted in SEQ ID NO 3457, by designing, as described in example 15, a suitable guide sequence targeting the frameshift and used it in combination with a base-editing technology such as described in WO2015089406. FIG. **12** shows suitable PAM motif and target sequence for CRISPR cas9 edition.

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SEQUENCE LISTING

The patent contains a lengthy sequence listing. A copy of the sequence listing is available in electronic form from the USPTO web site (https://seqdata.uspto.gov/?pageRequest=docDetail&DocID=US12398183B2). An electronic copy of the sequence listing will also be available from the USPTO upon request and payment of the fee set forth in 37 CFR 1.19(b)(3).

The invention claimed is:

1. A wheat plant restorer of fertility of *T. timopheevii* CMS cytoplasm comprising:

a Rf1 restorer allele comprising a nucleic acid encoding a Rf1 protein restorer of fertility of *T. timopheevii* CMS cytoplasm, wherein the corresponding amino acid sequence has at least 95% identity to SEQ ID NO:361,

a Rf3 restorer allele within the Rf3 locus, wherein the Rf3 locus comprises one or more of the following SNP allele(s):

SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP17	cfn1252000	3203	A
SNP18	IWB14060*	3204	G
SNP19	cfn1249269	3205	G
SNP20	219K1 166464	3206	T
SNP21	219K1 158251	3207	G
SNP22	219K1 111446	3208	A
SNP23	219K1 110042	3209	T
SNP24	219K1 110005	3210	C
SNP25	219K1 107461	3211	A
SNP26	219K1 99688	3212	T
SNP27	219K1 37	3213	C
SNP28	cfn1270524	3214	T
SNP29	136H5 3M5 7601	3215	T
SNP30	cfn1288811	3216	G
SNP31	136H5 3M5 89176	3217	A
SNP32	136H5 3M5 89263	3218	T
SNP33	136H5 3M5 138211	3219	T
SNP34	cfn0556874	3220	C
SNP35	136H5 3M5 64154	3221	C
SNP36	136H5 3M5 68807	3222	G
SNP37	136H5 3M5 77916	3223	A
SNP38	cfn1246088	3224	A
SNP39	cfn1287194	3225	G
SNP40	cfn1258380	3226	A
SNP41	IWB72107*	3227	A
SNP42	BS00090770	3228	T
SNP43	cfn1239345	3229	A

and

a Rf7 restorer allele within the Rf7 locus, wherein the Rf7 locus comprises one or more of the following SNP allele(s):

SNP#	Marker Name	Marker SEQ ID	Restorer Allele
SNP44	cfn0917304	3230	T
SNP45	cfn0919993	3231	G
SNP46	cfn0920459	3232	C
SNP49	cfn0915987	3445	G
SNP50	cfn0920253	3446	A
SNP51	cfn0448874	3447	T
SNP52	cfn0923814	3448	C
SNP53	cfn0924180	3449	G
SNP54	cfn0919484	3450	G.

2. The wheat plant according to claim **1**, wherein said Rf3 restorer allele is located within the chromosomal fragment between SNP markers cfn1249269 and BS00090770.

3. The wheat plant according to claim **2**, wherein the corresponding amino acid sequence of the Rf3 restorer allele has at least 95% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO: 684.

4. The wheat plant according to claim **1**, wherein said Rf3 locus comprises SEQ ID NO:1712, SEQ ID NO:3147, SEQ ID NO:2230, SEQ ID NO:3148 or SEQ ID NO:2238.

5. A method for producing the wheat plant according to claim **1**, said method comprising:

a. providing a first wheat plant comprising one or two restorer allele selected among Rf1, Rf3 and Rf7 restorer alleles,

b. crossing said first wheat plant with a second wheat plant comprising one or two restorer alleles selected among Rf1, Rf3 and Rf7 restorer alleles, wherein Rf1, Rf3 and Rf7 restorer alleles are represented at least once in the panel of restorer alleles provided by the first plant and the second plant,

c. collecting the F1 hybrid seed, and

d. obtaining homozygous plants from the F1 plants.

6. A method for producing a wheat hybrid plant, said comprising:

a. crossing a sterile female comprising the *T. timopheevii* cytoplasm with a fertile male wheat plant according to claim **1**; and

b. collecting the hybrid seed.

7. The method according to claim **6**, further comprising: detecting the presence of at least three of Rf locus chosen amongst Rf1, Rf3, and Rf7 in the hybrid seeds.

8. A wheat hybrid plant as obtained by the method according to claim **7**.

9. A method of identifying a wheat plant according to claim **1**, said method comprising:

identifying said wheat plant by detecting the presence of the restorer alleles Rf1, Rf3 and Rf7.

10. At least three nucleic acid probes or primers for the specific detection of the restorer alleles Rf1, Rf3 or Rf7 in a wheat plant, wherein the at least three nucleic acid probes or primers are chosen among:

SEQ ID NOs: 3254, 3259, 3261-3265, 3269-3275, 3279, 3280, 3282, 3293, 3294, 3309, 3318, 3323, 3325-3329, 3333-3339, 3343, 3344, 3346, 3357, 3358, 3373, 3382, 3387, 3389-3393, 3397-3403, 3407, 3408, 3410, 3421, 3422, and 3437 for the specific detection of restorer SNP marker(s) in locus Rf1,

SEQ ID NOs: 3253, 3255, 3260, 3266-3268, 3276-3278, 3281, 3283-3288, 3295-3308, 3310, 3311, 3313-3317, 3319, 3324, 3330-3332, 3340-3342, 3345, 3347-3352, 3359-3372, 3374, 3375, 3377-3381, 3383, 3388, 3394-3396, 3404-3406, 3409, 3411-3416, 3423-3436, 3438, 3439, and 3441-3444 for the specific detection of restorer SNP marker(s) in locus Rf3, and/or

SEQ ID NOs: 3256-3258, 3320-3322, and 3384-3386 for the specific detection of restorer SNP marker(s) in locus Rf7.

11. The wheat plant according to claim **3**, wherein the corresponding amino acid sequence of the Rf3 restorer allele has at least 98% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO: 684.

12. The wheat plant according to claim **3**, wherein the corresponding amino acid sequence of the Rf3 restorer allele has 100% identity to an amino acid selected from the group consisting of SEQ ID NO: 158, SEQ ID NO: 676 and SEQ ID NO:684.

13. The method according to claim **5**, further comprising: detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles in the hybrid seed and/or detecting the presence of the Rf1, Rf3 and Rf7 restorer alleles at each generation.

14. The method according to claim **6**, further comprising: detecting hybridity level of the hybrid seeds.

* * * * *